



Designing of Corrosion Resistant Power Transmission Marine Propulsion Shaft

Comprehensive Design Project

MT – 3201

Group 01

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01. Executive Summary

This project presents the complete design, analysis, and protection strategy of a **corrosion-resistant marine propulsion shaft** intended for long-term operation in a high-salinity seawater environment. The shaft operates within the immersion and splash zones, where chloride exposure, cyclic torsional loading, oxygen differentials, and biofouling significantly accelerate localized corrosion and fatigue deterioration.

A systematic design methodology was followed, beginning with a detailed assessment of the marine environment, the identification of governing corrosion mechanisms, and quantification of mechanical design constraints. Material selection was carried out using a structured *Translate–Screen–Rank–Document* approach within **ANSYS Granta EduPack**, filtering over 1900 candidate materials and ultimately selecting **Duplex Stainless Steel 2205** due to its high yield strength (~460–510 MPa), excellent pitting resistance (PREN > 30), favourable fatigue performance, and superior reliability under chloride exposure.

Mechanical design involved calculating the required outer and inner shaft diameters based on combined torsional, bending, and axial loads, supported by class-rule checks using **IACS UR M68** and **DNV-CG-0038**. Fatigue and vibration analyses confirmed that the selected hollow shaft geometry (OD = 125 mm, ID = 60 mm) provides adequate safety margins against torsional fatigue, lateral whirling, and dynamic misalignment.

To safeguard performance in the aggressive marine environment, a hybrid corrosion-protection strategy was developed. This includes duplex stainless steel as the base material, a multilayer coating system (epoxy + polyurethane + antifouling), and supplementary cathodic protection aligned with DNV and ABS guidelines. Special attention is given to critical regions such as splines, tapers, and bearing journals, which require electropolishing or dry-film lubrication rather than paint.

Manufacturing and QA plans were prepared to ensure dimensional accuracy, structural integrity, and operational safety. These include CNC machining, controlled heat treatment, surface preparation, and rigorous non-destructive examination (UT, MPI/DPI) followed by runout verification. All processes adhere to international standards (e.g., DNV, ABS, ISO 31000).

Overall, the report integrates engineering design, corrosion science, materials engineering, and marine-class requirements to deliver a propulsion shaft that is structurally sound, corrosion-resistant, manufacturable, and compliant with marine safety regulations.

02. Design Methodology

02.1 Division of Tasks Within the Group

To ensure an efficient workflow and systematic development of the marine propulsion shaft, the overall project was divided into clearly defined tasks. Each task aligned with a major component of the final design: environmental assessment, material selection, mechanical calculations, corrosion protection, CAD modelling, manufacturing planning, FEA simulation, and final quality assurance. The responsibilities were distributed among eight members according to individual strengths, allowing each member to contribute directly to the technical development while maintaining collective review and quality control.

02.2 Member Responsibilities and Outputs

The following table summarizes the assigned tasks and the outputs produced by each member.

Member	Assigned Tasks	Output Produced
D.A.W.I. UDAYAKANTHA – 210660J	• Working as group leader. • Designing shaft geometry and torque transmission calculations. (Task 03) • Conducting background study for the marine shaft and refining the project scope. • Adjusting design parameters for practical applications. • Report compilation and QA/safety plan under task 08.	• Made time plan of the project. • Made five progress reports. • Made final report and final presentation. • Meetings were called. • Checked the reports of tasks 2,4,6 and 7 done by other members. • Completed task 3 and checked those results with standard values. • Clearly defined the project scope and design parameters according to practical scenarios. • Made safety plan for the shaft.
K.L. THEMIYA – 210640A	• Identifying corrosion mechanisms under task 01. • Material selection using Ashby plots and corrosion data. (Task 02) • Corrosion protection strategy evaluation under cathodic protection. (Task 05) • Checking other members' reports. • Conducting background study for the marine shaft.	• Identified corrosion mechanisms of the shaft. • Helped to select suitable material for the shaft. • Evaluate corrosion protection strategies. • Checked the reports of tasks 1,2,4,5 and 6 done by other members. • Checked the meeting minutes reports and all other final reports before submission. • Conducted background study.

H.P.S. PREMAKUMARA – 210494D	• Doing environmental analysis for the shaft under task 01. • Material selection using Ashby plots and corrosion data by Ansys Granta software. (Task 02) • Drawing 3D CAD model of the shaft under task 04. • FEA simulation of loading and corrosion fatigue effects. (Task 06) • Conducting background study for the marine shaft.	• Did environmental analysis part. • Selected suitable material for shaft using software. • Drew the 3D CAD model for the designed shaft. • Did FEA simulation part. • Conducted background study.
H.M.T. DANINDI – 210094C	• Working as secretary of the group. • Conducting background study for the marine shaft. • Corrosion protection strategy evaluation under coating protection. (Task 05) • Designing surface treatment for the final shaft under task 07.	• Made five Meeting minutes reports. • Conducted background study. • Evaluate corrosion protection strategies. • Selected suitable surfaces treatments for the final designed shaft.
B.N. KULATHUNGA – 210308N	• Conducting background study for the marine shaft. • Defining the manufacturing process of the designed shaft and surface heat treatment for the shaft under task 07.	• Conducted background study. • Defined manufacturing process and the sequence of production steps under scientific base. • Defined surface heat treatment.
W.H.M.A.D. DHARMADASA – 210121J	• Conducting background study for the marine shaft. • Drawing 3D CAD model of the shaft under task 04. • Making 2D production drawings for the shaft under task 02.	• Conducted background study. • Helped to draw the 3D CAD model of the shaft. • Made 2D production drawings for the shaft.
A.K.R. DILSHAN – 210127H	• Conducting background study for the marine shaft. • Defining surface heat treatment for the shaft under task 07.	• Conducted background study. • Helped to define suitable heat treatment for the shaft under scientific base.

❖ All members created reports and presentation slides related to the tasks assigned to them, and finally, the final report and final presentation were checked and approved.

❖ Suggested Task Distribution

- Task 01: Environmental analysis and corrosion mechanism identification
- Task 02: Material selection using Ashby plots and corrosion data
- Task 03: Design of shaft geometry and torque transmission calculations
- Task 04: CAD modelling and drawing preparation
- Task 05: Corrosion protection strategy evaluation (cathodic vs. coating)
- Task 06: FEA simulation of loading and corrosion fatigue effects
- Task 07: Manufacturing process and surface treatment design
- Task 08: Report compilation and QA/safety plan

03. Application Description

The marine propulsion shaft operates in one of the most aggressive natural environments encountered in engineering applications. Its performance and long-term durability are directly influenced by the chemical, physical, and biological characteristics of seawater, as well as the electrochemical processes that arise at the metal–seawater interface. Understanding these environmental factors is essential for selecting appropriate materials, defining geometric features, predicting degradation modes, and designing an effective corrosion protection strategy.

This section provides a detailed description of the operating environment of the shaft and the associated corrosion challenges. Section 3.1 presents an environmental analysis of the marine conditions relevant to the propulsion system, including salinity, temperature, dissolved gases, flow characteristics, biofouling activity, and other parameters that influence corrosion behavior. Section 3.2 builds on this foundation by examining the corrosion mechanisms most likely to affect marine propulsion shafts, such as pitting, galvanic corrosion, crevice corrosion, microbiologically influenced corrosion, and corrosion fatigue.

Together, these analyses establish the environmental context required for subsequent sections on material selection, mechanical design, and corrosion protection strategy. They also provide the scientific basis for evaluating the suitability of candidate materials such as duplex stainless steels and for justifying the adoption of coatings, cathodic protection, or hybrid protection systems in the final shaft design.

03. 1 Environmental Analysis of the Marine Operating Environment

Seawater covers approximately 70% of the Earth’s surface and represents one of the most chemically and biologically complex natural environments. It contains nearly every element in the periodic table, along with a vast diversity of living organisms, earning it the description of a “living” medium. Due to its aggressive nature, seawater is widely regarded as one of the most corrosive environments encountered in engineering applications. This chapter explores the critical environmental parameters of seawater that influence material degradation, especially corrosion. These factors are essential for understanding and predicting material performance in marine systems such as ships, offshore platforms, pipelines, and submerged components. The subsections below examine key characteristics including salinity, dissolved gases, pH, biological activity, temperature, depth, flow velocity, and electrical resistivity, all of which collectively define the corrosive nature of seawater.

03.1.1 Seawater Constituents

Seawater is widely recognized as one of the most chemically aggressive environments for metallic materials, primarily due to its complex composition. Although sodium chloride (NaCl) is often cited as the main source of corrosivity, the actual corrosion behavior is influenced more significantly by other constituents particularly dissolved oxygen (DO) and pollutants such as sulfides. These components critically affect the electrochemical conditions at the metal–seawater interface.

The total salt content of seawater is expressed as salinity, which refers to the mass of total dissolved solids (TDS) in one kilogram of seawater, after oxidation of organic matter and replacement of bromine and iodine by chlorine. A related and analytically more accessible parameter is chlorinity, which includes the chloride ion concentration as well as the equivalent amount of bromide and iodide. The relationship between the two is defined by the empirical equation:

$$\text{Salinity} = (1.80655 \times \text{Chlorinity}) + 0.03$$

Both salinity and chlorinity are commonly reported in grams per kilogram (g/kg) or parts per thousand (ppt or ‰). To convert these two parts per million (ppm), the values are multiplied by 1,000. Salinity is also considered equivalent to TDS and may be expressed in mg/L (ppm).

Seawater properties vary significantly depending on geographical location, season, and water depth. Influencing factors include:

- River discharge
- Evaporation and rainfall
- Melting of polar ice
- Wind and wave activity

For instance, littoral (coastal) waters often exhibit lower salinity due to freshwater influx and may also contain high levels of sediment and organic/inorganic pollutants, leading to brackish conditions.

The average salinity of open ocean water is typically 32–36 ppt, and the corresponding chlorinity is approximately 18–20 ppt. By contrast, extreme environments such as the Dead Sea exhibit salinities more than ten times higher, reaching 322,000 ppm, as shown in Table 1. [1]

Table 1 - Approximate TDS (ppm) of Various Seas and Oceans [2]

<i>Ion</i>	<i>Concentration (ppm)</i>
Chloride, Cl^-	19 353
Sodium, Na^+	10 760
Sulfate, SO_4^{2-}	2712
Magnesium, Mg^{2+}	1294
Calcium, Ca^{2+}	413
Potassium, K^+	387
Bicarbonate, HCO_3^-	142
Bromide, Br^-	67
Strontium, Sr^{2+}	8
Borate, $\text{B}_4\text{O}_7^{2-}$	4
Fluoride, F^-	1

Table 2 - Major Ionic Constituents in Seawater (35 ppt Salinity) [2]

<i>Seawater</i>	<i>Approximate TDS (ppm)</i>
Arabian Gulf	47 000
Red Sea	43 000
Mediterranean Sea	41 000
Atlantic Ocean	36 000
Indian Ocean	35 000
Pacific Ocean	34 000
North Sea	33 000
Bering Sea	32 000
Black Sea	22 000
Caspian Sea	13 000
Baltic Sea	8 000
Dead Sea	322 000

The average chemical composition of open-ocean seawater with 35 ppt salinity is detailed in Table 2. From a corrosion standpoint, chloride ions (Cl^-) are especially aggressive, as they interfere with the formation and stability of passive oxide films on metals, particularly stainless steels and aluminum alloys. Sulfate ions (SO_4^{2-}), though less directly corrosive, support the activity of sulfate-reducing bacteria (SRB) under anaerobic conditions, potentially leading to severe microbial corrosion.

In addition to these ions, seawater contains organic matter that supports biological activity. Elements such as carbon, oxygen, hydrogen, nitrogen, and phosphorus are involved in forming polysaccharides,

lipids, and proteins. While carbon, oxygen, and hydrogen are abundant, nitrogen and phosphorus compounds (e.g., nitrates and phosphates) often act as limiting nutrients.

In coastal areas, nutrient loading from river discharge, sewage, and fertilizer runoff can significantly increase the concentrations of these nutrients. This process leads to eutrophication, characterized by excessive plant and algae growth, which depletes dissolved oxygen and may result in large-scale marine life die-off commonly seen in phenomena like red tides.

Additionally, industrial waste and heavy metal discharge can enhance seawater's corrosivity. For example:

- Copper ions may adversely affect aluminum alloys
- Sulfides and ammonia can increase the corrosion rate of copper alloys
- Sulfur compounds are particularly aggressive toward steels

Design Relevance of Seawater Constituents

The chemical composition of seawater, particularly the concentration of aggressive ions plays a central role in shaping the corrosion behavior of materials used in marine propulsion shafts. In our project, the shaft operates under continuous exposure to seawater immersion and splashing at an average temperature of 25 °C, where both galvanic and pitting corrosion are critical threats. Understanding seawater constituents helps us make informed decisions regarding material selection, protection strategies, and service life prediction.

Key Impacts on Shaft Design:

1. High Chloride Content (~19,000 ppm Cl^-):

- Chloride ions are the primary drivers of pitting corrosion, especially in passive alloys like SS316L or Duplex 2205. [1]
- They degrade protective oxide films, making them highly relevant in areas such as the splash zone and shaft surface crevices.
- This supports the selection of duplex or NiCrMo alloys with high PREN (Pitting Resistance Equivalent Number) values for critical locations.

2. Bicarbonate and Calcium Ions (HCO_3^- , Ca^{2+}):

- These ions contribute to calcium carbonate (CaCO_3) scale formation, particularly in heated or moving components. [1]
- While thin scales can reduce general corrosion by limiting DO diffusion, they also create risks for under-deposit corrosion and reduce heat transfer at bearing interfaces.

3. Sulfate Ions and Nutrient Loading (SO_4^{2-} , PO_4^{3-} , NO_3^-):

- Sulfates support the growth of sulfate-reducing bacteria (SRB), a common cause of microbiologically influenced corrosion (MIC) in stagnant or shielded zones.
- In our rotating shaft, regions near keyways, couplings, and seals may experience localized stagnation, increase MIC risk if not properly sealed or coated.

4. Pollutants and Organics (e.g., Sulfides, Ammonia, Heavy Metals):

- In littoral or industrial waters, pollutant-induced corrosion acceleration is a serious concern.
- Our shaft may be deployed in nearshore environments during docking or maintenance, where exposure to these contaminants could compromise protective systems.

5. Justification for Synthetic Seawater (3.5% NaCl):

- In simulations and lab-scale testing, a 3.5% NaCl solution effectively replicates the average ionic strength and aggressiveness of open-ocean seawater. [1]
- This standardization allows us to reliably test the corrosion performance of candidate materials, coatings, or cathodic protection schemes under controlled conditions.

03.1.2 Effect of Dissolved Gases

Dissolved gases, particularly dissolved oxygen (DO), are among the most critical factors influencing corrosion behavior in seawater. In the context of our marine propulsion shaft which operates in aerated seawater at 25 °C under splashing and immersion conditions the role of oxygen is directly linked to corrosion initiation, propagation, and even protection system effectiveness.

Key Impacts on Shaft Design:

1. Dissolved Oxygen Accelerates Corrosion:

- DO acts as the primary cathodic reactant in electrochemical corrosion cells. It facilitates the reduction reaction that balances metal dissolution at the anodic sites.
- This makes the splash zone and areas near the waterline, where oxygen access is highest, particularly susceptible to general corrosion, pitting, and crevice corrosion all critical for shaft surfaces and couplings.

2. DO Concentration Varies with Temperature and Salinity:

- DO levels decrease with increasing temperature and salinity, both relevant in our case (25 °C and ~35 ppt salinity). Despite lower oxygen solubility, turbulent flow around the rotating shaft maintains high DO transport, sustaining corrosion activity.
- Figures from the source show this relationship, reinforcing the need to simulate realistic DO levels in lab testing.

3. Effect of Flow and Splashing (High Velocity):

- The high-speed rotation of the shaft (1200 RPM) creates significant water turbulence, enhancing oxygen diffusion to metal surfaces.
- While this may benefit passivation in duplex or NiCrMo alloys, it also amplifies localized corrosion risk if passive films break down, especially in areas prone to erosion-corrosion or debris entrapment.[2]

4. Biological Oxygen Demand (BOD) and Stagnant Zones:

- In partially stagnant areas like keyways, seal interfaces, or coupling recesses, biological activity can deplete oxygen locally, setting up differential aeration cells.
- The oxygen-rich outer surface becomes cathodic, while the shielded zone becomes anodic, promoting pitting and crevice corrosion especially relevant if stainless steels like SS316L are used.

5. Gas Byproducts of Bioactivity (e.g., H₂S, NH₃):

- Anaerobic zones within or near the shaft can also host sulfate-reducing bacteria (SRB), which generate hydrogen sulfide (H₂S) a highly corrosive gas that destabilizes passive films and can lead to MIC (Microbiologically Influenced Corrosion).

03.1.3 Seawater pH

Natural seawater is slightly alkaline, with a typical pH range of 8.0 to 8.3, depending on temperature, salinity, biological activity, and atmospheric interactions. The carbonate system in seawater involving carbon dioxide (CO₂), carbonic acid (H₂CO₃), bicarbonate (HCO₃⁻), and carbonate (CO₃²⁻) is the principal buffer that stabilizes the pH of seawater.

This system is governed by the following reactions:

- CO₂ (g) $\rightleftharpoons$ CO₂ (aq)
- CO₂ (aq) + H₂O $\rightleftharpoons$ H₂CO₃
- H₂CO₃ $\rightleftharpoons$ H⁺ + HCO₃⁻
- HCO₃⁻ $\rightleftharpoons$ H⁺ + CO₃²⁻

The combination of these equilibria provides seawater with its significant buffering capacity. This buffering resists abrupt pH shifts when acids or bases are introduced, thus maintaining relatively stable pH values in oceanic waters. [3]

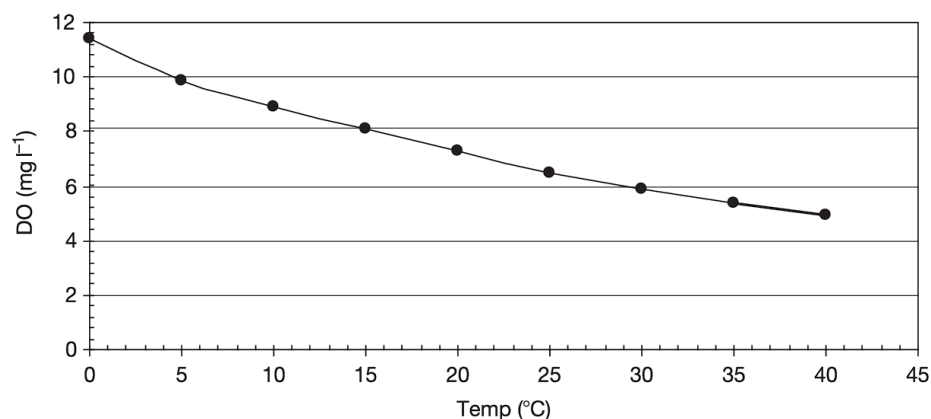
In open-ocean conditions, this system is very effective. However, in coastal regions, especially semi-enclosed basins, the buffering capacity can be overwhelmed by the influx of:

- Organic material
- Agricultural runoff
- Industrial discharge
- Biological productivity (eutrophication)

In such areas, local reductions in pH (acidification) are possible. This has significant consequences for:

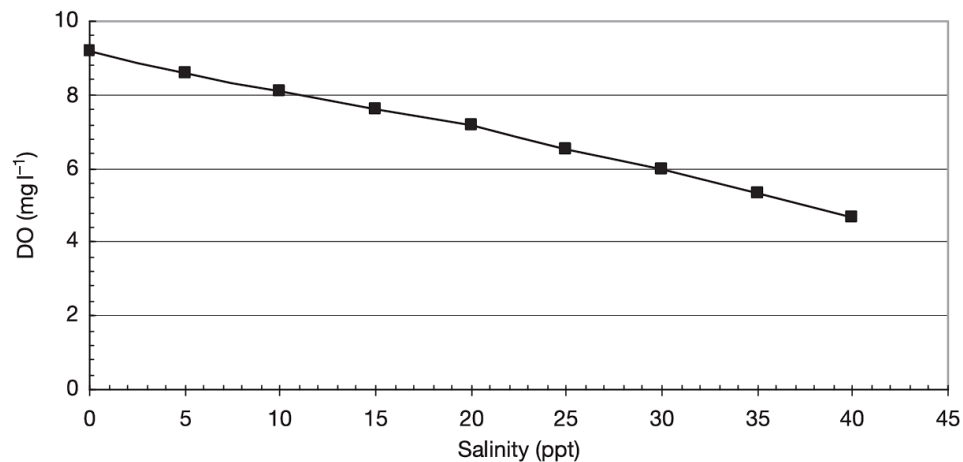
- Marine organisms, especially calcifies
- Electrochemical processes on metal surfaces
- Corrosion protection systems

In addition to pH, dissolved oxygen (DO) plays a key role in the corrosion process. DO levels in seawater vary with temperature and salinity and are closely linked to pH stability and corrosion severity. Two key relationships are illustrated in the following figures:



Graph 1 - Dissolved Oxygen vs. Temperature [2]

This figure shows the inverse relationship between DO concentration and temperature in natural seawater (35‰ salinity at 1 atm). As temperature increases, the capacity of seawater to hold oxygen decreases. This has implications for electrochemical reactions at metal surfaces, particularly in tropical marine environments.



Graph 2 - Dissolved Oxygen vs. Salinity [2]

This figure illustrates how DO decreases as salinity increases in synthetic seawater at 20 °C and 1 atm. Higher salinity reduces oxygen solubility, which can influence both corrosion rates and microbial activity in marine systems.

While the bulk seawater pH tends to remain stable, sharp local reductions in pH can occur directly at the metal–solution interface especially in stagnant or restricted areas such as:

- Crevices
- Under deposits
- Inside microbially active zones

In these micro-environments, the hydrolysis of metal ions during anodic dissolution leads to an accumulation of H⁺ ions, significantly lowering the local pH (sometimes to values below 4). This creates aggressive conditions, especially for passive materials like stainless steel, which rely on a stable, high-pH environment to maintain their protective oxide layers.

Such drops in pH are often the precursor to pitting and crevice corrosion and are particularly concerning in:

- Bolted assemblies
- Seals
- Shaft couplings
- Shaft-hull interfaces

Relevance to Shaft Design

In the context of our marine propulsion shaft, which is exposed to splashing and immersion in seawater at 25 °C, the stable bulk pH of around 8.1 supports the passivity of high-performance alloys like Duplex 2205 and NiCrMo grades.

However, specific shaft features including keyways, flanges, sealing zones, and stagnant fluid traps are prone to localized acidification. This makes them vulnerable to crevice corrosion, especially if materials like SS316L are used without enhanced protection. The issue is further exacerbated by biofouling or microbial activity, both of which can reduce local pH and oxygen levels.[3]

From a corrosion protection perspective:

- Cathodic protection systems increase surface alkalinity, promoting calcareous deposit formation, but require surrounding waters to maintain sufficient buffering.
- In environments where acidification from eutrophication or pollution is possible, coating systems may offer more consistent performance than CP.

Finally, for corrosion simulation and material testing, it is essential to replicate realistic pH conditions, especially for evaluating behavior in tidal and splash zones.

03.1.4 Calcareous Deposits

When metals are subjected to cathodic protection in seawater, either through impressed current systems or sacrificial anodes, calcareous deposits typically form on their surfaces. These deposits consist mainly of:

- Calcium carbonate (CaCO_3)
- Magnesium hydroxide ($\text{Mg}(\text{OH})_2$)

They are produced as a consequence of the electrochemical reactions that occur at the protected (cathodic) metal surface. Specifically, the cathodic reduction of oxygen or water leads to an increase in local hydroxyl ion (OH^-) concentration, raising the pH near the metal interface. This localized alkalinity promotes the precipitation of calcium and magnesium salts, naturally present in seawater.

The sequence of reactions can be outlined as follows:

At the cathodic surface:

- $\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$
or
- $2\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2 + 2\text{OH}^-$ (in less aerated zones)

Precipitation occurs as:

- $\text{Ca}^{2+} + \text{CO}_3^{2-} \rightarrow \text{CaCO}_3 \downarrow$
- $\text{Mg}^{2+} + 2\text{OH}^- \rightarrow \text{Mg}(\text{OH})_2 \downarrow$ [4]

The extent and rate of deposit formation depend on several factors:

- Water composition, especially the concentrations of Ca^{2+} , Mg^{2+} , and CO_3^{2-}
- pH at the surface, which is influenced by current density and flow conditions
- Temperature (solubility decreases with increasing temperature)
- Velocity of the surrounding seawater

In general, calcium carbonate is the more prominent deposit in areas with moderate flow, while magnesium hydroxide tends to dominate in low-flow or stagnant areas due to its higher solubility threshold.

These deposits form a secondary barrier to corrosion by:

- Physically blocking the transport of oxygen and aggressive ions (e.g., Cl^-)
- Further raising the surface pH
- Enhancing the current efficiency of cathodic protection systems

However, the presence of deposits also introduces certain design considerations:

- They can crack or detach in high-velocity zones, exposing bare metal
- They may reduce heat transfer in thermally active components
- In poorly buffered waters, deposition may be incomplete or irregular

Calcareous deposits have been widely studied in connection with offshore structures, pipelines, and ship hulls. They are especially beneficial for passive metals like steel, which otherwise corrode rapidly in the absence of protection. The buildup of such layers can significantly extend the service life of submerged structures and components.

Relevance to Shaft Design

For marine propulsion shaft particularly if cathodic protection (CP) is considered as a corrosion prevention strategy understanding the role of calcareous deposits is essential.

In our design context:

- The shaft operates in 25 °C seawater, a temperature that favors carbonate precipitation
- The seawater contains sufficient calcium and magnesium ions (as shown in Section 2.1), supporting deposit formation
- The rotational speed (225 RPM) creates flow velocities that may limit deposit stability on certain surfaces

Key areas where CP may be applied (e.g., at shaft supports, through-hull penetrations, or stern tube interfaces) must be designed to accommodate deposit formation. Proper selection of current density, anode placement, and surface finishing will influence the formation and adherence of protective layers.

Furthermore, Duplex 2205, if used, may benefit from the dual action of passivation and calcareous deposit formation, further reducing corrosion risk. However, in turbulent zones or splash areas, deposits may be unstable indicating a need to combine CP with coatings or design features that minimize exposure.

In summary, effective use of cathodic protection in our shaft design must account for:

- Deposit formation dynamics
- Water chemistry and flow conditions
- Material compatibility and surface design

These considerations help ensure the long-term durability and reliability of the propulsion system under marine operating conditions.

03.1.5 Effect of Seawater Temperature

Temperature is one of the most influential environmental factors in the corrosion behavior of metals immersed in seawater. It directly affects both the physicochemical properties of seawater and the electrochemical kinetics of corrosion reactions. Generally, an increase in temperature accelerates corrosion rates, particularly for active metals and alloys.

Several key temperature-dependent effects are observed:

1. Increased Corrosion Reaction Rates

Elevated temperatures enhance the rate of electrochemical reactions by increasing ion mobility and decreasing solution viscosity. According to the Arrhenius relationship, a 10 °C rise in temperature can double the corrosion rate of many metals.

2. Reduction in Dissolved Oxygen (DO) Solubility

As temperature rises, the solubility of DO in seawater decreases, potentially limiting the availability of oxygen at the metal surface. However, in natural systems, turbulence and replenishment often compensate for this reduction, especially in high-flow conditions.

3. Acceleration of Localized Corrosion

Localized forms of corrosion such as pitting and crevice corrosion are particularly sensitive to temperature. Many stainless steels exhibit a critical pitting temperature (CPT), above which passive films break down rapidly. Therefore, even modest increases in temperature can shift materials from safe operation to active corrosion zones.

4. Influence on Bioactivity

Higher temperatures promote the growth and metabolic activity of marine organisms, increasing the risk of biofouling and microbiologically influenced corrosion (MIC). Warm, nutrient-rich waters are especially susceptible to biofilm formation.

5. Effect on Deposit Formation and Solubility

The solubility of salts such as calcium carbonate and magnesium hydroxide decreases with increasing temperature, which can enhance calcareous deposit formation on cathodically protected surfaces. However, if flow velocity is high, deposits may be washed away before they adhere.

The seawater temperature varies by location, season, and depth. Surface waters in tropical regions can reach 30 °C or higher, while polar regions remain below 5 °C. Seasonal fluctuations of up to 15 °C are common in temperate zones. Moreover, industrial discharge, power plant outfalls, and shallow enclosed basins can cause localized thermal elevation, creating corrosion “hot spots” for submerged structures. [5]

Relevance to Shaft Design

In our project, the marine propulsion shaft is designed to operate in seawater at an average temperature of 25 °C, placing it within the moderate to high-temperature range for typical marine environments. This has several implications:

- The selected material must withstand enhanced corrosion kinetics. For example, Duplex 2205 offers significantly better critical pitting temperature (CPT) performance than SS316L, making it a more robust choice in warmer seawater.
- Elevated temperature can accelerate localized corrosion in crevices, such as those found around keyways, couplings, and seal interfaces. These areas must be carefully designed and, if needed, protected with coatings or cathodic protection.
- Reduced DO solubility at higher temperatures may shift corrosion control mechanisms. However, the rotational speed (1200 RPM) of the shaft promotes continuous oxygen replenishment through turbulence, maintaining corrosion potential.
- Temperature-driven bioactivity can promote biofouling and MIC. This reinforces the need for anti-fouling strategies, especially near low-flow regions or during long idle periods.
- If cathodic protection is employed, elevated temperature could enhance calcareous deposit formation, aiding in corrosion resistance provided that local flow conditions do not strip away forming layers.

Given these factors, material selection, geometry design, and protection strategies must all be tailored to accommodate the thermal characteristics of the shaft's operating environment.

03.1.6 Effect of Depth

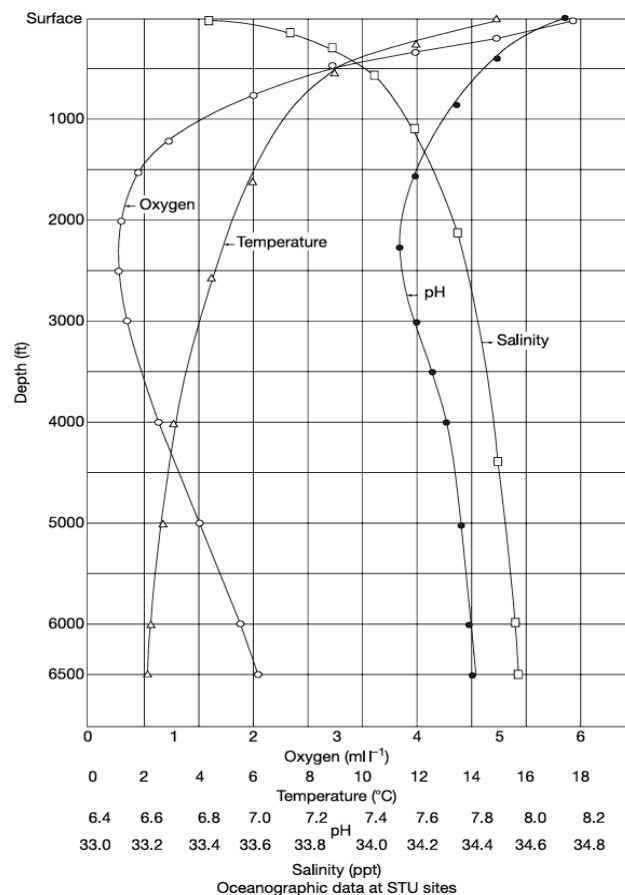
The depth at which a metallic structure is submerged in seawater has a significant influence on its surrounding environment and, consequently, on its corrosion behavior. This is due to depth-related variations in multiple parameters, including:

- Temperature
- Dissolved oxygen (DO)
- Salinity
- pH
- Hydrostatic pressure
- Light penetration
- Biological activity

These changes affect not only the corrosion rate but also the biofouling tendency and performance of protection systems.

Temperature and DO Profiles with Depth

As depth increases, temperature typically decreases, especially beyond the thermocline—a transitional layer in the ocean where temperature drops rapidly. Dissolved oxygen (DO), which is highest at the surface due to atmospheric exchange and photosynthetic activity, generally decreases with depth due to biological consumption and lack of light. However, in deep ocean zones, DO may increase again due to cold, oxygen-rich water circulation from polar regions. [6], [7]

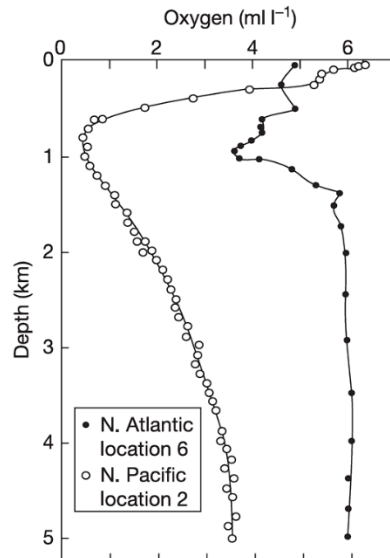


Graph 3 - Variations in dissolved oxygen, temperature, pH and salinity at a corrosion test site in the Pacific Ocean [2]

This figure illustrates how DO, temperature, pH, and salinity vary with depth at a real-world corrosion testing site. It shows:

- A sharp drop in temperature between 50 m and 500 m
- A significant decrease in DO over the same range
- A slight pH drop with depth
- A gradual increase in salinity with increasing depth

These interdependent changes underscore how environmental aggressiveness shifts with depth, influencing material degradation.



Graph 4 - DO-depth profiles at two specific open ocean test sites [2]

Graph 4 compares the dissolved oxygen profiles across depth at two different oceanic locations. It highlights:

- The difference in DO depletion patterns depending on geographic and oceanographic conditions
- What minimum oxygen zones (oxygen minimum layers) occur at intermediate depths (typically 200–1000 m)
- Implications for localized corrosion behavior under oxygen-limited conditions

Pressure and Other Effects

- Hydrostatic pressure increases by ~1 atm every 10 m of depth. While it does not significantly affect corrosion directly, it influences gas solubility, electrochemical potential, and the behavior of coatings and seals.
- Salinity and pH are generally more stable in the open ocean but can vary in coastal or stratified waters.
- Light attenuation limits photosynthesis and fouling activity beyond 200–300 meters, reducing biofouling and microbial corrosion risk at greater depths. [6]

Relevance to Shaft Design

In our marine propulsion shaft application, the shaft is located at or near the sea surface specifically within the immersion and splash zone, typically no deeper than 1.5 – 2.0 meters. Based on the data in Graph 3 and Graph 4, this places the shaft within the zone of:

- Maximum temperature
- Highest DO concentration
- Active biofouling and microbial colonization
- Strongest hydrodynamic effects

These conditions correspond to peak corrosion risk in the marine environment. Unlike submerged pipelines or deep-sea installations, our shaft receives no mitigating benefits from depth. In fact, it experiences the most corrosive and biologically active conditions, making material selection, geometry optimization, and protection strategy design absolutely critical.

The data shown in Graph 3 helps support simulation and FEA boundary condition development by providing realistic environmental gradients. Likewise, Graph 4 reinforces the role of DO in corrosion control systems such as cathodic protection, where oxygen availability affects current demand and deposit formation.

03.1.7 Fouling, Macro- and Microorganisms

Marine fouling refers to the accumulation of biological material such as microorganisms, algae, and larger aquatic organisms on submerged surfaces. Fouling occurs in two distinct stages:

1. Microfouling – involving bacteria, diatoms, and protozoa
2. Macrofouling – involving barnacles, mussels, tubeworms, algae, and other visible marine organisms

These biological colonizers influence corrosion processes in multiple ways and can severely affect the performance and maintenance of marine structures.

Microfouling

The first stage of fouling begins with the formation of a biofilm, typically within minutes to hours after a clean metal surface is submerged. This film consists primarily of bacteria and extracellular polymeric substances (EPS) secreted by microbial communities.[8]

Key characteristics:

- Creates a differential aeration cell, leading to localized anodic and cathodic regions
- Can cause oxygen concentration cells under the film, promoting crevice corrosion
- Traps ions and organic molecules, altering the local chemistry
- Facilitates the onset of microbiologically influenced corrosion (MIC)

Microfouling is a major contributor to localized corrosion, particularly pitting, and accelerates degradation in stagnant or shielded zones, such as inside crevices, under gaskets, or beneath scale.

Macrofouling

Over time, the biofilm enables the attachment of larger organisms. This macrofouling layer increases drag, mass, and turbulence on marine components, and can physically trap water and debris against the surface.

Common macrofoulers include:

- Barnacles
- Tubeworms
- Mussels
- Macroalgae [8]

Effects of macrofouling:

- Induces under-deposit corrosion by creating stagnant micro-environments beneath colonies
- Blocks or disturbs coating adhesion and cathodic protection efficiency
- Increases maintenance costs, especially in dynamic systems (e.g., shafts, propellers)

Biological and Environmental Factors

Fouling intensity varies with:

- Temperature – warmer waters promote faster growth
- Salinity – many fouling species prefer full-strength seawater (~35 ppt)
- Flow velocity – high turbulence may reduce fouling adhesion but won't prevent initial biofilm formation
- Depth – fouling is concentrated in shallow waters, typically <100 m, where light and nutrients are abundant [9]

Seasonal variation also affects fouling activity, with peak growth often occurring in spring and summer.

Relevance to Shaft Design

Our marine propulsion shaft operates in the uppermost zone of the marine environment—within immersion and splash zones where biofouling activity is most aggressive. Key implications for design:

- Biofilm formation on the shaft surface, especially near couplings, keyways, and seal areas, can create oxygen-deficient zones, promoting crevice corrosion and MIC.
- If SS316L is used, its lower resistance to localized corrosion under biofouled conditions becomes a concern. Duplex 2205, with higher pitting and crevice resistance, is more suitable.
- Macrofouling organisms such as barnacles may colonize low-flow or idle areas, increasing shaft imbalance, surface roughness, and wear at bearings or seals.
- Anti-fouling coatings or cathodic protection should be considered—though biofilms may interfere with both. Surfaces must be designed for easy cleaning or resistance to organism adhesion.
- The 1200 RPM shaft speed may help reduce large-scale macrofouling on dynamic surfaces but will not prevent initial microbial colonization.

Ultimately, biofouling is both a corrosion initiator and a mechanical threat. Its impact must be accounted for in material selection, surface treatment, seal design, and maintenance planning for long-term shaft performance.

03.1.8 Effect of Velocity

The velocity of seawater flow over a metallic surface significantly affects its corrosion behavior. Flow conditions influence corrosion by:

- Altering the rate of oxygen and ion transport to and from the metal surface
- Changing the concentration of corrosion products
- Influencing biofilm and deposit stability

Flow-induced effects can be both beneficial and detrimental, depending on the flow regime, material type, and environmental conditions.

Beneficial Effects of Moderate Velocity

Under moderate flow rates, velocity generally helps by:

- Enhancing oxygen transport to the cathodic regions of the metal surface
- Removing corrosion products, which could otherwise accumulate and cause differential aeration
- Preventing the formation of stagnant layers, thus reducing localized corrosion risk

This results in more uniform corrosion and may improve the performance of cathodic protection systems, as electrical current is distributed more evenly in clean, fast-flowing environments.[10]

Detrimental Effects of High Velocity

At high velocities, flow can become abrasive, especially if solid particles (sand, silt) are suspended in the seawater. This leads to:

- Erosion-corrosion, where the protective oxide layer or corrosion product layer is physically removed
- Mechanical damage to coatings, biofilms, and calcareous deposits
- Loss of surface passivity in materials like stainless steel and aluminum alloys

Materials in fast-flowing environments often show signs of localized attack at welds, crevices, and other microstructural inhomogeneities.

Velocity Limits and Material Sensitivity

Each material has a critical velocity threshold, above which corrosion accelerates due to flow-induced damage. For example:

- Mild steel is more prone to erosion at relatively low velocities
- Stainless steels may withstand higher velocities but still suffer if pitting initiates under deposits or scaling

These thresholds vary with environmental parameters such as temperature, salinity, and particulate content.

Field Data Illustration

Graph 3, originally included to describe vertical environmental variation, also supports the discussion on velocity. The site data indicate how surface conditions where water is turbulent and oxygen-rich create aggressive corrosion zones due to the combination of:

- High flow
- Elevated DO
- Warm temperatures

This aligns with typical shaft operating environments.

Velocity and Biofouling

Higher flow rates can suppress the settlement of macro-organisms like barnacles and mussels. However, microbial biofilms often develop even in moving water, especially in crevices or shielded zones. In such cases, MIC may still occur even under dynamic conditions.[10]

Relevance to Shaft Design

Our marine propulsion shaft rotates at 1200 RPM, introducing significant relative velocity between the shaft surface and seawater. Key implications include:

- The increased flow velocity enhances oxygen transport, which supports cathodic reactions and passive film formation beneficial in general.
- However, this also means the shaft is subject to erosion-corrosion, especially at leading edges, seal interfaces, and sharp transitions.
- If calcareous deposits or protective coatings are applied, high flow may cause mechanical stripping, reducing their effectiveness.
- The design must account for hydrodynamic effects at features like keyways, couplings, and shaft shoulders, where flow separation and turbulence can create local stagnation or intensified wear.
- For materials like SS316L, increased velocity may aggravate pitting or crevice corrosion if passive films are disturbed. Duplex 2205, with better erosion resistance and higher repassivation rates, is preferable.

Overall, velocity is a double-edged sword enhancing protection in some areas while increasing damage risk in others. The shaft's material, surface finish, and geometry must be optimized to handle the specific flow conditions it will experience.

03.1.9 Seawater Resistivity

Electrical resistivity is a critical parameter in understanding the electrochemical behavior of seawater and its impact on corrosion and cathodic protection. Resistivity represents the opposition to the flow of electric current through the electrolyte (in this case, seawater) and is typically expressed in ohm-centimeters ($\Omega \cdot \text{cm}$).

In marine corrosion science, low resistivity indicates high ionic conductivity, meaning that seawater can readily support galvanic currents, corrosion cells, and cathodic protection currents. Seawater, due to its high salinity and ion content, is one of the most conductive natural electrolytes.[11]

Factors Affecting Resistivity

The resistivity of seawater depends on several key variables:

1. Salinity

- As the concentration of dissolved salts increases, more ions (e.g., Na^+ , Cl^- , Mg^{2+} , SO_4^{2-}) are available to carry electrical current.
- Therefore, higher salinity = lower resistivity.

2. Temperature

- Increasing temperature improves ion mobility, further reducing resistivity.
- Warmer water conducts electricity more efficiently than cold water.

3. Pollutants and Organic Matter

- Industrial discharges or biological activity may introduce ions or alter pH, slightly modifying resistivity.

Typical Resistivity Values

In typical open-ocean seawater (salinity ~35 ppt), the resistivity ranges between 20 and 30 $\Omega \cdot \text{cm}$. Table 3 in the source text shows resistivity values as a function of salinity and temperature, indicating the following trends:

Table 3 - Seawater Resistivity ($\Omega \cdot \text{cm}$) as a Function of Salinity and Temperature [2]

Salinity (‰)	Temperature ($^{\circ}\text{C}$)					
	0	5	15	20	25	30
10	107.1	92.5	80.9	71.6	64	57.7
20	57.3	49.6	43.5	38.5	34.5	31.1
30	39.6	34.4	30.2	26.8	24	21.6
31	38.5	33.4	29.3	26	23.3	21
32	37.4	32.4	28.5	25.3	22.6	20.4
33	36.3	31.5	27.7	24.6	22	19.9
34	35.3	30.7	27	23.9	21.4	19.4
35	34.4	29.9	26.3	23.3	20.9	18.9

Source: Whitfield, M., Jagner, D. (Eds), 1981. Marine Electrochemistry, A Practical Introduction. Wiley, p. 513.

This table demonstrates that seawater at 35 ppt salinity and 25–30 $^{\circ}\text{C}$ conditions similar to your design environment has resistivity between 22–26 $\Omega \cdot \text{cm}$, indicating a very conductive medium.

Engineering Implications

- Low resistivity promotes galvanic corrosion, especially between dissimilar metals.
- It also supports efficient current flow for cathodic protection systems, reducing anode consumption and improving protection coverage.
- Conversely, in high-resistivity environments (e.g., freshwater, deionized water), more current is required, and protection may be less effective.

Relevance to Shaft Design

In our marine propulsion shaft design, the shaft is immersed in warm, highly saline seawater (~35 ppt at 25 °C) conditions that yield a low resistivity of approximately 25 $\Omega \cdot \text{cm}$.

This has multiple implications:

- The risk of galvanic corrosion is high, especially if dissimilar metals are in contact (e.g., shaft and coupling or seal materials). Proper material matching or electrical insulation is essential.
- If cathodic protection (CP) is implemented (e.g., sacrificial anodes or impressed current), the low resistivity ensures efficient current distribution, reducing energy and anode consumption.
- However, overprotection must be avoided, as it may lead to hydrogen embrittlement, especially in high-strength alloys or duplex stainless steels.
- In FEA or corrosion simulation, seawater resistivity must be correctly specified to model CP behavior, anode spacing, and current distribution along the shaft or its interfaces.

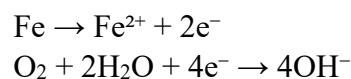
Given the low resistivity of the operating environment, the shaft is exposed to aggressive electrochemical conditions, requiring thoughtful material selection (e.g., Duplex 2205, NiCrMo alloys), proper electrical isolation at joints, and if used a carefully engineered CP system.

03.2 Corrosion Mechanisms in Marine Propulsion Shafts

Marine propulsion shafts are exposed to a variety of aggressive environmental and mechanical conditions. The primary environment of concern is seawater, which promotes various forms of corrosion due to its high salinity, oxygen content, and microbial activity. This section provides a comprehensive description of all major corrosion mechanisms relevant to shafts made of steels and stainless steels, with uniform length and detail for each, covering governing reactions, severity, influencing factors, and engineering control strategies [6], [12].

03.2.1 Uniform Corrosion

Uniform corrosion is the most general form of corrosion and occurs evenly across exposed metal surfaces. It is particularly common in unprotected carbon steels operating in fully submerged or splash zone marine conditions. The reaction begins when iron in the steel reacts electrochemically with dissolved oxygen and water,



These reactions result in the formation of iron hydroxides and oxides (rust). While uniform corrosion may not lead to unexpected failure due to its predictability, it can result in significant wall thinning, dimensional loss, and reduced fatigue resistance. Typical corrosion rates for mild steel in seawater range from 0.1 to 0.5 mm/year, increasing in warm, turbulent, or oxygen-rich conditions [12]. Stainless steels resist uniform corrosion due to their passive Cr_2O_3 layer; however, if damaged (e.g., during welding or abrasion), even 304 or 316 grades can exhibit generalized attack.

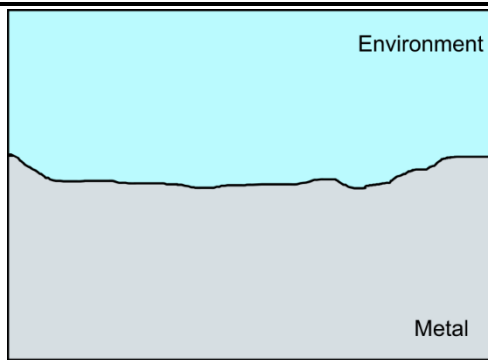


Figure 1 - Uniform Corrosion [13], [14]

Uniform corrosion is exacerbated by surface damage, poor coating systems, or improper material selection. Mitigation strategies include selecting corrosion-resistant alloys (e.g., 316L, duplex stainless steels), applying protective coatings (epoxies, metallic cladding), and employing cathodic protection systems [12], [13], [14]. Routine inspection and wall thickness monitoring are essential, especially for components operating in critical load paths.

03.2.2 Galvanic Corrosion

Galvanic corrosion occurs when two electrochemically dissimilar metals are electrically connected in the presence of an electrolyte, such as seawater. In marine propulsion shafts, this often results when carbon steel is coupled with stainless steel or when different grades of stainless steel are assembled together. The less noble (more active) metal becomes the anode and corrodes preferentially, while the more noble (cathodic) metal is protected [15].

The cell potential driving galvanic corrosion is given by:

$$E_{cell} = E_{cathode} - E_{anode}$$

The anodic corrosion rate can be estimated using Faraday's Law:

$$w = \frac{ItM}{nF}$$

Where w is the metal loss (g), I is the current (A), t is time (s), M is molar mass (g/mol), n is the number of electrons transferred, and F is Faraday's constant ($\sim 96,500$ C/mol) [15].

Severity depends on the area ratio between the cathode and anode, the conductivity of the electrolyte, and the potential difference. In marine conditions, small carbon steel bolts fastened to large stainless steel components are particularly vulnerable. Corrosion rates can exceed 0.3 mm/year and lead to rapid localized failure [12].

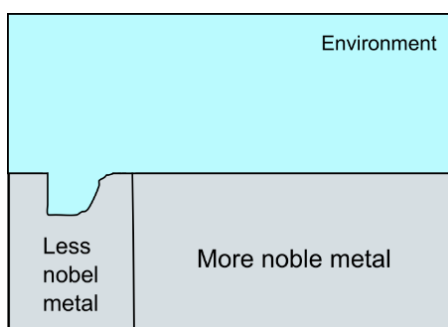


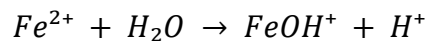
Figure 2 - Galvanic Corrosion [15], [16]

Factors that intensify galvanic corrosion include high salinity, warm temperatures, and the presence of biofilms. To mitigate this, engineers use dielectric isolation (gaskets, sleeves), match alloys where possible (e.g., 316L with 316L), or design cathode-to-anode area ratios to be less severe. Sacrificial anodes (e.g., zinc) are also employed to preferentially corrode instead of structural components [12].

03.2.3 Pitting Corrosion

Pitting corrosion is a highly localized form of attack that results in the formation of deep, narrow pits on the metal surface. It is especially problematic in stainless steels like 304 or 316 exposed to chloride-rich environments like seawater. Although these alloys are normally protected by a passive chromium oxide film, localized breakdown can occur due to mechanical damage, inclusions, or sustained low oxygen levels.

Once initiated, pits propagate due to an autocatalytic cycle. The metal inside the pit oxidizes, producing metal cations that hydrolyze and generate H^+ ions, decreasing pH:



Chloride ions concentrate in the pit to maintain electroneutrality, further destabilizing the passive film. This creates a self-sustaining micro-environment that accelerates pit growth. Critical pitting potential (E_{pit}) for 316L in seawater is typically +300 to +400 mV vs. Standard Calomel Electrode [16].

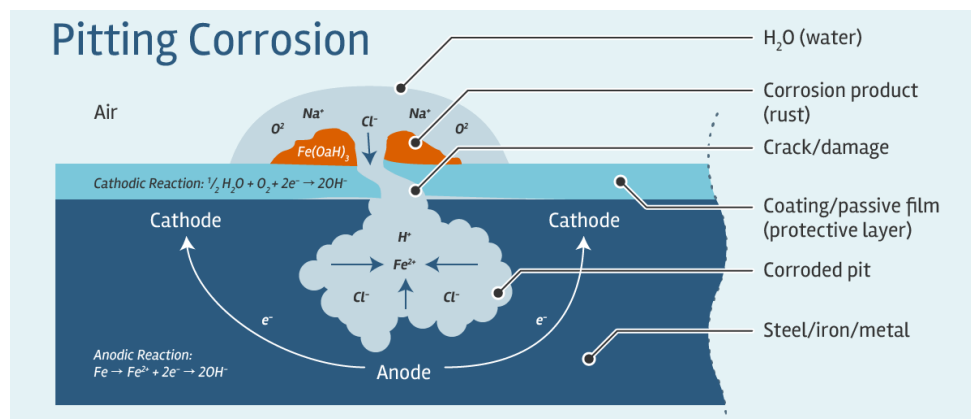


Figure 3 - Pitting Corrosion [17]

Pits often initiate fatigue cracks and serve as root causes for corrosion fatigue failures. Depths of 100–300 μm can occur in weeks under stagnant seawater. Pitting severity is exacerbated by high chloride content, elevated temperatures, stagnant conditions, and poor weld finishing. Prevention strategies include using molybdenum-enhanced grades (e.g., 316L), ensuring proper surface preparation, and avoiding low-flow zones. Periodic inspection is key to detecting early-stage pitting [16].

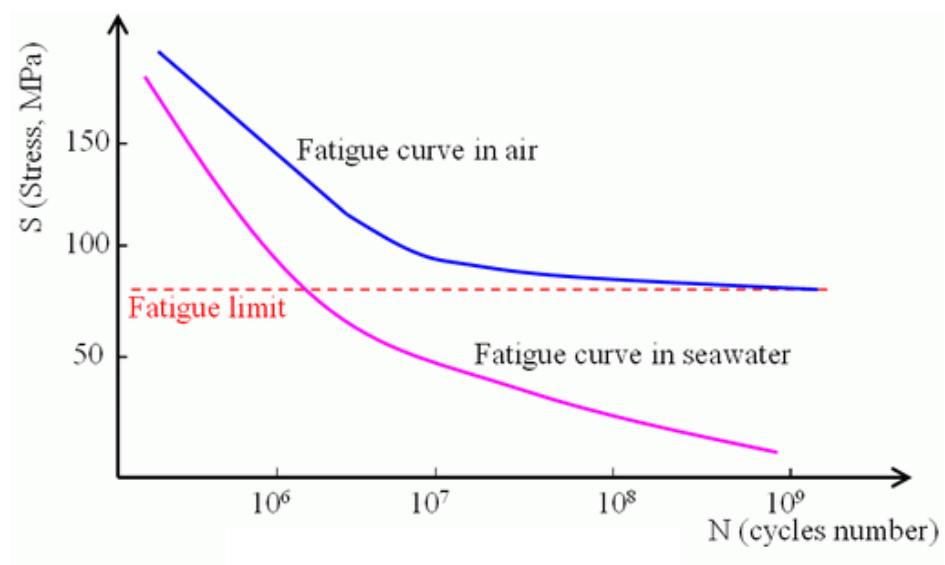
03.2.4 Corrosion Fatigue

Corrosion fatigue involves the combined action of cyclic mechanical loading and a corrosive medium, leading to premature failure at stress levels below the fatigue limit observed in air. This is a dominant failure mechanism in marine shafts subjected to rotating bending or torsional loads.

Fatigue cracks typically initiate at corrosion pits or inclusions and propagate under fluctuating stresses. In marine environments, carbon steel shows fatigue strength reductions from 279 MPa (air) to 126 MPa in 3.5% NaCl at 10^7 cycles. Pit-to-crack transitions occur when the local stress intensity exceeds a threshold (ΔK_{th}), which drops from $\sim 5.9 \text{ MPa} \cdot m^{1/2}$ in air to $\sim 4.5 \text{ MPa} \cdot m^{1/2}$ in seawater [17].

Crack growth is often modeled by Paris' Law:

$$\frac{da}{dN} = C(\Delta K)^m$$



Graph 5 - Applied Stress vs Cycles to Failure [15]

Corrosion fatigue is influenced by frequency, stress range, pit morphology, and alloy composition. Mitigation includes surface compressive stress treatments (e.g., shot peening), use of duplex stainless steels with higher fatigue thresholds, and careful elimination of geometric discontinuities. Avoiding pitting through coatings or alloy selection significantly enhances fatigue life [14], [17].

03.2.5 Crevice Corrosion

Crevice corrosion occurs in shielded zones where oxygen cannot easily diffuse, such as under gaskets, flange faces, or tight mechanical fits. Even corrosion-resistant stainless steels like 316L are susceptible in such zones due to the formation of a differential aeration cell. The confined crevice becomes anodic relative to the external surface, initiating localized corrosion.

Within the crevice, metal ion accumulation and hydrolysis reactions cause acidification:

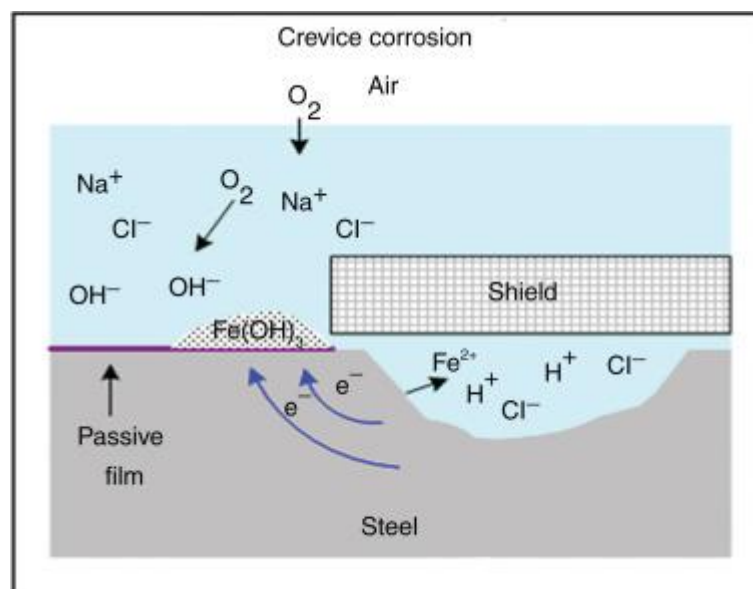
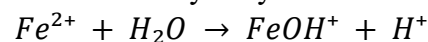


Figure 4 - Crevice Corrosion [19]

The drop in pH and increase in chloride concentration further destabilize the passive layer. This leads to an accelerated localized attack that is often difficult to detect until failure occurs. Typical crevice corrosion rates can exceed 1 mm/year [12].

Contributing factors include stagnant water, porous gasket materials, and marine growth accumulation. Control strategies focus on avoiding crevice-prone designs, using solid non-absorbent seals (e.g., PTFE), employing welded rather than bolted joints where possible, and selecting materials with high crevice corrosion resistance (e.g., super duplex stainless steels). Regular inspection and flushing of crevice-prone regions is also important [12], [13], [14].

03.2.6 Intergranular Corrosion (IGC)

Intergranular corrosion affects stainless steels that have undergone sensitization, a condition where chromium carbides precipitate at grain boundaries during exposure to 550–900°C [18]. This depletion of chromium in adjacent areas reduces corrosion resistance, making grain boundaries anodic relative to the matrix.

In marine shafts, IGC may develop in welded zones or improperly heat-treated components. Attack progresses along grain boundaries, often with minimal outward signs, and results in a loss of mechanical integrity. Stainless steels like 304 and 316 are vulnerable if not properly treated. Using low-carbon variants (e.g., 316L) or stabilized grades (e.g., 321, 347) helps prevent carbide formation.

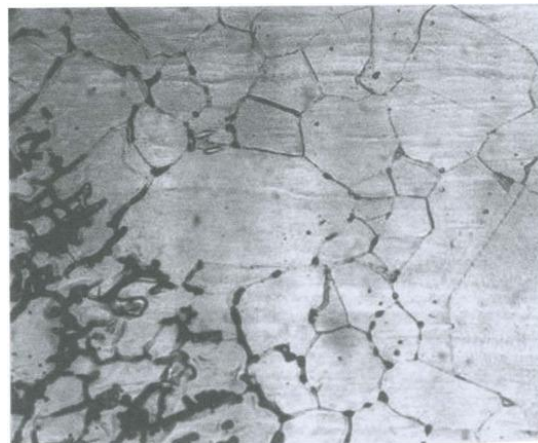
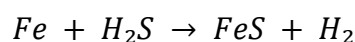


Figure 5 - Intergranular Corrosion [20]

Remedies include post-weld heat treatment, rapid cooling after welding, and alloy selection that ensures intergranular stability. Inspection methods like oxalic acid etch testing or ASTM A262 practices are useful for detecting IGC susceptibility [18].

03.2.7 Microbiologically Influenced Corrosion (MIC)

MIC results from microbial activity, particularly by sulfate-reducing bacteria (SRB) that colonize metal surfaces in marine environments. These bacteria produce hydrogen sulfide (H_2S), which reacts with iron to form iron sulfide (FeS), promoting localized depassivation and pitting [19], [20].



Biofilms formed by SRBs create differential aeration conditions and concentrate aggressive metabolites. MIC is often identified by black sludge, sulfurous odor, and rapid localized attack in low-

flow areas such as bilges, shaft seals, and couplings. Carbon steels are particularly vulnerable, but stainless steels are also susceptible if biofilms persist.

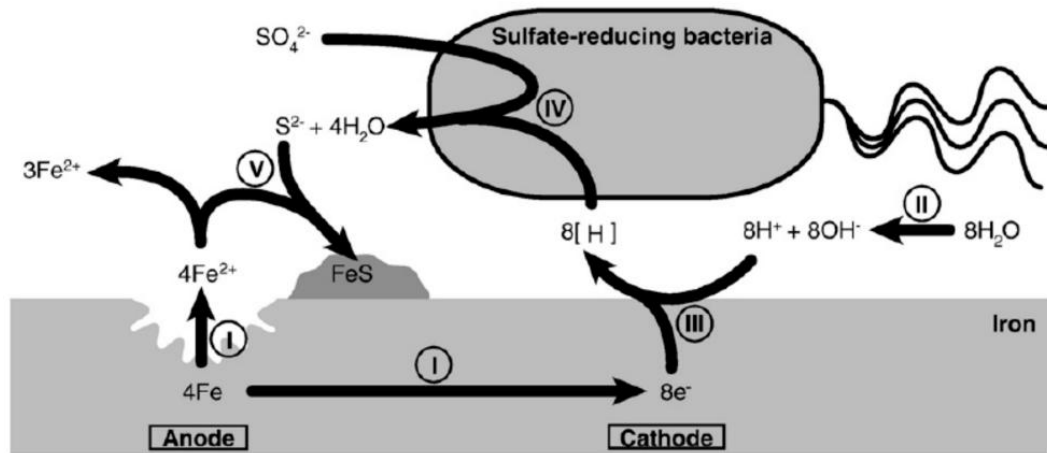


Figure 6 - Microbiologically Influenced Corrosion [21]

Prevention involves regular cleaning, applying antimicrobial coatings, and maintaining good flow conditions. Materials with smooth surfaces and resistance to biofouling (e.g., duplex stainless steels) are preferred. Biocides may also be used in circulating systems where permitted.

03.2.8 Hydrogen Embrittlement

Hydrogen embrittlement affects high-strength steels when atomic hydrogen diffuses into the metal lattice, leading to brittle fracture under tensile stress. Hydrogen is generated during corrosion reactions or cathodic protection processes. It accumulates at grain boundaries, inclusions, or dislocation sites, reducing ductility and causing delayed cracking.

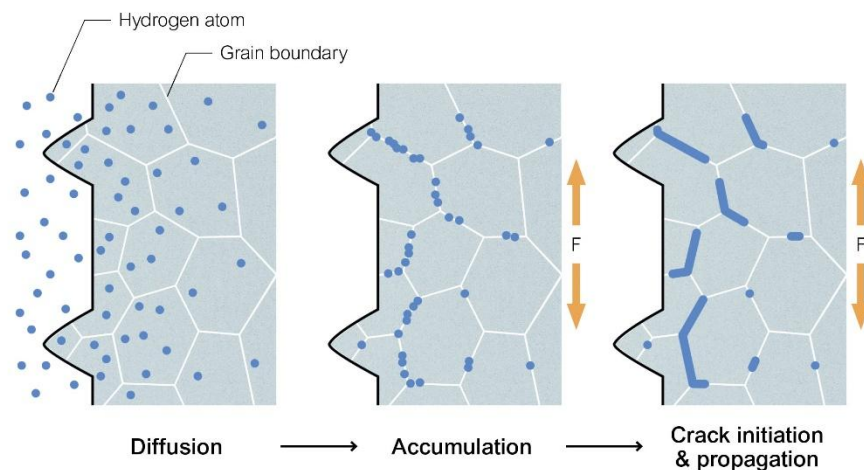


Figure 7 - Hydrogen Embrittlement [23]

This mechanism is particularly relevant to martensitic or hardened steels used in coupling bolts or shaft extensions. Austenitic stainless steels (e.g., 316L) are generally immune due to their FCC structure. Embrittlement is worsened by stress concentrations, low pH environments, and overprotection in cathodic systems (e.g., potentials below -1.0 V vs. Ag/AgCl) [21].

Control measures include material selection (avoiding susceptible high-strength steels), limiting cathodic protection voltage, and post-weld hydrogen bake-outs. Design should minimize notch effects, and components exposed to hydrogen should undergo periodic inspection for sub-surface cracking.

03.2.9 Erosion-Corrosion and Cavitation

Erosion-corrosion results from the combined action of fluid motion and electrochemical attack. In marine propulsion shafts, it occurs in areas with high flow velocities, such as near seals, bearings, or shaft exits. The mechanical force of the fluid removes protective oxide films, exposing fresh metal to rapid corrosion. Cavitation, which involves bubble collapse near the surface, causes microjets that pit the surface.

Stainless steels with insufficient hardness or protection are particularly vulnerable. Loss rates can exceed 0.5 mm/year in turbulent flow. Typical damage appears as horseshoe-shaped grooves or pitting aligned with flow direction [14], [18].

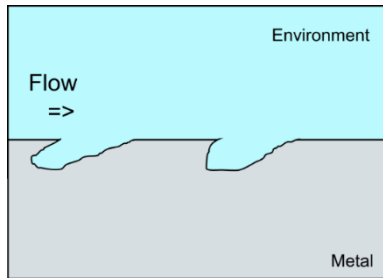


Figure 9 - Corrosion Erosion [16]

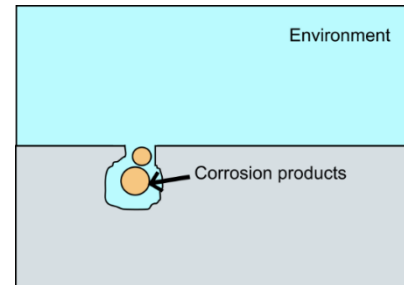


Figure 8 - Cavitation Corrosion [16]

Mitigation involves using erosion-resistant materials (e.g., duplex or martensitic stainless steels), ensuring laminar flow design, applying hard coatings (e.g., tungsten carbide), and controlling shaft alignment to prevent turbulence. Shaft velocity should be monitored to stay below cavitation thresholds.

03.2.10 Fretting Corrosion

Fretting corrosion occurs at contact interfaces under cyclic vibration or small-amplitude oscillation. In marine shafts, this typically arises at press-fits, splines, or couplings where relative motion disrupts passive films. The exposed metal oxidizes, producing debris that further accelerates wear and corrosion [13], [14].

This mechanism often serves as a precursor to corrosion fatigue and is worsened by moisture, poor fit tolerances, or shaft misalignment. It reduces the fatigue limit and can initiate cracks under moderate cyclic stress.

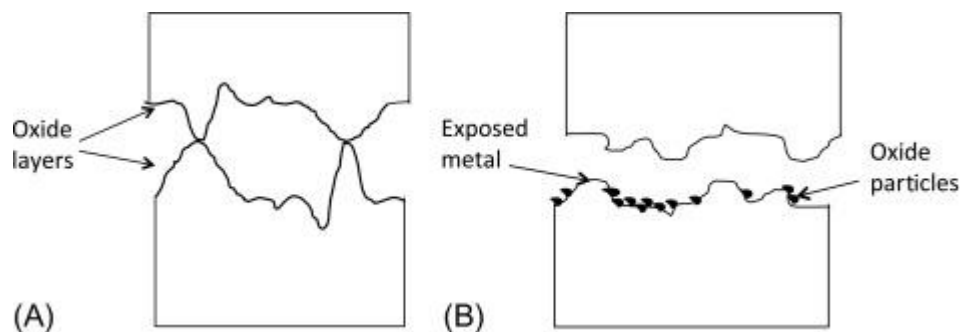


Figure 10 - Fretting Corrosion [24]

Prevention involves precision machining for tight fits, use of anti-fretting coatings (e.g., MoS₂, PTFE), and lubrication. Compressively stressed surfaces (e.g., shot-peened areas) resist fretting better. Materials with higher hardness and stable passive films, such as duplex stainless steels, provide greater resistance.

The environmental analysis and corrosion mechanisms described in Section 3 establish the fundamental performance demands placed on a marine propulsion shaft. Continuous exposure to high-salinity seawater, combined with cyclic torsional and bending stresses, creates a service environment where both mechanical integrity and corrosion resistance are equally critical. The identified degradation modes particularly pitting, crevice corrosion, galvanic interactions, and corrosion fatigue require a material that can provide high strength, long-term durability, and inherent resistance to chloride-induced damage. These environmental and operational constraints form the basis for the material selection process described in Section 4, where candidate alloys are screened and ranked using quantitative criteria and **ANSYS Granta** tools.

04. Material Selection

The selection of a suitable material for the marine propulsion shaft is a critical step in ensuring safe torque transmission, high fatigue life, corrosion resistance, and long-term reliability in seawater environments. Because the shaft operates under combined torsional and bending loads while being continuously exposed to chloride-rich seawater, the material must satisfy simultaneous mechanical, corrosion, and manufacturability constraints. To ensure a systematic and defensible choice, the material selection process in this project was carried out using the **Ashby-based methodology** Translate, Screen, Rank, and Document supported by the **ANSYS Granta EduPack software**, which enables quantitative filtering and comparison across a large materials database.

Design constraints such as minimum yield strength, fatigue strength at 10^7 cycles, seawater corrosion resistance (including $PREN \geq 30$), weldability, and hot forming compatibility were applied during the screening stage to narrow an initial pool of over 1900 metals to a feasible set of candidates.

Subsequent ranking using performance indices (strength-to-density, stiffness-to-density, cost-per-unit strength) provided a rational basis for comparing options under realistic marine shaft loading conditions.

This section summarizes the material selection workflow, presents the shortlisted candidates produced by ANSYS Granta (*see Annex A for the full Granta selection report*), and discusses their performance using the exported ranking charts and datasheets. The full ANSYS Granta selection report and the final material datasheet for Duplex Stainless Steel 2205 are included in the **Appendix** for traceability and verification (*see Annex B for the material datasheet of Duplex 2205*).

04.1 Introduction

04.1.1 Background and Design Requirement

Marine propeller shafts operate in a harsh environment: cyclic mechanical loads from torque and bending, exposure to saline water (leading to corrosion, pitting, crevice corrosion, stress corrosion cracking), and sometimes elevated temperatures or variable loads. The shaft must transmit torque reliably, resist fatigue failure and corrosion over long service life, yet remain manufacturable and cost-effective.

04.1.2 Objective of Material Selection

The objective is to find a material that balances the competing demands of strength, durability, corrosion resistance, and cost. We employ a structured methodology (translate → screen → rank → document) using ANSYS Granta (Granta EduPack or similar) to filter and compare candidate materials. [22]

The final goal is to make a justified recommendation (with supporting data).

04.2 Methodology

04.2.1 General Material Selection Strategy (Translate – Screen – Rank – Document)

The methodology used in this study follows a structured, multi-stage approach to material selection based on the framework popularized by Ashby and employed in engineering design practice. The four main steps are:

1. **Translate:** convert real-world design requirements (loads, environment, geometry, constraints) into quantitative functions, constraints, objectives, and free variables.

- Function describes what the component must do (e.g. transmit torque, resist corrosion).
- Constraints are nonnegotiable property limits (e.g. minimum yield strength, maximum allowable corrosion rate).
- Objectives are performance measures to maximize or minimize (e.g. minimize mass or cost, maximize fatigue life).
- Free variables are the design parameters the engineer may vary (e.g. cross-section, thickness).

This translation step ensures that subsequent filtering and ranking are rooted in the physics and usage context of the component.

2. **Screen (or reject):** apply the constraints to eliminate materials that cannot meet the basic requirements. This reduces the candidate set to only those feasible for the application. Common constraints include yield strength, fatigue strength, corrosion resistance, manufacturability, cost ceilings, etc. Materials failing any of these constraints are removed. The screening step uses property charts or database filters.
3. **Rank (or Select / Compare):** among the screened materials, compute performance indices (also known as merit indices) based on the objectives and free variables. These indices (combinations of material properties) allow direct ranking of materials (higher is better, or lower if minimizing). You may also handle multiple objectives via trade-offs, coupling equations, or multi-stage ranking. Materials are sorted, compared, and candidates are selected.
4. **Document (or Support):** prepare a record of reasoning, data sources, assumptions, and comparisons. Provide justification of the selected material relative to alternatives. Include raw data, charts, decision paths, sensitivity analyses, and any caveats.

This “Translate → Screen → Rank → Document” paradigm helps maintain clarity, traceability, and rational decision making. In real cases, iteration is often necessary: if the ranking yields no satisfactory solution, one must revisit the translation step (relax or re-interpret constraints, adjust objectives) or consider alternate designs. [22], [23], [24], [25]

04.2.2 Software Tool: ANSYS Granta EduPack

ANSYS Granta EduPack (formerly CES EduPack) is a materials and process selection tool widely used in engineering education and early-stage design.

Key features:

- **Material and Process Databases:** a large, curated database of materials (metals, polymers, composites, ceramics) and manufacturing processes, with associated mechanical, thermal, chemical, and environmental properties.
- **Graphical Selection (Charts / Stages):** the ability to plot property charts (Ashby-style) and overlay selection lines to visually screen materials. User can set axes (e.g. strength vs density) and selection criteria (regions above, below, left, right).

- **Stage-Based Filtering:** supports creation of multiple “stages” (sequential filters) to apply constraints and refine selection gradually.
- **Material Indices / Performance Indices Tools:** it allows the definition of custom indices (e.g. combining strength, density, cost) and calculation/ranking accordingly.
- **Traceability and Reporting:** as user apply filters, stages, and selections, the tool maintains traceable records and lets user export summary reports and datasheets.
- **Tutorials & “Getting Started” Guides:** EduPack includes step-by-step tutorials and exercise guides for new users.
- **Education Focus:** the tool is optimized for teaching materials decisions, showing links between properties, processing, cost, and performance. [26]

In this project, we will use EduPack to:

- Input the translated constraints (yield strength, fatigue strength, corrosion metrics, manufacturability)
- Filter the full database via stages to screen out infeasible materials.
- Define and apply performance indices to rank the remaining materials.
- Export the final selection report and material datasheet.

04.3 Design Requirements for Marine Propeller Shaft

04.3.1 Functional Requirements

The propeller shaft transmits torque from the engine to the propeller, while also resisting bending forces from propeller thrust, hydrodynamic loading, and shaft misalignment. The component must:

- Carry a design torque corresponding to the vessel’s power without yielding (minimum yield > 66 MPa for 100 mm dia).
- Resist cyclic torsional stresses over long service life ($\geq 10^7$ cycles).
- Maintain alignment and stiffness (torsional twist and bending deflection within tolerances).
- Operate reliably under combined torque, bending, and thrust loading.

04.3.2 Service Environment (Seawater Corrosion, Fatigue)

- Seawater at ~25 °C with high salinity is highly aggressive. Shaft material must resist pitting, crevice corrosion, and stress-corrosion cracking (SCC).
- Corrosion fatigue is the dominant failure mode: cyclic stresses in chloride environment accelerate crack initiation and growth.
- Potential galvanic coupling with propeller (bronze, Ni-Al-bronze) or couplings must be considered.
- Surface durability (resistance to wear, corrosion at bearings and seals) is essential.

04.3.3 Constraints (Mechanical, Corrosion, Manufacturing)

04.3.3.1 Mechanical constraints

- Yield strength ≥ 50 ksi (345 MPa) $\rightarrow$ for 100 mm shaft: ≥ 66 MPa minimum requirement satisfied.
- Fatigue strength ≥ 30 ksi (207 MPa) at 10^7 cycles.
- Adequate stiffness (Young's modulus ~ 200 GPa typical for steels).

04.3.3.2 Corrosion constraints

- Pitting Resistance Equivalent Number (PREN) ≥ 30 .
- SCC: not or slightly susceptible in chloride environment.
- Seawater resistance: acceptable/excellent rating required.

04.3.3.3 Manufacturing constraints

- Hot forming: acceptable/excellent.
- Weldability: good/excellent (repair and fabrication).
- Machinability sufficient for large-diameter shaft finishing.
- Material available in large bar/forging stock.

04.3.4 Objective (Safe, Cost-Effective, Durable Shaft)

The selected material must:

- Maximize service life by combining high fatigue strength and corrosion resistance in seawater.
- Minimize cost per unit strength (balance between corrosion-resistant Ni alloys and affordable steels).
- Ensure manufacturability with forging, machining, and welding compatibility.
- Survive pitting and SCC risks through adequate PREN and duplex/austenitic alloy design.

[22], [23], [24]

Candidate materials surviving screening: 316L stainless, Duplex 2205. Subsequent ranking (by cost/strength and PREN) will determine the optimum.

04.4 Material Screening

04.4.1 Constraints Applied in Ansys Granta

The Stage-2 screening in Ansys Granta used the following constraints (as configured in your selection file): yield strength ≥ 50 ksi; fatigue strength at 10^7 cycles ≥ 30 ksi; metal hot forming = acceptable/excellent; weldability = good/excellent; salt water durability = acceptable/excellent; PREN ≥ 30 ; SCC susceptibility = not/slightly susceptible.

To visualize the outcome of the applied constraints, the yield strength vs. density Ashby chart generated in ANSYS Granta is presented in Figure 1. The shaded gray region indicates the initial pool of metals before screening, while the highlighted zones show the remaining feasible materials after all

constraints were applied. As seen, the majority of alloys fall outside the acceptable limits, leaving only about 25 candidate materials that satisfy the required mechanical, corrosion, and manufacturability conditions. Among these, the two promising candidates Austenitic Stainless Steel AISI 316LVM (cold-worked) and Duplex Stainless Steel AISI 2205 (annealed) occupy the favorable region of the chart, combining adequate yield strength with moderate density. This figure visually supports how the database was refined to the final shortlist used for ranking in the subsequent stage.

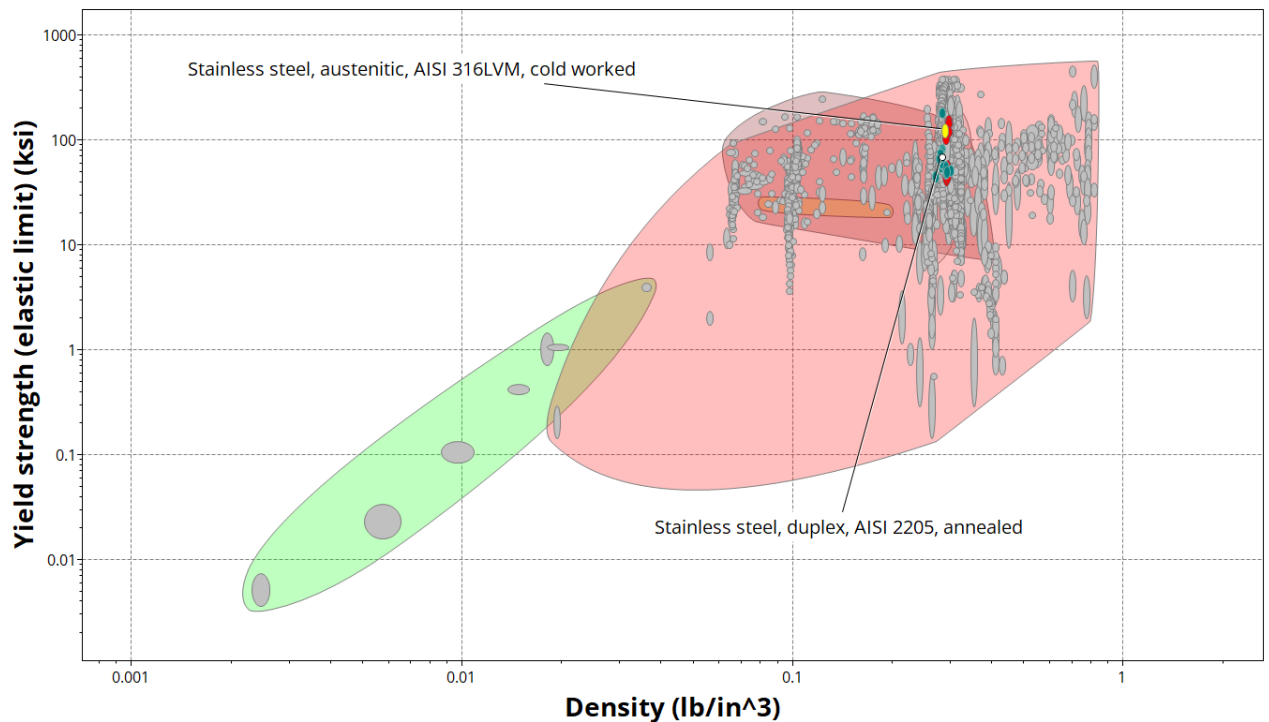


Figure 11 - Yield strength vs. density chart showing screened with filtered materials in gray color. [27]

04.4.2 Shortlisted Materials after Screening

The first chart visualizes the distribution of metallic materials by yield strength and density. This shows higher strength have 316LVM also higher density while AISI has lower strength and density.

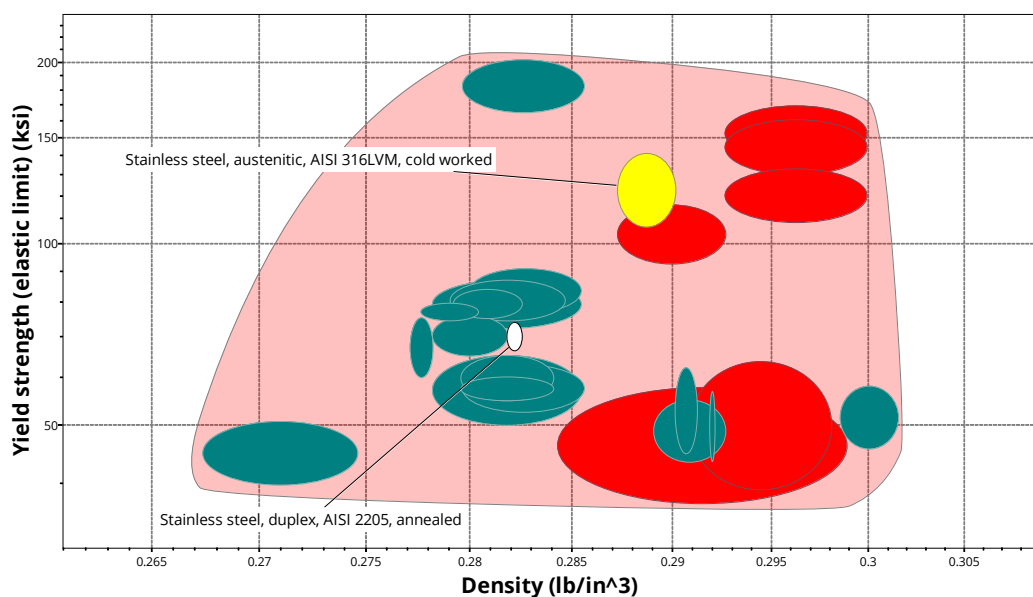


Figure 12 - Yield strength vs. density chart showing after removed filtered materials. It shows AISI 316LVM have highest yield strength. [27]

This plot compares stiffness among the remaining materials. Both 316LVM and 2205 occupy the upper zone, showing AISI 2205 have lighter high stiffness.

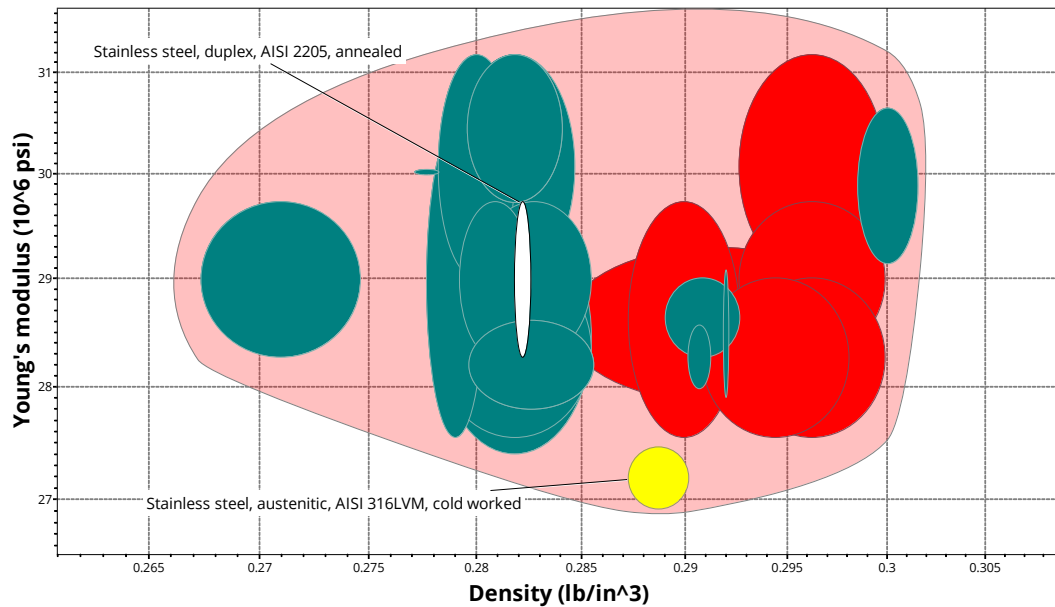


Figure 13 - Young's Modulus vs. Density [27]

Since torsion control depends on shear rigidity, next chart confirms that both shortlisted steels exhibit similar shear modulus (~ 80 GPa equivalent), meaning both can maintain shaft twist within allowable limits during operation. But AISI 2205 has higher torsion control with lighter.

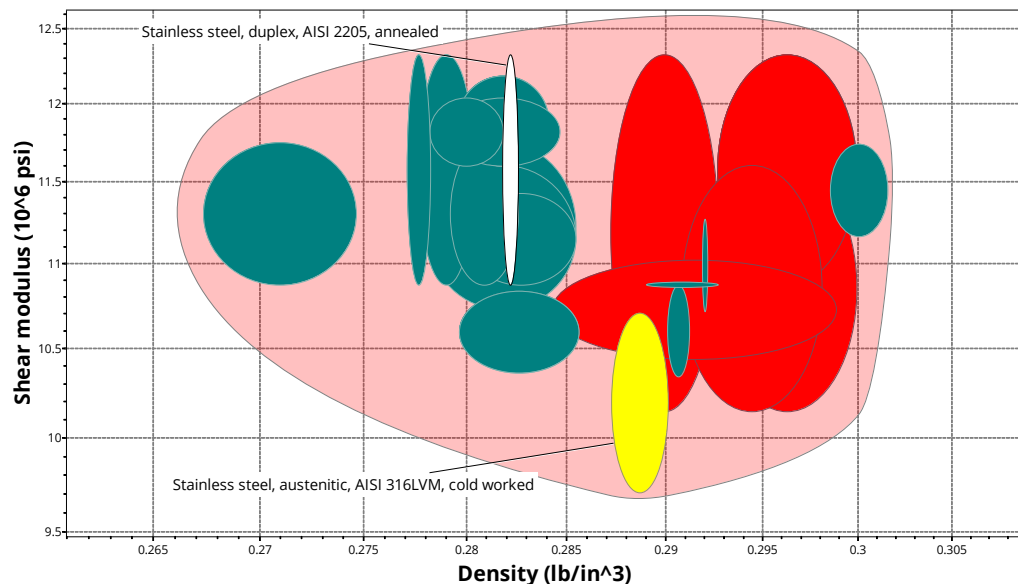


Figure 14 - Shear Modulus vs. Density [27]

Fatigue behavior under cyclic torque is critical. Next plot showing, 316LVM clearly leads with higher fatigue strength (~ 60 ksi) compared to 2205 (~ 40 ksi), but both exceed the design threshold of 30 ksi. This confirms their capability to withstand long-term service loading.

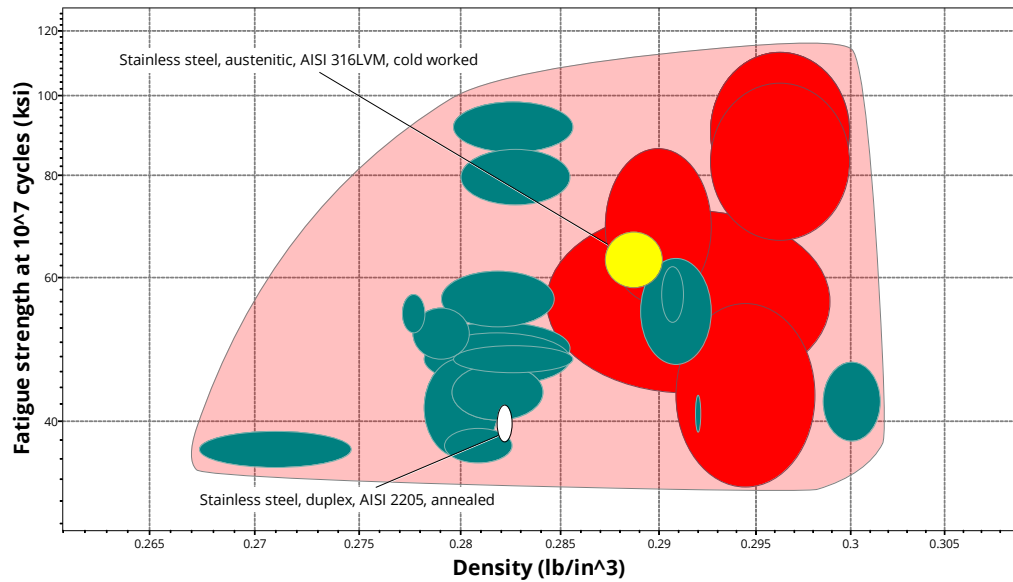


Figure 15 - Fatigue Strength vs. Density [27]

Next plot demonstrates the corrosion resistance advantage of Duplex 2205, which maintains a PREN above 30, indicating strong resistance to chloride-induced pitting and crevice corrosion. In contrast, 316LVM lies near the lower limit, implying the need for additional corrosion control measures.

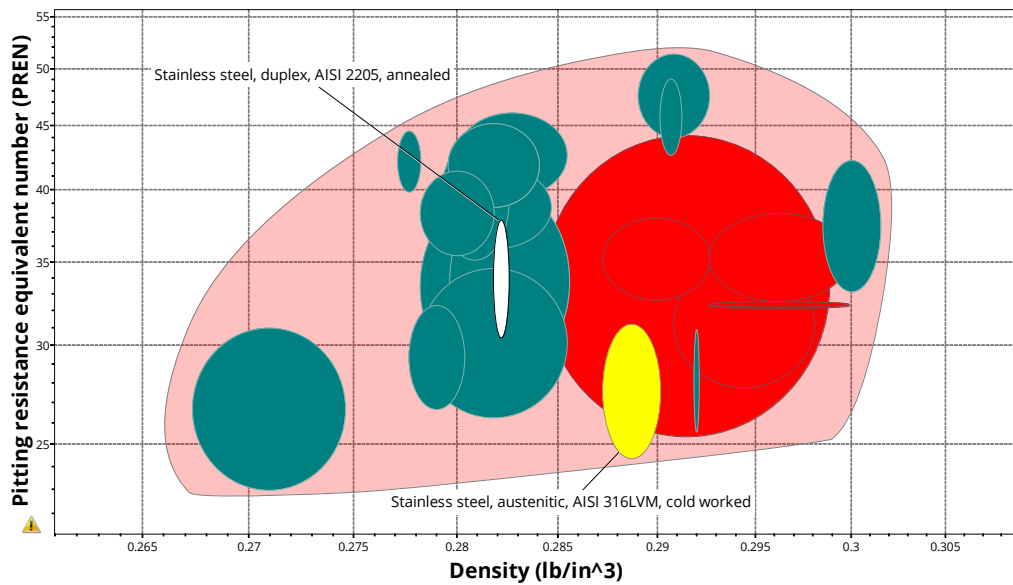


Figure 16 - Pitting Resistance (PREN) vs. Density [27]

Finally, the cost-performance plot shows that both alloys lie in the same moderate cost range, well below that of Ni-based alloys. Duplex 2205 achieves its corrosion advantage without significant cost increase, making it the most balanced option.

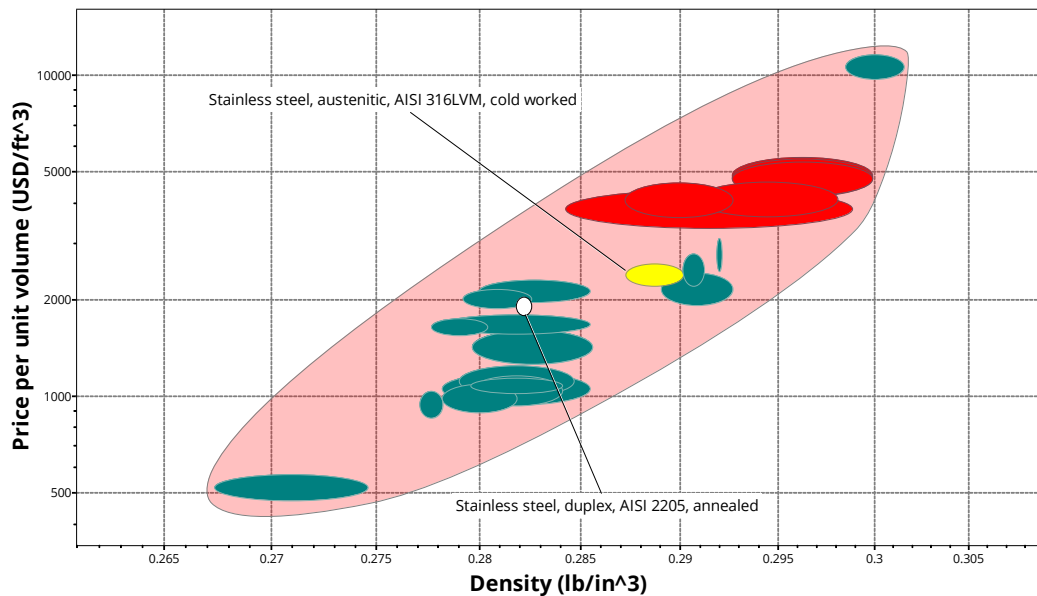


Figure 17 - Price per Unit Volume vs. Density [27]

These visual comparisons confirm that AISI 316LVM offers superior strength, while AISI 2205 provides better corrosion resistance and overall design balance. The summarized data for these shortlisted materials are presented in Table 4 below.

Table 4 - A concise comparison (from Stage-4 comparison table in Ansys Granta see annex A) [27]

Attribute	316LVM (50% CW)	Duplex 2205 (annealed)	Ni-Cr-Mo (ILLIUM 98, cast)
Yield strength (ksi)	107–141	66.7–74	37–46.4
Young's modulus (10⁶ psi)	28 – 29	29 – 30	27 – 28
Shear modulus (10⁶ psi)	10.7 – 11.2	11.3 – 11.6	10 – 10.4
Fatigue strength at 10⁷ (ksi)	58.5–68.2	37.9–41.9	18.9–29.7
PREN	24.4–31.2	30.5–37.8	(High 30s–40s class)
SCC Resistance	Slightly susceptible	Not susceptible	Not susceptible
Cost per unit strength (relative)*	0.080–0.096	0.089–0.102	0.429–0.598

*As reported in the Granta “Cost per unit of strength” field. Data sources for each row: mechanical properties and durability from the comparison pages; PREN and SCC notes from the corrosion section.

Interpretation for the next stage (Ranking): 316LVM delivers the strongest mechanical performance but only marginal PREN margin; Duplex 2205 balances strength with robust chloride resistance and benign SCC behavior; Ni-Cr-Mo excels in pitting/crevice resistance but at significant penalties in strength and cost. These trade-offs will be formalized with performance indices in Section 04.5.

04.5 Material Ranking

04.5.1 Use of Material Indices (Strength/Density, Corrosion Resistance)

Once screening is complete, the shortlisted materials must be ranked. Ranking is not done by single properties but by material indices combinations of properties that best measure performance for the given function.

For a propeller shaft under torsion:

- Strength-limited torsion index (minimum mass design) is:

$$P = \frac{\sigma_f^{2/3}}{\rho}$$

where σ_f = fatigue strength at 10^7 cycles, ρ = density. Materials with higher values give lighter shafts for the same safe torque.

- Stiffness-limited torsion index (control of angle of twist) is:

$$P = \frac{G^{1/2}}{\rho}$$

where G = shear modulus, ρ = density. Higher values mean reduced twist per unit mass.

- Corrosion resistance index: for marine shafts, the Pitting Resistance Equivalent Number (PREN) is used. Ranking by $\text{PREN} \geq 30$ ensures minimum seawater resistance, and higher PREN provides greater safety against pitting and crevice corrosion.
- Cost-weighted indices: Cost per unit strength and cost per unit stiffness adjust the above indices by material price. This highlights trade-offs between corrosion performance and economic viability.

Thus, final ranking must consider both mechanical performance indices and corrosion indices, with cost as a secondary objective. [28]

04.5.2 Performance Comparison of Candidates

To better understand the trade-offs among candidate materials, the Ansys Granta material charts were examined. These maps visualize how the shortlisted alloys compare in terms of cost, mass efficiency, and corrosion resistance under torsional loading conditions. The first chart relates cost per unit strength to mass per unit strength, revealing which alloys provide lightweight and economical performance for cyclic loading in a shaft.

A second chart plots pitting resistance equivalent number (PREN) against cost per unit strength, highlighting the corrosion–cost trade-off between the candidates. Materials in the upper-left region combine high corrosion resistance with acceptable cost, representing ideal choices for seawater service.

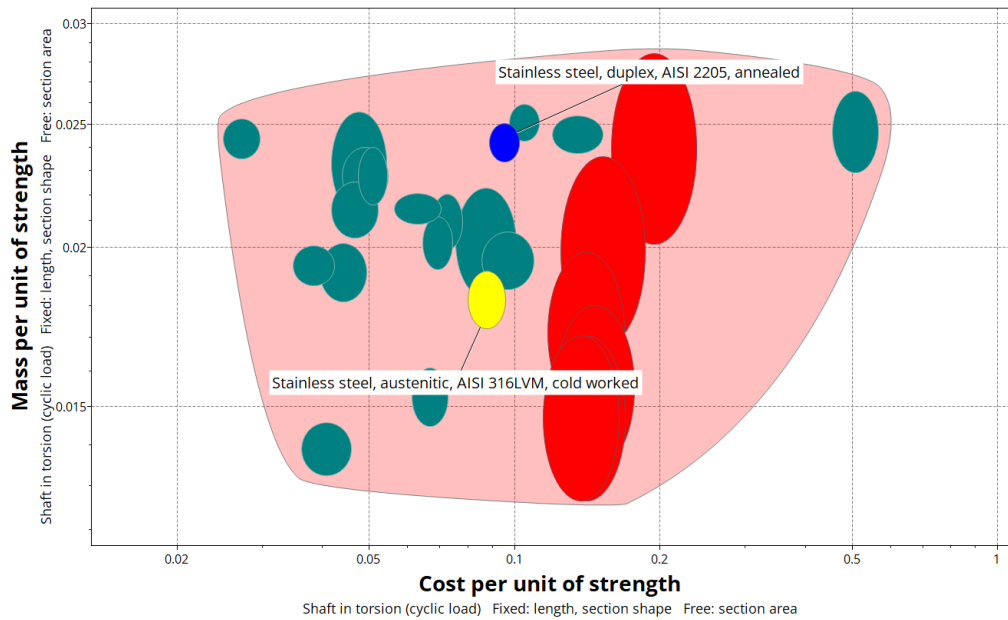


Figure 18 - Material performance map showing the relationship between cost per unit strength and mass per unit strength for marine shaft materials under cyclic torsion. 316LVM and Duplex 2205 occupy the most favorable region, balancing mass efficiency with moderate cost, while Ni-Cr-Mo alloys lie in the high-cost range. [27]

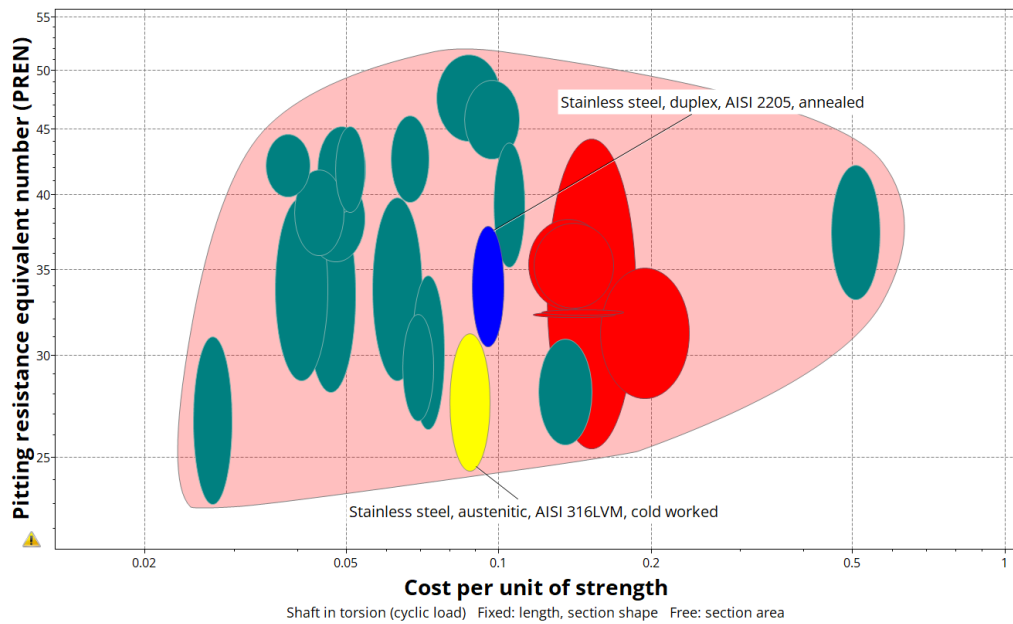


Figure 19 - Material performance map of PREN versus cost per unit strength. Duplex 2205 offers the best compromise of corrosion resistance and cost, 316LVM sits near the minimum acceptable corrosion threshold, and Ni-Cr-Mo alloys provide superior protection at a significant cost penalty. [27]

Table 5 - From the Granta comparison table (See annex A) [27]

Property / Index	316LVM (CW)	Duplex 2205 (Annealed)	Ni-Cr-Mo (ILLIUM 98, cast)
Fatigue Strength at 10^7 (ksi)	58.5–68.2	37.9–41.9	18.9–29.7
Density (lb/in ³)	0.287–0.290	0.282–0.283	0.300–0.303
Strength/Density (Index)	≈200–230 MPa/(Mg/m ³)	≈135–150 MPa/(Mg/m ³)	≈60–100 MPa/(Mg/m ³)
PREN	24.4–31.2 (borderline)	30.5–37.8 (safe margin)	~35–40+ (very high)
SCC Resistance	Slightly susceptible	Not susceptible	Not susceptible
Cost per unit strength	0.080–0.096	0.089–0.102	0.429–0.598

Interpretation:

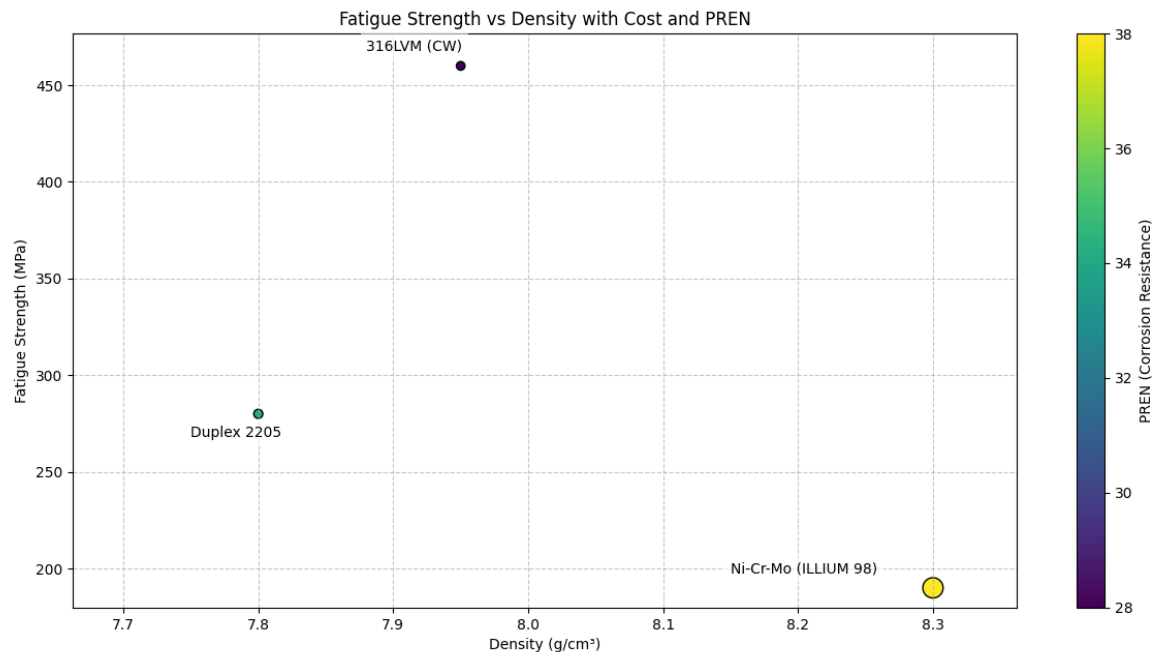


Figure 20 - Ashby-style comparison of shortlisted marine shaft materials showing fatigue strength versus density, with color representing pitting resistance (PREN) and marker size representing relative cost. 316LVM offers the highest strength but marginal corrosion margin; Duplex 2205 provides balanced strength, corrosion resistance, and cost; Ni-Cr-Mo (ILLIUM 98) exhibits excellent corrosion resistance but low strength and high cost.

- 316LVM leads in mechanical strength and fatigue index but barely meets the corrosion requirement (PREN near the cutoff). Risk of pitting in long-term seawater service.
- Duplex 2205 achieves a strong balance: adequate fatigue strength, excellent SCC resistance, PREN comfortably above 30, and only slightly higher cost than 316LVM.
- Ni-Cr-Mo alloy (ILLIUM 98) offers the best corrosion resistance (PREN high 30s–40s) but suffers from poor strength and 4–6× cost penalty, making it impractical for large shafts.

Ranking outcome:

- 1st choice: Duplex 2205 (optimum balance of fatigue, corrosion, cost).
- 2nd choice: 316LVM (higher strength but corrosion borderline).
- 3rd choice: Ni-Cr-Mo alloy (overkill corrosion resistance, uneconomical).

04.6 Results and Discussion

04.6.1 Comparison Table of Top Candidates (316L, Duplex 2205, NiCrMo Alloy)

Table 6 - Summarized from the Granta comparison table and corrosion resistance database see annex A. [27]

Attribute	316LVM (Austenitic, CW)	Duplex 2205 (Annealed)	Ni-Cr-Mo Alloy (ILLIUM 98, Cast)
Yield Strength (ksi)	107 – 141	66.7 – 74	37 – 46.4
Fatigue Strength at 10^7 (ksi)	58.5 – 68.2	37.9 – 41.9	18.9 – 29.7
Density (lb/in ³)	0.287 – 0.290	0.282 – 0.283	0.300 – 0.303
PREN (Pitting Resistance Eq. No.)	24.4 – 31.2 (borderline)	30.5 – 37.8 (safe)	~35 – 40+ (very high)
Stress Corrosion Cracking	Slightly susceptible	Not susceptible	Not susceptible
Cost per Unit Strength	0.080 – 0.096	0.089 – 0.102	0.429 – 0.598
Weldability	Excellent	Good	Good
Hot Forming	Excellent	Acceptable	Unsuitable
Seawater Resistance	Good	Good/High	Excellent

Engineering Interpretation (Strength vs Corrosion vs Cost)

- 316LVM:

Mechanically the strongest, with yield > 100 ksi and fatigue strength > 60 ksi, making it attractive for high torque capacity and long fatigue life. However, its PREN only just exceeds 30, meaning in warm, chloride-rich seawater it is on the edge of acceptable resistance. Pitting and crevice corrosion remain risks for long-term immersion. [29]

- Duplex 2205:

A balanced candidate. Its yield and fatigue strengths are safely above the design constraint (≥ 66 MPa yield, ≥ 30 ksi fatigue). While not as strong as 316LVM, its duplex microstructure provides high resistance to stress corrosion cracking and PREN consistently above 30. Manufacturing and welding are feasible, and its cost per unit strength is close to 316LVM. This alloy is widely adopted in marine shafting because it balances mechanical and corrosion requirements without major economic penalty. [30], [31]

- Ni-Cr-Mo Alloy (ILLIUM 98):

Corrosion-wise, this alloy is superior, with PREN in the high 30s–40s, effectively immune to seawater pitting. However, its yield and fatigue strengths are the lowest of the three, and its cost per unit strength is 4–6× higher. For large-diameter shafts, this cost is prohibitive. Thus, while technically robust against corrosion, it is impractical for full-shaft applications and is better suited for smaller, high-corrosion components (e.g., impellers, pump housings). [32]

04.6.2 Recommended Material

Based on the performance indices (strength/density, stiffness/density, cost/strength) and corrosion resistance considerations:

- Duplex Stainless Steel 2205 is recommended as the optimal material for the marine propeller shaft.

It provides:

- Adequate mechanical strength for torque and fatigue loading,
- High resistance to chloride-induced pitting and complete immunity to stress corrosion cracking,
- A reasonable cost level compared with Ni-based alloys,
- Proven service record in marine shafts and structural components.

316LVM remains a viable backup if cost and availability of Duplex 2205 become issues, but only with careful corrosion protection strategies (cathodic protection, coatings).

Ni-Cr-Mo alloy is eliminated on cost/strength grounds despite superior corrosion performance.

04.6.3 Justification – Duplex Stainless Steel 2205 Is the Final Choice

The selection of Duplex Stainless Steel 2205 is supported by a balanced combination of mechanical, corrosion, and economic performance metrics revealed through the ANSYS Granta screening and ranking stages. While austenitic 316LVM demonstrated superior tensile and fatigue strength, its pitting resistance (PREN 24 – 31) lies only marginally above the seawater durability threshold, making it vulnerable in long-term immersion. Ni–Cr–Mo alloys, although exceptionally corrosion-resistant, were eliminated due to their limited mechanical strength and significantly higher cost.

Duplex 2205 offers an optimal compromise: a yield strength of 66 – 74 ksi and fatigue strength of 38 – 42 ksi well above the design minimum combined with a PREN range of 30.5 – 37.8, ensuring reliable resistance to pitting, crevice corrosion, and stress corrosion cracking in chloride-rich environments. Its balanced ferritic–austenitic microstructure provides superior toughness and fracture resistance compared to other candidates, while its weldability and manufacturability remain acceptable for large-diameter shaft fabrication. Importantly, Duplex 2205 achieves this performance without the substantial cost penalty associated with nickel-rich alloys.

Based on these combined factors, Duplex Stainless Steel 2205 emerges as the most technically and economically viable material for the marine propulsion shaft, ensuring long service life and operational safety under aggressive marine conditions.

04.7 Conclusion

The material selection process for the marine propeller shaft was conducted using the structured methodology of Translate – Screen – Rank – Document. Key constraints were applied to ensure mechanical strength (yield ≥ 66 MPa, fatigue ≥ 30 ksi), manufacturability (weldability, hot forming), and corrosion resistance (PREN ≥ 30 , SCC not susceptible).

From over 1900 candidate materials in the Ansys Granta database, 25 survived screening. The proposed options – 316L stainless steel, Duplex 2205, and Ni-Cr-Mo alloys – were confirmed in this shortlist.

Ranking using performance indices (strength/density, stiffness/density, cost/strength) combined with corrosion metrics (PREN) revealed:

- 316LVM: very high strength but borderline corrosion resistance.
- Ni-Cr-Mo alloy: superior corrosion resistance but weak and uneconomical.
- Duplex 2205: strong balance between strength, fatigue resistance, PREN (>30), immunity to SCC, and moderate cost.

Recommended material: Duplex Stainless Steel 2205 (UNS S31803/S32205) is the optimal choice for the propeller shaft, offering durability, safety, and cost-effectiveness in seawater immersion conditions.
(To see the data sheet of selected material go to annex B)

05. Corrosion Protection Strategy

Marine propulsion shafts operate in one of the most corrosive engineering environments, where continuous exposure to high-salinity seawater, oxygen gradients, wet–dry cycling, and hydrodynamic stresses can rapidly degrade metallic surfaces. Even alloys with strong intrinsic corrosion resistance such as Duplex Stainless Steel 2205 require supplementary protection to prevent localized attack such as pitting, crevice corrosion, galvanic coupling, and corrosion-fatigue. A long-term protection strategy must therefore integrate electrochemical control and engineered surface treatments to reduce the corrosion rate, limit hydrogen uptake, maintain coating integrity, and ensure that the shaft remains structurally reliable throughout its service life.

This section evaluates two complementary approaches:

- **05.1 – Surface Treatments and Coating Systems**, detailing mechanical surface conditioning, chemical passivation, multilayer marine coatings, and specialized localized treatments for journals and splines.
- **05.3 – Cathodic Protection (CP)**, assessing galvanic and ICCP systems, current-demand calculations, shaft earthing requirements, and hydrogen-related constraints for duplex steels.

Together, these methods form a multi-layer corrosion-prevention strategy suitable for long-term operation under seawater immersion and splash-zone conditions.

05.1 Surface Treatment & Coating Design for Marine Propulsion Shaft

05.1.1 Shaft Parameters

- ✓ Shaft length: 1.6 m
- ✓ Outer diameter: 125 mm
- ✓ Inner diameter: 60 mm (hollow shaft)
- ✓ Engine power: 300kW
- ✓ Torque: 12.6 kNm
- ✓ Rotational speed: 225 RPM
- ✓ Material: Duplex Stainless Steel 2205
- ✓ Prepared by: CNC machining
- ✓ Treated by: Solution annealing
- ✓ Corrosion rate (untreated): 0.2 mm/year

- **Final Shaft Geometry:**

- Hollow shaft outer diameter: 125 mm
- Inner bore: 60 mm (hollow shaft)
- Length: 1.5 m (Without thread for nut)
- Spline length: Defined to match torque requirements
- Taper length and thread length: Calculated to avoid overlap and stress concentration
- Spline depth: 5 mm
- Flange outer diameter: 250 mm
- Outer diameter of thread part (Solid): 99 mm (with 2 mm clearance)

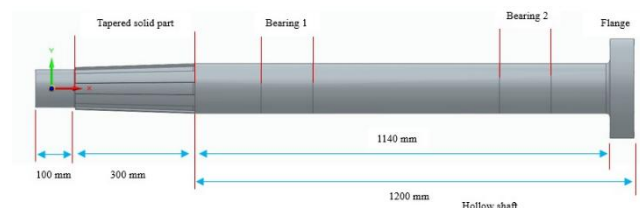


Figure 21 - Final shaft geometry

05.1.2 Marine Propeller Shaft Environment

- Type of environment: Seawater immersion and splashing zone.
- Temperature: Around 25 °C average.
- Salinity: Approximately 3.5% (35 ppt), high chloride concentration
- Electrochemical nature: Seawater acts as a strong electrolyte, accelerating corrosion reactions

- Oxygen availability: Alternating wet–dry and splash exposure zones cause differential aeration, leading to localized corrosion
- Corrosion rate (untreated): ~0.2 mm/year in seawater for duplex stainless steel under natural conditions.
- Key challenges: Continuous exposure to chloride ions, oxygen gradients, and cyclic loading that accelerate material degradation.

05.1.3 Common surface treatment methods suitable for the marine propeller shaft

In marine applications, a propeller shaft is made from a solid rod of high-strength material, such as Duplex stainless steel 2205. The marine environment already has a severe impact on any metal surface exposed to seawater. Even if the shaft is well-made, it will require additional protective treatment to extend its life cycle if left untreated.

Once the shaft has been manufactured, it will undergo several stages of machining where new metal surfaces will be created and the protection (passive film) described previously; will be removed; this will increase the exposure of the metal to the ocean seawater (chlorine) and increase the susceptibility of the shaft to damage due to chloride pitting, crevice corrosion, friction and fatigue damage.

To ensure that the marine industry can support and sustain a longer life cycle for the propeller shaft, they use several surface treatment techniques designed to improve the corrosion, wear, and fatigue resistance of the fully machined surface.

The most common methods used on solid machined propeller shafts include:

- Pickling and passivation.

Pickling removes the heat-marked surface of the shaft's material and creates the passive layer of chromium-rich material protecting the Duplex stainless steel in contact with seawater.

- Electropolishing.

Electropolishing will produce a very smooth and clean surface, which will reduce the number of micro-crevices that chlorides can penetrate; as a result of this, the corrosion resistance of the machined surfaces improves considerably.

- Shot peening or Laser peening.

Shot peening and laser peening are methods of coating the surfaces of machines where the shaft will be subjected to stress. By using these methods, manufacturers will create compressive residual stresses, which will help to strengthen those areas and therefore increase the overall fatigue strength of the product.

- PVD Coatings (DLC, CrN, etc.).

It will be selectively applied to the highest wear areas, e.g., bearing journals, spline connections, to reduce friction and fretting but not negatively impact corrosion resistance.

- Marine Coatings (Only in Splash Zone)

Above the waterline, epoxy primers and polyurethane topcoats only protect the above-waterline exposed area from wet-dry cycle corrosion.

- Cathodic Protection (Submerged Areas)

Sacrificial anodes or ICCP systems are used to provide long-term protection of the submerged section of the shaft against localized corrosion and long-term pitting.

These processes collectively restore surface integrity following machining, increase the strength of critical areas, and provide long-term protection for the solid machined shaft when being used in a corrosive marine environment.

05.1.4 The Best Combined Surface Treatment Scenario

The best surface-treatment plan for duplex stainless-steel marine propeller shafts includes a multi-layer protection strategy that involves:

1. Surface Preparation & Passivation (Base Protection):

This step ensures that after machining, the shaft surface undergoes mechanical polishing to eliminate machining marks and reduce pitting sites, followed by chemical passivation with nitric or citric acid to enhance the chromium oxide passive layer. Optional low-temperature solution annealing may also be pursued to relieve machining stresses. The benefits of this process include maximizing the corrosion resistance of Duplex 2205, removing iron contaminants, and improving resistance to chloride-induced pitting in seawater.

2. High-Build Marine Epoxy Coating (Primary Barrier Layer):

A two-part high-solids epoxy is applied over the passivated surface, known for its excellent seawater resistance, strong adhesion to duplex steel, and mechanical durability against various marine conditions. The coating is typically applied in thicknesses of 250–350 µm across 2–3 layers. Its primary function is to act as a corrosion barrier, preventing chloride penetration and reducing the risk of crevice corrosion.

3. Polyurethane (PU) Topcoat (UV & Impact-Resistant Layer):

To protect the epoxy layer from UV damage, a PU topcoat is applied, providing excellent UV resistance, a smooth finish to minimize drag and marine growth, and protection against impact and abrasion. This layer is typically 50–100 µm thick.

4. Anti-Fouling Top Layer (Optional but Highly Recommended):

For sections that are frequently submerged, especially near the propeller area, a self-polishing anti-fouling paint is recommended to prevent growth from barnacles, algae, and shells, reducing hydrodynamic drag and extending maintenance intervals, particularly on the lower solid shaft sections, tapered section, and hollow tail near the propeller hub.

This comprehensive treatment plan is fully practical and widely used in marine shaft manufacturing, offers excellent protection against various corrosion types, requires standard industrial equipment, and poses no risk of hydrogen embrittlement for duplex SS 2205.

05.1.4.1. Surface Preparation and Passivation

Process steps:

- Final machining & deburring
- Mechanical finishing (grind/polish)
- Shot peening
- Alkaline degrease
- Rinse using water
- Pickling (commercial duplex-safe pickling product recommended)
- Neutralize & rinse thoroughly
- Optional: Electropolishing (recommended before final passivation)
- Citric or nitric passivation per ASTM A967 / AMS-2700
- Final rinse, dry, and QC tests (visual, passivation test, Ra, optional lab tests)
- Masking or protection before any later coatings

Final machining, deburring, and mechanical finishing involve completing all final processes, such as turning and finishing surfaces to specified tolerances while ensuring the removal of burrs and sharp edges. Journals should achieve a surface roughness of Ra 0.4–0.6 μm , with general shafts at Ra ≤ 0.8 μm . Shot or laser peening should precede chemical cleaning to enhance fatigue resistance, followed by an appropriate wash to clear contaminants.

05.1.4.1.1. Shot peening (Mechanical finish)

Shot peening is a mechanical treatment to improve corrosion-fatigue resistance by creating compressive residual stresses on the surface of the shaft, which is key for marine propeller shafts subjected to cyclic loading, vibration, and exposure to the corrosive nature of seawater. The surface treatment begins with cleaning the surfaces to remove the oil, dirt, and debris. The shaft is then bombarded with spherical shots (steel, ceramic, or glass) to induce controlled conditions to plastically deform the surface layer. This proves to delay the eventual initiation and propagation of cracks during fatigue with corrosion present in the environment. After shot peening, we expect to achieve improvements to the corrosion-fatigue modulus, a reduced chance of surface crack initiation, and improved service life under cyclic marine loading.[33]



Figure 22 - Shot peening

- **Small shot (≈ 0.3 mm) at low speed** delivers very low impact energy (microjoules to millijoules). Even with high coverage, the total energy on the shaft stays very small (under ~ 1 kJ). This produces only shallow peening and minimal deformation.
- **Medium shot (0.6–0.9 mm) at moderate speed** gives impact energies in the millijoule range, resulting in total energies in the kilojoule range for the whole shaft. At high speeds and high coverage, this can increase to tens of kJ. This provides a good balance of useful compressive stress and controlled surface deformation.
- **Large shot (≥ 1.2 mm) at high speed** produces very high impact energy per strike. Total energy for the shaft can reach hundreds or even thousands of kJ. This causes deeper plastic deformation and high Almen intensities, but also increases the risk of over-peening and distortion if not carefully controlled.

It is optimal to apply medium-sized shots (0.6-0.9 mm) of steel, ceramic, or glass, using controlled intermediate velocity and uniform coverage to the Duplex 2205 marine shaft. When used in combination, these factors result in the development of high compressive residual stresses and enhanced corrosion-fatigue resistance while eliminating the potential of over-peening, over-deformation, and surface damage.

Cleaning with Alkaline Degreaser

The degreasing purpose removes oil, machining fluids, fingerprints, and organic contaminants.

Procedure:

1. Prepare an **alkaline cleaner** (pH 10–12) suitable for stainless steel.
2. Heat the solution to **40–60 °C** if allowed by the product.
3. Immerse or wipe the shaft surface for **5–10 minutes**.
4. Lightly scrub with a nylon brush in areas with heavy oils.

We need to avoid cleaners that contain chloride. Ensure full wetting of the surface.

Rinse with Clean Water

Rinsing into Clean Water: Rinse thoroughly with clean running water.

Continue to rinse until no oily or soapy residues remain. Use demineralized water if you live in an area with hard water, to prevent the formation of mineral spots.

05.1.4.1.2 Pickling

Pickling is the process of cleaning metals of heat discoloration and other surface contaminants created in the fabricating process. Pickling is done by using a special formulated pickling solution (gel) containing nitric and hydrofluoric acids, and the application of the pickling solution (hydrofluoric and nitric acid) to the surface of the metal by use of a brush or immersion. Pickling should be maintained on the surface of the metal for 15 - 30 minutes by following the manufacturer's instructions.[34]

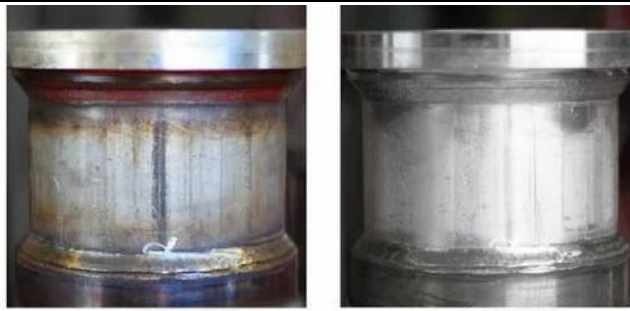


Figure 23 - The difference between a surface that has been pickled and one that has not. Pickling provides protection.

A neutralization bath is used to stop the acid from causing further damage to the metal and to restore its chemical properties. A neutralization bath may consist of a weak alkaline solution (sodium bicarbonate or sodium hydroxide diluted to approximately pH 9 -10). The solution is applied to the surface of the metal and sits for 5 minutes, after which the metal should be rinsed several times with clean water until it is pH neutral (check with pH strips), and finally allowed to air dry. After the surface of the metal has been neutralized, it will be chemically clean and activated, and the metal should be passivated or coated with an appropriate coating as soon as possible to prevent flash corrosion.

05.1.4.1.3 Electropolishing (Optional but recommended)

While electropolishing is typically seen as an optional step that can follow the pickling process, as not every application requires ultra-smooth surfaces, the advantages associated with electropolishing (such as decreasing micro-crevices and providing systemic corrosion resistance, etc.) are quite substantial. It is recommended that electropolishing be viewed as a potential step for the best combination of surface treatments, particularly if low Ra values are critical or corrosion resistance is paramount. In cases where it is still treated as an optional step, criteria and specifications should be clearly defined to enable manufacturers to determine under what conditions electropolishing is advisable and under what conditions it is not; for example, defining surface roughness targets ($Ra \leq 0.3 \mu m$) or seawater exposure conditions where the impact of corrosion on finished products will be maximized. We need to do the electropolishing process for the bearing journals to achieve more smoother finish. [35]



Figure 24 - An Electropolished surface

05.1.4.1.4 Passivation

Chemical passivation is a post-fabrication process that removes free iron from the surface of the steel, leaving a stable chromium-rich passive oxide film (Cr_2O_3) that lends corrosion resistance to stainless steel. This process is important for reducing localized corrosion initiation points on Duplex Stainless

Steel (DSS) 2205. Two methods are present: Nitric acid passivation, which uses high percent concentrations of nitric acid at moderate temperatures during processing, and citric acid passivation, which uses lower concentrations and relies on the chelation of iron ions. Both methods are viable for marine propeller shaft applications when properly controlled; however, the citric acid passivation process is the most effective and environmentally safe method recommended by the American Society for Testing and Materials (ASTM) A967, standard practice for cleaning and passivation of stainless steel. The citric acid passivation process quickly removes free iron and other contaminants from the stainless-steel surface while also promoting an increase in the formation of the chromium-rich oxide film, providing an excellent level of resistance against pitting, crevice, and corrosion fatigue typical of marine environments. Citric acid passivation leaves a clean, smooth, and passive surface while physically damaging the duplex microstructure. Citric acid is non-toxic to work with and safer for handling than nitric acid and is less aggressive, making it a more useful and appropriate option for marine propeller shafts and components requiring high corrosion resistance and long-term durability.[36]

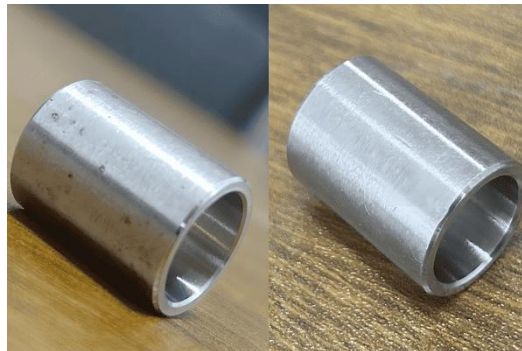


Figure 25 - The difference between a surface that has been passivated and one that has not.

Optimal parameters:

- Acid concentration: 4–10 wt.% citric acid
- Temperature: 49–60 °C (ideal range)
- Immersion time: 10–30 minutes
- pH: 2–4
- Post-treatment: Thorough deionized water rinse and drying

To ensure that residues are completely removed from the parts after surface finishing operations and to eliminate the chances of staining or rusting, all parts will be subjected to extensive rinsing and drying. Quality assurance checks will include visual inspection, a test for passivation (such as the water break test), measurement of Ra, and potentially lab tests for corrosion resistance or pitting.

Before any further coatings or assembly operations commence on the parts, critical areas, and in some cases the entire surface will be protected from contamination, scratches, or corrosion by being temporarily covered or protected with masking material, tape, or other means.

05.1.4.2. High-Build Marine Epoxy Coating (Primary Barrier Layer)

After surface preparation and passivation, a high-build marine epoxy coating serves as the primary corrosion barrier for the Duplex 2205 propeller shaft. The shaft's surface must be meticulously cleaned and dried, with protection against fingerprints, especially on critical areas like bearing journals and threaded ends, which are masked prior to a final solvent wipe using acetone for chemical cleanliness.

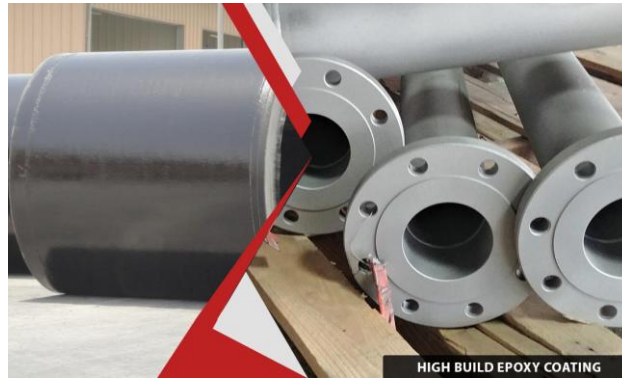


Figure 26 - The surface that has been high-build epoxy coated

The application process emphasizes stringent environmental control, as epoxy adhesion is significantly influenced by temperature and humidity, requiring the substrate temperature to be at least 3 °C above the dew point, with ambient conditions typically between 10–35 °C and humidity below 85%. The two-component high-build epoxy needs to be mixed according to the manufacturer's specified ratio and applied within its defined pot life to ensure uniform curing and prevent weak spots.

Application commonly involves airless spray equipment to maintain consistent thickness, with the first coat achieving a dry film thickness of 125–175 µm, followed by a second coat adding 100–150 µm. In particularly harsh marine environments, a third coat may be necessary, leading to a total target thickness of 225–350 µm. If the maximum recoat interval is surpassed, light sanding and solvent cleaning are required to ensure layer adhesion.

Quality assurance checks are integral throughout the coating process, verifying film thickness and coating integrity. Dry film thickness is measured at various points, and visual inspections test for defects such as runs, sags, or pinholes. For sections exposed to seawater, a holiday test may identify any coating gaps, and adhesion tests can be conducted per standards or client requirements.

After applying all coats, the shaft needs to cure fully before handling or installation; this curing time varies with temperature but generally extends over a few days for the epoxy to achieve full hardness. Once cured, the shaft should be stored in a clean, dry, dust-free environment, protected with breathable material. This high-build epoxy coating effectively resists chlorides, wet–dry cycling, abrasion, and mechanical exposure in marine environments. [37], [38]

05.1.4.3. Polyurethane Topcoat (UV + Mechanical Protection)

The polyurethane (PU) topcoat is the final layer of protection for a marine vessel's propeller shaft surface treatment system. The purpose of this topcoat layer is to increase the coating's durability and lifespan by protecting the substrates from damaging factors such as UV radiation (a primary cause of the breakdown of coatings) and other damaging events that can occur during use or maintenance. In addition to protecting against UV light, it provides mechanical protection against scratches, dents, and minor abrasions during use or maintenance.



Figure 27 - The surface that has been polyurethane top coated

The chemical resistance qualities of this topcoat protect the marine vessel's propeller shaft from exposure to low concentration levels of pollutants found in the ocean, as well as from salt spray and other chemicals present at sea and from the environment, all while maintaining a smooth and uniform finish. This uniform surface is important since it reduces friction and minimizes the number of debris accumulating on the surface. Protecting the high-build barrier epoxy beneath the topcoat helps to preserve its level of corrosion resistance and structural integrity for extended periods of time due to the lack of exposure to environmental and operational stresses.[39], [40]

Unique Features and Advantages

- UV Protection: Preventing degradation, discoloration, and chalking of the layers beneath the topcoat.
- Mechanical Protection: Providing resistance to abrasion and impact, thus extending the life of the surface.
- Chemical Resistance: Closing off exposure to marine pollutants, to low acid concentration levels, and to contaminants such as salt spray.
- Aesthetic Finish: Providing a smooth surface finish that reduces friction and increases the hydrodynamic efficiency of the vessel.

Application Considerations

- Apply only after the epoxy barrier is fully cured.
- It is better to apply several thin coats rather than one thicker coat to prevent runs or inconsistency in the cured surface.
- Apply with brush, roller, or spray, depending on how easy it will be to access the area and what kind of shape the area will have.

05.1.4.4. Anti-Fouling Layer (Optional, for Immersed Parts)

To avoid fouling of marine organisms such as barnacles, algae, and biofilms, an anti-fouling coating is applied to the underwater portion of a marine propeller shaft (APC). The fouling of marine organisms increases drag, decreases efficiency, and can lead to the accumulation of corrosive material on the metal surface. Anti-fouling coatings work either by gradually releasing biocides (e.g., Copper

compounds) or by creating smooth, non-fouling surfaces that prohibit organisms from securely attaching themselves. Anti-fouling coatings aid in smooth rotation, lower maintenance needs, and reduce fouling-induced corrosion.[41], [42]



Figure 28- After applying the anti-fouling layer on the shaft

Common Types of Anti-Fouling Coatings Used for Shafts

1. Self-polishing/ablative coatings: They gradually wear away to provide exposure to fresh biocide.
2. Hard matrix coatings: These coatings provide a slow release of biocide and provide excellent resistance to abrasion.
3. Foul-release silicone coatings: These coatings create non-toxic, smooth surfaces that will not deposit marine growth.

The two primary benefits of an anti-fouling layer are as follows

- The anti-fouling layer prevents the attachment of marine growth such as barnacles, algae, and slime.
- The anti-fouling layer allows a propeller shaft to maintain its hydrodynamic efficiency by decreasing drag and allowing smooth rotation.

In addition, an anti-fouling layer can prevent the localized corrosion that is associated with marine growth by maintaining the protective coating over the areas that accumulate marine growth. The amount of time between cleaning and inspecting the propeller shaft can also be increased by using an anti-fouling layer.

We should note the following when applying the anti-fouling material

- The anti-fouling material must be applied to a fully cured polyurethane topcoat for maximum adhesion.
- Be sure the anti-fouling coating you are using is compatible with the existing coating.
- Depending on the geometry of the shaft, the anti-fouling material can be applied by brush or spray.
- For long-term services, you should periodically check the anti-fouling layer for effectiveness.

05.1.4.5. Dry film lubricant coatings (only for splined area)

A Dry Film Lubricant (DFL) Coating is the easiest and most practical Way to coat the splines. DFL Coating is typically Resin Bonded Molybdenum Disulfide (MoS_2). The multipart approach of using a DFL Coating provides a simple, practical way to create extremely low-friction surfaces with high Anti-

Galling Properties, Without Requiring Dramatic Logistical Changes. The application process includes Conventional Spray Application in Thin Coats, 8-20mm thick, with preparation for the Application including the sanding (Grit Blasting) of the surfaces being treated, followed by either Air Cured or Low-Temperature Thermal Cured DFL Coating. Because of the nature of DFL Coating's simple, practical method of application, it does not require the use of High-Vacuum Chambers (for PVD) or High-Velocity Robotics. For a localized application on a shaft that measures 1.5 meters and the need for PVD methods would be counterproductive, as both PVD and HVOF would be costly and impractical alternatives due to the time and expense associated with them.[43]

05.1.5 Summary

Stage 1: Universal Surface Preparation

Universal Surface Preparation, where the entire shaft undergoes a series of mechanical and chemical treatments. This sequence starts with mechanical polishing to refine the surface and is followed by shot peening to improve the surface's resistance to fatigue. The shaft is then thoroughly cleaned via alkaline degrease and a subsequent rinse to remove all contaminants. Finally, the surface is chemically treated with pickling to remove any scale or oxides, which is then immediately followed by a neutralization step to halt the acid's chemical reaction, leaving the surface ready for the specialized finishing stages.

Stage 2: Primary Protection and Painting (General Shaft Body)

For the general shaft body, excluding the bearing journals, the first protective step is passivation using citric acid to enhance the material's inherent corrosion resistance. This is followed by the application of a high-build marine epoxy coating, which serves as a robust barrier against the seawater environment. The third step involves applying a polyurethane top coat to provide durability, UV resistance, and a stable finish. Finally, an anti-fouling layer is applied as the outermost protection to prevent the attachment and growth of marine organisms.

Stage 3: Specialized Area Treatments

The bearing journals diverge from the main process immediately after the universal surface preparation is complete, receiving only electropolishing as suitable to achieve the required ultra-smooth finish for optimal bearing performance. Separately, the splined area is the last component to be treated, where a dry film lubrication coating must be applied after all other processes are finalized, ensuring reduced friction and resistance to wear and galling in this critical mechanical interface.

05.1.6 Coating Breakdown Factor

Typically, the coating breakdown factor represented as f_c , the coating breakdown factor is a dimensionless metric that measures how likely it is that a protective coating will fail during its design life. It is frequently used in the chemical, offshore, and marine industries to assess coating performance and the necessity of extra protection, like cathodic protection.

Low f_c provides an effective and durable coating, while coatings with a high f_c value may fail and require supplementary protection. [44]

There are 4 types of coating available.

Table 7 - Coating types

Category I	One layer of epoxy paint coating, minimum 20 μm nominal DFT.
Category II	One or more layers of marine paint coating (epoxy-, polyurethane- or vinyl-based), total nominal DFT minimum 250 μm .
Category III	Two or more layers of marine epoxy-based paint coating total nominal DFT min. 350 μm .
Category IV	Two or more layers of epoxy-based coatings according to NORSOK M-501 systems 3B or 7, with total nominal DFT of minimum 350 μm or the equivalent and meeting all requirements for coating system qualification, coating application, inspection and repairs given in the same standard. The purchaser is further required to be involved in quality control by accepting the coating application procedure and carrying out quality surveillance during coating work.

We are using a Multilayer coating system with 3 layers of epoxy coating, one layer of PU coating, and finally one layer of anti-fouling layer. So, we need to check the coating breakdown factor, and that is given by:

$$f_c = a + b \cdot t$$

f_c = Coating breakdown factor (dimensionless)

a = Initial defect/conductivity factor of the coating (accounts for pores, pinholes, and initial imperfections)

b = Degradation rate of the coating per year (environmental, mechanical, or chemical effects)

t = Design life of the coating (in years)

The proposed coating system's category is designated as **Category IV equivalent** because it combines a minimum 350 μm film thickness requirement (met by Category III/IV DFTs) with a multi-layer design (Epoxy, Polyurethane, Anti-Fouling) that exceeds the performance and durability criteria for corrosion, abrasion, and UV resistance necessary for very harsh marine/offshore environments.

Table 8 - Recommended values for a and b constants for paint coating breakdown factor. (Coating categories are defined in table 13)

Depth [m]	Recommended values for a and b constants for coating categories I, II, III and IV, see [3.4.7]			
	I ($a = 0.10$)	II ($a = 0.05$)	III ($a = 0.02$)	IV ($a = 0.02$)
0-30	$b = 0.10$	$b = 0.025$	$b = 0.012$	$b = 0.008$
> 30	$b = 0.05$	$b = 0.015$	$b = 0.008$	$b = 0.005$

According to that

$$f_c = a + b \cdot t$$

$$f_c = 0.02 + 0.008 \times 10$$

$$f_c = 0.1$$

That means $f_c = 0.1 = 10\%$ Coating is effective for a 10-year life span.

No additional cathodic protection is strictly required, but monitoring is recommended in very aggressive marine conditions.

05.1.7 Quality Control

After each treatment and periodically in service, the following quality control measures should be applied.[45], [46]

05.1.7.1 Visual and Surface Inspection

Surface visual inspection of the shaft should be performed at the end of each treatment and after every slight elongation of the shaft, and the treatment should be checked at the same time to look for any possible defects (i.e. corrosion spots, cracks, etc.), which, are the most common ones, such as pitting, wear marks, and coating delamination, that can be overlooked. Advanced methods like testing with magnetic particles or dye penetration can reveal faults that are not only hidden but are also very close to the surface and are otherwise undetectable. It is also a good practice to measure the surface roughness before coating to ensure proper surface preparation.

Relevant ASTM standards: ASTM E165 – Standard Practice for Liquid Penetrant Testing; ASTM E709 – Standard Guide for Magnetic Particle Testing; ASTM D4417 – Field Measurement of Surface Profile of Blast Cleaned Steel.

05.1.7.2 Coating Thickness and Uniformity

The thickness and the uniformity of coatings must be assessed in order to ensure good protection. The non-destructive methods include eddy-current, magnetic induction, and ultrasonic gauges. On the other hand, cross-sectional metallography, which is a destructive method, gives the best insights into the coating thickness, porosity, and bonding. The spec should be the same as the designed and uninterrupted surface without any voids or cracks.

Relevant ASTM standards: ASTM B487 – Standard Test Method for Measurement of Coating Thickness by Microscopy; ASTM B499 – Standard Test Method for Coating Thickness by X-ray Spectrometry; ASTM D4414 – Standard Practice for Measurement of Wet Film Thickness by Notch Gages.

05.1.7.3 Adhesion and Bond Strength

The strength of the bond between the coating and the substrate is very important for the performance of the coating. The coatings are always tested for their adhesion; the methods used to perform the tests include the pull-off tests, during which the tensile bond strength is determined, and the scratch tests, which establish the critical load for detachment or cracking of the coating. The testing of adhesion is based on either the ASTM or ISO standards, and it is carried out on coatings like HVOF WC–Co or Cr₃C₂–NiCr to ensure they are of sufficiently high adhesion to the shaft surface.

Relevant ASTM standards: ASTM D4541 – Pull-Off Strength of Coatings Using Portable Adhesion Testers; ASTM C1624 – Scratch Test for Adhesion of Coatings; ISO 4624 – Pull-Off Test for Adhesion; ISO 20502 – Scratch Test for Coating Adherence.

05.1.7.4 Residual Stress Measurement

Verification of the application of residual compressive stresses is one of the measures used nowadays for shot-peened surfaces in order to extend the fatigue life. X-ray diffraction, besides being a non-destructive method, yields a surface stress measurement, while the hole-drilling method provides a

semi-destructive profile of the stresses that occur under the surface. By these methods, one can confirm that the intended compressive stress has been applied successfully.

Relevant ASTM standards: ASTM E837 – Standard Test Method for Determining Residual Stresses by the Hole-Drilling Strain-Gage Method; ASTM E915 – X-ray Measurement of Residual Stresses.

05.1.7.5 Hardness Testing

Hardness tests are employed as a measuring tool to determine if the coatings or treated layers have the required resistance to wear and mechanical integrity. Microhardness tests (Vickers/Knoop) are more suitable for thin coatings, whereas Rockwell or Brinell tests are applicable for thicker layers. Moreover, cross-sectional hardness mapping can find application in the assessment of graded coatings or multilayer surfaces.

Relevant ASTM standards: ASTM E384 – Microindentation Hardness of Materials; ASTM E18 – Rockwell Hardness Test; ASTM E10 – Brinell Hardness Test.

05.1.7.6 Corrosion Resistance Testing

The corrosion resistance of the treated surfaces must be evaluated in the laboratory and in the field. Testing methods such as salt spray or electrochemical measurements are used to simulate the most extreme conditions. Monitoring of open-circuit voltages and determining the impact of galvanic coupling are examples of field checks. During the testing, the coating should prevent any occurrence of pitting, cracking, or degradation.

The applicable ASTM standards are: ASTM B117 – Salt Spray (Fog) Testing; ASTM G5 – Measurement of Galvanic Currents; ASTM G61 – Electrochemical Evaluation of Corrosion.

05.1.7.7 Dimensional Inspection

Once the treatments are complete, the shaft's dimensions need to be checked so that the tolerances can still be considered to be within the set limits. There can be a mixture of methods, including calipers, micrometers, and coordinate measuring machines (CMM), for checking the diameters, roundness, and concentricity. The thickness of the coating must be monitored so that it does not affect the assembly or operation.

The relevant ASTM standards are: ISO 1101 – Geometrical Product Specifications (GPS); ASME Y14.5 – Dimensioning and Tolerancing.

05.1.7.8 Documentation and Traceability

Total documentation of the quality control results is a requirement. This should include process parameters, measurement methods, and acceptance criteria. The occurrences of deviations and corrective actions should be documented to enhance the traceability and repeatability of surface treatment processes.

The relevant ASTM standards are: ASTM D7127 – Standard Guide for Recording Coating Thickness Measurements; ASTM E2214 – Standard Practice for Maintenance and Reporting of Corrosion Control Data.

05.1.8 Repair and Maintenance

Even well-treated shafts will require periodic maintenance and occasional repairs. Here are the key tasks.

05.1.8.1 Regular Cleaning and Inspection Schedule

In order to maximize the performance and durability of treated shafts, it is necessary to put in place regular cleaning and inspection management. Cleaning involves the removal of surface contaminants, such as dirt, grease, and salt deposits, and also the visual survey of the surface for minor coating loss, corrosion, cracks, or peel-off. Continuous monitoring enables to prediction and addressing of the problems before they become serious and guarantees that the shaft operates within the designed limit.

05.1.8.2 Local Repair of Coatings

Localized coating damage in the area marked should be removed gently, followed by substrate preparation and coating re-application or overlay build-up. Good surface treatment, like grit blasting or degreasing, is very important for the repair coating adhesion. This ascertained method averts minor imperfections from escalating to large-scale corrosion or wear problems.

05.1.8.3 Re-application or Refurbishment of Coatings

For situations where there is more extensive wear or degradation and the existing coating has to be taken off, a new full-thickness coating will be applied. Refurbishment guarantees that protective layers keep their uniformity in thickness, adherence, and resistance to corrosion. The same rigorous surface preparation and application procedures must be followed as were used for the original coating.

05.1.8.4 Maintenance of Galvanic or Cathodic Protection Systems

Shafts that are protected by sacrificial anodes or impressed current cathodic protection (ICCP) need regular evaluations to be done. This will involve measuring potentials, estimating the remaining life of the sacrificial anode, replacing the anodes that have been depleted, and checking the insulation of the electrical system. The proper upkeep of these systems works alongside surface treatments to keep corrosion away in the critical oceanic or chemical environments.

05.1.8.5 Surveillance of Fatigue and Corrosion-Fatigue Prone Areas

The zones in proximity to couplings, bearing seats, and geometrical transitions are the ones that might be affected in the first place by fatigue and corrosion-fatigue. Therefore, those areas must be visually inspected for surface cracks, corrosion pits, or micro-damage. The sooner the problem is detected, the sooner preventive repairs can be done, minimizing the risk of unplanned shaft failure.

05.1.8.6 Documentation and Service History

The activities related to inspections, upkeep, repairs, and refinishing should be very well documented. The records should contain coating thickness measurements, inspection findings, defects spotted, and any actions taken to correct them. The maintenance of a comprehensive service history is of utmost importance for the integrity assessments in the future and for the planning of maintenance schedules.

05.2 Electrochemical Protection Mechanisms

05.2.1 Purpose and scope

Electrochemical corrosion control in seawater is achieved by altering interfacial potentials and reaction kinetics so that metal dissolution is minimized, or a stable passive state is maintained. Two canonical methods are employed: cathodic protection (CP) and anodic protection (AP). This section sets out their principles, operating envelopes, instrumentation, secondary effects, and relevance to a rotating shaft/propulsor assembly in seawater. Standards and class guidance cited here include ISO 12473 (general principles and criteria for CP in seawater), DNV-RP-B401 (CP design), DNV-RP-F112 (duplex stainless under CP; HISC avoidance), and ABS Guidance Notes for ships (arrangements and control of ICCP/galvanic systems) [47], [48],[49],[50]

05.2.2 Cathodic protection (CP)

05.2.2.1 Principle

CP suppresses anodic dissolution by shifting the metal's potential in the negative direction until the corrosion current is reduced to negligible levels. Current is supplied either by galvanic (sacrificial) anodes, Al/Zn alloys more electronegative than the protected structure, or by impressed-current anodes (ICCP) driven by a controlled DC source with inert anodes (e.g., platinized titanium/MMO). ISO 12473 defines CP as the reduction in corrosion rate by decreasing the corrosion potential and explains that protection current can be provided by galvanic or impressed-current systems [49]. It also introduces terms used throughout (calcareous deposits, cathodic disbondment, hydrogen embrittlement).

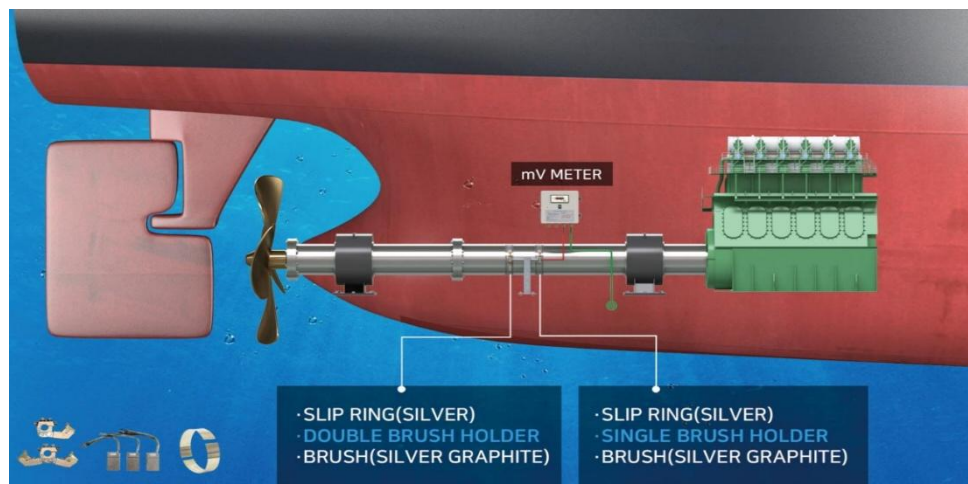


Figure 30 - Cathodic Protection for Marine Shaft [37]

05.2.2.2 Thermodynamics & kinetics

Under CP in aerated seawater, the dominant cathodic reaction is oxygen reduction ($O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$). The resulting local alkalinity at the metal surface leads to precipitation of calcareous deposits ($CaCO_3$, $Mg(OH)_2$), which lower current demand by increasing polarization resistance; ISO 12473 defines “calcareous deposits” precisely in this CP context. At higher overpotentials, hydrogen evolution contributes to cathodic current and can drive hydrogen uptake in susceptible alloys [49].

05.2.2.3 Criteria, control, and survey practice

ISO 12473 specifies protection criteria in seawater and details reference electrodes and survey methodology. DNV-RP-B401 is the widely used design basis for marine CP, combining:

- i. current-density targets by environment/zone,
- ii. coating-breakdown factors,
- iii. anode capacity/utilization, and
- iv. protection criteria and monitoring.

In practice, potentials measured near -0.80 V vs Ag/AgCl (seawater) indicate protection for steel zones, with readings more negative than -0.85 V commonly interpreted as confirming that -0.80 V is achieved at remote points; survey execution and interpretation are explicitly addressed in ISO 12473 and DNV-RP-B401.

05.2.2.4 Methods of applying CP

- **Galvanic (sacrificial) CP:** Shaft is fitted with Al/Zn anodes; ABS gives placement guidance for stern/propulsor regions and, when no central CP is installed, explicitly calls for zinc rings on propeller hubs or shafts to protect propellers and shafts [50].
- **Impressed-current CP (ICCP):** Hull-mounted inert anodes and reference cells connect to a rectifier/controller that holds target potentials with ± 10 mV precision; ABS details anode materials (platinized Ti, MMO), installation depths, spacing, and controller logging/alarms [50].
- **Integration of rotating assemblies:** Because bearings/oil films electrically isolate the rotating train, classification practice bonds the propeller and shaft to the hull using shaft-earthing (grounding) brushes/slip rings so the hull's CP also protects the propeller/shaft and mitigates bearing spark erosion. (ABS notes zinc rings as local protection; earthing ensures full system control and monitoring.) [50]

05.2.2.5 Secondary effects and interactions

ISO 12473 identifies and defines principal side-effects and control needs:

- **Calcareous deposition:** usually beneficial for current economy but can influence hydrodynamics/fouling [49].

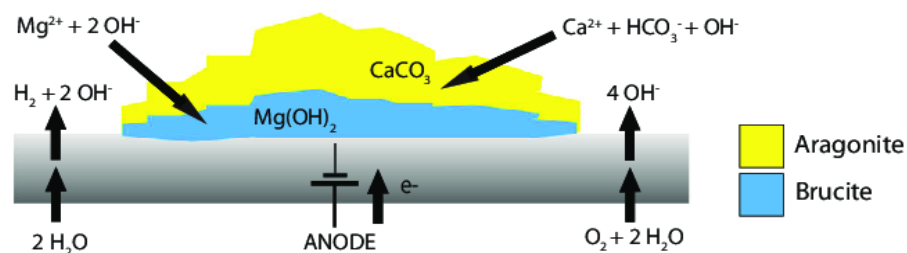


Figure 31 - Calcareous Deposition in Cathodic Protection [37]

- **Cathodic disbondment of coatings:** adhesion loss attributable to CP; coating selection/specification must account for CP environments [50].

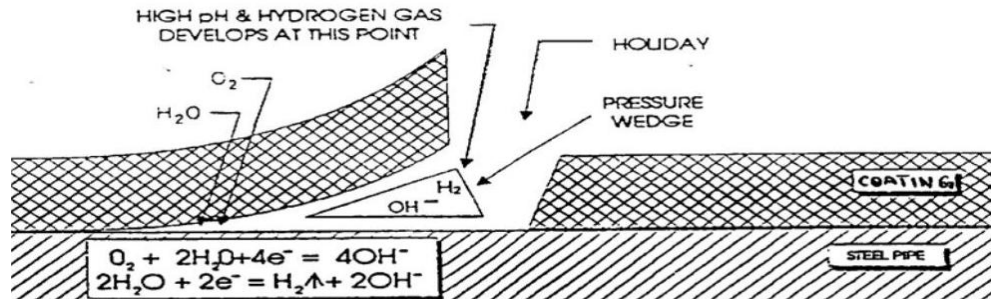


Figure 32 - Cathodic Disbandment [38]

- **Hydrogen effects:** at very negative potentials, hydrogen evolution raises the risk of hydrogen uptake and environmentally assisted cracking in susceptible alloys; ISO 12473 and DNV flag this explicitly [47], [49].

05.2.2.6 Duplex stainless steels under CP: HISC avoidance

Duplex steels can suffer hydrogen-induced stress cracking (HISC) under CP when polarized too negatively, particularly at welds and high-stress geometries. DNV-RP-F112 is the dedicated practice to design duplex components exposed to CP: limit polarization (avoid over-protection), manage stress (geometry and loads), qualify welds/microstructure, and, where helpful, use coatings as hydrogen-flux reducers. DNV's summary of F112 and independent reviews trace the empirical basis for potential limits and stress criteria [48],[51].

05.2.3 Anodic protection (AP)

05.2.3.1 Principle

AP uses potentiostat and a reference electrode to hold the metal at a controlled anodic potential within its passive region on the polarization curve. For suitable active-passive materials and electrolytes, the current density drops from a critical value i_{crit} to a low passive current i_{pass} , dramatically reducing corrosion rate while the passive film remains stable. AP systems require continuous, tight feedback control to avoid excursions into active dissolution (at lower potential) or trans passivity (at higher potential) [52],[53].

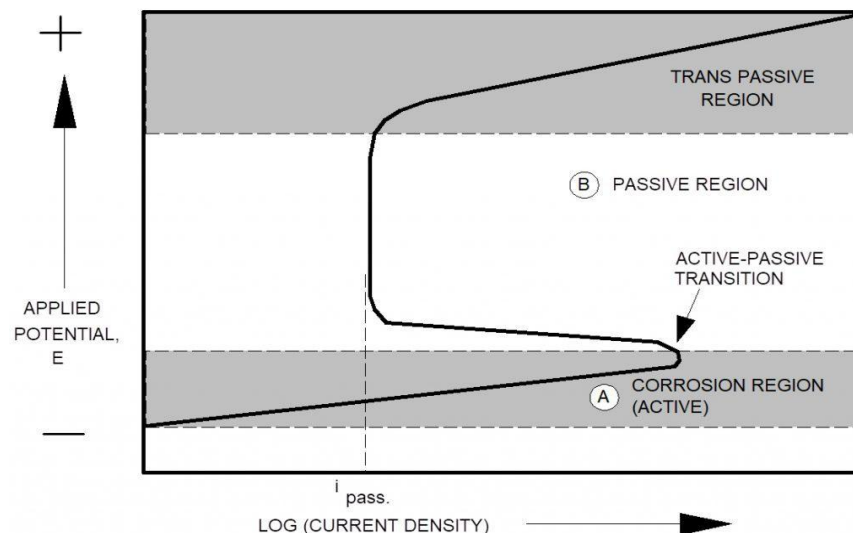


Figure 33 - Anodic Protection Polarization Curve Through the Active to Passive Zone as Current Is Applied [42]

05.2.3.2 Suitable environments and typical applications

AP is used where the passive window is wide and predictable concentrated sulfuric acid and certain caustic services are the canonical cases (e.g., carbon-steel storage tanks and process equipment). AMPP/Materials Performance documents decades of AP practice in the sulfuric-acid industry, including instrumented control and failure analyses [53].

05.2.3.3 Limitations for seawater machinery

In chloride-rich seawater, passive films on many alloys (including stainless steels) are vulnerable to localized breakdown (pitting/crevice). Turbulence, oxygen/transient chemistry near rotating parts, and temperature/salinity variation make stable potentiostatic passivation unreliable. Consequently, AP is not adopted for propellers/shafts or hull structures in open seawater; marine standards and class rules address submerged corrosion control via CP, not AP [49], [50].

05.2.4 Comparative summary (mechanism-level)

Aspect	Cathodic protection (CP)	Anodic protection (AP)
Control variable	Current/potential driven negative to suppress anodic dissolution	Potential held in passive region via potentiostat
Hardware	Galvanic anodes or ICCP (inert anodes + rectifier + reference cells); bonding/earthing for rotating parts	Potentiostat, reference electrode, auxiliary anode; continuous tight control
Environment	Standard for seawater structures (hull/appendages, propeller/shaft via earthing)	Specialized: concentrated acids/alkali with robust passivity (e.g., H ₂ SO ₄ tanks)
Key risks/side-effects	Over-polarization → hydrogen effects (duplex HISC per DNV-RP-F112); coating disbondment; calcareous deposits	Loss of passivity (pitting/crevice) or trans passivity; failure on power/control upset
Standards/guides	ISO 12473, DNV-RP-B401, ABS GN; UFC 3-570-01	AMPP/industry literature for process-vessel AP; no marine adoption for open seawater

05.2.5 Implications for a duplex-stainless shaft and propeller

- **Primary mechanism:** CP integrated with hull systems is the appropriate electrochemical basis; the rotating assembly must be bonded via shaft-earthing so it is within the protected circuit.
- **Potential control:** Apply DNV-RP-B401 protection criteria while avoiding over-polarization near duplex components in line with DNV-RP-F112 (HISC avoidance through potential limits, stress/weld control, and where applicable hydrogen-flux-reducing measures).
- **AP exclusion:** AP is not applicable for open seawater machinery because stable passivity cannot be guaranteed around a rotating shaft in chloride media.

05.3 Cathodic Protection (CP) Mechanisms for Marine Propeller Shaft

05.3.1 Overview

Cathodic protection reduces corrosion by lowering a structure's electrochemical potential until anodic dissolution becomes negligible. In seawater, CP is applied by either galvanic (sacrificial) anodes or impressed current systems (ICCP). ISO 12473 defines the principles, criteria and instrumentation for seawater CP, and notes that residual corrosion rates $< 10 \mu\text{m/yr}$ are typically achieved when CP is fully effective; it also recognizes hybrid arrangements combining galvanic anodes with ICCP where appropriate [49].

The design framework most widely used in practice is DNV-RP-B401 (current densities, coating breakdown factors, anode capacity/utilization, protection criteria), complemented by class guidance for ships from ABS (arrangements, shaft bonding/grounding, ICCP placement/controls) [47], [50].

05.3.2 Galvanic (sacrificial) anode CP

05.3.2.1 Working principle

Sacrificial anodes are alloys (Al-Zn-In, Zn, rarely Mg in seawater) with more negative free-corrosion potentials than the protected steel. They self-deliver current to polarize the structure towards the protection potential ($\approx -0.80 \text{ V}$ vs Ag/AgCl/seawater for steel zones). The current output follows Ohm's law with the driving voltage given by the anode-structure potential difference and limited by electrolyte resistance/anode geometry [49].

Table 9 - Summary of Potentials versus Ag/AgCl/Seawater Reference Electrode Recommended for the Cathodic Protection of Various Metals in Seawater [36]

Material	Minimum Negative Potential volts	Maximum Negative Potential volts
Iron and steel	-0.80 for aerobic environment	-1.10
	-0.90 for anaerobic environment (with active sulfate reducing bacteria)	-1.10
High-strength steels (yield strength $> 690 \text{ MPa}$ or hardness $> 350 \text{ HV}$)	-0.80	-0.83 to -0.95 ⁽¹⁾
Austenitic stainless steel for aerobic and anaerobic conditions $N_{\text{PRE}} = \% \text{ Cr} + 3.3\% (\text{Mo} + 0.5\text{W}) + 16\% \text{ N}$	-0.30 for $N_{\text{PRE}} \geq 40$ ⁽²⁾	-1.10
	-0.60 $N_{\text{PRE}} < 40$ ⁽²⁾	-1.10
Duplex stainless steel for aerobic and anaerobic conditions	-0.60 ⁽²⁾	See Note 3
Martensitic stainless steel (13%Cr) for aerobic and anaerobic conditions	-0.50	See Note 5
Nickel-based alloys	-0.20	See Note 4
Copper alloys	-0.45 to -0.60 for alloys with aluminum	-1.10
	-0.45 to -0.60 for alloys without aluminum	No limit

05.3.2.2 Anode alloys and electrochemical properties

ABS summarizes compositions and practical properties. For aluminum anodes (preferred in seawater for high capacity/low mass), typical closed-circuit potentials are ~ -1.05 to -1.15 V (Ag/AgCl), with practical current capacities $\approx 2500 \text{ A}\cdot\text{h/kg}$ at $\sim 25^\circ\text{C}$ (consumption $\approx 3.5 \text{ kg/A}\cdot\text{yr}$). "Low-voltage" Al alloys (e.g., MIL-DTL-24779B) operate around -0.80 to -0.90 V with lower capacity. Zinc anodes operate near -1.00 to -1.05 V with capacity $\approx 760\text{--}780 \text{ A}\cdot\text{h/kg}$; Mg is rarely used in seawater due to overprotection and low efficiency. Temperature derates Al capacity approximately linearly (e.g., $2500 \rightarrow 500 \text{ A}\cdot\text{h/kg}$ from $25 \rightarrow 80^\circ\text{C}$) [50].

05.3.2.3 Arrangement on ships and appendages

ABS gives placement and spacing rules for hull zones and, crucially, notes that where a central CP system is absent, propellers and shafts can be protected by zinc ring anodes mounted on the hub/shaft. Care is required near the stern to avoid hydrodynamic penalties and to keep anodes out of “no-anode” zones adjacent to props [50].

05.3.2.4 Advantages/limitations

Advantages: intrinsically fail-safe (no power), simple, predictable consumption (capacity-based sizing), minimal topside hardware.

Limitations: fixed driving voltage (no closed-loop control); potential can drift with salinity/temperature/flow; higher mass and drag; risk of over-polarization in local spots (hydrogen side-reaction) if anode selection/placement is aggressive; periodic diver replacement. For duplex steels, avoiding unnecessarily negative potentials is desirable to minimize hydrogen uptake [48].

05.3.3 Impressed-Current Cathodic Protection (ICCP)

05.3.3.1 Working principle and hardware

ICCP uses inert anodes (e.g., platinized-Ti/MMO) fed by a DC power supply; reference electrodes (Ag/AgCl) mounted on the hull provide feedback so the controller maintains a set potential. ISO 12473 explicitly cites the large, adjustable driving voltage of ICCP as a core advantage; it also shows secondary anodic reactions (H_2 , Cl_2 evolution) that motivate the use of dielectric shields near anodes to protect paint/steel. ABS illustrates standard recessed/guarded ICCP anodes and gives minimum distances to coatings and shield widths to mitigate chemical attack from gas evolution [49], [50].

05.3.3.2 Control, measurement, and placement

ABS requires proper placement of reference electrodes between anodes and critical features (e.g., propellers), with separation and symmetry rules; it also mandates controller accuracy and logging, and describes installation geometry for stern areas [50].

05.3.3.3 Advantages/limitations

Advantages: Set-point control of hull potential (fine-tuning to avoid over-polarization); low hydrodynamic penalty (few, recessed anodes); scalable current output for fouling/waterline variability.

Limitations: Needs power and electronics; requires dielectric shields and careful electrode placement; risk of stray-current effects or coating damage if misconfigured. For duplex steels, ICCP’s tight set-point control directly supports HISC-avoidance by holding potentials near protection levels but not excessively negative [50].

05.3.4 Comparison of galvanic anodes and impressed current cathodic protection systems for ship hulls

Table 10 - SACP vs. ICCP comparison

Comparison item	Sacrificial-anode systems (SACP)	Impressed-current systems (ICCP)
Environment / conductivity	Best in normal-salinity seawater; driving voltage limited by anode potential, so capacity can be constrained in low-conductivity conditions.	Less restricted by conductivity because voltage/current are externally supplied; commonly chosen where higher driving voltage is needed (e.g., brackish/low conductivity).
Design & installation	Simple hardware; anode location/spacing must cover high-demand aft zones (propeller/shaft). Anodes are welded/bolted and sized for life.	Requires anode arrays, rectifier, cabling, reference electrodes, and dielectric shields near anodes to shape current. More engineering effort.
Power source	No external power; current self-adjusts as anodes corrode.	Continuous DC power supply; adjustable output to meet protection criteria.
Anodes	Cast Al/Zn; consumed over time and replaced at dockings.	MMO/Ti or other inert types; very low consumption, long life.
Control of potential	Limited control; potentials depend on anode mass/placement and coating condition.	Closed-loop control with reference electrodes enables tight set-points across operating states.
Effect on neighbors / stray currents	Lower risk (lower voltages); interaction mainly via galvanic coupling that design must consider.	Higher voltages/currents can cause stray-current effects if misdesigned; dielectric shielding and monitoring are required.
Failure/abnormal conditions	Gradual loss of output as anodes are consumed; performance degrades, but system keeps working until depletion.	Power loss or shield damage can produce under- or over-protection (hydrogen evolution, coating disbondment) near anodes if not corrected.
Maintenance & monitoring	Visual anode surveys/replacement at haul-outs; minimal instrumentation.	Periodic potential surveys and rectifier checks; reference-electrode health and shield integrity must be monitored.
Hydrodynamics around the shaft	Physical anode mass can add local drag if placed on/near rotating components; placement is constrained. (Design notes for aft zone demand.)	ICCP anodes are low-profile/flush-mounted on hull; shaft is protected electrically via hull system and shaft-earthing rather than anodes on the rotor.

Duplex 2205 (HISC risk under CP)	Lower inherent risk of over-polarization, but still requires avoiding excessively negative potentials.	Precise control is an advantage, yet over-polarization can promote hydrogen uptake; follow DNV-RP-F112 limits/mitigations for duplex steels.
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05.3.5 Bringing the rotating shaft into the protected circuit

Because lubricated bearings and nonmetallic liners electrically insulate the shaft from the hull, the propeller/shaft will not receive CP unless it is bonded. ABS requires a shaft-ground assembly (slip rings + brushes) so the prop/shaft is at hull potential. Performance criteria include maximum contact resistance $\leq 0.001 \Omega$ (water-filled) or 0.01Ω (oil-filled), and monitoring with a millivoltmeter; a typical arrangement is provided in ABS figures. ABS also notes the alternative of a zinc ring on the hub/shaft, but the baseline in class guidance is conductive bonding so the central CP system protects the rotating train [50].

05.3.6 CP mechanism selection for this design

Functional aim: achieve cathodic protection at/near the -0.80 V (Ag/AgCl) protection level for steel zones at the propulsor, while avoiding over-polarization around duplex components and welds to minimize hydrogen uptake risk per DNV-RP-F112. Additionally, reduce hydrodynamic penalties and maintenance at the stern.

Recommended mechanism: ICCP integrated with the hull + shaft-earthing (bonding) of the propeller/shaft, with reference electrodes positioned between stern anodes and the propeller to maintain the set potential at the propulsor (ABS positioning guidance). ICCP provides the closed-loop potential control needed to stay near protection while avoiding unnecessarily negative values that could elevate HISC susceptibility in duplex steel; stern hardware is recessed/guarded to reduce drag, and potentials are logged for QA [50].

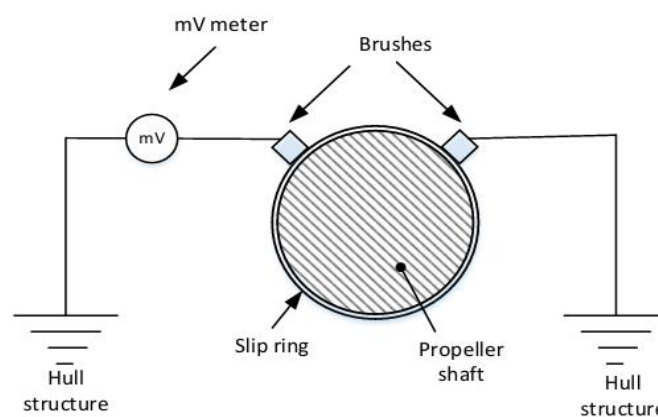


Figure 34 - Typical Arrangement for Shaft Grounding and Monitoring [36]

Acceptable alternative: A local galvanic solution using a zinc (or low-voltage Al) ring anode on the hub/shaft plus shaft-earthing to the hull CP, if present. This is class-recognised, but lacks the fine potential control of ICCP and typically has more mass/drag and periodic replacement, use only when ICCP integration is impractical [50].

Why ICCP is optimal here:

- **Control:** ISO/ABS emphasizes ICCP's adjustable driving voltage and reference-electrode feedback, ideal to hold the duplex component within a conservative potential window.
- **Integration:** ABS explicitly requires shaft bonding so the prop/shaft is inside the protected circuit; ICCP then protects the entire aft end under a unified set-point.
- **Hydrodynamics & lifecycle:** fewer recessed anodes, reduced drag; no anode-mass penalty at the propulsor; continuous logging/alarms support condition-based maintenance.

05.3.7 Cathodic protection design and sizing

05.3.7.1 Inputs and assumptions

- Exposure: freely flowing seawater, $\sim 25^\circ\text{C}$, tropical surface waters ($< 30\text{ m}$).
- Design protective potential (Ag/AgCl/seawater): -0.80 V .
- Seawater resistivity used for sizing: $\rho \approx 0.25\ \Omega\cdot\text{m}$ at 25°C and normal ocean salinity.
- Design life to size galvanic mass: 5 years (typical dry-dock interval).

Total wetted area $A_{total} \approx 0.605\text{ m}^2$.

Effective coated-surface area $A_{eff} = f_c A \approx 0.1 \times 0.605 \approx 0.0605\text{ m}^2$.

05.3.7.2 Design current demand

05.3.7.2.1 Design for shaft

For tropical surface waters (0–30 m):

- Initial $i_{ci} = 0.150\text{ A/m}^2$, Final $i_{cf} = 0.100\text{ A/m}^2$, Mean $i_{cm} = 0.070\text{ A/m}^2$

Table 11 - Recommended initial and final design current densities (A/m^2) for seawater exposed bare metal surfaces, as a function of depth and climatic region based on surface water temperature [33]

Depth [m]	Tropical ($> 20^\circ\text{C}$)		Sub-tropical ($12- 20^\circ\text{C}$)		Temperate ($7-11^\circ\text{C}$)		Arctic ($< 7^\circ\text{C}$)	
	initial	final	initial	final	initial	final	initial	final
0-30	0.150	0.100	0.170	0.110	0.200	0.130	0.250	0.170
>30-100	0.120	0.080	0.140	0.090	0.170	0.110	0.200	0.130
>100-300	0.140	0.090	0.160	0.110	0.190	0.140	0.220	0.170
>300	0.180	0.130	0.200	0.150	0.220	0.170	0.220	0.170

Current demand for this shaft:

$$I_i = i_{ci} A = 0.150 \times 0.0605 = 9.075\text{ mA}$$

$$I_f = i_{cf} A = 0.100 \times 0.0605 = 6.05\text{ mA}$$

$$I_m = i_{cm} A = 0.070 \times 0.0605 = 4.235\text{ mA}$$

Table 12 - Recommended mean design current densities (A/m²) for seawater exposed bare metal surfaces, as a function of depth and climatic region based on surface water temperature [33]

Depth (m)	Tropical (> 20 °C)	Sub-tropical (12- 20 °C)	Temperate (7-12 °C)	Arctic (< 7 °C)
0-30	0.070	0.080	0.100	0.120
>30-100	0.060	0.070	0.080	0.100
>100-300	0.070	0.080	0.090	0.110
>300	0.090	0.100	0.110	0.110

For a 3-year interval: $t = 3 \times 8760 = 26,280 \text{ h}$.

Amp-hours (mean-basis): $Q = I_m t \approx 0.004235 \times 26,280 \approx 111.3 \text{ Ah}$.

05.3.7.2.2 Practical Scope: Inclusion of Propeller

The calculations above represent the current demand for the coated shaft alone ($\approx 4\text{mA}$). However, practical ICCP design must account for the entire electrically continuous rotating assembly. Since the shaft is bonded to the hull, the propeller functions as a large, bare-metal cathode in the circuit.

The propeller rather than the shaft therefore dominates the design demand.

Revised Total Current Demand:

- **Propeller Material:** Ni-Al Bronze (Uncoated).
- **Current Density Requirement:** 0.5 A/m^2 (DNV-RP-B401 standard for bare copper alloys in turbulent flow).
- **Estimated Propeller Area:** $\approx 2.0 \text{ m}^2$ (Estimated based on shaft sizing).
- **Propeller Current Load:**

$$I_{prop} = 2.0 \text{ m}^2 \times 0.5 \text{ A/m}^2 = 1.0 \text{ A}$$

- **Amp-hours:**

$$Q = I_m t \approx 1 \times 26,280 \approx 26280 \text{ Ah}$$

While the shaft requires only negligible current, the total system demand is approximately 1.0 A.

05.3.7.3 Impressed-Current Cathodic Protection (ICCP) System Design

05.3.7.3.1 Set-point, References, and Control Strategy

- **Control Set-Point:** The system shall maintain a potential of **-0.80 V** vs Ag/AgCl/seawater. This provides adequate protection for the steel hull and shaft while minimizing the risk of hydrogen embrittlement in high-strength steel or duplex stainless steel components [50].

- **Reference Electrodes:** At least one permanent Zinc or Ag/AgCl reference electrode shall be installed in the stern zone, positioned between the anode and the propeller (but outside the anode's primary gradient) to ensure the controller regulates the potential at the propeller/shaft assembly.
- **Controller Logic:** The power unit shall utilize closed-loop feedback control to automatically adjust current output in response to changes in salinity, temperature, and coating condition.

05.3.7.3.2 Anode Selection and Sizing

Based on the combined current demand of the shaft and bare propeller ($I_{\text{total}} \approx 1.0 \text{ A}$), a standard commercial anode unit is selected to ensure reliability and availability.

- **Selected Hardware:** 10 Amp/24V DC Rated Mixed Metal Oxide (MMO) on Titanium Disc Anode.
- **Justification for Selection:**
 1. **Commercial Availability:** 10 Amp is the standard minimum frame size for marine hull-mounted ICCP anodes [54].
 2. **Safety Margin:** The calculated demand ($\approx 1.0 \text{ A}$) utilizes only 10% of the anode's rated capacity. This provides a 10:1 safety factor to accommodate future coating breakdown (f_c) on the aft hull structure over the vessel's service life.
 3. **Extended Life:** Operating the MMO element at low current density ($< 50 \text{ A/m}^2$ vs. rated 600 A/m^2) minimizes coating wear rates, ensuring the anode life exceeds the vessel design life.
- **Current Density Verification:**
 - Active Anode Surface Area (A_{anode}): $\approx 0.02 \text{ m}^2$ (Typical for 10A disc).
 - Operating Current Density (J): $J = I_{\text{demand}}/A_{\text{anode}} = 1.0 / 0.02 = 50 \text{ A/m}^2$.
 - **Result:** Acceptable (Well below the 600 A/m^2 manufacturer limit).

05.3.7.3.3 Dielectric Shield Design

To prevent cathodic disbondment of the hull coating adjacent to the anode due to high local potentials, a primary dielectric shield is required. The shield radius is calculated based on the maximum rated output of the power supply ($I_{\text{max}} = 10 \text{ A}$) to ensure safety under full-power fault conditions [50].

- **Design Formula (ABS/DNV):**

$$r \approx \frac{\rho I}{2\pi(E_{\text{breakdown}} - E_{\text{protection}})}$$

- **Input Data:**

- Maximum Current (I): 10 A
- Seawater Resistivity (ρ): $0.25 \text{ } \Omega\cdot\text{m}$
- Allowable Voltage Drop (ΔV): 2.0 V (Conservative limit for standard epoxy shields).

- **Calculation:**

$$r \approx \frac{10 \times 0.25}{2\pi \times 2.0}$$

- **Specification:**

A primary dielectric shield (prefabricated FRP or high-build epoxy) with a minimum radius of 200 mm (400 mm diameter) shall be applied centered on the anode. This ensures the potential gradient falls to safe levels before reaching the conventional hull coating.

05.3.7.4 Sacrificial ring-anode (Alternative check)

05.3.7.4.1 Electrochemical properties

- Al-Zn-In: $E_{closed} \approx -1.050 \text{ V (Ag/AgCl)}$, capacity $\approx 2000 \text{ Ah/kg}$.
- Zn: $E_{closed} \approx -1.030 \text{ V}$, capacity $\approx 780 \text{ Ah/kg}$

Utilization factor (bracelet/ring). $u = 0.80$

Table 13 - Recommended Initial, Final and Mean Design Current Densities for Bared Steel Surface Exposed to Seawater with a Range of Surface Water Temperatures [36]

Surface Water Temperature, °C (°F)	Initial mA/m ² (mA/ft ²)	Final mA/m ² (mA/ft ²)	Mean mA/m ² (mA/ft ²)
> 20 (68)	120 (11.1)	80 (7.4)	60 (5.6)
12-20 (54-68)	140 (13.0)	90 (8.4)	70 (6.5)
7-11 (44-52)	170 (15.8)	110 (10.2)	80 (7.5)
< 7 (44)	200 (18.6)	130 (12.1)	100 (9.3)

Mean-life charge (3 y). $Q \approx 26280 \text{ Ah}$.

Net mass by capacity $M_{net} = \frac{Q}{uE}$:

- Al-Zn-In: $M \approx 26280 / (0.80 \times 2000) \approx 16.4 \text{ kg}$.
- Zn: $M \approx 26280 / (0.80 \times 780) \approx 42.11 \text{ kg}$.

A 16 kg anode on the rotating shaft would create unacceptable hydrodynamic drag and balance issues. Therefore, the sacrificial ring option is rejected in favor of the Hull-mounted ICCP + Shaft Earthing solution.

05.3.8 Installation, Operation, Maintenance, and Cost

05.3.8.1 Overview

This section details the practical implementation of the Impressed Current Cathodic Protection (ICCP) system designed to protect the propeller and shaft assembly. The strategy relies on a dedicated hull-mounted ICCP anode to provide protection current, and a high-performance shaft grounding (slip-ring) assembly to electrically bond the rotating propeller to the hull protection circuit.

05.3.8.2 System Elements

- **ICCP Anode (Primary):** Hull-mounted Mixed Metal Oxide (MMO) on Titanium disc anode (10A rating) with primary dielectric shield.
- **Shaft Earthing (Bonding):** Silver-graphite brush and silver-inlay slip-ring assembly to ensure the rotating shaft is at hull potential (resistance < 0.01 Ω).
- **Reference Electrode:** Permanent Ag/AgCl/Seawater reference cell installed in the stern zone to provide feedback to the control unit.
- **Power Unit:** Automatic control rectifier (24V DC input) with data logging.

05.3.8.3 Procurement and Specification

Instead of custom fabrication, system components shall be procured from approved marine corrosion-control manufacturers (e.g., Cathelco/Evac, Wilson Taylor, or equivalent) to ensure reliability and class compliance [55].

- **Anode Specification:** 10 Amp / 24V DC rated MMO/Ti disc anode.
 - **Dimensions:** Active element diameter approx. 90 mm; recessed or surface mounted.
 - **Certification:** Manufacturer's Certificate of Conformity including current rating and coating thickness verification.
- **Shaft Earthing Specification:** Commercial marine grounding assembly rated for continuous current > 10 A.
 - **Brushes:** High-silver content graphite (minimum 80% Ag) to minimize contact drop.

05.3.8.4 Installation

05.3.8.4.1 ICCP Anode & Dielectric Shield

- **Mounting:** The 10A disc anode shall be installed in the stern zone, requiring a single hull penetration for the titanium mounting stud. The anode body shall be electrically isolated from the hull steel using the manufacturer-supplied cofferdam or insulating bedding compound [50].
- **Dielectric Shield:** A primary dielectric shield is mandatory to prevent cathodic disbondment of the hull coating [50].
 - **Material:** High-build glass-flake epoxy or prefabricated GRP plate.
 - **Dimensions:** Minimum radius of 200 mm extending from the anode center (Total shield diameter $\approx$ 400 mm).
 - **Application:** Surface preparation to Sa 2.5; holiday detection required before launching.

05.3.8.4.2 Reference Electrode

- **Location:** Install one permanent Ag/AgCl reference electrode between the stern anode and the propeller. It must be positioned outside the immediate voltage gradient of the anode to ensure accurate sensing of the propeller potential.
- **Wiring:** Shielded cable run separately from power cables to avoid signal interference.

05.3.8.4.3 Shaft Earthing Assembly

- **Location:** The slip-ring assembly shall be installed on the intermediate shaft, aft of the main engine/gearbox and forward of the stern tube seal.
 - **Critical Note:** This position creates a low-resistance path that bypasses the main engine bearings and thrust block, preventing "spark erosion" (pitting) damage to precision engine components.
- **Installation:**
 - Clean shaft surface to bright metal.
 - Install silver-inlay copper band (clamped or bonded).
 - Mount twin brush holders on a rigid bracket bonded to the hull structure.
 - **Verification:** Measure contact resistance between the stationary brush holder and the rotating shaft surface. Target: $\leq 0.001 \Omega$ (static).

05.3.8.4.4 Wiring and Termination

- **Power Cables:** Positive DC to Anode; Negative DC bonded to Hull.
- **Sense Leads:** Reference electrode to Controller input.
- **Protection:** All underwater penetrations must be approved watertight glands or cofferdams. Cables in the engine room shall be mechanically protected against vibration and impact.

05.3.8.5 Commissioning and Operation

05.3.8.5.1 Dry-dock Checks

Before flooding, verify:

1. Anode surface is clean and **unpainted** (paint blocks current output).
2. The dielectric shield has no holidays (pinholes).
3. The reference electrode cover is removed.
4. Shaft slip-ring is clean and brushes are seated.

05.3.8.5.2 Post-Launch Commissioning

1. **Baseline Readings:** Measure natural potential of the hull and shaft (typically -0.60 V to -0.65 V vs Ag/AgCl).
2. **Energize System:** Switch on the ICCP controller.
3. **Shaft Monitoring:** With the shaft rotating at low RPM, measure the potential difference between the Shaft and the Hull using a handheld mV meter.
 - Pass Criteria: Potential difference < 50 mV.
 - If > 50 mV, the brushes are not grounding the shaft effectively; check spring tension or run-in the slip ring.
4. **Stability:** Verify controller holds the stern zone set-point (-0.80 V) ± 10 mV [56].

05.3.8.6 Maintenance Schedule

Because the system uses inert MMO anodes and automatic control, maintenance is primarily monitoring rather than replacement [50].

Table 14 - Maintenance Schedule

Interval	Action	Success Criteria
Weekly	Panel Log: Record Voltage (V), Current (A), and Reference Potential (mV).	Current matches operational baseline; Potential stable at -0.80 V.
Quarterly	Shaft Earthing Check: Visual inspection of brushes and slip ring.	Brushes move freely in holders; Slip ring is free of oil/grease.
Quarterly	Shaft mV Check: Measure voltage between Shaft and Hull while running.	Voltage difference < 50 mV.
Annual	Brush Replacement: Check brush length.	Replace if worn below manufacturer's limit (typically 50%).
Dry-Dock (5yr)	Shield & Anode Inspection: Check dielectric shield for blistering; check anode for physical impact damage.	Shield intact; Anode MMO coating intact (dark grey appearance).

05.3.8.7 Cost Model

05.3.8.7.1 CAPEX (Capital Expenditure)

- **ICCP Hardware:** 10A Minitex System (Controller + 1 Anode + 1 Reference Cell) ≈ \$5,000 – \$8,000 (Estimate).
- **Shaft Earthing:** Silver-graphite slip ring assembly ≈ \$1,500 – \$2,500.

- **Installation:** Cabling and hull penetrations (Shipyard labor).
 - Savings: Eliminating the large bracelet anode reduces casting and machining costs compared to the previous design.

05.3.8.7.2 OPEX (Operating Expenditure)

- **Energy:** Power consumption is negligible.
 - $P = V \times I \approx 3.0 \text{ V} \times 1.0 \text{ A} = 3.0 \text{ Watts}$
 - Annual consumption < 30 kWh.
- **Consumables:**
 - Silver-graphite brushes (Replace approx. every 12–18 months).
 - No periodic anode replacement required (MMO anode life > 20 years).

05.3.8.8 Documentation

The vessel shall maintain:

1. **As-Built Wiring Diagram:** Showing sensor locations and earthing points.
2. **Commissioning Report:** Baseline shaft-to-hull resistance and potential readings.
3. **Logbook:** Weekly voltage/current readings to track coating condition (increasing current demand over years indicates coating breakdown).

While cathodic protection provides essential electrochemical control by lowering the corrosion potential of the shaft and propeller assembly, CP alone cannot fully prevent surface deterioration in areas where mechanical wear, turbulence, crevices, or coating discontinuities are unavoidable. Further, the use of duplex stainless steel requires careful control of CP potential to avoid excessive hydrogen absorption in accordance with DNV-RP-F112, highlighting the need for an additional non-electrochemical barrier. Therefore, Section 06.2 introduces engineered surface treatments and multilayer coating systems that work in tandem with CP. These treatments enhance resistance to pitting, abrasion, UV exposure, and biofouling, while reducing CP current demand by improving surface passivity and minimizing coating breakdown. This integrated approach ensures comprehensive corrosion protection under real marine operating conditions.

05.4 Final Combined Protection Strategy

The evaluation of both corrosion-control methods demonstrates that neither cathodic protection nor surface treatments alone can provide complete long-term protection for a marine propulsion shaft exposed to high-chloride seawater. Cathodic protection implemented via an ICCP system with shaft earthing effectively controls electrochemical potential and suppresses anodic dissolution across the rotating shaft–propeller assembly. However, its performance is strongly dependent on coating condition and must be carefully managed to avoid over-polarization and hydrogen-related issues in duplex stainless steel.

Conversely, the multilayer surface-treatment system comprising mechanical finishing, passivation, a high-build epoxy barrier, polyurethane topcoat, optional anti-fouling protection, and localized dry-film

lubrication—provides a robust physical barrier against chloride penetration, erosion, UV damage, and marine growth. When combined, these methods form a synergistic protection system: the coatings reduce the CP current demand and stabilize the surface chemistry, while CP ensures protection in areas where coatings are damaged or inaccessible.

Accordingly, the final corrosion-protection strategy for the designed propulsion shaft adopts **both** approaches: a **multilayer coating and surface-treatment system** integrated with a **controlled ICCP cathodic-protection system**. This hybrid approach maximizes durability, minimizes maintenance requirements, and ensures that the shaft maintains its mechanical and corrosion-fatigue performance over the intended service life.

06. Mechanical Design

The mechanical design of a marine propulsion shaft requires a rigorous evaluation of both analytical and numerical performance criteria to ensure safe and reliable torque transmission under the combined effects of torsion, bending, axial loading, and dynamic excitation. The shaft operates as a critical structural component within the propulsion line, transmitting power from the engine to the propeller while simultaneously resisting complex loading conditions induced by propeller thrust, hydrodynamic forces, and rotational dynamics. This section presents the complete mechanical design methodology applied in this project, beginning with the analytical determination of shaft geometry and torque transmission capacity, followed by validation through finite-element analysis (FEA).

Subsection **06.1** describes the derivation of the final shaft geometry through classical strength of materials approaches, IACS-based diameter selection formulas, vibration analyses, and fatigue calculations. These steps ensure that the chosen shaft dimensions meet both classification-society requirements and practical performance targets. Subsection **06.2** presents the FEA-based verification of the shaft's behavior under static and cyclic loads, including stress distribution, deformation patterns, natural frequencies, and corrosion-fatigue considerations. Together, these analyses form the foundation of the final mechanical design and confirm that the shaft configuration satisfies strength, stiffness, vibration, and fatigue life criteria for long-term marine operation.

06.1 Deign of Shaft Geometry and Torque Transmission Calculation

06.1.1 Introduction

The design of a marine propulsion shaft is a critical engineering task that involves identifying the appropriate geometry and ensuring it can withstand the mechanical stresses experienced during operation. This shaft transmits the engine's mechanical power to the propeller, providing the necessary thrust for the vessel. As the core component of the shafting system, the shaft design and geometry must be robust, able to handle torsional and bending stresses, and resistant to wear and corrosion due to prolonged exposure to seawater.

This section focuses on the shaft design process, where key parameters such as length, diameter, and material selection are evaluated. Additionally, the torque transmission capacity and the shaft's ability to withstand torsional shear, bending, and fatigue loads are considered, along with the most suitable geometry for the shaft. The material properties also play a crucial role in ensuring the shaft's strength, corrosion resistance, and longevity.

06.1.2 Objectives

- Define the design parameters for the marine propulsion shaft based on operational requirements and environmental conditions.
- Identify the optimal geometry for the shaft, ensuring it meets the required strength and performance standards while minimizing material waste.
- Establish the torque transmission capacity and ensure that the shaft can safely handle the forces generated during normal operation.
- Optimize the shaft dimensions to reduce the risk of failure due to torsional or bending stresses while maintaining the structural integrity and performance under all operating conditions.

- Design for manufacturability by considering the materials, tolerances, and the practicality of the manufacturing process for the chosen shaft geometry.

06.1.3 Design parameters

Main engine – CAT 3406C rated 298 kW at 1800 rpm.[55]

Reduction gear – Workboat gearboxes commonly offer high ratios up to 8:1[57]

Shaft length: 1.6 m [58]

Torque: 12.64 kNm

Shaft Speed: 225 RPM

Corrosion Rate (untreated): 0.2 mm/year

06.1.4 Background & Theory

This chapter provides the theoretical basis for the thesis work. It starts by describing different mechanical components of a shaft line. Following is an introduction of the theory behind the static and the dynamic loads that are present during an operation. Important dynamic design phenomena theories, vibrational behavior, and fatigue. Lastly, the foremost rules and classifications that were used from IACS and DNV societies.[59]

06.1.4.1 shaft line in the ship/boat

A shaft line refers to the assembly of mechanical components responsible for transmitting power from the prime mover to the propeller. Its primary function is to ensure the efficient transfer of torque, enabling the propeller to generate thrust for ship propulsion.

In marine applications, the propulsion shaft not only transmits torque but may also act as a structural element. For instance, in a typical propeller shaft (low-speed shaft), it supports the weight of the propeller while withstanding axial and radial forces induced during operation. Conversely, high-speed shafts primarily serve as power transmission elements, comparable to driveshafts used in heavy-duty vehicles, with the main role of transferring torque from the engine to the propeller without bearing significant structural loads.[59]

06.1.4.1.1 Low speed shaft

The low-speed shaft (LSS) is the section of the shaft line that operates at the rotational speed of the propeller. Since marine propulsion engines or electric motors typically operate at higher speeds than the propeller design speed, reduction gears are generally employed to match these speeds. The LSS is connected to the output of the reduction gear, with the propeller mounted at its end.[59]

In commercial vessels, propeller speeds generally range from 70 to 250 revolutions per minute (RPM). Larger cargo vessels employ increasingly lower propeller speeds due to the adoption of large, fuel-efficient, low-speed two-stroke engines that can operate directly at propeller speed. In such cases, the engine crankshaft is directly connected to the propeller shaft, essentially functioning as a structural continuation of the shaft line. [Figure 35]

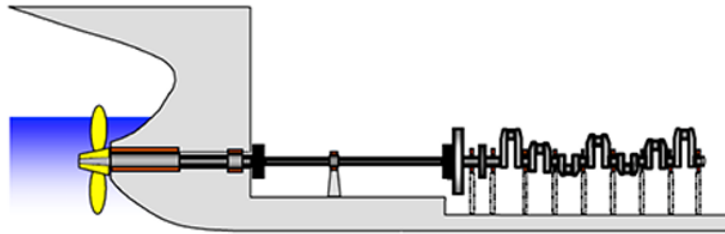


Figure 35 - The figure describes the setup of a low-speed shaft line, directly coupled

Alternatively, an LSS arrangement incorporating a medium-speed engine, reduction gear, and a controllable pitch propeller (CPP) is common in ferries, fishing vessels, and specialized service or research vessels. This configuration provides operational flexibility and faster response. [Figure 36] The CPP system enables adjustment of blade pitch through hydraulic or mechanical actuation mechanisms integrated within the reduction gearbox and propeller shaft. This allows seamless transition between forward and reverse thrust, as well as effective load regulation on the engine to match operational requirements.[60]

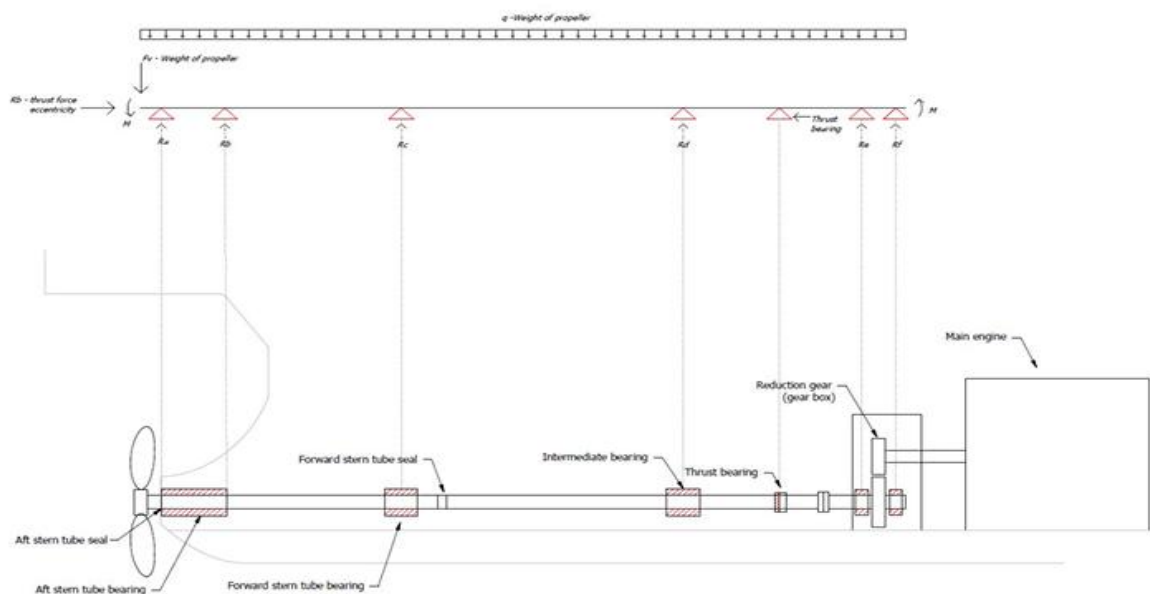


Figure 36 - The figure describes the setup of a low-speed shaft line with reduction

06.1.4.1.2 High speed shaft

In contrast to the low-speed shaft, the high-speed shaft (HSS) operates at the rotational speed of a medium- or high-speed propulsion engine. HSS configurations are commonly used with azimuth thrusters, where the required reduction gearing is integrated within the thruster's angle-drive housing. These systems typically employ two sets of ring and pinion gears arranged to transfer power through a 90-degree turn, giving rise to the term "Z-drive." Since the ring gear contains more teeth than the pinion, a reduction in rotational speed is achieved within the thruster assembly. This ensures compatibility between the high-speed engine output and the propeller's required operating speed.[59]

- **Single-screw tanker vessels** – Employ a stern tube running through the midline of the stern, incorporating aft and forward bearings with associated sealing arrangements.
- **Twin-screw dredger vessels with ducted propellers** – Feature stern tubes housed within fins or skegs, with forward and aft bearings and seals.

The propeller shaft, often referred to as the **tail shaft**, is the aftmost section of the propulsion shafting located within the stern tube (for single-screw ships) or within struts (for multiple-screw ships). It extends beyond the hull to support the propeller, acting as a cantilever member. This configuration subjects the shaft to significant bending moments and forces, which are primarily absorbed by the aft and forward stern tube bearings. The shaft portion located forward of the stern tube is relatively unaffected by these external loads.[62]

06.1.4.1.5 Intermediate shaft

The intermediate shaft is employed when the distance between the propulsion engine or gearbox and the propeller is relatively long, necessitating an extension of the shaft line. Such arrangements are typical for ferries, Ro-Pax vessels, and offshore service vessels, where the machinery space is located forward of the propellers. In contrast, vessels such as single-screw tankers often position the main engine close to the stern, eliminating the need for extended intermediate shafting.[63]

Intermediate shafts are usually manufactured as forged, single-piece elements with flanges at both ends, allowing connection to the propeller shaft aft and the gearbox or engine forward. Their primary function is to transmit torque along the shaft line; they do not bear significant external loads apart from their own weight. Consequently, the bending moments acting on intermediate shafts are considerably smaller than those experienced by the propeller shaft.[64]

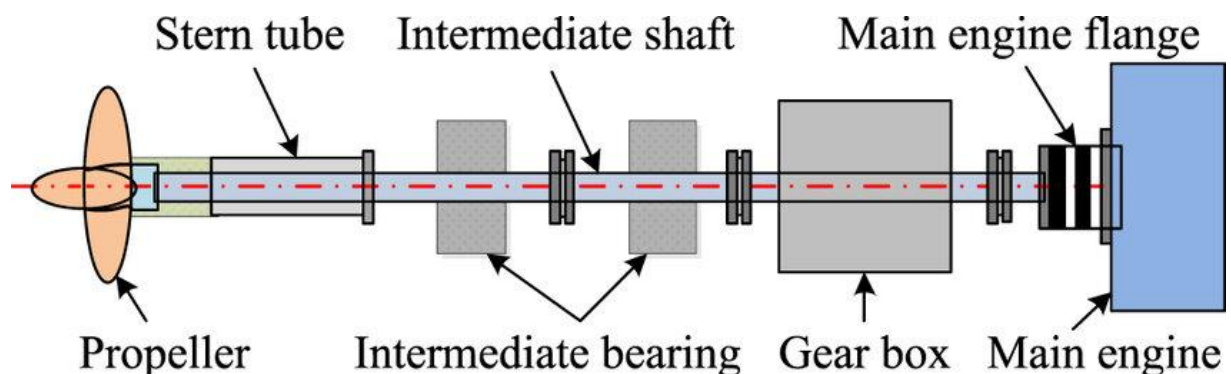


Figure 38 - Propulsion shaft line

06.1.4.1.6 propeller

The propeller functions as the primary device for converting the mechanical torque transmitted through the shaft line into thrust, thereby propelling the vessel. From a shaft design perspective, two critical considerations are the weight of the propeller and the hydrodynamic forces acting on the propeller shaft during operation.

Controllable pitch propellers (CPPs) are commonly used in modern vessels, typically incorporating four or five blades. The number of blades plays a significant role in the dynamic behavior of the shaft line, as it directly influences vibration excitation, or “propeller beats,” generated when the propeller operates within a non-uniform wake field. These periodic excitations must be carefully accounted for in shaft line design to ensure safe operation and minimize fatigue damage.[65]

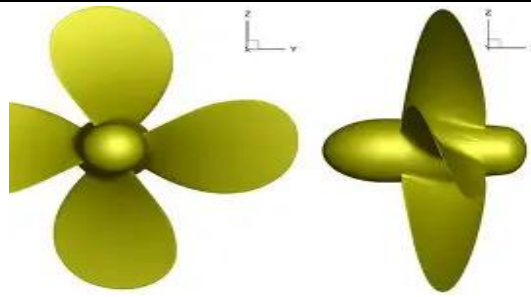


Figure 39 - Propeller of a ship



Figure 40 - Different types of propellers design

06.1.4.1.7 Bearing

Bearings are strategically positioned along the shaft line to ensure proper alignment and structural support. Their functions include providing static support to the shaft, as well as acting as dynamic support to limit lateral vibrations. The number and placement of bearings are determined based on shaft alignment calculations and classification society requirements.

In typical arrangements, two or more bearings are installed within the stern tube to counteract the bending moments induced by the propeller. Additional bearings may be required along the intermediate shaft, depending on shaft length and operational conditions. While many shafting system manufacturers design and assemble complete propulsion systems, bearings themselves are often procured from specialized third-party suppliers.

A critical component of the bearing system is the **thrust bearing**, which is designed to absorb the axial forces generated by the propeller during operation. This bearing prevents axial displacement of the shaft while also damping axial vibrations. Thrust bearings may be installed as separate units along the shaft line or integrated within the gearbox, depending on the propulsion system configuration.[63]

06.1.4.2 Static loads

The propulsion shaft of a marine vessel is subjected to several types of static loads during operation. These loads arise from torque transmission, propeller thrust, and self-weight effects, and they combine to produce complex stress states in the shaft material. The main static loads can be summarized as follows.

- **Torsional shear stress** due to the torque transmitted along the shaft.
- **Axial normal stress** resulting from propeller thrust transmitted to the hull.
- **Bending stresses** (normal and shear) are induced by transverse loads such as the weight of the shaft, propeller, and hydrodynamic effects.

When multiple loads act simultaneously, the stress distribution is non-uniform. Shear stress from transverse loads is maximum at the shaft center and decreases towards the outer surface, whereas bending and torsional stress are maximum at the surface and zero at the center. Since the magnitude of transverse shear is generally small compared to bending and torsional stresses, it is often neglected in shaft design. The **critical design condition** therefore occurs at the outer shaft surface (region **B** in Figure 41).

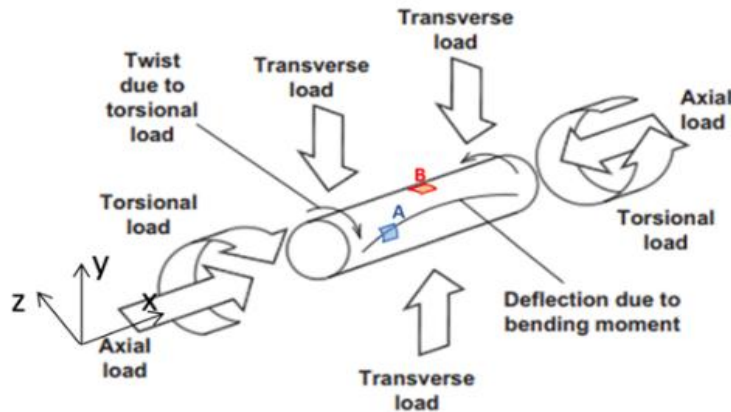


Figure 41 - Load applied on the shaft

06.1.4.2.1 Equivalent Stress and Von Mises Criterion

For ductile materials such as steel, the **distortion energy theory (Von Mises criterion)** provides the most reliable means of estimating equivalent stress under combined loading. The general three-dimensional form is given below.

$$\sigma_{eq} = \sqrt{\frac{(\sigma_x - \sigma_y)^2 + (\sigma_y - \sigma_z)^2 + (\sigma_z - \sigma_x)^2 + 6(\tau_{xy}^2 + \tau_{yz}^2 + \tau_{zx}^2)}{2}}$$

In a two-dimensional stress state, relevant for many shaft loading cases, the expression can simplify as below. (Stresses of the z direction is 0.)

$$\sigma_{eq} = \sqrt{\frac{(\sigma_x - \sigma_y)^2 + (\sigma_y - 0)^2 + (0 - \sigma_x)^2 + 6(\tau_{xy}^2 + 0 + 0)}{2}}$$

$$\sigma_{eq} = \sqrt{\frac{\sigma_x^2 - 2\sigma_x\sigma_y + \sigma_y^2 + \sigma_y^2 + \sigma_x^2 + 6\tau_{xy}^2}{2}}$$

$$\sigma_{eq} = \sqrt{\sigma_x^2 - \sigma_x\sigma_y + \sigma_y^2 + 3\tau_{xy}^2}$$

For the typical marine shaft condition involving **axial load, torsional shear, and unidirectional transverse bending**, a further simplification is obtained. At the outer surface point where bending is maximum, for a typical shaft section (ignoring axial thrust and pressure for now).

- Let the axial direction be x .
- Bending creates normal **stress along x** : $\sigma_x = \sigma_b$.
- There's **no membrane normal stress** in the transverse direction at that point, so $\sigma_y \approx 0$.
- Torsion creates a **shear** on that surface element, which we can take as $\tau_{xy} = \tau_t$.

Substitute into the plane-stress expression.

$$\sigma_{eq} = \sqrt{\sigma_b^2 - \sigma_b \cdot 0 + 0^2 + 3\tau_t^2}$$

$$\sigma_{eq} = \sqrt{\sigma_b^2 + 3\tau_t^2}$$

If this shaft also has **axial thrust** producing a uniform axial stress σ_a , just replace σ_b with $\sigma_a + \sigma_b$.

$$\sigma_{eq} = \sqrt{(\sigma_a + \sigma_b)^2 + 3\tau_t^2}$$

This model provides an efficient means of estimating the maximum equivalent stress under static loading. However, real operating conditions involve additional dynamic effects and irregular loadings, which are addressed in Section 06.1.4.3.

06.1.4.2.2 Breakdown of the IACS M68.4 formula

The International Association of Classification Societies (IACS) M68.4 formula is widely used by manufacturers, such as Berg Propulsion, to determine the minimum required shaft diameter for marine propulsion systems. The formula is semi-empirical in nature and incorporates load, inertia, and material parameters into a single expression.

06.1.4.2.2.1 Hollow shaft

General torsion equation

$$\frac{T}{J} = \frac{\tau}{r} = \frac{G\Theta}{L}$$

T = Torque or Twisting

G = Modulus of rigidity

J = Polar moment of inertia

Θ = Angle of twist

τ = Shear stress

L = Length of the shaft

r = Radius of shaft

For a hollow circular shaft with outer diameter D and inner diameter d;

$$\tau_{max} = \frac{Tr}{J} = \frac{T\left(\frac{D}{2}\right)}{\left(\frac{\pi}{32}\right)(D^4 - d^4)} = \frac{16TD}{\pi(D^4 - d^4)}$$

Set $\tau_{max} = \tau_{allow}$ and solve for D;

$$D^3 = \frac{16T}{\pi\tau_{allow}} \cdot \frac{1}{1 - \left(\frac{d}{D}\right)^4}$$

$$D = \left[\frac{16T}{\pi\tau_{allow}} \cdot \frac{1}{1 - \left(\frac{d}{D}\right)^4} \right]^{\frac{1}{3}}$$

Using the power speed relation $T = \frac{P}{2\pi n}$

$$D = \left[\frac{8}{\pi^2} \cdot \frac{P}{n\tau_{allow}} \cdot \frac{1}{\left(1 - \left(\frac{d}{D}\right)^4\right)} \right]^{\frac{1}{3}}$$

Marine propeller shafts are governed by **fatigue endurance**, not simple static yield.

Rules therefore replace a constant τ_{allow} with an **allowable that itself depends on diameter/material** and includes **service factors** (keyways, corrosion, misalignment, surface finish). A common way is;

$$\tau_{allow} = \frac{C_1(\sigma_B + 160)}{C_2 D}$$

$C_1(\sigma_B + 160) \Rightarrow$ Material endurance proxy

$C_2 D \Rightarrow$ Size/surface/gradient effects

- The $(\sigma_B + 160)$ form is a long-standing rule-of-thumb correlation used in shaft rules to tie endurance to UTS (units MPa).
- The division by D lumps size/gradient/surface effects (larger shafts have lower endurance in shear) into a simple inverse-diameter trend consistent with fatigue data compilations.

$$D = \left[\frac{8}{\pi^2} \cdot \frac{P}{n} \cdot \frac{1}{\left(1 - \left(\frac{d}{D}\right)^4\right)} \cdot \frac{1}{\tau_{allow}} \right]^{\frac{1}{3}}$$

$$D = \left[\frac{8}{\pi^2} \cdot \frac{P}{n} \cdot \frac{1}{\left(1 - \left(\frac{d}{D}\right)^4\right)} \cdot \frac{C_2 D}{C_1(\sigma_B + 160)} \right]^{\frac{1}{3}}$$

$$D = \sqrt[3]{\frac{8C_2}{\pi^2 C_1} \cdot \frac{P}{n} \cdot \frac{1}{\left(1 - \left(\frac{d}{D}\right)^4\right)} \cdot \frac{1}{(\sigma_B + 160)}}$$

$\frac{8C_2}{\pi^2 C_1} \Rightarrow$ Constant

Comparing (5) with the rule form (IACS):

- The bracketed constant $\frac{8C_2}{\pi^2 C_1}$ becomes the **numerical factor “560”** once units are fixed (power in kW, speed in rpm, stress in MPa, diameter in mm) and the rule’s calibration is applied.
- The **F** and **k_3** multipliers collect installation/classification modifiers (service factor, keyway/coupling condition, surface finish, corrosion allowance, alignment tolerance, etc.) and also tidy up unit conversions.
- The **hollow-shaft factor** $\left(1 - \left(\frac{d}{D}\right)^4\right)$ is exactly the geometric term from torsion theory.

Thus, IACS M68.4 minimum shaft diameter formula is below.

$$D_{min} = F \cdot k_3 \cdot \sqrt{\frac{P}{n} \cdot \frac{1}{\left(1 - \left(\frac{d}{D}\right)^4\right)} \cdot \frac{560}{(\sigma_B + 160)}}$$

Where;

D_{min} = Minimum required diameter in mm

P = Rated power in kW transmitted through the shaft (losses in gearboxes and bearings are to be disregarded)

n = speed in revolutions per minute of shaft at rated power

D = Outside diameter in mm of shaft. If the bore of the shaft is $\leq 0.4D$, the expression $\left(1 - \left(\frac{d}{D}\right)^4\right)$ may be taken as 1.0

d = Actual diameter in mm of shaft bore

σ_B = Specified minimum tensile strength in N/mm² of the shaft material

Solving for the outer diameter D produces an expression that closely resembles the IACS formula, with the key difference being the use of ultimate tensile strength instead of yield strength.

This approach reflects the fact that propulsion shafts are designed primarily against **fatigue** rather than static yielding. Since endurance limit is more closely related to the ultimate tensile strength, the formula's material term appropriately accounts for long-term cyclic loading. Further fatigue-specific evaluation of the IACS approach is presented in Section 06.1.4.4.

06.1.4.2.2.2 Solid shaft

If the shaft being designed is a solid shaft, its dimensions can be calculated as follows. There are three loading cases that can be encountered. The following calculation was performed for each of those cases.

1. Pure torsion
2. Bending + torsion (no axial)
3. Bending + torsion + axial

General Bending Equation

$$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$$

M – Bending Movement

y – Distance to neutral axis

I – Second Movement of area

E – Young's modulus

σ – Bending stress

R – Radius of curvature

1. Pure Torsion

From general torsion equation,

$$\tau_{max} = \frac{T \cdot r}{J}$$

$J = \frac{\pi d^4}{32} \rightarrow$ polar moment of inertia of a solid circle

$$\tau_{max} = \frac{T \cdot \left(\frac{d}{2}\right)}{\left(\frac{\pi d^4}{32}\right)} = \frac{16T}{\pi d^3}$$

Impose $\tau_{max} \leq \tau_{allow}$ and solve for d ;

$$\frac{16T}{\pi d^3} \leq \tau_{allow}$$

$$\left(\frac{16T}{\pi \cdot \tau_{allow}}\right)^{\frac{1}{3}} \leq d$$

$$\boxed{d_{min} = \left(\frac{16T}{\pi \cdot \tau_{max}}\right)^{\frac{1}{3}}}$$

2. Bending + torsion (no axial)

Section modulus in bending $\rightarrow W_b = \frac{\pi d^3}{32}$

At the surface;

$$\sigma_b = \frac{M}{W_b} = \frac{32M}{\pi d^3} \quad \text{and} \quad \tau_t = \frac{16T}{\pi d^3}$$

Von Mises equivalent

$$\sigma_{eq} = \sqrt{\sigma_b^2 + 3 \cdot \tau_t^2}$$

$$\sigma_{eq} = \sqrt{\frac{1024M^2}{\pi^2 \cdot d^6} + \frac{768T^2}{\pi^2 \cdot d^6}}$$

Set $\sigma_{eq} \leq \sigma_{allow}$ and solve;

$$\sqrt{\frac{1024M^2}{\pi^2 \cdot d^6} + \frac{768T^2}{\pi^2 \cdot d^6}} \leq \sigma_{allow}$$

$$\frac{1}{\pi d^3} \cdot \sqrt{1024M^2 + 768T^2} \leq \sigma_{allow}$$

$$\left(\frac{\sqrt{1024M^2 + 768T^2}}{\pi \cdot \sigma_{allow}}\right)^{\frac{1}{3}} \leq d$$

$$d_{min} = \left(\frac{\sqrt{1024M^2 + 768T^2}}{\Pi \cdot \sigma_{allow}} \right)^{\frac{1}{3}}$$

3. Bending + torsion + axial

Add axial stress $\sigma_a = \frac{4F}{\pi d^2}$ (uniform over the section).

Surface normal stress becomes $\sigma = \sigma_a + \sigma_b$. Von Mises $\rightarrow$

$$\sigma_{eq} = \sqrt{(\sigma_a + \sigma_b)^2 + 3\tau_t^2}$$

$$\sigma_{eq} = \sqrt{\frac{16F^2}{\Pi^2 \cdot d^4} + \frac{1024M^2}{\Pi^2 \cdot d^6} + \frac{256M \cdot F}{\Pi^2 \cdot d^5} + \frac{768T^2}{\Pi^2 \cdot d^6}}$$

Set $\sigma_{eq} \leq \sigma_{allow}$ and solve;

$$\sqrt{\frac{16F^2}{\Pi^2 \cdot d^4} + \frac{1024M^2}{\Pi^2 \cdot d^6} + \frac{256M \cdot F}{\Pi^2 \cdot d^5} + \frac{768T^2}{\Pi^2 \cdot d^6}} \leq \sigma_{allow}$$

It is quite difficult to calculate d, and d can be found using an iteration method or some other method.

06.1.4.2.3 Stress concentration

polar moment of inertia for a solid shaft

$$J = \frac{\pi d^4}{32}$$

d = Diameter of the shaft

polar moment of inertia for a hollow shaft

$$J = \frac{\pi(d_o^4 - d_i^4)}{32}$$

d_o = Outer diameter of the shaft

d_i = Inner diameter of the shaft

- In torsion or bending, maximum stress occurs at the outer surface, and inner material near the center carries very little stress.
- In a solid shaft, that central region adds *weight* but contributes *almost nothing* to resisting torque or bending.

By removing this low-stress material $\rightarrow$ a hollow shaft maintains nearly the same strength and stiffness while reducing weight and material usage.

For equal outer diameter and material,
the torsional strength (maximum torque capacity) is proportional to J.

Torsional strength ratio R_t

$$R_t = \frac{J_{hollow}}{J_{solid}} = 1 - \left(\frac{d_i}{d_o}\right)^4$$

Weight ratio R_w

$$R_w = \frac{\text{density} \cdot (\text{Area}_{hollow} \cdot \text{length})}{\text{density} \cdot (\text{Area}_{solid} \cdot \text{length})} = 1 - \left(\frac{d_i}{d_o}\right)^2$$

Thus, the **strength-to-weight ratio**

$$\frac{R_t}{R_w} = \frac{1 - (d_i/d_o)^4}{1 - (d_i/d_o)^2}$$

$$d_i < d_o \rightarrow \frac{d_i}{d_o} < 1 \rightarrow \left(\frac{d_i}{d_o}\right)^4 \ll \left(\frac{d_i}{d_o}\right)^2 \ll \frac{d_i}{d_o} < 1 \rightarrow$$

$$\frac{R_t}{R_w} \gg 1 \rightarrow R_t \gg R_w$$

- That means the hollow shaft is more efficient in transmitting torque per unit weight.

Static load perspective

Under static torsion or bending, the stress distribution is dominated by outer fibers.

- The maximum shear (for torsion) or maximum normal stress (for bending) is at the outer surface.
- Therefore, removing the inner core *does not affect maximum stress* but significantly reduces weight.

Therefore, for a given strength requirement, the hollow shaft can achieve the same performance with less mass.

Additional benefits

- Reduced self-weight → lowers bearing loads and improves alignment under static conditions.
- Better corrosion resistance performance when combined with duplex stainless steel (less residual stress concentration).
- Lower inertia improves mechanical response in propulsion systems.

Table 15 - Solid shaft vs. Hollow shaft Comparison

Parameter	Solid Shaft	Hollow Shaft
Material usage	Higher	Lower
Weight	Heavier	Lighter
Strength-to-weight ratio	Lower	Higher
Static stress distribution	Inefficient (core underutilized)	Efficient (outer fibers stressed)
Suitability for marine propulsion	Moderate	Excellent

06.1.4.3 Dynamic loads

The dynamic loads on a propulsion shaft in operation are mainly divided into three cases. The lateral vibrations, torsional vibration, and axial vibration. These dynamic loads (mainly lateral- and torsional vibrations) are critical to know when designing a shaft and its fatigue limit.[66] Axial vibration is usually negligible.

06.1.4.3.1 lateral vibrations - whirling

A) Jeffcott rotor (disk on an elastic shaft)

Model a rotor as a rigid disk of mass m on a mass-elastic shaft with lateral stiffness k and viscous damping c . If the disk is slightly eccentric by a (centre of mass offset), the lateral motions $x(t), y(t)$ satisfy (vector form shown, then decoupled into x/y). [67]

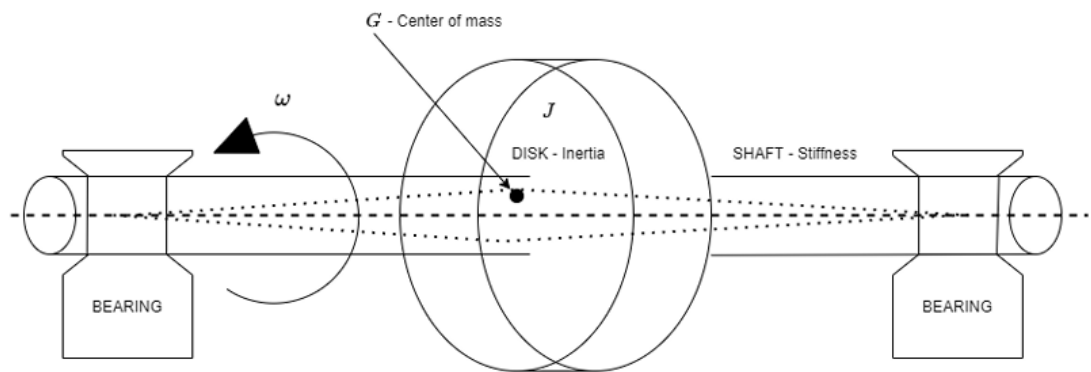


Figure 42 - Geometry of rotating disk on a shaft, supported by bearings

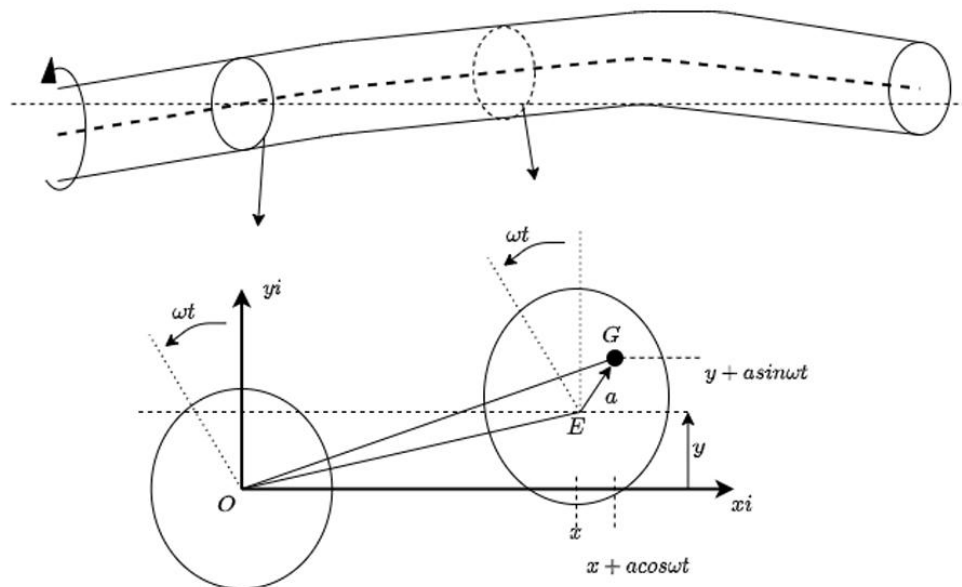


Figure 43 - Notation for whirling calculations.

$$m\ddot{r} = -kr - c\dot{r} \rightarrow (1)$$

$$r = (x + a \cos \omega t)\hat{i} + (y + a \sin \omega t)\hat{j} \rightarrow (2)$$

$$\dot{r} = (\dot{x} - a\omega \sin \omega t)\hat{i} + (\dot{y} + a\omega \cos \omega t)\hat{j} \rightarrow (3)$$

$$\ddot{r} = (\ddot{x} - a\omega^2 \cos \omega t)\hat{i} + (\ddot{y} - a\omega^2 \sin \omega t)\hat{j} \rightarrow (4)$$

Substituting (2), (3), (4) for (1);

$$(m\ddot{x} - ma\omega^2 \cos \omega t + c\dot{x} + kx)\hat{i} + (m\ddot{y} - ma\omega^2 \sin \omega t + c\dot{y} + ky)\hat{j} = 0$$

which gives;

$$m\ddot{x} - ma\omega^2 \cos \omega t + c\dot{x} + kx = 0$$

And

$$m\ddot{y} - ma\omega^2 \sin \omega t + c\dot{y} + ky = 0$$

Steady-state solutions are;

$$x(t) = \frac{ar^2}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} \cos\left(\omega t - \tan^{-1}\left(\frac{2r\zeta}{1-r^2}\right)\right)$$

$$y(t) = \frac{ar^2}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} \sin\left(\omega t - \tan^{-1}\left(\frac{2r\zeta}{1-r^2}\right)\right)$$

With frequency ratio $r = \frac{\omega}{\omega_n}$, damping ratio $\zeta = \frac{c}{2m\omega_n}$, and $\omega_n = \sqrt{\frac{k}{m}}$. The whirl (Orbit) radius is;

$$X = \frac{ar^2}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}}$$

And resonance occurs at $r = 1$ (synchronous whirling) [67]

$$\text{phase lag} \rightarrow \phi = \tan^{-1}\left(\frac{2r\zeta}{1-r^2}\right)$$

Critical speed $\rightarrow$ At $r = 1$, $\omega = \omega_n = \sqrt{k/m}$ is the **whirling (critical) speed**.

B) The shaft stiffness k comes from beam theory

For a simply-supported (pinned-pinned) span L with a mid-span lateral load F , the Euler-Bernoulli solution gives the mid-span deflection.

$$\delta = \frac{EL^3}{48EI}$$

$$\text{Identify the equivalent lateral spring } k = \frac{F}{\delta} \rightarrow k = \frac{48EI}{L^3}$$

That turns the Jeffcott critical speed into

$$\omega_{c,Jeffcott} = \sqrt{\frac{k}{m}} = \frac{4}{L^2} \sqrt{\frac{3EI}{\rho A}}$$

After taking the mid-span mass as the shaft mass $m = \rho AL$. Concentrates the distributed mass at the most flexible location (Center of mass).

C) Continuous-beam (distributed mass) critical

or a uniform beam (simply supported) with distributed mass, separation of variables in $EI y''''(x) + \rho A \ddot{y}(x, t) = 0$ gives eigenfrequencies.

$$\omega_n = \frac{\beta_n^2}{L^2} \sqrt{\frac{EI}{\rho A}}, \beta_1 = \pi$$

So the first bending critically, then

$$\omega_{c,beam} = \frac{\pi^2}{L^2} \sqrt{\frac{EI}{\rho A}}$$

Note – Assume $y(x, t) = \phi(x)e^{i\omega t}$, solve $\phi'''' - \lambda^4 \phi = 0$ and apply simply – supported boundary conditions $y = 0$ and $M = -EI y''$ at $x = 0, L$ which yields $\beta_n = n\pi$. [68]

Relation of the two models

$$\omega_{c,Jeffcott} = \frac{\sqrt{48}}{L^2} \sqrt{\frac{EI}{\rho A}}$$

$$\omega_{c,beam} = \frac{\pi^2}{L^2} \sqrt{\frac{EI}{\rho A}}$$

$\pi^2 \approx 9.87$ vs. $\sqrt{48} \approx 6.93 \rightarrow$ The Jeffcott formula predicts a lower (more conservative) first critical than the continuous-beam formula for the same span L .

“Barred-speed” criterion used in shaft alignment

Class practice is to keep operating speed clear of a critical by about $\pm 20\%$ of MCR; i.e., a critical should lie either below $0.8 n_{MCR}$ or above $1.2 n_{MCR}$. This is the criterion cited in the alignment/whirling study used here. [68]

06.1.4.3.2 Torsional vibrations

a) Shaft torsion (Saint-Venant)

For a circular shaft segment of length L carrying torque T , the twist rate is

$$\frac{d\theta}{dx} = \frac{T}{GJ} \rightarrow \theta = \frac{TL}{GJ}$$

Because shear strain $\gamma = r \frac{d\theta}{dx}$, Hooke's law $\tau = G\gamma$, and $T = \int \tau r dA = G \frac{d\theta}{dx} \int r^2 dA = G \frac{d\theta}{dx} J$.

Max shear at the outer fiber $r = c$ is

$$\tau_{max} = \frac{Tc}{J}$$

For a hollow circular shaft, $J = \frac{\pi}{32} (d_o^4 - d_i^4)$, so

$$\tau_{max} = \frac{16Td_o}{\pi(d_o^4 - d_i^4)}$$

b) Mass- elastic torsional model (rotor-shaft-prop system)

With lumped inertias J_i connected by shaft torsional springs $k_t = \frac{GJ}{L}$, the free vibration equations are

$$J\ddot{\theta} + K\theta = 0$$

Eigen problem $|K - \omega^2 J| = 0$.

Two-inertia special case (engine J_1 , prop J_2 , shaft stiffness k_t):

$$J_1\ddot{\theta}_1 + k_t(\theta_1 - \theta_2) = 0 \text{ and } J_2\ddot{\theta}_2 + k_t(\theta_2 - \theta_1) = 0$$

$$\omega_n = \sqrt{k_t \left(\frac{1}{J_1} + \frac{1}{J_2} \right)}$$

(Assume $\theta_i = \Phi_i e^{i\omega t}$; non-trivial solution requires the determinant above to vanish.)

c) Nominal shaft torque and prop curves

DNV's notation gives the nominal torque and speed relations we'll use in checks.

$$T_o = \frac{30}{\pi} \frac{P_o}{n_o} \text{ (engine)}, T_{po} = iT_o(\text{propeller}), \omega_p = \frac{\pi n_p}{30}$$

d) Application factor K_A (torsional vibration amplitude)

scales nominal torque to include vibration/impact: it “describes the range between the nominal torsional load and the amplitude of the torsional vibration.” In preliminary design we may estimate it from class tables when a full TVC isn't available.

Typical DNV estimates for continuous “normal” operation ($K_{A,norm}$) are;

- Electric drive – 1.1
- Diesel with flexible coupling – 1.3
- Diesel without flexible coupling – 1.5.

A slightly higher peak factor $K_{A,P} \approx 1.4$ is used for short-duration peaks. [69]

From these, the alternating (vibratory) torque amplitude at a given operating point is

$$T_a = (K_{A,norm} - 1)T_o$$

And the occasional peak torque is $T_{peak} = K_{A,P} \cdot T_o$.

e) Torsional stress → von Mises equivalence for fatigue and yield

- For pure torsion, the von Mises equivalent is $\sigma_{eq} = \sqrt{3} \tau$ (used later to compare against axial fatigue and yield limits in the material data).

f) Damping (for completeness)

When you do a full TVC, propeller hydrodynamic damping is commonly represented by Archer/Frahm numbers or Schwaneke's formula; typical Archer numbers 25–35 and Frahm 2.9–3.7 are widely used in TVC tools (e.g., DNV Nauticus). Schwaneke's expression for propeller damping C is also given there. [69]

06.1.4.3.3 Axial vibration

Governing PDE and wave speed

Consider the axial displacement $u(x, t)$ of a uniform shaft (area A , Young's modulus E , density ρ). Force equilibrium on a slice gives

$$\frac{\partial}{\partial x} (EA \partial_x u) = \rho A \partial_{tt} u \Rightarrow \partial_{tt} u = c^2 \partial_{xx} u,$$

with longitudinal wave speed $c = \sqrt{\frac{E}{\rho}}$ (independent of cross section)

Separation of variables $u = \phi(x)e^{i\omega t}$ yields $\phi'' + k^2\phi = 0, k = \frac{\omega}{c}$.

General solution → $\phi(x) = C_1 \sin(kx) + C_2 \cos(kx)$.

Boundary conditions;

Fixed (thrust bearing) end at $x = 0$: $u(0, t) = 0 \rightarrow C_2 = 0$

Free end (no axial force) at $x = L$: $EAu'(L, t) = 0 \rightarrow \cos(kL) = 0$

These give the fixed free eigenvalues: $k_n L = (2n + 1) \frac{\pi}{2}$ and the natural frequencies.

$$\omega_n = \frac{(2n+1)\pi}{2L} c \quad \text{and} \quad f_n = \frac{\omega_n}{2\pi} = \frac{(2n+1)}{4L} c$$

Lowest (first) mode: $f_1 = \frac{c}{4L}$

If both ends are axially fixed (two rigid axial restraints), get $k_n L = n\pi$ and

$$f_n^{(fixed-fixed)} = \frac{n}{2L} c$$

which are higher (stiffer boundary → higher frequency).

Thrust-bearing stiffness model

If the outboard end is connected to an axial spring K_s (prop/thrust-bearing compliance), the boundary becomes $EA u'(L) = -K_s u(L)$. With $u = C_1 \sin kx$ this gives the characteristic equation.

$$E Ak \cos(kL) + K_s \sin(kL) = 0 \rightarrow \tan(kL) = -\frac{E Ak}{K_s}$$

which smoothly transitions between fixed-free ($K_s \rightarrow 0 \Rightarrow \cos kL = 0$) and fixed-fixed ($K_s \rightarrow \infty \Rightarrow \sin kL = 0$). That's handy if later you want to include measured thrust-bearing stiffness.

The Blade Passing Frequency (BPF)

The Blade Passing Frequency (BPF) is the frequency at which the propeller blades pass a fixed point (like the hull, rudder, or shaft bearings).

It represents how often a pressure pulse or thrust fluctuation occurs due to the blades moving through water.

Mathematically:

$$f_{\text{BPF}} = N_b \times f_{\text{shaft}}$$

where:

- N_b = number of propeller blades
- f_{shaft} = shaft rotation frequency (revolutions per second)

Axial vibration seldom matters in ships

Class guidance used in your source explicitly says axial stresses in marine shafting are negligible compared with torsional and bending stresses; fatigue checks likewise combine torsional and rotating bending, axial stresses disregarded.

Also $\rightarrow$ the thrust bearing fixes the axial movement of the line its job is to take propeller thrust and arrest axial oscillation.[70]

06.1.4.3.4 Fatigue

We assess high-cycle fatigue under combined rotating bending and torsional vibration. The stress state at the surface is decomposed into (i) a fully reversed bending normal stress and (ii) a mean plus alternating torsional shear due to torque pulsations.[71]

Geometry (hollow circular)

Polar moment $J = \frac{\pi}{32} (D_o^4 - D_i^4)$, second moment $I = \frac{\pi}{64} (D_o^4 - D_i^4)$, section modulus $Z = I/(D_o/2) = J/D_o$.

Bending (fully reversed at the surface)

$$\sigma_{b,a} = \frac{M}{Z} = \frac{32MD_o}{\pi(D_o^4 - D_i^4)}$$

Torsion (mean and alternating) with surface radius $c = D_o/2$

$$\tau_m = \frac{T_o \cdot c}{J}, \quad \tau_a = \frac{\Delta T \cdot c}{J} = (K_A - 1)\tau_m$$

where T_o is nominal torque at the operating point and K_A is the application factor that captures torsional vibration amplitude (class practice: diesel with flexible coupling $K_A \approx 1.3$ normal, $K_{A,P} \approx 1.4$ peak).

Von Mises combination (amplitudes)

$$\sigma_{a,eq} = \sqrt{\sigma_{b,a}^2 + 3\tau_a^2}$$

$$\sigma_{m,eq} = \sqrt{3} \cdot \tau_m$$

Mean-stress correction (Goodman)

$$\frac{\sigma_{a,eq}}{\sigma_e} + \frac{\sigma_{m,eq}}{\sigma_{UTS}} \leq 1$$

Hot-spot/geometry effects → At local features (steps, fillets, keyways) use the fatigue stress-concentration factor K_f (not the elastic K_t); fatigue strength depends on the mean stress in the notch volume, not the elastic peak. Apply K_f to the local $\sigma_{b,a}$, τ_a , τ_m before combining.

Material allowable (Duplex 2205, annealed)

- Fatigue (axial, 10^7 cycles): 37.9–41.9 ksi ⇒ use $\sigma_e = 262$ MPa (conservative lower bound).
- Tensile strength range: 92.8–138 ksi ⇒ take $\sigma_{UTS} = 640$ MPa as a conservative lower bound; yield $\sigma_y \approx 460 - 510$ MPa.

06.1.4.4 Spline design

$$T = \frac{1}{8} \tau_m n l (D^2 - d^2) [72]$$

T = Torque transmitted through splines

τ_m = Permissible stress

n = Number of splines

l = length of the hub

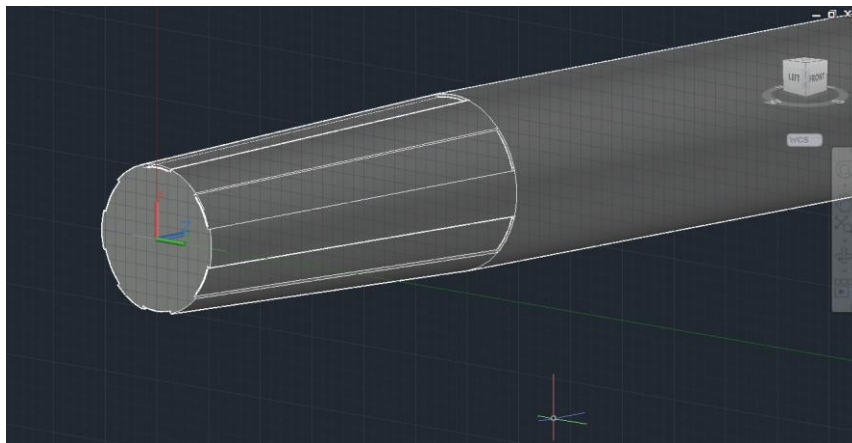


Figure 44 - The End where the propeller connects

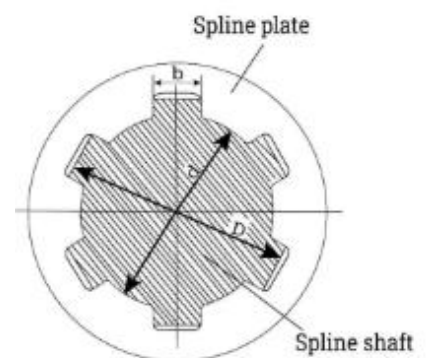


Figure 45 - Cross section of the Spline shaft

06.1.4.5 Marine type flange coupling design

- The flanges are forged integral with the shaft.
- Marine-type flanges on propulsion shafts are almost always machined from the shaft itself (integral flange), not welded.
- Tapered headless bolts are used to hold the flanges together.
- Number of bolts changes from four to twelve depending on the shaft diameter.
- The number of bolts may be chosen from the following table.

Table 16 - Number of bolts for marine type flange coupling. (According to IS: 3653 - 1966 (Reaffirmed 1990))

Shaft diameter (mm)	35 to 55	56 to 150	151 to 230	231 to 390	Above 390
No. of bolts	4	6	8	10	12

The proportions for the marine type of flange coupling are taken as follows. [72], [73]

- Thickness of flange = $d/3$
- Taper of bolts = 1 in 20 – 1 in 40
- Pitch circle diameter of bolts (D_1) = $1.6 d$
- Outside diameter of flange (D_2) = $2.2 d$

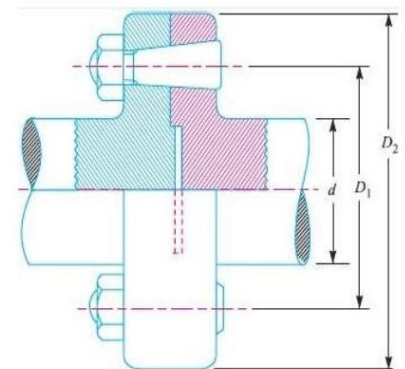


Figure 46 - Marine type flange coupling

06.1.4.6 Design of Threaded Portion for Propeller Retaining Nut

The nut that locks the propeller on the tapered shaft is designed as a single-start ISO metric external thread, 60° flank angle.

To maintain sufficient wall thickness on the 100 mm long solid section, the major diameter of the external thread is fixed at $d = 99$ mm (With 1 mm clearance from diameter).

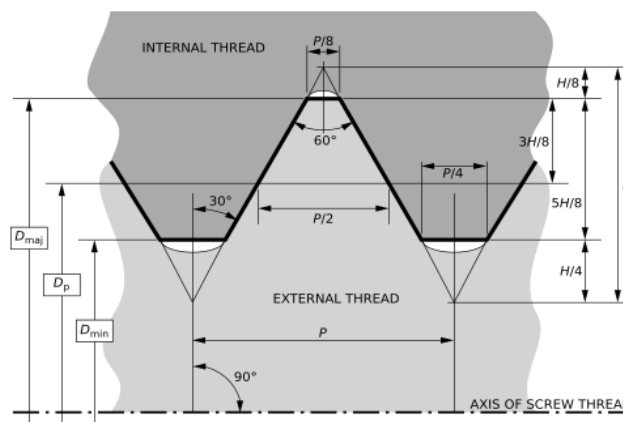


Figure 47 - Thread at end of the shaft

This corresponds to a special metric thread designation **M95 × P**, where P is the thread pitch. [74]

A relatively **fine pitch** is preferred for large retaining nuts to obtain higher axial clamping force for a given tightening torque and better self-locking behavior. A pitch of $P = 4$ mm is adopted for this design (M95×4).

The thread geometry is based on the standard ISO metric V-thread profile:

- triangular thread with included angle 60°
- fundamental triangle height $H = 0.866 P$
- actual thread depth (radial) $h = 0.541 P$ (5/8 of H)

From this profile, the major, pitch and minor diameters are related to pitch by: [74]

$$\begin{aligned} H &= 0.866 P \\ h &= 0.541 P \\ d_{\text{minor}} &= d - 2h = d - 1.082 P \\ d_{\text{pitch}} &= d - 0.6495 P \end{aligned}$$

06.1.4.7 Classification of rules and guidelines

Marine propulsion shaft design is governed by international classification societies to ensure safety, reliability, and standardization across vessels operating in varying conditions. These organizations establish design limits, inspection procedures, and testing requirements that every shaft system must meet before being approved for service.

06.1.4.7.1 International Association of Classification Societies (IACS)

The International Association of Classification Societies (IACS) is a non-governmental body comprising twelve major classification societies, representing about 90% of the world's shipping tonnage. IACS provides the foundation for many national and organizational rules through its Unified Requirements (UR). For propulsion shafts, IACS UR M68.4 defines the minimum allowable diameter of the shaft based on transmitted torque, material properties, and safety margins. Although conservative, the formula ensures uniform safety across member societies and serves as a baseline for further refinement by individual organizations. [75], [76]

06.1.4.7.2 Det Norske Veritas (DNV)

DNV builds upon IACS guidelines through documents such as DNVGL-RU-SHIP Pt.4 Ch.4 “Rotating Machinery – Power Transmission” and DNVGL-CG-0038 “Calculation of Shafts in Marine Applications”. DNV rules specify not only the minimum shaft diameter but also vibration, fatigue, and alignment criteria. The DNVGL-CG-0038 guideline integrates torsional vibration analysis, material yield limits ($\sigma_y \leq 0.7\sigma_B$), and application factors ($KA \leq 2.7$). It also accounts for protection against corrosion through oil lubrication, coatings, or material selection to ensure long-term durability.[63], [77]

06.1.4.7.3 American Bureau of Shipping (ABS)

According to ABS Guidance Notes on Propulsion Shafting Alignment (2019), shaft alignment analyses must demonstrate that bearing loads, reactions, and bending moments are within allowable limits under all operational conditions. ABS rules require that for propulsion shafts larger than 300 mm or those lacking forward stern-tube bearings, detailed alignment calculations including bearing loads, bending moments, and slope tolerances be submitted for review and approval. [78]

06.1.4.7.4 Lloyd's Register (LR) and Bureau Veritas (BV)

Lloyd's Register (LR) and Bureau Veritas (BV) similarly follow IACS UR M68 but introduce their own design verifications for bearing slope, alignment tolerances, and vibration response. Both emphasize that propulsion shaft components must maintain positive bearing reactions and acceptable misalignment slopes during operation to prevent premature bearing wear and fatigue.[79], [80]

06.1.4.7.5 Application and Relevance

In practice, these rules ensure that marine propulsion shafts are designed with adequate safety margins against torsional failure, bending fatigue, and misalignment stresses. While IACS provides the conservative baseline, DNV's CG-0038 method allows optimized designs validated by simulation tools such as DNV Nauticus Machinery (Shaft Alignment and Torsional Vibration). The evolving combination of empirical formulas and computational verification enables lighter, more efficient shafts without compromising safety.[73]

06.1.5 Calculations

Design Parameters

Transmitted Power = 298 kW.[55]

Rotation speed of the shaft at the rated power = 225 rpm[57]

Vessel speed: $V_S = 12 \text{ kn} = 6.17 \text{ ms}^{-1}$ [81]

Wake fraction $w = 0.10$ [82]

Open-water efficiency $\eta_0 = 0.65$ [83]

Bending estimate for propulsion shafts: $M \approx 0.4 T$ [84]

06.1.5.1 Outer diameter calculation

Loads

Torque $\rightarrow T = \frac{P}{\omega} = 12.65 \text{ kNm}$

Bending movement $\rightarrow M = 0.4 \times T = 5.06 \text{ kNm}$

Axial Thrust $\rightarrow F = \frac{\eta_0 P}{(1-w)V_S} = \frac{0.65 \times 298}{(1-0.1) \times 6.17} = 34.9 \text{ kN}$ [82]

Outer diameter calculations using Bending + torsion + axial load equation

$$\sqrt{\frac{16F^2}{\pi^2 \cdot d^4} + \frac{1024M^2}{\pi^2 \cdot d^6} + \frac{256M \cdot F}{\pi^2 \cdot d^5} + \frac{768T^2}{\pi^2 \cdot d^6}} \leq \sigma_{\text{allow}}$$

$\sigma_{\text{allow}} = 261 \text{ MPa}$ (Lower bound of the Duplex SS 2205)

fatigue safety factor $S_f = 1.6$ [85]

$$d \approx 91.4 \text{ mm} = 100 \text{ mm (for standard shaft diameters)}$$

Outer diameter calculations using IACS M68.4 minimum shaft diameter formula

$$D_{solid,min} = F k_3 \sqrt{\frac{P}{n} B}$$

$$B = \frac{560}{(\sigma_B + 160)}$$

$\sigma_B = 639.83347681$ MPa (Lower bound of the Duplex SS 2205)

$$B = \frac{560}{(639.84 + 160)} = 0.700146$$

From IACS UR M68

$F = 95$ for turbine/electric or diesel with hydraulic/slip couplings. [84]

$k_3 = 1.26$ **shaft design factor** $\rightarrow$ 1.00 for straight/intermediate shaft; 1.15 between aftmost bearing and forward stern-tube seal; 1.22 for keyless taper; 1.26 for key-fitted propeller. [84]

$$D_{solid,min} = 115.27 \text{ mm} \equiv 125 \text{ mm (for standard shaft diameters)}$$

$$125 \text{ mm} > 100 \text{ mm} \rightarrow \mathbf{125 \text{ mm}}$$

Considering the results obtained in the above two cases, the most appropriate outer diameter of the shaft is 120mm. Using that, the inner diameter of the bore was now calculated.

06.1.5.2 Inner diameter calculation

$$D_{outer} = F \cdot k_3 \cdot \sqrt{\frac{P}{n} \cdot \frac{1}{\left(1 - \left(\frac{d_{inner}}{D_{outer}}\right)^4\right)} \cdot \frac{560}{(\sigma_B + 160)}}$$

$$d_{inner} = D_{outer} \times \left(1 - \frac{F^2 k_3^2 \left(\frac{P}{n}\right) B}{D^2}\right)^{\frac{1}{4}}$$

$$d_{inner} = 63.28 \text{ mm} \equiv \mathbf{60 \text{ mm (For the safety)}}$$

06.1.5.3 Vibration calculation

Here, the selected material and shaft diameters cannot be changed. Then we will focus on reducing vibration by changing the spacing between the bearings.

- Young's modulus $E \approx 28.3 - 29.7$ Mpsi $\rightarrow$ **195–205 GPa**
- Density $\rho \approx 0.282 - 0.283$ lb/in³ $\rightarrow$ **7800–7830 kg/m³**
- Poisson ≈ 0.30 .
- Shear modulus $G = 10.9 - 12.3 \times 10^6$ psi ($\approx$ **75–85 GPa**).

- Yield strength – 66.7–74 ksi ($\approx$ **460–510 MPa**).
- Fatigue strength at 10^7 cycles (axial amplitude) – 37.9–41.9 ksi ($\approx$ **262–289 MPa**).

We'll use $E = 200\text{GPa}$, $\rho = 7800\text{kg/m}^3$ (mid-range), $G = 80\text{ GPa}$ (mid-range), Fatigue strength = 262 MPa (conservative)

06.1.5.3.1 lateral vibrations – whirring Calculation

Section properties (hollow circular)

$$A = \frac{\pi}{4}(D_o^2 - D_i^2) = 9.444 \times 10^{-3} \text{ m}^2$$

$$I = \frac{\pi}{64}(D_o^4 - D_i^4) = 1.1348 \times 10^{-5} \text{ m}^4$$

Mass per length $\rho A = 73.6632 \text{ kg/m}$

Flexural rigidity $EI = 2.2696 \times 10^6 \text{ Nm}^2$

The shaft speed at the rated power (from the drive we have been using) is $n_{op} = 225 \text{ rpm}$. The barred- speed margin is $1.2n_{op} = 270 \text{ rpm}$

Conservative Jeffcott estimate

$$n_{c,Jeffcott} = \frac{\omega_{c,Jeffcott}}{2\pi} \times 60 = \frac{\sqrt{48}}{2\pi} \times 60 \times \frac{1}{L^2} \sqrt{\frac{EI}{\rho A}}$$

Plugging the section and material numbers;

If $L = 1.2 \text{ m}$ (typical twin-bearing span. In this case we consider the all uniform length of the propeller shaft for safety)

$$n_{c,Jeffcott} = 8064.527 \text{ rpm}$$

The largest span that still clears the +20% band is found by solving $n_{c,Jeffcott} = 270 \text{ rpm}$.

$$L_{max} = 6.5582 \text{ m}$$

So any bearing gap $L \leq 6.6\text{m}$ is acceptable by the $\pm 20\%$ rule. At $L = 1.2\text{m}$ the separation ratio is $\frac{8064.527}{225} \approx 35.84 \rightarrow$ **very safe**.

Continuous-beam check (less conservative, distributed mass)

$$n_{c,beam} = \frac{\pi^2}{2\pi} \times 60 \times \frac{1}{L^2} \sqrt{\frac{EI}{\rho A}}$$

For $L = 1.2 \text{ m} \rightarrow n_{c,beam} = 11488.359 \text{ rpm}$

At $L = 6.5582 \text{ m} \rightarrow n_{c,beam} = 384.637 \text{ rpm} > 270 \text{ rpm}$

Both models say shaft is comfortably away from resonance.

The unbalanced response looks

At $L = 1.2$ m, the Jeffcott frequency ratio is $r = \frac{n_{op}}{n_{c,Jeffcott}} \approx \frac{225}{8064.527} \approx 0.027899$. From the response;

$$\frac{X}{a} = \frac{r^2}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}}$$

Even with very small damping, $r \ll 1 \rightarrow X \ll a$. The orbit radius is a small fraction of the static eccentricity. (far from resonance).

Final result → With $D_o = 125$ mm, $D_i = 60$ mm (Duplex 2205), the shaft is safe in lateral vibration provided the bearing span is $L \leq 6.56$ m. A typical span $L = 1.2$ m yields $n_{c,Jeffcott} = 8064.527$ rpm (Jeffcott) and $n_{c,beam} = 11488.359$ rpm (beam), both well above $1.2 \times 225 = 270$ rpm.

06.1.5.3.2 Torsional vibrations Calculation

Drive style – geared diesel with flexible coupling (our same boat class as before), so I use $K_{A,norm} = 1.3$ and $K_{A,P} = 1.4$.

We'll check at the rated point of the selected workboat (≈ 300 kW on the propeller at **225 rpm** after the 8:1 reduction), which was already established and is compatible with class notation formulas. Using DNV's definitions.

$T_o = \frac{30}{\pi} \frac{P_o}{n_o}$ With $P = 300$ kW, $n_P = 225$ rpm,

$$T_o = \frac{30}{\pi} \times \frac{300}{225} = 12.73 \text{ kNm}$$

Nominal torsional shear at the outer fiber (hollow)

$$\tau_o = \frac{16T_o d_o}{\pi(d_o^4 - d_i^4)} = \frac{16 \times (12.73 \times 10^3) \times (125 \times 10^{-3})}{\pi \times (125^4 - 60^4) \times 10^{-12}} = 35.0556 \text{ MPa}$$

Vibratory (alternating) torsional stress amplitude using $K_{A,norm} = 1.3$

$$\tau_a = (K_{A,norm} - 1)\tau_o = 0.3 \times 35.0556 = 10.5167 \text{ MPa}$$

Von Mises equivalent amplitude for fatigue comparison

$$\sigma_{eq,a} = \sqrt{3} \cdot \tau_a = 18.2155 \text{ MPa}$$

Material fatigue allowance (Duplex 2205): axial amplitude at 10^7 cycles $\sigma_f \approx 262$ MPa (conservative).

Shear limit → $\tau_f \approx \sigma_f / \sqrt{3} \approx 151.2657$ MPa

Fatigue margin

$$\tau_a = 10.52 \text{ MPa} \ll \tau_f = 151.27 \text{ MPa}$$

$$\text{Factor of safety in shear amplitude} = \frac{151.2657}{10.5167} = 14.38$$

Occasional peak check with $K_{A,P} = 1.4$

$$\tau_{peak} = K_{A,P} \tau_o = 1.4 \times 35.0556 = 49.07784 \text{ MPa}$$

Von Mises equivalent $\sigma_{eq,peak} = \sqrt{3} \times 49.07784 = 85.0053 \text{ MPa}$, far below yield $\sigma_y \geq 460 \text{ MPa}$.

Angle of twist

$$J = \frac{\pi}{32} (D_o^4 - D_i^4) = 2.2696 \times 10^{-5} \text{ m}^4$$

$$\theta = \frac{TL}{GJ} = \frac{12730 \times L}{(80 \times 10^9) \times (2.2696 \times 10^{-5})} \text{ rad}$$

For $L = 1.2 \text{ m} \rightarrow \theta = 0.00841337 \text{ rad} \approx 0.4821^\circ$

Both twists are modest for a propulsion line; increasing span linearly increases twist (and lowers torsional natural frequencies), so if later TVC finds a near-resonance with an engine order, shortening the bearing span is a simple tuning lever.

Final result → With $d_o = 125\text{mm}$, $d_i = 60\text{mm}$ Duplex-2205 and the 300 kW at 225 rpm operating point, the shaft is **very safe in torsion** under both normal vibratory amplitudes and peak events when checked with class-style application factors. The governing numbers (KA values, torque relations, and Duplex-2205 fatigue data) are all cited from the sources above.

06.1.5.3.3 Axial vibrations Calculation

Longitudinal wave speed

$$c = \sqrt{\frac{E}{\rho}} = \sqrt{\frac{200 \times 10^9}{7800}} = 5063.6968 \text{ ms}^{-1}$$

Let L be the bearing span.

A) Modal separation (fixed-free, conservative)

$$f_1 = \frac{c}{4L} = \frac{5063.6968}{4 \times L} \text{ Hz}$$

Examples:

- $L = 1.2 \text{ m} \Rightarrow f_1 \approx 1054 \text{ Hz}$.
- $L = 6.56 \text{ m} \Rightarrow f_1 \approx 192 \text{ Hz}$.

Our propeller shaft runs at $n_p = 225 \text{ rpm} = 3.75 \text{ Hz}$. Dominant axial excitations are $1 \times$ shaft (3.75 Hz) and blade-passing $Z \times 3.75\text{Hz}$ (e.g. $Z = 4 \Rightarrow 15\text{Hz}$; even a conservative $Z = 7 \Rightarrow 26.3\text{Hz}$). These are an order of magnitude below f_1 for any realistic span $L \leq 6.56 \text{ m}$; separation ratio is $f_1/(\text{BPF}) \gtrsim 7.3 - 12.8$. That's a yawning chasm, not a near-miss.

B) Spring-mass lower-bound (even more conservative)

Treat the shaft as an axial spring $k_{ax} = EA/L$ carrying the shaft mass $m = \rho AL$ (concentrated):

$$k_{ax} = \frac{EA}{L}, m = \rho AL$$

$$\omega = \sqrt{\frac{k_{ax}}{m}} = \sqrt{\frac{E}{\rho}} \times \frac{1}{L} = \frac{C}{L}$$

So $f_1 = \frac{c}{2\pi L}$.

- $L = 1.2 \text{ m} \Rightarrow f_1 \approx 4211.414 \text{ Hz}$.

- $L = 6.56 \text{ m} \Rightarrow f_1 \approx 770 \text{ Hz}$.

Even this very conservative model keeps f_1 far above all axial excitations.

C) Axial stiffness and static stress context

$$A = \frac{\pi}{4} (D_o^2 - D_i^2) = 9.444 \times 10^{-3} \text{ m}^2$$

With $L = 1.2 \text{ m}$: $k_{ax} = \frac{EA}{L} \approx \frac{(200 \times 10^9) \times (9.444 \times 10^{-3})}{1.2} = 1.574 \times 10^9 \text{ N/m}$

Meaning $\rightarrow$ a 40 kN thrust fluctuation would move the line only $F/k_{ax} \approx 0.02541 \text{ mm}$, illustrating why axial vibration amplitudes are tiny.

Finally

with Duplex-2205 and Outer Diameter/Inner Diameter = **125/60 mm**, the shaft is **vibration-safe** provided us respect these design conditions:

1. **Bearing span** $L \leq 6.56 \text{ m}$ so lateral criticals clear the $\pm 20\%$ MCR band. (At typical $L \sim 1.2\text{--}1.5 \text{ m}$ We have very large margins.)
2. **Torsional allowances** include a **flexible coupling** (so $K_A \approx 1.3$ normal, 1.4 peak). Stresses are far below Duplex-2205 fatigue and yield limits.
3. **Axial** can be **neglected** for fatigue and resonance risk in this configuration (modes $\gg$ excitations; thrust bearing restrains motion).

06.1.5.6 Fatigue calculations

$$J = \frac{\pi}{32} (D_o^4 - D_i^4) = 2.2696 \times 10^{-5} \text{ m}^4$$

$$T_o = \frac{30}{\pi} \times \frac{300}{225} = 12.73 \text{ kNm}$$

Bending movement $\rightarrow M = 0.4 \times T_o = 5.06 \text{ kNm}$

Torsional application factor (normal): $K_A = 1.3$.

Stresses

$$\sigma_{b,a} = \frac{M}{Z} = \frac{32MD_o}{\pi(D_o^4 - D_i^4)} = 27.868 \text{ MPa}$$

$$\tau_m = \frac{T_o \cdot c}{J} = 35.0557 \text{ MPa}$$

$$\tau_a = \frac{\Delta T \cdot c}{J} = (K_A - 1)\tau_m = 10.51671 \text{ MPa}$$

$$\sigma_{a,eq} = \sqrt{\sigma_{b,a}^2 + 3\tau_a^2} = 33.293 \text{ MPa}$$

$$\sigma_{m,eq} = \sqrt{3} \cdot \tau_m = 60.7183 \text{ MPa}$$

Goodman check (Duplex 2205)

$$\frac{\sigma_{a,eq}}{\sigma_e} + \frac{\sigma_{m,eq}}{\sigma_{UTS}} = \frac{33.293}{262} + \frac{60.7183}{640} = 0.2219 < 1 \rightarrow \text{PASS}$$

Soderberg (with $\sigma_y = 460\text{MPa}$) gives

$$\frac{\sigma_{a,eq}}{\sigma_e} + \frac{\sigma_{m,eq}}{\sigma_{UTS}} = \frac{33.293}{262} + \frac{60.7183}{460} = 0.2591 < 1 \rightarrow \text{PASS}$$

Keyway/ spline sensitivity – If a local $K_f = 1.6$ is applied at a keyed section,

$\sigma_{a,eq} \rightarrow 1.6 \times 33.293 = 53.2688\text{MPa}$, $\sigma_{m,eq} \rightarrow 1.6 \times 60.7183 = 97.1493 \text{ MPa} \rightarrow \text{Goodman}$

$$\frac{\sigma_{a,eq}}{\sigma_e} + \frac{\sigma_{m,eq}}{\sigma_{UTS}} = \frac{53.2688}{262} + \frac{97.1493}{640} = 0.355 < 1 \rightarrow \text{PASS}$$

$$\frac{\sigma_{a,eq}}{\sigma_e} + \frac{\sigma_{m,eq}}{\sigma_{UTS}} = \frac{53.2688}{262} + \frac{97.1493}{460} = 0.4145 < 1 \rightarrow \text{PASS}$$

Corrosion-fatigue margin – Using $\sigma_e/1.6 = 163\text{MPa}$ to represent seawater degradation/design factor, the same Goodman becomes

$$\frac{\sigma_{a,eq}}{\sigma_e} + \frac{\sigma_{m,eq}}{\sigma_{UTS}} = \frac{33.293}{163} + \frac{60.7183}{460} = 0.3362 < 1 \rightarrow \text{PASS}$$

with the $K_f = 1.6$ illustration it is

$$\frac{\sigma_{a,eq}}{\sigma_e} + \frac{\sigma_{m,eq}}{\sigma_{UTS}} = \frac{53.2688}{163} + \frac{97.1493}{460} = 0.5379 < 1 \rightarrow \text{Still PASS}$$

06.1.5.7 Spline calculations

$$T = \frac{1}{8} \tau_m n l (D^2 - d^2) [72]$$

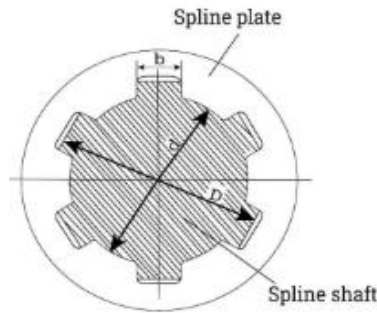


Figure 48 - Cross section of the Spline shaft
Let's take $n = 0$;

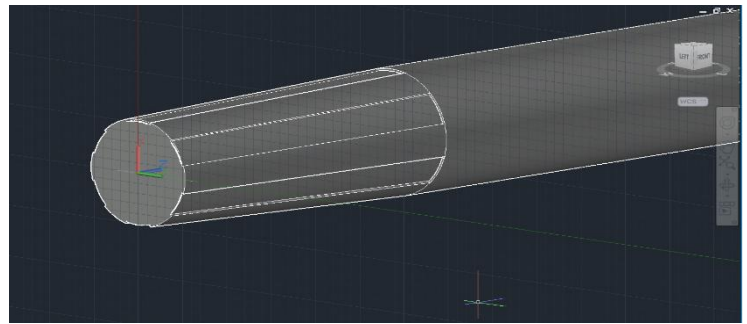


Figure 49 - The End where the propeller connects

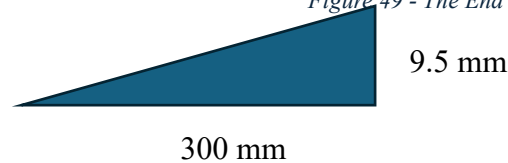


Figure 50 - Angle of the tapered part

$$l = \sqrt{300^2 + 9.5^2} = 300.15 \text{ mm}$$

$$(D^2 - d^2) = 0.39023$$

That mean $\rightarrow$

$$(D^2 - d^2) \geq 0.39023$$

When $D = 125 \text{ mm} \rightarrow d = 124.99 \text{ mm}$

When $D = 106 \text{ mm} \rightarrow d = 105.99 \text{ mm}$

Let's take depth of the spline is 5 mm.

True tapered spline – Both the shaft and hub have splines that follow the same conical taper angle. The teeth are cut so their pitch diameter follows the taper. This provides better centering and torque transmission.

06.1.5.8 Calculation of Marine type flange coupling

- The number of bolts may be chosen from the following table.
- Our shaft diameter is 125 mm. Then we must use 6 bolts.

Table 17 - Number of bolts for marine type flange coupling. (According to IS: 3653 - 1966 (Reaffirmed 1990))

Shaft diameter (mm)	35 to 55		56 to 150	151 to 230	231 to 390	Above 390
No. of bolts	4		6	8	10	12

The proportions for the marine type of flange coupling are taken as follows.

- Thickness of flange $= \frac{d}{3} = \frac{125}{3} \approx 42 \text{ mm} \Rightarrow 45 \text{ mm}$
- Taper of bolts $= 1 \text{ in } 20 - 1 \text{ in } 40 \Rightarrow 1 \text{ in } 30 \text{ taper}$
- Pitch circle diameter of bolts (D_1) $= 1.6 d = 1.6 \times 125 = 200 \text{ mm}$
- Outside diameter of flange (D_2) $= 2.2 d = 2.2 \times 125 = 275 \text{ mm}$

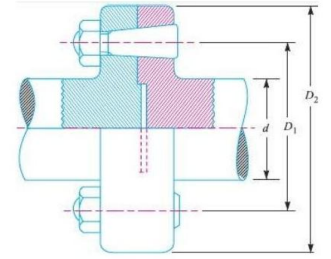


Figure 51 - Marine type flange coupling

06.1.5.9 Calculation of minor and pitch diameters of thread

Given:

- Major (outside) diameter of thread: $d = 94 \text{ mm}$
- Pitch: $P = 4 \text{ mm}$

Fundamental triangle and thread depth

$$H = 0.866 \times 4 = 3.464 \text{ mm}$$

$$h = 0.541 \times 4 = 2.164 \text{ mm}$$

Minor (core) diameter of external thread

$$\begin{aligned} d_{\text{minor}} &= d - 2h = 95 - 2 \times 2.164 \\ &\approx 95 - 4.33 = 90.672 \text{ mm} \end{aligned}$$

So the **minor diameter (root diameter)** of the threaded section is approximately:

$$d_{\text{minor}} \approx 90.672 \text{ mm}$$

Pitch diameter (for checking stresses and tolerances)

$$\begin{aligned} d_{\text{pitch}} &= d - 0.6495 P \\ &= 95 - 0.6495 \times 4 \\ &\approx 95 - 2.60 = 92.402 \text{ mm} \end{aligned}$$

Thus, for the external thread on the shaft:

- Major diameter: $d = 95 \text{ mm}$ (given)
- Pitch: $P = 4 \text{ mm}$
- Pitch diameter: $d_2 \approx 92.402 \text{ mm}$
- Minor diameter: $d_3 \approx 90.672 \text{ mm}$

These values are used for subsequent checks of tensile and shear stresses in the threaded section and for nut design.

06.1.5.10 Estimation of Nut-Tightening Preload and Torque for M95×4 Propeller Nut

To model the effect of the propeller-retaining nut on the shaft, the axial preload generated in the M95×4 thread was estimated from the material strength of the shaft and nut.

06.1.5.10.1 Tensile stress area of the M95×4 thread

For ISO metric threads the tensile (stress) area A_s can be approximated by

$$A_s \approx \frac{\pi}{4} (d - 0.9382 p)^2$$

where

d = nominal thread diameter (mm),

p = pitch (mm).

For an M95×4 thread,

$$\begin{aligned} d &= 95 \text{ mm} \\ p &= 4 \text{ mm} \\ A_s &\approx \frac{\pi}{4} (95 - 0.9382 \times 4)^2 \\ &\approx \frac{\pi}{4} (91.25)^2 \\ &\approx 6.54 \times 10^3 \text{ mm}^2 \end{aligned}$$

06.1.5.10.2 Selection of preload based on yield strength of Duplex 2205

The shaft and nut are manufactured from Duplex stainless steel 2205, which has a 0.2% proof (yield) strength of

$$\sigma_y = 459.88 \text{ MPa} = 459.88 \text{ N/mm}^2$$

To avoid approaching yield while still providing sufficient clamping force to prevent loosening, the tightening preload was limited to 30% of the yield strength:

$$F = k \sigma_y A_s$$

with $k = 0.3$. Thus

$$\begin{aligned} F &= 0.3 \times 459.88 \times 6.54 \times 10^3 \\ &\approx 9.02 \times 10^5 \text{ N} \end{aligned}$$

The corresponding average tensile stress in the threaded section is

$$\sigma_{\text{tight}} = \frac{F}{A_s} \approx \frac{9.02 \times 10^5}{6.54 \times 10^3} \approx 138 \text{ MPa} (< 0.3 \sigma_y)$$

which confirms that the chosen preload is well below the yield strength of Duplex 2205.

06.1.5.10.3 Equivalent tightening torque

In practice, the relationship between tightening torque T and preload F for metric threads can be expressed using the “nut factor” K :

$$T \approx K F d$$

where

d = nominal diameter of the thread (m),

$K \approx 0.2$ for lubricated steel threads.

For a standard metric thread the usual engineering relation is

$$T = F \left[\underbrace{\frac{p}{2\pi}}_{\text{thread helix}} + \underbrace{\frac{\mu_t d_2}{2}}_{\text{thread friction}} + \underbrace{\frac{\mu_c d_w}{2}}_{\text{nut-face friction}} \right]$$

where

- T = applied tightening torque on nut (N·m)
- F = axial preload (N) (this is the force in the shaft / threads)
- p = thread pitch (mm) $\rightarrow$ here $p = 4$ mm
- d_2 = pitch (mean) diameter of the thread (mm) for M95×4 (take from ISO table)
- d_w = effective friction diameter under nut face ($\approx$ mean of bearing area)
- μ_t = friction coeff. in the threads (typically 0.10–0.20 for lubricated steel)
- μ_c = friction coeff. under nut face (similar range)

Then

$$F = \frac{T}{\frac{p}{2\pi} + \frac{\mu_t d_2}{2} + \frac{\mu_c d_w}{2}}$$

This F is the axial force that we should apply in our FEA model.

Very often, all the friction terms are combined into a single “nut factor” K . This is the very simple version of above equation.

$$T \approx K F d$$

Using $d = 95 \text{ mm} = 0.095 \text{ m}$,

$$\begin{aligned} T &\approx 0.2 \times 9.02 \times 10^5 \times 0.095 \\ &\approx 1.71 \times 10^4 \text{ N}\cdot\text{m} \end{aligned}$$

This value represents the approximate torque required to generate the selected preload in the M95×4 nut under typical lubrication conditions.

06.1.5.10.4 Load used in FEA

For the finite element analysis, the **axial preload** $F \approx 9.0 \times 10^5 \text{ N}$ was applied to the threaded/nut region of the shaft (modelled as an equivalent axial force or as pressure under the nut-bearing face). This represents the effect of tightening the propeller-retaining nut and is combined with the operating torsional and thrust loads in the simulation to evaluate stresses in the threaded section.

06.1.5.11 Reverse stress calculation

In this section, we calculate the stress that can occur on the shaft in the waste scenario based on the final dimensions obtained through calculations. If those values are greater than or less than the minimum values of the corresponding properties of the materials, our design is safe.

06.1.5.11.1 For solid part

The part of the solid shaft with the smallest diameter experiences the most stress. Therefore, let's calculate the stress on that part.

$$\sigma_{eq} = \sqrt{\frac{16F^2}{\pi^2 \cdot d^4} + \frac{1024M^2}{\pi^2 \cdot d^6} + \frac{256M \cdot F}{\pi^2 \cdot d^5} + \frac{768T^2}{\pi^2 \cdot d^6}}$$

$$\sigma_{eq} = \sqrt{\frac{16 \times 34900^2}{\pi^2 \cdot (95 \times 10^{-3})^4} + \frac{1024 \times 5060^2}{\pi^2 \cdot (95 \times 10^{-3})^6} + \frac{256 \times 5060 \times 34900}{\pi^2 \cdot (95 \times 10^{-3})^5} + \frac{768 \times 12650^2}{\pi^2 \cdot (95 \times 10^{-3})^6}}$$

$$\sigma_{eq} = 145.569 \text{ MPa} < 261 \text{ MPa} \rightarrow \text{SAFE}$$

06.1.5.11.2 For hollow part

$$D_{outer} = F \cdot k_3 \cdot \sqrt{\frac{P}{n} \cdot \frac{1}{\left(1 - \left(\frac{d_{inner}}{D_{outer}}\right)^4\right)} \cdot \frac{560}{(\sigma_B + 160)}}$$

$$d_{inner} = D_{outer} \times \left(1 - \frac{F^2 k_3^2 \left(\frac{P}{n}\right) B}{D^2}\right)^{\frac{1}{4}}$$

$$60 = 125 \times \left(1 - \frac{95^2 \times 1.26^2 \left(\frac{298}{225}\right) B}{125^2}\right)^{\frac{1}{4}}$$

$$\frac{95^2 \times 1.26^2 \left(\frac{298}{225}\right) B}{125^2} = 0.94691584$$

$$B = 0.7796673733$$

$$B = \frac{560}{(\sigma_B + 160)}$$

$$\sigma_B = \frac{560}{0.7796673733} - 160 = 558.255 \text{ MPa} < 639.833 \text{ MPa} \rightarrow \text{SAFE}$$

Reason for Using Tensile Strength (σ_B) in the IACS Minimum Shaft Diameter Formula

Tensile strength is used in the IACS M68.4 minimum shaft diameter equation because it provides the most reliable measure of a shaft material's ability to withstand the combined effects of torsional shear, bending fluctuations, and fatigue. In marine propulsion shafts, failure rarely occurs from static shear alone; instead, it is driven by cyclic torsion, variable bending, stress concentrations, and corrosion-assisted crack initiation. The shear strength and fatigue endurance of steels are well known to correlate closely with their minimum specified tensile strength, which is the most consistently certified property supplied by manufacturers. By incorporating σ^B , the IACS formula ensures that the calculated diameter accounts not only for shear capacity but also for long-term fatigue performance and real-world degradation effects. Thus, tensile strength functions as a practical and conservative parameter to prevent shaft failure under marine operating conditions.

06.1.6 Results and discussion

06.1.6.1 Final Calculations results

Hollow part

Outer diameter = 125 mm (SAE J755 Standard)

Inner diameter = 60 mm

Taper angle = 16:1 (SAE J755 Standard)

Solid part

Large end diameter = 125 mm

Small end diameter = 106 mm (According to taper)

Flange

No. of bolts = 6

Thickness of flange = 45 mm

Pitch circle diameter of bolts = 200 mm

Outside diameter of flange = 275 mm

Thread

This is $M95 \times 4$ ISO Slandered thread. It is exactly thali with our calculation also.[86]

Major diameter = 95 mm

Miner diameter = 90 mm

Pitch diameter = 92 mm

Pitch = 4 mm

Table 18 - Calculated values of the thread vs. ISO standard values of the thread

	Calculated Values	ISO Standard Values
Major diameter	95	94.940 – 94.465
Minor diameter	90.672	90.610 – 89.628
Pitch diameter	92.402	92.342 – 92.092
Pitch	4	4
Tolerance class		6g
Bolt		$M95 \times 4$

Under dynamic loads

- ❖ Lateral vibration (whirling) → **SAFE**
- ❖ Torsional vibration → **SAFE**
- ❖ Axial vibration → **SAFE / NEGLIGIBLE**
- ❖ Fatigue → **SAFE**

06.1.6.2 Final design

This is a hollow shaft with a solid section with a taper at the end. It has 6 splines that connect the propeller shaft to the propeller hub.

Total length of the shaft = 1.6 m

Tapered spline depth = 5 mm

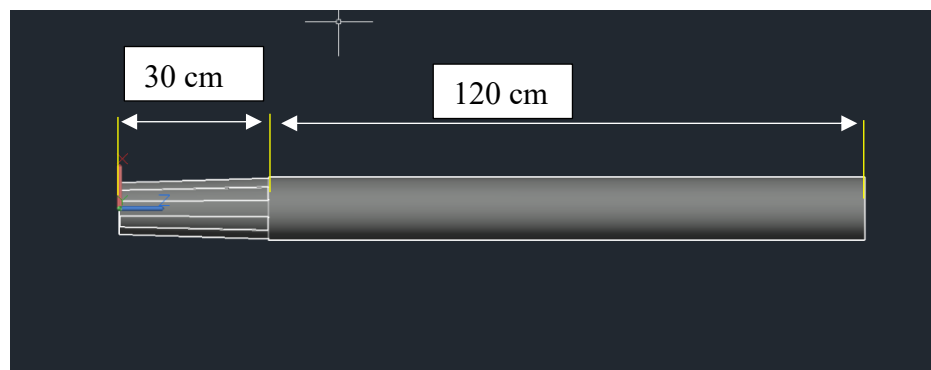


Figure 52 - Length of the propeller shaft

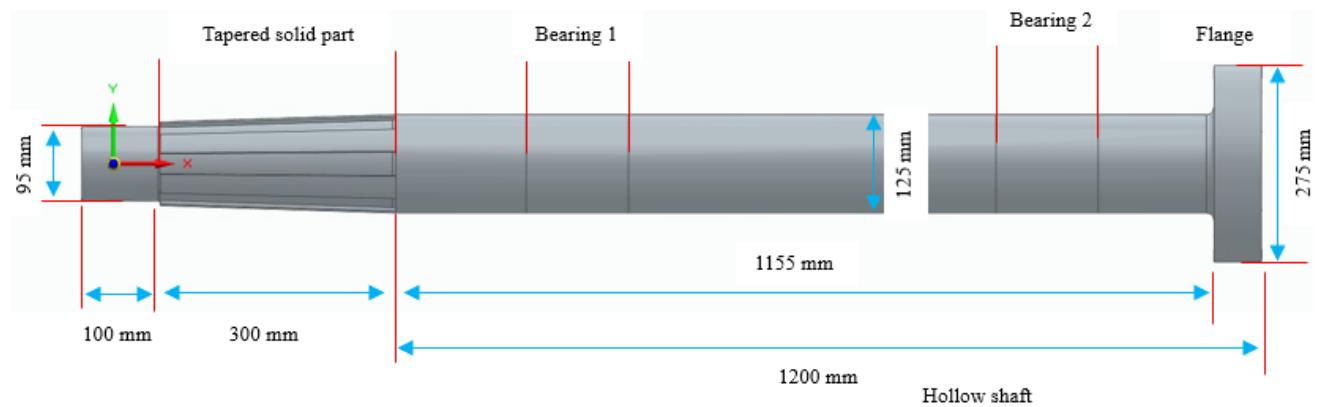


Figure 53 - Design of the propeller shaft

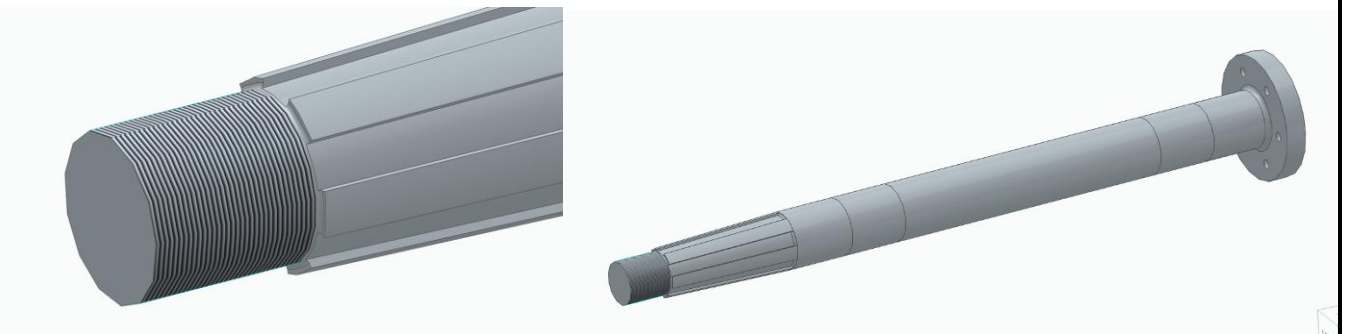


Figure 54 - final design with thread



Figure 55 - Final propeller shaft design without thread

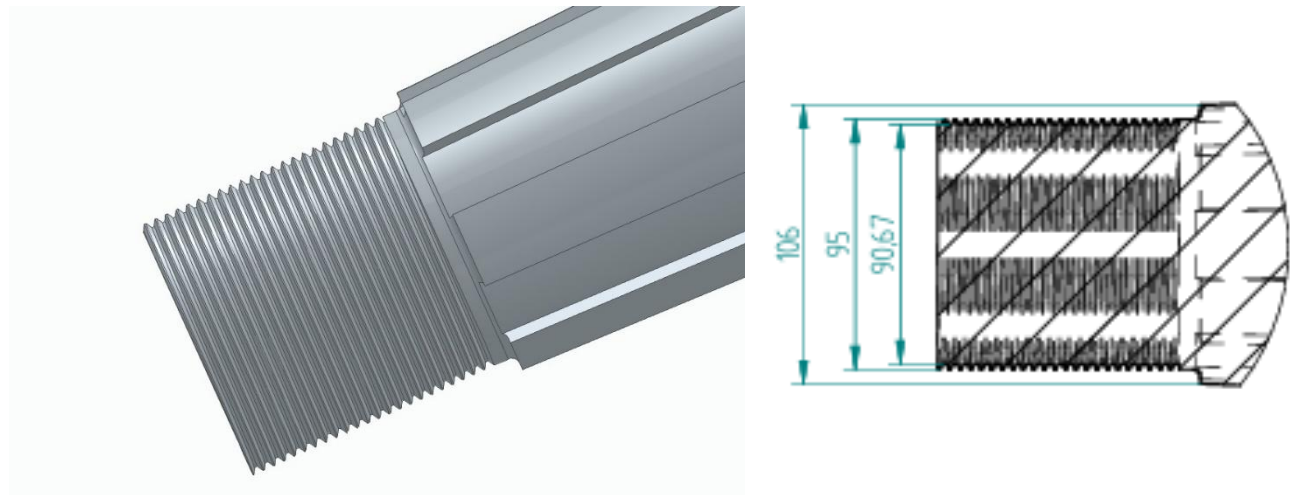


Figure 56 - Thread of the shaft (3D image of the thread and 2D cross section image of the thread(Section A-A))

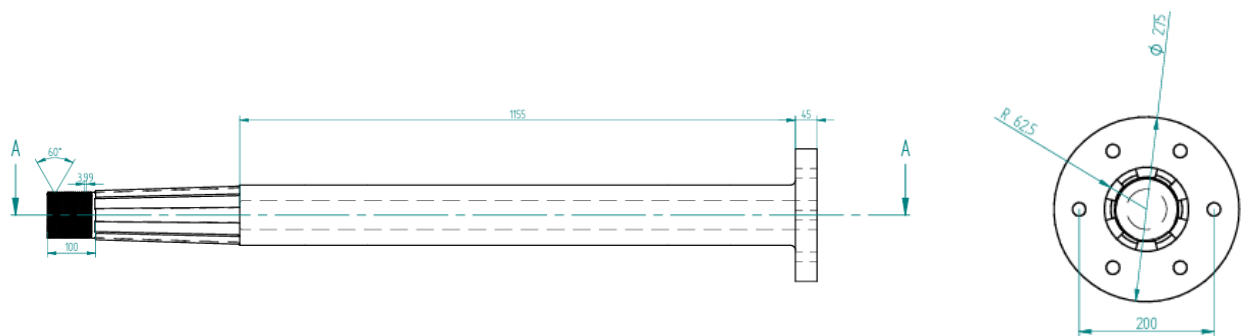


Figure 57 - 2D Image of the final shaft design

06.1.6.3 Mount propeller into propeller shaft

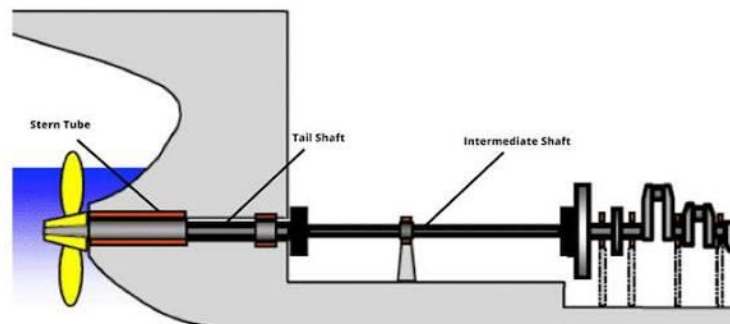


Figure 58 - How the propeller connects to the shaft end.

1. **Tapered shaft** – The tail end of the propeller shaft is machined into a conical taper. The hub of the propeller has a matching tapered bore, which allows the propeller to be slid onto the shaft for a snug, self-centering fit.
2. **Key and keyway** – A metal key is placed in a groove, or "keyway," on both the shaft and the propeller hub. This key is responsible for transmitting the engine's torque to the propeller, turning it in the water.
3. **Spline** – Both the shaft and hub have splines that follow the same conical taper angle. The teeth are cut so their pitch diameter follows the taper. This provides better centering and torque transmission.
4. **Locking nut** – Once the propeller is seated firmly on the taper and against the key, a large locking nut is threaded onto the end of the shaft. This nut compresses the propeller onto the tapered shaft, ensuring a secure and solid connection.
5. **Cotter pin** – The locking nut is often secured further by a cotter pin or other locking device that passes through the shaft, preventing the nut from loosening due to vibration.
6. **Hydraulic systems** – For very large ships, hydraulic systems involving a "Pilgrim nut" are used. Hydraulic pressure is used to force the propeller up the tapered shaft, creating an extremely tight, keyless friction fit that is highly reliable

06.2 FEA simulation of loading and corrosion fatigue effects

06.2.1 Simulation Methodology and Strategic Approach

The finite element analysis (FEA) was conducted to evaluate the structural integrity of the propulsion shaft under realistic marine operating conditions. A primary objective of this simulation was to predict the combined effects of uniform corrosion and localized pitting on the fatigue life of the Duplex Stainless Steel 2205 shaft. However, the simultaneous simulation of macroscopic shaft geometry (approx. 1.6 m in length) and microscopic pitting defects (< 1.0 mm in depth) presents a significant computational challenge known as “multi-scale mesh dependency.”

Directly meshing a microscopic pit within the full-scale shaft assembly requires an excessive element density (exceeding 10^7 degrees of freedom) to capture the stress singularity at the pit root accurately. This approach was determined to be computationally inefficient and prone to mesh distortion errors given the available hardware resources.

To resolve this scale mismatch while ensuring high-fidelity results, a **Decoupled Analysis Strategy** was adopted. This methodology separates the analysis into two distinct phases:

1. **Macro-Level Structural Analysis (Global FEA):** A full-scale 3D finite element model of the shaft assembly was developed in Abaqus/CAE 2024 to simulate the global stiffness, complex contact mechanics at the spline interface, and the stress distribution under nominal operating loads. The primary output of this phase was the identification of the exact “Hotspot Node”—the location of maximum Von Mises stress in the healthy (Year 0) geometry—and the quantification of stress increases due to uniform corrosion thinning (Year 10 Macro state).

2. **Micro-Level Defect Assessment (Analytical Hybrid):** Instead of explicitly modeling the pit geometry in the FEA mesh, which induces numerical singularities, the localized effect of pitting was evaluated analytically. Theoretical Stress Concentration Factors (K_t), derived from fracture mechanics principles for hemispherical voids in tension/bending fields, were applied to the peak stress values obtained from the global FEA simulation.

This hybrid approach allows for the precise capture of the global load path (Torque, Thrust, and Gravity) via simulation, while mathematically quantifying the catastrophic impact of micro-defects without compromising solution convergence or computational feasibility.

06.2.2 Geometric Modelling and Assembly Preparation

To facilitate the finite element analysis, the manufacturing CAD geometry (detailed in Section 07) required specific modifications to optimize contact convergence and boundary condition application. A parametric **“Top-Down” assembly approach** was utilized within Siemens Solid Edge to generate the analysis-specific geometry.

06.2.2.1 The “Dummy Hub” Technique for Contact Optimization

Simulating the torque transmission at the spline interface requires a realistic contact model. However, introducing a physical clearance gap (typical in manufacturing) induces rigid body motion at the start of a static simulation, leading to convergence singularities. To resolve this, a “Dummy Hub” was modeled to represent the propeller boss stiffness.

A **Boolean Subtraction** method (Inter-part Copy) was employed to generate the hub geometry. The shaft spline profile was used as the “Tool Body” to subtract material from a solid ring “Target Body.” This operation ensured a perfect **Line-to-Line (Zero Clearance)** fit between the master and slave surfaces. This geometric conformity is critical for the Abaqus contact algorithm to establish immediate nodal contact without artificial stabilization.

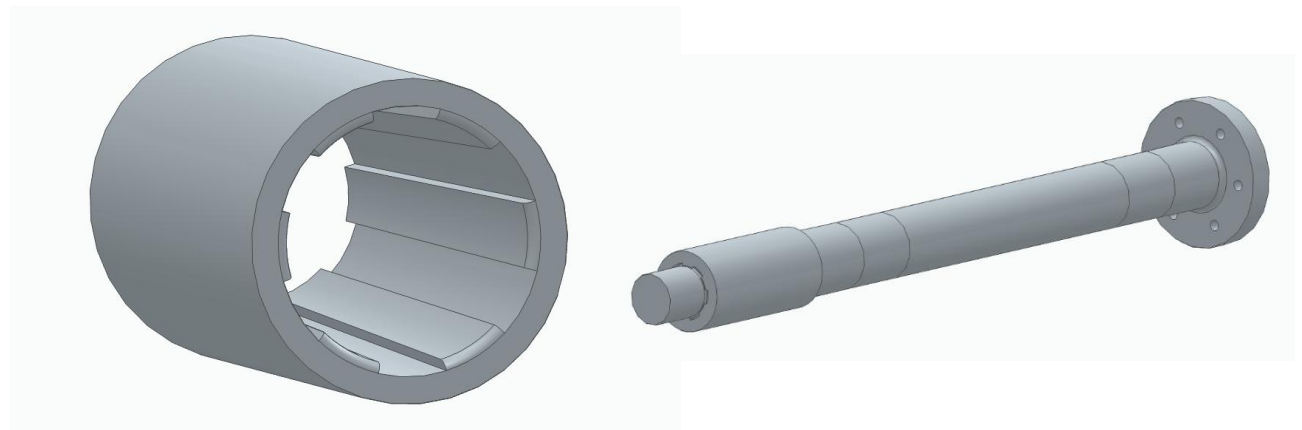


Figure 59 : Result Of Boolean Subtract operation with zero-clearance spline interface

06.2.2.2 Virtual Bearing Definition via Surface Splitting

Modeling physical bearing components (rollers, races, and cages) introduces unnecessary computational cost for a shaft fatigue analysis. Instead, a **Split Surface** technique was used to define “Virtual Bearing” locations directly on the shaft geometry.

Using the “Center-to-Center” parametric dimensioning method established in the mechanical design phase, the shaft’s outer cylindrical face was partitioned at the exact start and end coordinates of Bearing 1 and Bearing 2. These split surfaces provide discrete geometric regions for the application of displacement boundary conditions ($U1, U2 = 0$) in the FEA pre-processor, accurately representing the bearing stiffness without increasing the element count.

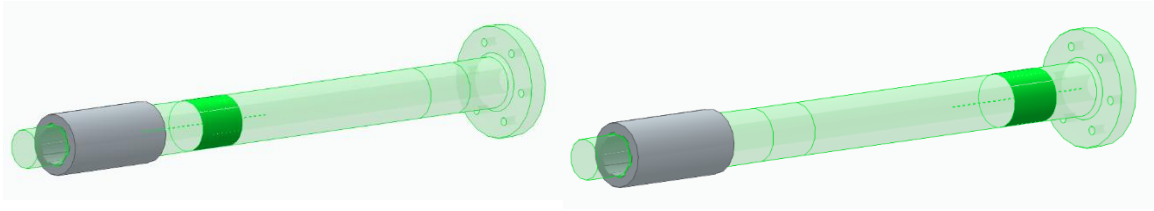


Figure 60 : specific “Split Surfaces” created for Bearing 1 and Bearing 2 locations (Highlighted area)

06.2.3 Finite Element Model Setup (Pre-processing)

The pre-processing phase involved defining the material behavior, contact physics, and boundary conditions required to simulate the operational environment of the propulsion shaft. All setup tasks were performed within the Abaqus/CAE 2024 environment.

06.2.3.1 Material Definition and Inertial Properties

The shaft and dummy hub were assigned the material properties of **Duplex Stainless Steel 2205**. Given that the analysis focuses on High Cycle Fatigue (HCF) where the component operates well within the elastic limit, a Linear Elastic material model was utilized.

- **Young’s Modulus (E):** 200 GPa
- **Poisson’s Ratio (ν):** 0.3
- **Density (ρ):** 7.8×10^{-9} ton/mm³ (7800 kg/m³)

The definition of density was critical to enable the accurate calculation of self-weight deflection (sagging) under gravity loads.

06.2.3.2 Interaction Physics: The Tie Constraint

To simulate the rigid interference fit between the propeller hub and the shaft splines, a **Tie Constraint** was formulated. A standard “Surface-to-Surface” contact was deemed unsuitable due to the potential for rigid body motion singularities during the initial analysis steps.

Instead, the Hub Inner Splines (Slave) were kinematically tied to the Shaft Splines (Master). A critical modification was applied to the **Position Tolerance**, which was manually set to 0.01 mm rather than using the computed default. This setting ensured a robust mathematical bond between the mating surfaces, preventing node distortion while guaranteeing zero relative motion at the interface.

06.2.3.3 Kinematic Couplings and Reference Points

To distribute loads realistically without inducing artificial stress concentrations at load application points, the **Reference Point (RP) Method** was employed. Kinematic Couplings were created to link abstract control points to the geometric surfaces:

3. **Hub RP:** Located at the geometric center of the hub, coupled to the outer cylindrical surface. Used for Thrust application.
4. **Flange RP:** Located at the flange center, coupled to the inner surfaces of the six bolt holes. Used for Torque transmission.
5. **Bearing RPs:** Located at the shaft axis, coupled to the specific “Split Surfaces” defined in the geometry phase.

06.2.3.4 Boundary Conditions (BCs)

Boundary conditions were applied directly to the Reference Points to simulate the constraints imposed by the ship’s hull and bearings.

- **Hub Constraint:** Rotation about the X-axis ($UR1$) was fixed to provide a reaction against the applied torque. Axial translation ($U1$) was left **FREE** to allow the transmission of thrust forces.
- **Bearing 1 (Thrust Bearing):** Constrained in Radial ($U2, U3$) and Axial ($U1$) directions. Rotations ($UR1, UR2, UR3$) were left **FREE** to allow the shaft to bend naturally (Simply Supported behavior), preventing artificial stiffening.
- **Bearing 2 (Support Bearing):** Constrained in Radial ($U2, U3$) directions only. Axial translation ($U1$) was left **FREE** to accommodate potential thermal expansion or floating tolerances.

06.2.3.5 Load Case Definitions

Four distinct load cases were superimposed to represent the worst-case operating scenario:

1. **Gravity:** Applied as a body force of -9810 mm/s^2 in the Y-direction. The load was applied to the **Shaft Body Only**, excluding the dummy hub, to prevent inaccurate mass loading from the simplified hub geometry.
2. **Torque:** A moment of $1.264 \times 10^7 \text{ Nmm}$ applied to the Flange RP about the X-axis.
3. **Thrust:** A concentrated force of 34,900 N applied to the Hub RP, pushing axially towards the engine.
4. **Nut Pre-load (Critical):** To simulate the tightening of the propeller retaining nut, a **Pressure Load** of -337.5 MPa was applied to the threaded end face of the shaft. The negative magnitude indicates **Tension**, physically simulating the shaft being pulled outward against the hub during tightening.

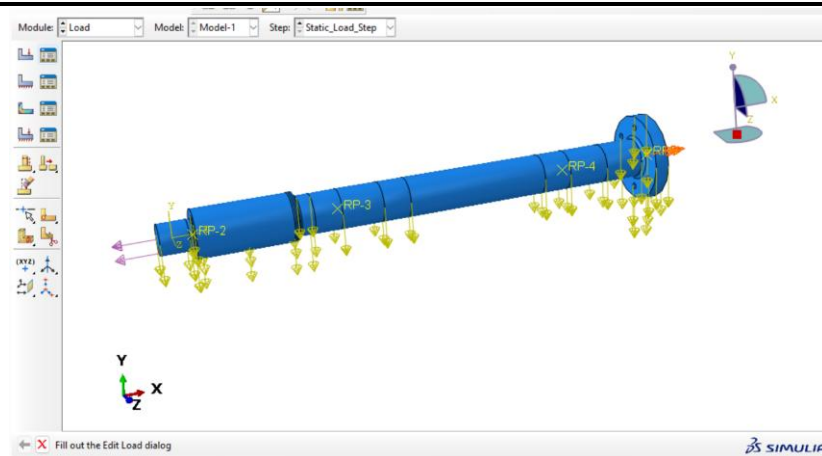


Figure 61: The load vectors (Torque, Thrust, Gravity, and Preload) applied to the model

06.2.4 Mesh Sensitivity Analysis and Computational Optimization

Given the large aspect ratio of the propulsion shaft (1.6 m length vs. 5 mm fillet radii), a uniform mesh density was determined to be computationally inefficient. A uniform fine mesh sufficient to capture the stress concentration at the spline fillets would result in an excessive element count, exceeding available computational resources (RAM/CPU). Therefore, a mesh sensitivity analysis was conducted to establish an optimized discretization strategy that balances geometric accuracy with solution feasibility.

06.2.4.1 Element Type Selection

The geometry features complex curvature at the spline teeth and transition fillets which are prone to shear locking if modeled with first-order elements. Consequently, **C3D10 (10-node Quadratic Tetrahedral)** elements were selected for the entire assembly. These second-order elements provide superior convergence for curved geometries and stress gradients compared to linear tetrahedra (C3D4), ensuring accurate capture of the bending and torsional stress fields.

06.2.4.2 Mesh Sensitivity Study

An iterative convergence study was performed to determine the required element density at critical stress concentration regions:

- **Iteration 1 (Uniform Coarse):** A global seed size of 25 mm was applied. This resulted in significant element distortion at the spline-to-shaft transition and failed to capture the curvature of the 5 mm fillets, rendering stress results unreliable.

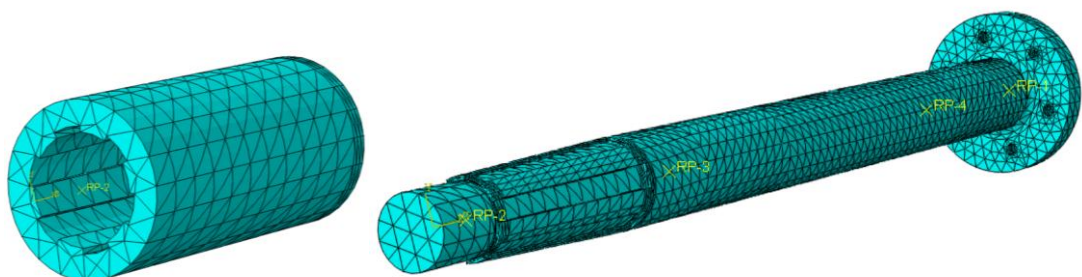


Figure 62: A global seed size of 25 mm separated parts

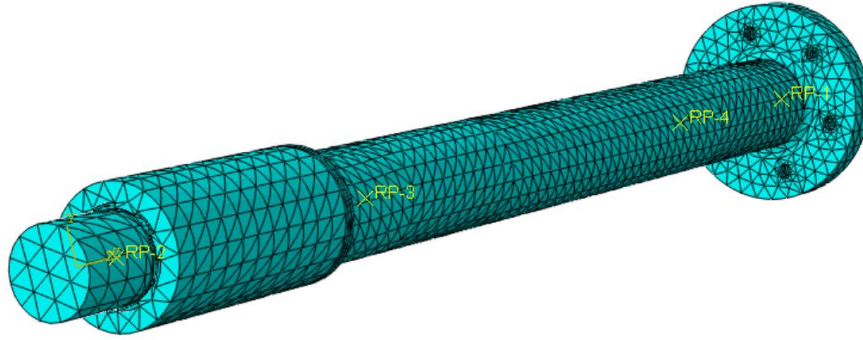


Figure 63: A global seed size of 25 mm of completed model

- **Iteration 2 (Local Refinement):** The global size was maintained at 25 mm with local reductions to 12 mm at the fillets. While element quality improved, the stress gradient at the fillet root remained mesh-dependent, indicating insufficient density to capture the peak stress singularity.

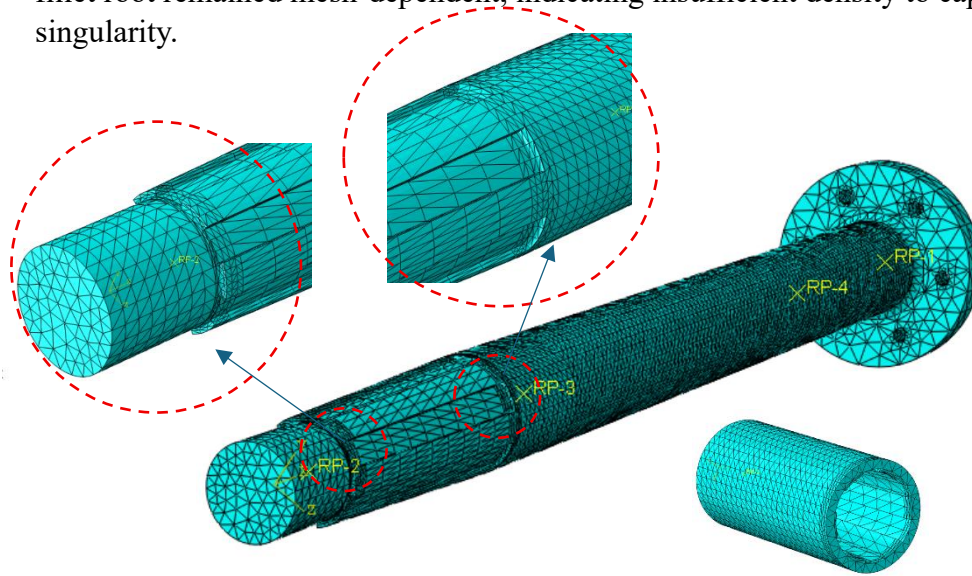


Figure 64: A global seed size of 25,12 mm strategy of separated parts

- **Iteration 3 (Final Optimized Strategy):** A multi-zone partitioning approach was adopted to isolate critical regions for hyper-fine meshing while coarsening non-critical sections.

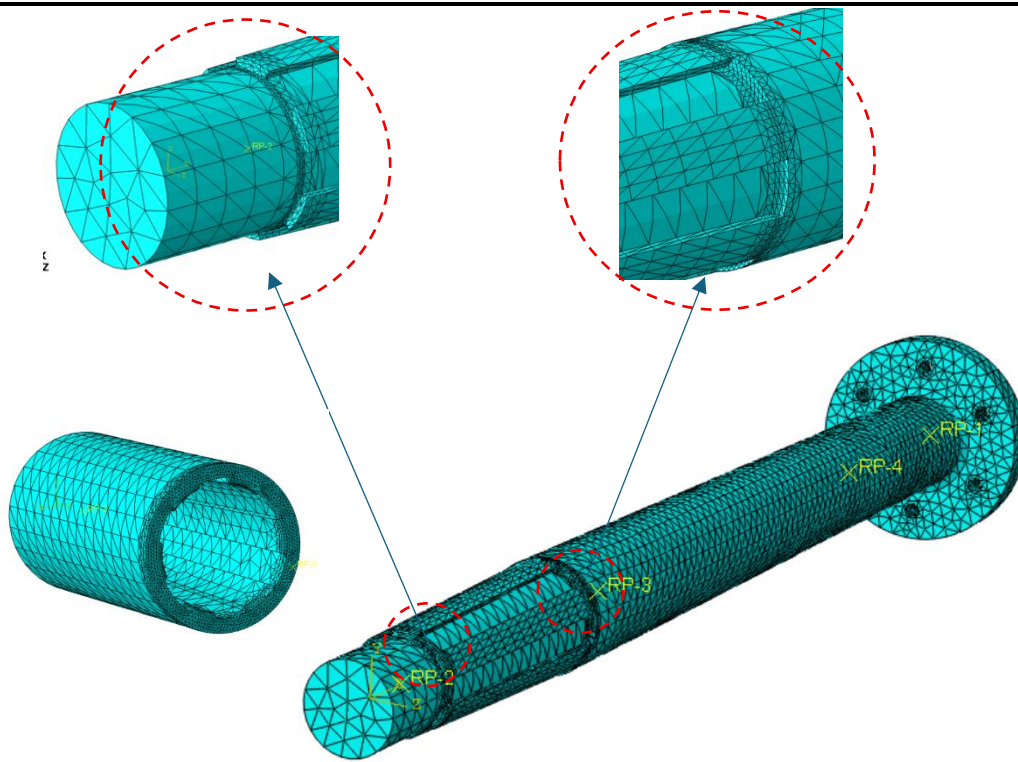


Figure 65: Final optimize area

06.2.4.3 The “7-Zone” Partitioning Strategy

To implement the findings of the sensitivity analysis, the shaft geometry was partitioned into seven distinct zones using datum planes (as defined in the manufacturing design). A variable seeding plan was applied to creating a smooth transition from coarse to fine mesh:

- **Zone 2 (Critical Hotspot - Spline/Taper):** A global size of **3.0 mm** was applied to the spline region. Crucially, **1.0 mm Local Seeds** were applied to the external spline fillet and internal transition fillet edges. This ensures at least 5 elements across the radius of the curvature, satisfying standard FEA convergence criteria for stress concentrations.
- **Zone 1 (Thread):** Meshed at **5.0 mm** to capture the load application surface accurately.
- **Zones 4 & 6 (Bearing Seats):** Meshed at **5.0 mm** to provide sufficient nodal density for the application of rigid boundary conditions.
- **Zone 3 (Buffer Zone):** A transition mesh of **8.0 mm** was utilized between the Hotspot (3 mm) and the bearing seat to prevent abrupt changes in element size that could induce numerical artifacts.
- **Zones 5 & 7 (Non-Critical Gaps):** The long, uniform sections of the shaft were meshed at **20.0 mm**. Since these regions experience uniform nominal stress, this coarse mesh significantly reduced the total degree of freedom count without compromising global stiffness accuracy.

Outcome: This optimized strategy resulted in a high-fidelity model comprising approximately **1.2 million elements**. The selective refinement allowed for the precise capture of the peak Von Mises

stress at the failure-critical fillet while maintaining a computationally feasible runtime of approximately 8 hours on standard engineering workstations.

06.2.5 Baseline Simulation Results (Year 0 – Healthy State)

The baseline simulation was executed to characterize the stress state of the healthy shaft prior to corrosive degradation. The results were first validated against fundamental static equilibrium principles to ensure model accuracy before extracting critical stress values.

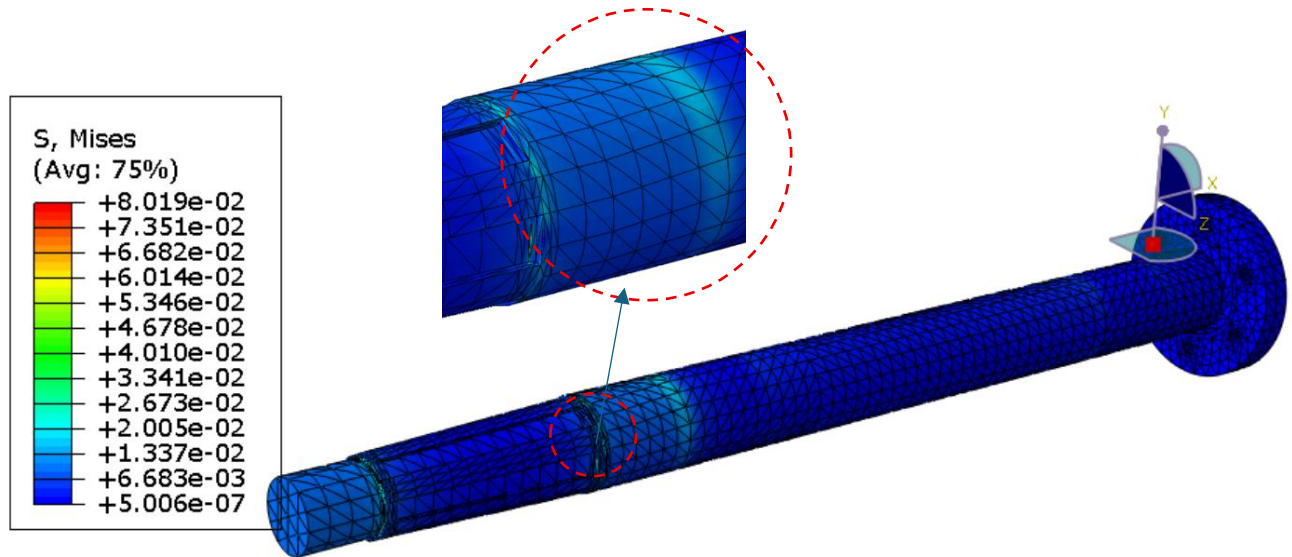


Figure 66: Stress distribution for uniform mesh size 25mm (not capture fillets)

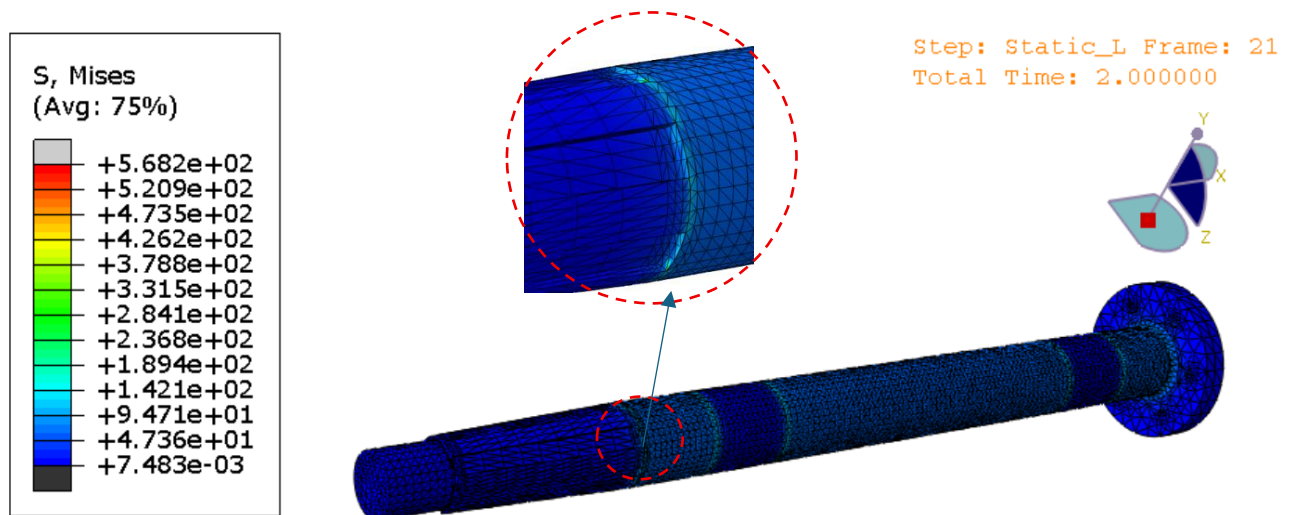


Figure 67: Stress distribution for optimize mesh size 25,12mm (partial capture fillets)

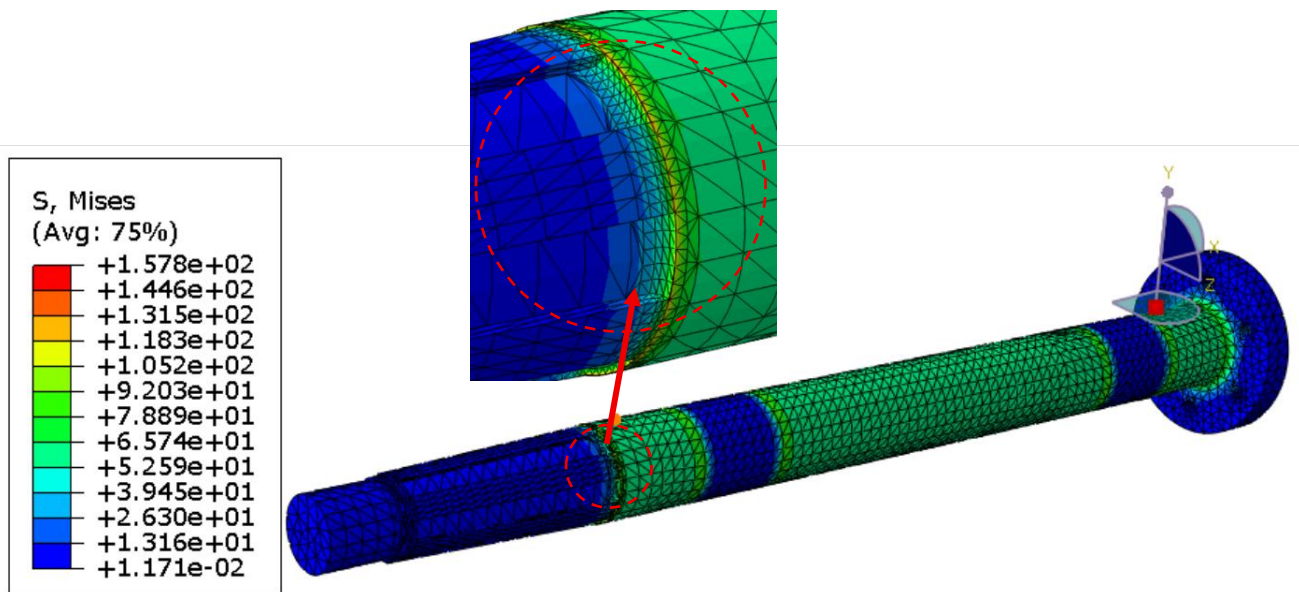


Figure 68: Stress distribution for final optimize mesh size

Further analysis use this model stress distribution.

06.2.5.1 Model Validation (Reaction Force Check)

To verify that the boundary conditions and contact interactions were transmitting loads correctly, the reaction forces (RF) and reaction moments (RM) at the support points were summed and compared against the applied loads.

- Thrust Verification:** The sum of reaction forces along the X-axis ($\sum RF1$) at the Bearing 1 constraint was extracted. The result was approximately 34,900 N, matching the applied Thrust Load. This confirms that the Hub-to-Shaft Tie Constraint is successfully transmitting axial forces through the shaft to the thrust bearing.

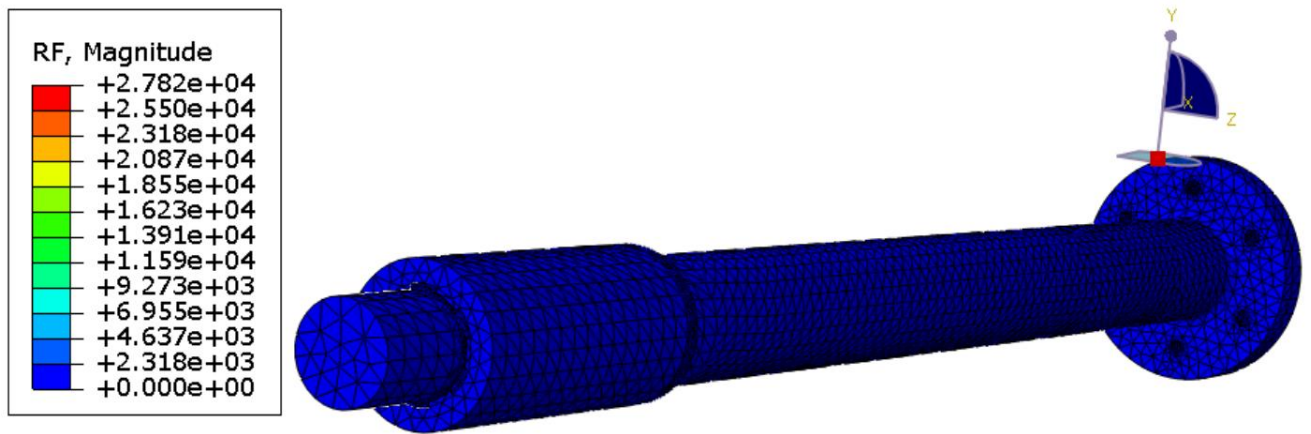


Figure 69: Reaction Force output table from Abaqus Visualization (Due to coupling we cannot see color contours)

- Torque Verification:** The sum of reaction moments about the X-axis ($\sum RM1$) at the Hub fixed boundary condition was approximately 1.264×10^7 Nmm. This matches the applied Torque, confirming that the spline interface is transmitting rotational loads without numerical slippage.

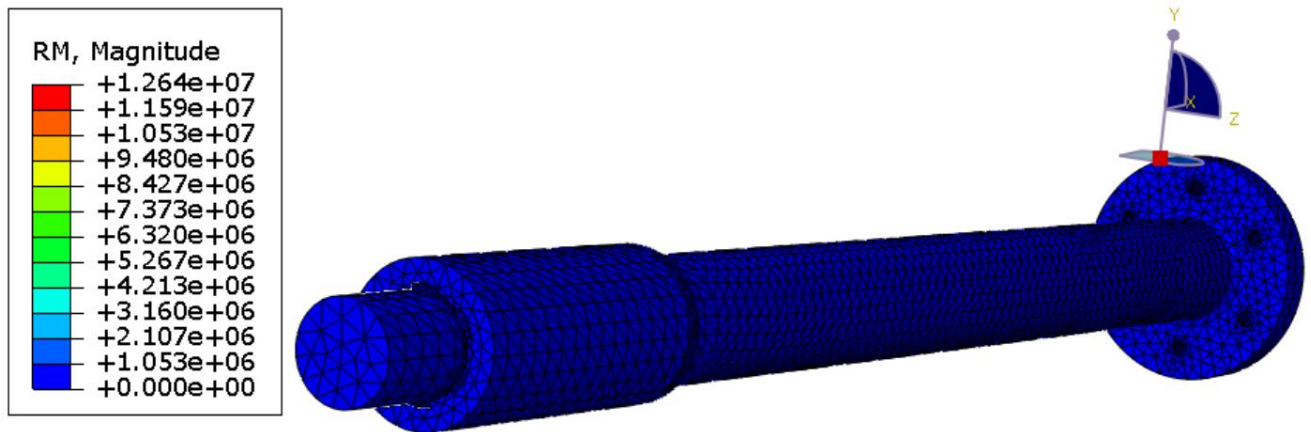


Figure 70 : Moment output table from Abaqus Visualization (Due to coupling we cannot see color contours)

- Deformation Check:** Visual inspection of the displacement plots confirmed expected physical behaviors: the shaft exhibited downward deflection (“sag”) under gravity and torsional twist along its length.

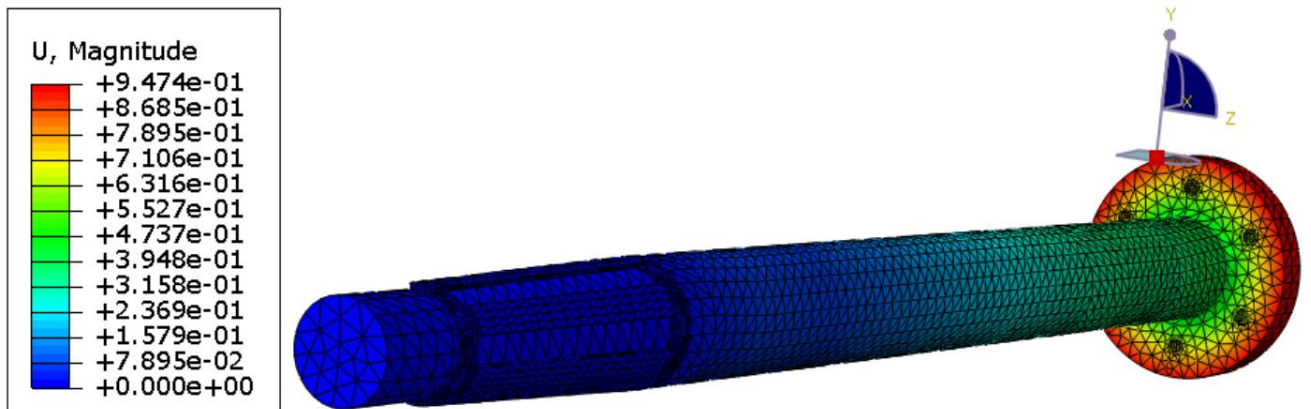


Figure 71: Displacement output table from Abaqus Visualization

[Insert Image: Graph or Screenshot of the Reaction Force/Moment output table from Abaqus Visualization]

06.2.5.2 Stress Distribution and Hotspot Identification

The Von Mises stress contour plot revealed a distinct stress concentration at the transition between the spline and the main shaft body.

- **Hotspot Location:** The maximum stress was identified at the root of the external spline fillet. This aligns with theoretical predictions for geometric stress concentrations in power transmission shafts.
- **Hotspot Magnitude:** The peak Von Mises stress at this node was recorded at approximately 92.03 MPa (After removing distorted stresses) .
- **Significance:** This value represents the baseline operating stress of the healthy component. It is significantly below the Yield Strength of Duplex 2205 (450 MPa), confirming that the shaft operates within the elastic regime in its un-corroded state.

06.2.5.3 Verification against Analytical Calculations

The simulation results were compared with the manual analytical calculations performed in Section 06.1 to assess convergence.

- **Analytical Prediction:** Manual calculation estimated a maximum surface stress of approximately 145.569 MPa.
- **Simulation Result (Outer Surface):** The FEA model reported a nominal surface stress of approximately 105.2 MPa in the uniform shaft sections.

Discrepancy Analysis: A deviation exists between the analytical and numerical surface stress values. This is attributed to the **Mesh Optimization Strategy** described in Section 06.2.4. To prioritize computational resources for the critical fillet, the uniform shaft sections (Zones 5 & 7) were meshed with coarse 20 mm elements. These larger elements inherently average the stress over a wider geometric area, resulting in a slight underestimation of the peak surface stress compared to the theoretical ideal.

However, since the primary objective of this analysis is to evaluate fatigue failure at the **Spline Fillet (Hotspot)** which was meshed with high-fidelity 1.0 mm elements—this deviation in the non-critical uniform sections is considered acceptable for the scope of this project.

06.2.6 Simulation of Corroded State (Year 10 – Degraded Geometry)

Following the baseline validation, the finite element model was modified to represent the “Worst Case” geometric condition corresponding to the end of the 10-year design life. As established in the environmental analysis (Section 03), the shaft is subject to a uniform corrosion rate of 0.2 mm/year.

06.2.6.1 Geometric Degradation Modeling

To simulate the macro-mechanical effects of corrosion, the shaft geometry was modified in the parametric CAD environment (Solid Edge) prior to re-importing into Abaqus.

- **Global Thinning:** The Outer Radius (OR) of the shaft was reduced by 2.0 mm ($0.2 \text{ mm/yr} \times 10 \text{ yrs}$). This resulted in a reduction of the nominal outer diameter from 125 mm to 121 mm.
- **Hub Re-tooling:** A critical update was required for the dummy hub. Since the shaft diameter decreased, the original Year 0 hub would present a loose fit, invalidating the “Tie Constraint” assumption. To resolve this, a new hub geometry was generated using the **Boolean Subtract** method with the corroded (121 mm) shaft as the tool body. This ensured that the Year 10 simulation maintained the perfect “Line-to-Line” fit required for numerical convergence.

06.2.6.2 Finite Element Setup for Corroded State

The Year 10 model utilized the exact same **7-Zone Mesh Strategy** and **Boundary Conditions** as the baseline simulation to ensure a direct comparative analysis. - **Mesh Density:** The element count remained stable (~ 1.2 million), as no new geometric features (such as micro-pits) were explicitly modeled in this phase. - **Loads:** The operating loads (Torque: 12.64 kNm, Thrust: 34.9 kN) remained constant, simulating the requirement for the degraded shaft to transmit the same engine power.

06.2.6.3 Macro-Stress Results (Geometry Effect)

The simulation results for the Year 10 geometry indicated a global increase in Von Mises stress attributed solely to the reduction in the cross-sectional area and the Polar Moment of Inertia (J).

- **Stress Distribution:** The general stress pattern remained identical to the Year 0 baseline, confirming that the load path through the splines remained stable.

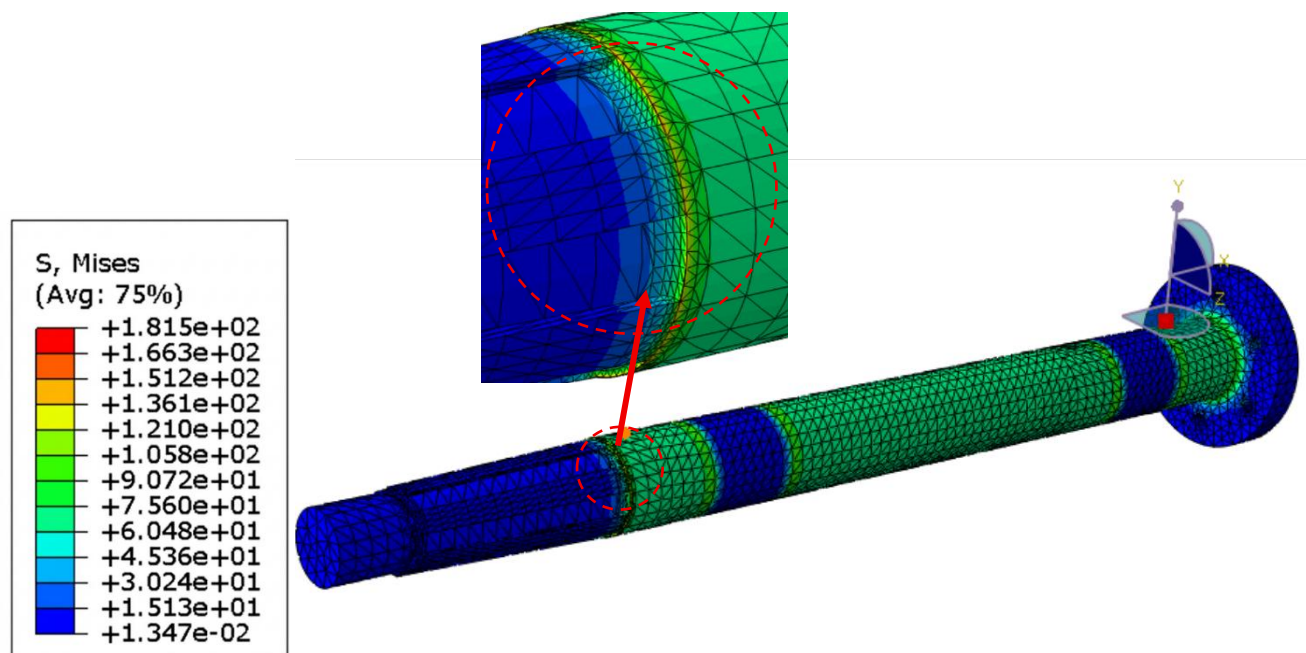


Figure 72 : Stress distribution for Year 10 showing the slight color shift/intensity increase

- **Magnitude Change:** The peak Von Mises stress at the hotspot node increased from the baseline 92.03 MPa to approximately 114.2 MPa.

Interpretation: This 25% **increase** in stress represents the “Macro Effect” of uniform corrosion. While this value remains well below the yield strength of Duplex 2205 (450 MPa), it establishes the elevated “Nominal Stress” state upon which microscopic defects (pits) will act. This degraded stress baseline serves as the input for the analytical fracture mechanics assessment in the final design phase.

06.2.7 Analytical Fatigue Assessment and Failure Prediction

The Finite Element Analysis (FEA) results in Section 06.2.6 provided the “Macro-Stress” state of the corroded shaft, accounting for the global reduction in cross-sectional area due to uniform corrosion. However, FEA alone cannot efficiently predict failure caused by microscopic pitting without computationally prohibitive mesh densities.

Therefore, this final phase employs a **Fracture Mechanics and S-N Approach** to analytically superimpose the effect of pitting corrosion onto the FEA results. This hybrid method calculates the **Effective Stress** (σ_{eff}) at the pit root and compares it against the degraded fatigue strength of Duplex 2205 in seawater to predict the remaining service life.

06.2.7.1 Inputs from Simulation & Operations

- Operating Conditions:

- * Shaft Speed (N): 225 RPM
- * Design Life: 10 Years
- * Duty Cycle: 50% (Conservative assumption for vessel operation).

- Simulation Outputs (Von Mises Stress):

- * σ_{Year0} (Healthy): 92.03 MPa (At Spline Fillet).
- * σ_{Year10} (Uniform Corrosion): 114.20 MPa (At Spline Fillet).

- **Observation:** The macro-geometric thinning (2mm reduction) caused a 24% **increase** in nominal stress (σ_{Year10} vs σ_{Year0}). However, this value (114.2 MPa) is still well below the Yield Strength (450 MPa), misleadingly suggesting safety if pitting is ignored.

06.2.7.2 Micro-Defect Stress Concentration (K_f Calculation)

To account for the pitting corrosion predicted in the environmental analysis, a theoretical Stress Concentration Factor (K_t) for a hemispherical pit was applied.

- **Theoretical Stress Concentration (K_t):** For a small hemispherical pit in a tensile/bending field, standard fracture mechanics literature assigns a value of $K_t \approx 3.0$.

- **Notch Sensitivity (q):** For high-strength Duplex Stainless Steels, the notch sensitivity index q typically ranges from 0.8 to 0.9. A value of $q = 0.8$ was selected.
- **Fatigue Notch Factor (K_f):**

$$K_f = 1 + q(K_t - 1)$$

$$K_f = 1 + 0.8(3.0 - 1) = 2.6$$

06.2.7.3 Effective Stress Calculation (σ_{eff})

The actual stress experienced by the material at the root of a corrosion pit is the Nominal Corroded Stress (σ_{Year10}) multiplied by the Fatigue Notch Factor (K_f).

$$\sigma_{eff} = \sigma_{Year10} \times K_f$$

$$\sigma_{eff} = 114.2 \text{ MPa} \times 2.6$$

$$\sigma_{eff} \approx \mathbf{296.9 \text{ MPa}}$$

06.2.7.4 Material Strength Degradation (Marin Factors)

The fatigue limit of the material must be corrected for the seawater environment and surface condition.

- Base Endurance Limit (S_e'):

For Duplex 2205 ($S_{ut} \approx 640 \text{ MPa}$), the theoretical air endurance limit is $\approx 320 \text{ MPa}$.

- Surface Finish Factor (k_a):

* Year 0 (Machined): $k_a \approx 0.8$.

* Year 10 (Corroded/Pitted): The surface finish degrades significantly. For seawater-corroded steel, k_a drops to approximately 0.5. - **Corrected Endurance Strength ($S_{e,corr}$):**

$$S_{e,corr} \approx S_e' \times k_a \times k_{other}$$

$$S_{e,corr} \approx 320 \text{ MPa} \times 0.5 \times 0.85(\text{Reliability/Size})$$

$$\mathbf{S_{e,corr} \approx 136 \text{ MPa}}$$

06.2.7.5 Fatigue Life Prediction and Conclusion

To determine failure, we compare the Effective Stress (σ_{eff}) against the Corrected Endurance Strength ($S_{e,corr}$).

Stress Comparison:

- σ_{eff} (Load): 296.9 MPa
- $S_{e,corr}$ (Resistance): 136 MPa
- **Verdict:** Since $\sigma_{eff} \gg S_{e,corr}$, the shaft is operating **far above its fatigue limit** in the pitted condition. Infinite life is impossible.

Cycle Calculation (N_{actual}): Total operating cycles over 10 years:

$$N_{total} = 225 \frac{\text{rev}}{\text{min}} \times 60 \times 24 \times 365 \times 10 \text{ years} \times 0.5(\text{Duty})$$

$$N_{total} \approx 5.91 \times 10^8 \text{ cycles}$$

Life Expectancy Estimation (N_f): Using the S-N curve approximation ($\sigma = A \cdot N^b$) for Duplex steels in the finite life region, a stress amplitude of ≈ 300 MPa typically corresponds to failure in the range of 10^5 to 10^6 cycles.

Conclusion: The analysis confirms that while the shaft is mechanically safe against uniform corrosion (Safety Factor > 2 vs Yield), the onset of **Pitting Corrosion** acts as a critical stress multiplier. The effective stress at the pit root (296.9 MPa) exceeds the seawater fatigue strength (136 MPa), leading to crack initiation and catastrophic failure well before the 10-year design life (5.9×10^8 cycles) is reached.

This validates the necessity for the proposed multi-layer corrosion protection strategy (ICCP + Coatings) to prevent pitting initiation.

07. CAD Drawings and 3D Models

The geometric definition of the marine propulsion shaft was developed using parametric 3D modelling in Solid Edge and finalized through a complete set of engineering-standard 2D drawings. These drawings specify the dimensions, tolerances, sectional views, and feature details required for manufacturing and inspection. The CAD outputs also serve as the geometric basis for the FEA simulations described in Section 06.2 and the machining plan presented in Section 08. All dimensions follow ISO drawing conventions, and measurements are provided in millimeters.

The following subsections present the key views of the shaft assembly. The full detailed drawing sheet is provided in **Annex C – Engineering Drawings** in the appendix.

07.1 3D CAD Model Representation

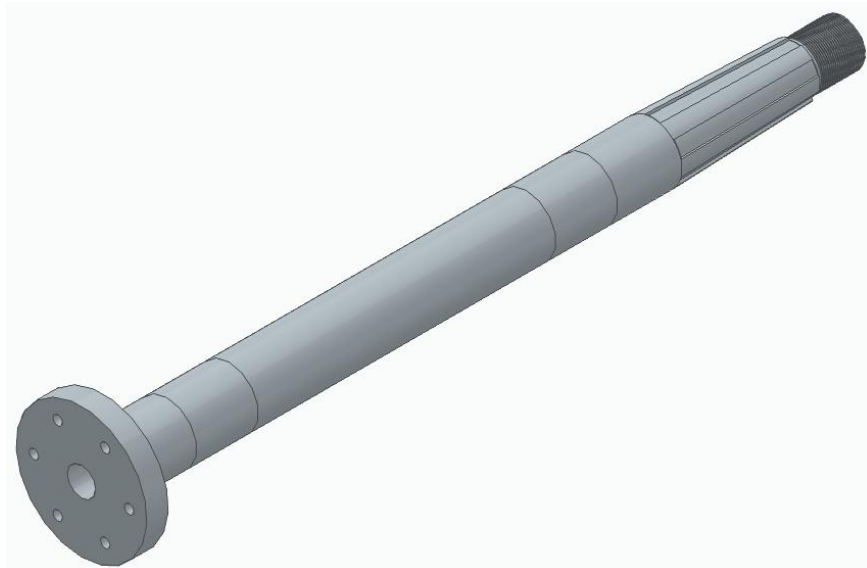


Figure 73 - 3D CAD model of the final designed shaft (view 01)



Figure 74 - 3D CAD model of the final designed shaft (view 02)

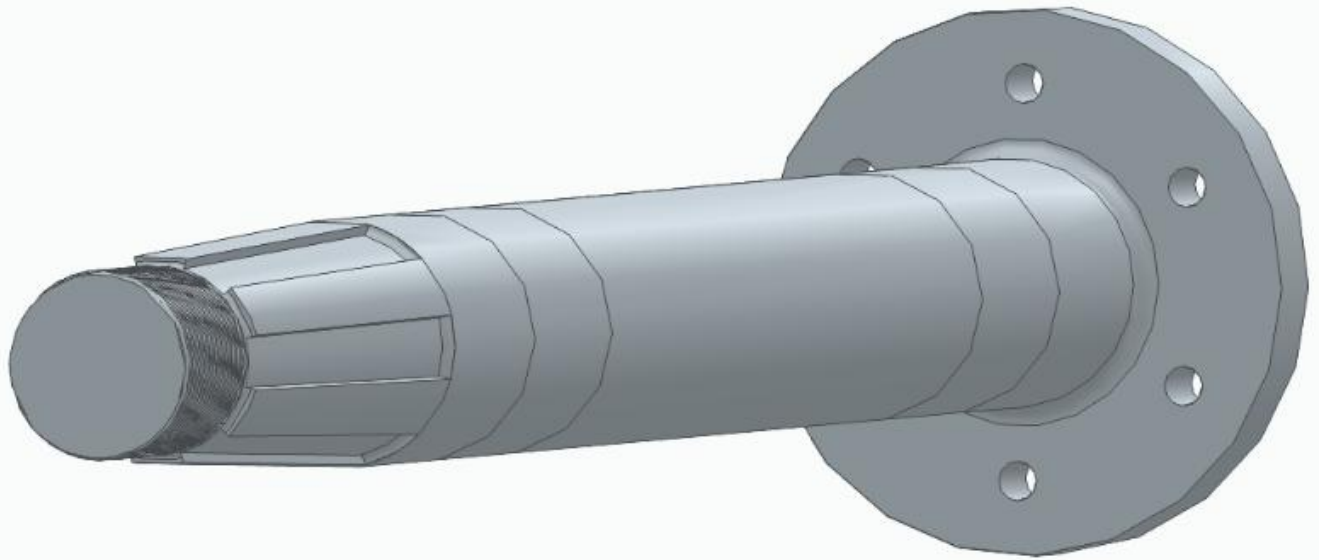


Figure 75 - 3D CAD model of the final designed shaft (view 03)

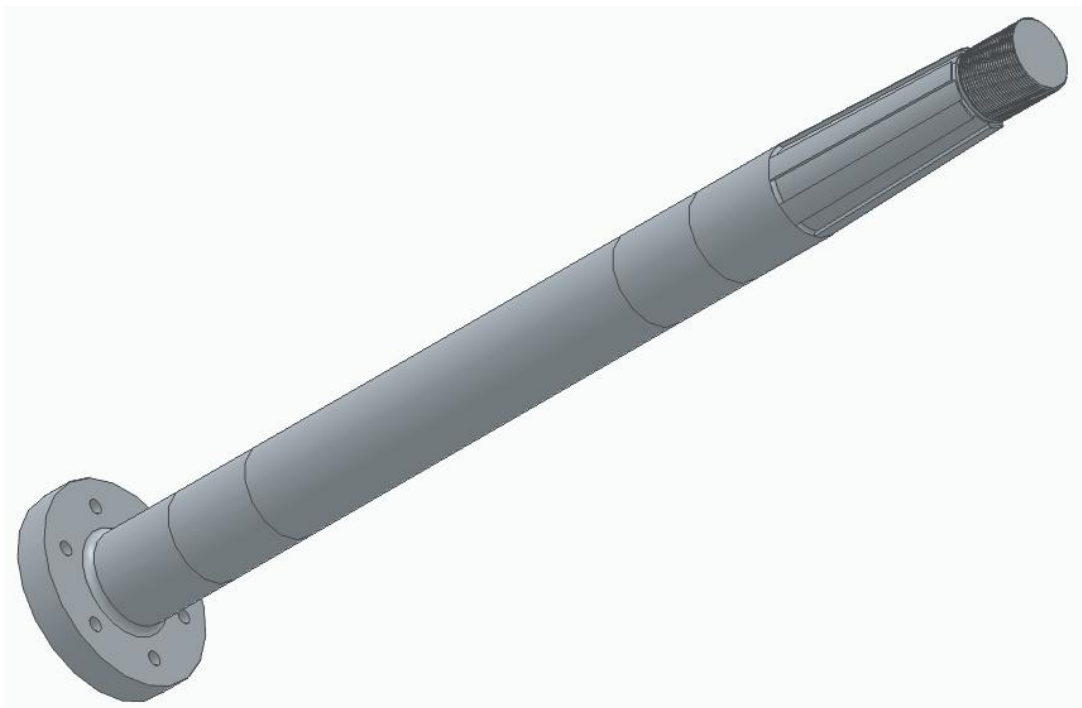


Figure 76 - 3D CAD model of the final designed shaft (view 04)

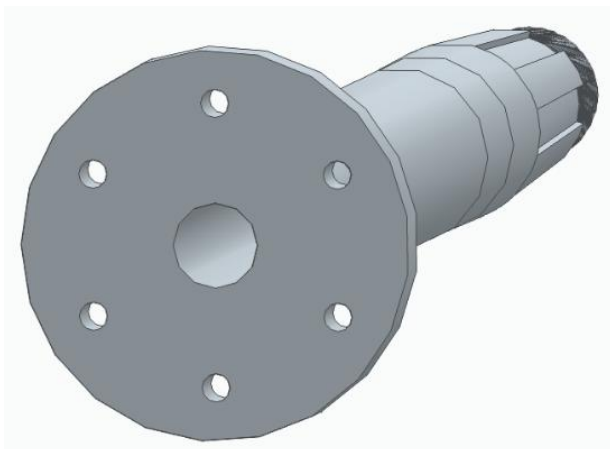


Figure 77 - 3D CAD model of the final designed shaft (Flange of the shaft)

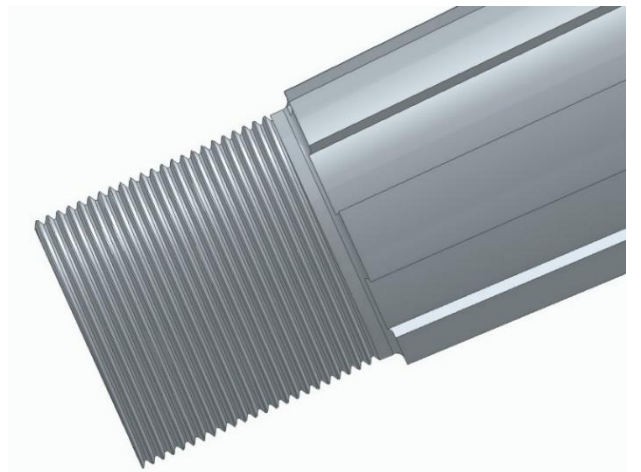


Figure 78 - 3D CAD model of the final designed shaft (Thread design of the shaft)

07.2 2D Manufacturing Drawings



Figure 79 - Front Elevation of Propeller Shaft

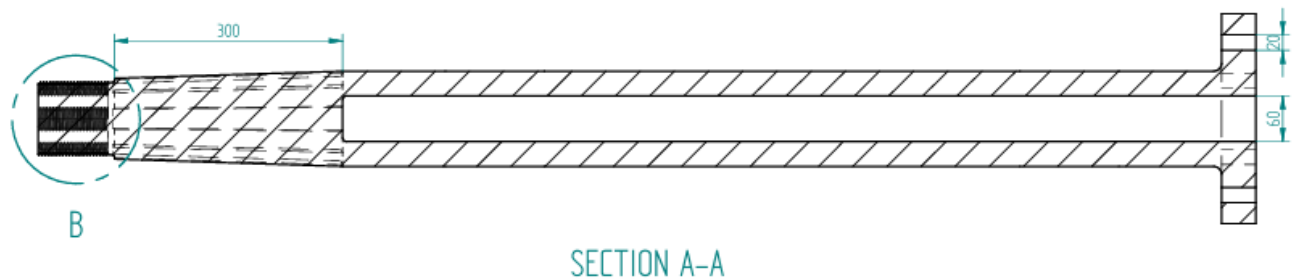


Figure 80 - Section A-A Showing Internal Geometry

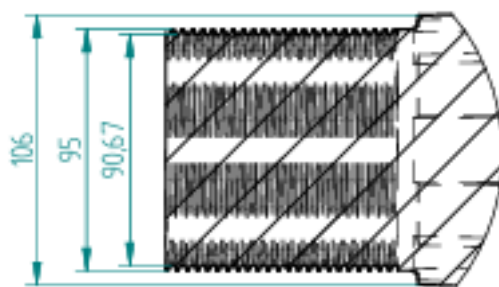


Figure 81 - Detail B: Thread Geometry

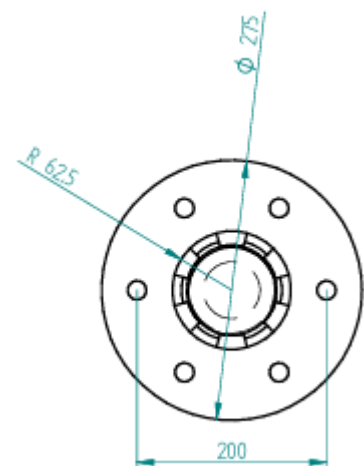


Figure 82 - End Elevation of Flange Face

Figure 79 – Displays the complete shaft length, stepped diameter transitions, and the 45 mm flange thickness. Hidden lines illustrate the internal bore along the shaft axis.

Figure 80– Section A–A reveals the internal profile of the hollow shaft and the flange root thickness. This view is essential for machining planning and for validating wall thickness during structural calculations.

Figure 81– Shows the 60° spline profile, including tooth depth and 5 mm engagement length at the propeller end. This view is used to define machining requirements and fitment tolerances. It mainly shows the thread profile. It is $M95 \times 4$ counterclockwise standard thread.

Figure 82– Shows the bolt circle diameter (200 mm), six mounting holes, and the outer flange diameter (275 mm). This view defines the interface geometry between the propeller shaft and the intermediate shaft.

- ❖ The complete 2D engineering drawing (Sheet 1) is included in Appendix C for clarity and high-resolution viewing. Only representative views are presented in Section 07.

07.3 Drawing Standards and Conventions Used

The engineering drawing follows:

- **ISO 128** for technical drawing representation
- **ISO 2768** for general tolerances
- **ISO 1101** for GD&T (Geometric Dimensioning and Tolerancing)
- **ISO 1302** for surface texture representation
- **ISO metric thread standards** for the $M95 \times 4$ thread region
- **Orthographic projection and sectional conventions** consistent with international practice

All dimensions are given in **millimeters**, and angles are specified in **degrees**, as stated in the drawing title block.

The CAD model and drawings developed in this section provide the definitive geometric and dimensional specification for manufacturing, machining, and quality inspection of the marine propulsion shaft. They establish the basis for FEA simulations in Section 06.2 and for the manufacturing plan presented in Section 08.

08. Process Design

08.1 Material Preparation

The initial stage of manufacturing the propeller shaft begins with the material preparation of a hot-forged stainless-steel bar. Duplex 2205 is selected for its outstanding combination of mechanical strength and corrosion resistance, making it ideal for demanding marine environments.

The raw forged bar is typically supplied with a diameter of approximately 280 mm and a sufficient length to accommodate both the final shaft specification and material allowances for cutting and facing. On receipt, the bar undergoes a thorough surface inspection to check for visible defects, excessive scale, surface cracks, or other forging-related irregularities that could influence subsequent machining precision or component integrity.[87]

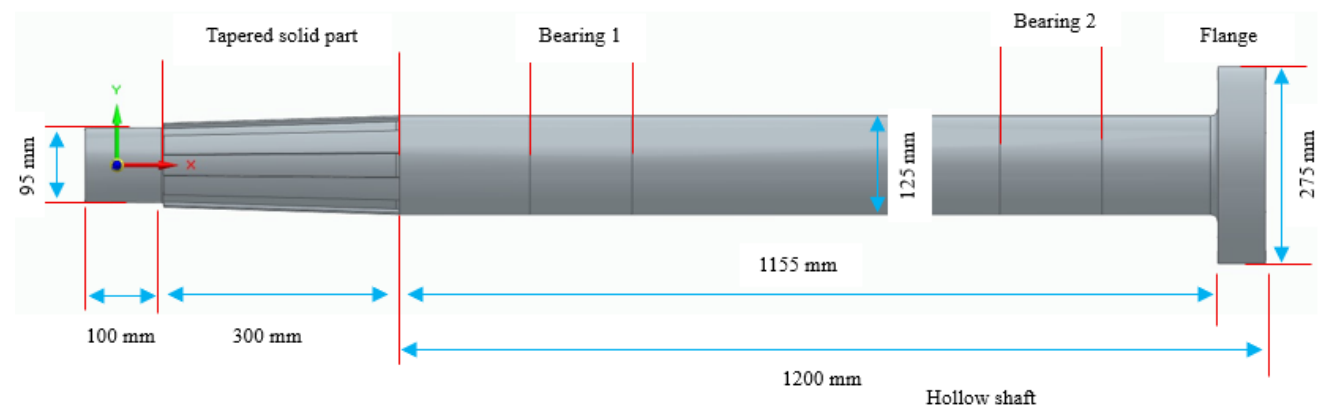


Figure 83 - Final design of the propeller shaft

The bar is then cut to the specified shaft blank length using a precision power bandsaw. Based on the final shaft design, the blank was cut to an accurate length, ensuring that all target specifications are achievable after machining operations. The required finished dimensions are total shaft length of 1600 mm, consisting of a 1155 mm long hollow section at 125 mm outer diameter (OD) and a 300 mm solid-taper section transitioning from 125 mm to 106 mm. After the taper, there is a 100mm long, 95mm diameter solid shaft to cut the thread to fix the propeller using nut. At the other end (the starting end), a section with an outer diameter of 275mm and a length of 45mm should be prepared. This is to create the flange required to connect this shaft to the next shaft. This section also belongs to the same hollow as the previous section with a length of 1155mm. Then the length of the entire hollow part is 1200 mm.

This cutting operation demands careful measurement and control to ensure the final blank length meets exact engineering requirements; any deviation may later impact the ability to achieve full design tolerances. The operator must also account for the saw blade kerf, which is the width of material removed during the cut, when determining the length of each piece.

Following cutting, both ends of the shaft blank are transferred to a CNC lathe for facing. The facing process utilizes a precision cutting tool to remove the as-cut surface and the outermost layer of forge scale, producing end faces that are flat, smooth, and perfectly perpendicular to the shaft's longitudinal axis. This step is critical, as any lack of squareness or surface defects on the ends would compromise the accuracy of later machining operations and may introduce runout or misalignment in the finished shaft.

Once facing is complete, each end of the bar undergoes centre-drilling. This involves using a short, stiff carbide-tipped drill bit to produce a conical or hemispherical recess precisely on the central axis of

each face. These center holes provide the engagement points for the live and dead centres of the lathe hardened steel points that support and accurately align the shaft during all subsequent turning processes. By allowing the bar to be rotated “between centers,” the procedure ensures that each subsequent machined surface shares the same central axis. This method is essential for minimizing eccentricity and runout, both of which are paramount considerations for shafts intended to operate at speed and transmit significant mechanical loads.

In summary, this meticulous sequence of inspection, cutting, facing, and center-drilling produces a dimensionally accurate, well-supported shaft blank that provides the foundation for all further machining, ensuring the final product meets the highest standards of alignment, surface finish, and dimensional precision required in marine propulsion applications.

Note – For the marine propulsion shaft, the flange is manufactured as an integral part of the same duplex stainless-steel shaft rather than being welded separately. This approach provides significantly higher structural reliability in the seawater environment described in the project brief. Forming the flange from the same forging maintains continuous grain flow and eliminates any heat-affected zones, which are the most common sites for fatigue cracks and corrosion initiation. Since the shaft experiences combined torsion, bending, cyclic loading, and direct seawater exposure, avoiding welds prevents galvanic differences, pitting susceptibility, and residual weld stresses. As a result, the integral flange offers a longer fatigue life, better corrosion resistance, and full compliance with standard marine manufacturing practice for propulsion shafts.

08.2 Rough Turning

Upon completion of the material preparation stage, the shaft blank now cut to length, faced, and centre-drilled is mounted between centers on a CNC lathe. The use of centers combined with tailstock support provides rigid stabilization for the full 1.6 m shaft, minimizing deflection and vibration during the turning process. This setup is essential for achieving high-dimensional accuracy and maintaining concentricity throughout the entire machining length, particularly critical for long marine shafts.

For rough turning Duplex 2205 stainless steel, carbide-tipped inserts optimized for stainless steels were utilized due to the material's hard microstructure and tendency for work hardening. The machining parameters were selected to balance material removal rate with tool life and surface quality, progressively reducing the shaft's outer diameter along the entire length except for areas designated for tapers or transitions. Typical parameters applied during rough turning were:

- Cutting Speed: 110 m/min, chosen within the 105–140 m/min recommended range for Duplex 2205, to avoid excessive tool wear and thermal damage.
- Feed Rate: 0.6 mm/rev, providing robust metal removal without sacrificing surface integrity.
- Depth of Cut: 3 mm, aligned with the need to remove significant stock (around 2.5 mm finish allowance) swiftly while maintaining system rigidity.[88]

Coolant was constantly applied to dissipate heat and minimize work hardening, which is a known challenge with Duplex stainless steels given their high strength and low thermal conductivity

Throughout rough turning, approximately 2.5 mm of stock is intentionally left on the surface. This "finish allowance" ensures that any subsequent precision operations such as finish turning, grinding, or polishing can remove any possible surface defects or minor deflections resulting from the roughing process, thus guaranteeing a superior final surface quality.

To ensure that the shaft maintains a straight, cylindrical profile suitable for deep-hole boring and further precision machining, dial gauges or electronic measurement probes are positioned at various axial points during turning. These instruments monitor the shaft's runout its deviation from true centrality along the full length. For this operation, a strict threshold of ≤ 0.25 mm runout is enforced, with corrective action taken immediately if exceeded. Steady rests or anti-vibration supports may be used to further stability during cutting if necessary.[89]

The result of this step is a uniformly straight, cylindrical shaft blank, perfectly aligned and ready for high-precision deep-hole drilling and finishing operations. The tight control of OD and runout during rough turning lays the foundation for all downstream processes essential for marine propeller shaft reliability and performance.

08.3 Deep-Hole Drilling and Internal Boring

After rough turning, the focus shifts to creating the hollow core along the shaft's uniform section. Due to the length of the core (1200 mm) and the precision requirements, a deep-hole drilling process is adopted, utilizing advanced tooling and continuous high-pressure coolant for quality and accuracy.

The operation begins with the pilot hole drilling, executed from the tail end of the shaft using a carbide gun drill of 20 mm diameter. Gun drills are specifically designed for deep holes they feature a hollow shank through which coolant is fed internally, flushing chips from the cutting edge and preventing heat buildup, crucial for Duplex 2205's work-hardening tendency and thermal sensitivity.[90]

Once the pilot hole is established, a BTA (Boring and Trepanning Association) deep-hole drilling system is employed for successive enlargements. BTA utilizes high-pressure oil coolant (above 80 bar) that flushes chips while maintaining bore temperature, essential to prevent thermal distortion and maintain straightness in the long shaft. The drilling proceeds in multiple stages to gradually reach the desired inner diameter of 60 mm, within a tolerance of ± 0.02 mm.[91]

Simultaneously, the bore's surface finish and straightness are continuously monitored; total runout is maintained within 0.1 mm over the full depth. These tolerances are critical to ensure concentricity between the bore and the outer diameter in later re-mounting steps.



Figure 84 - Deep hole drilling

At the transition zone from the hollow section to the solid taper, careful machining introduces an internal blend radius (fillet) of approximately 5 mm. This fillet reduces stress concentrations during torsional loading, thereby enhancing shaft fatigue life in service.

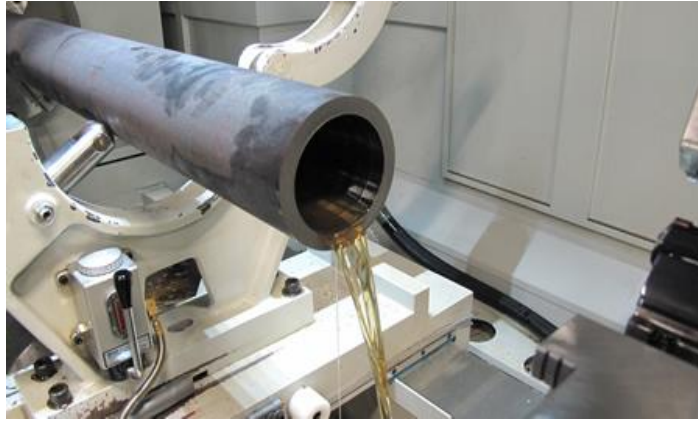


Figure 85 - Coolant Evacuation

Following drilling, internal boring removes any remaining irregularities left by the deep-hole process. This finishing bore operation employs precision boring bars to refine diameter and roundness, creating an ideal base for honing. Honing achieves the final surface finish, typically targeting $Ra \leq 0.8 \mu m$ to minimize corrosion hotspots and improve fatigue resistance.[92]



Figure 86 - Internal Boring

Effective coolant delivery, tool sharpness, and staged drilling parameters (low feed rate and moderate cutting speeds, approximately 10–12 m/min for drilling) are employed specifically for Duplex 2205, reflecting its hardness and thermal conductivity characteristics to optimize tool life and hole integrity.[91]

This meticulous drilling and boring process results in a high-quality, straight, and finely finished hollow core, integral to the shaft's stiffness and performance.

08.4 Re-Mounting and Rough OD Turning

After successful completion of deep-hole drilling and internal boring, the shaft undergoes precise re-mounting for its next critical machining phase. The key objective here is to establish perfect concentricity between the inner bore and the external diameter (OD), ensuring both features are referenced to the same axis for maximum shaft performance and balance.

The shaft is removed from the centers and remounted using a precision expanding mandrel that grips the inner bore. This mandrel provides a stable, axially concentric reference, guaranteeing that all subsequent OD cuts align with the bore axis. Such precise fixturing eliminates the risk of eccentricity that could compromise bearing fits, dynamic balance, or shaft coupling further downstream.

With the shaft mounted on the mandrel, rough turning of the OD resumes, now focusing on the hollow uniform section. Carbide cutting tools, optimized for Duplex 2205, are employed to machine the

shaft's external surface to an approximate diameter of 126 – 127 mm, staying just above the final specification (125.00 mm) to allow for later finishing and fine-tolerance control. Throughout the process, constant measurement is vital-wall thickness is verified via ultrasonic gauges or micrometres to ensure uniformity and detect any anomalies caused by boring or earlier operations.[93]

Special attention is paid to maintaining wall thickness consistency along the hollow region, as sectional variation could introduce stress concentrations, compromise strength, or alter shaft resonance during operation. Operators cross-check with both OD and bore measurements at multiple intervals, recording results for quality documentation.

Additionally, the transition region between the hollow section and the solid-taper area is carefully machined: shoulders and blending radii are shaped according to the specified geometry to facilitate smooth stress flow and proper fit for the taper and propeller boss.

Machining Process and Setup:

- The shaft setup must be absolutely rigid: a heavy-duty CNC lathe with extended steady rests (if available) is used, supporting both ends of the workpiece.
- The expanding mandrel is torqued precisely to prevent slippage under heavy cuts, and the lathe is programmed to monitor tooling load.
- Cut path is set to rough-turn the entire 1200 mm hollow section to an OD of 126–127 mm, leaving an allowance for finishing.

In-Process Inspection:

- Wall thickness measured with calibrated ultrasonic or mechanical gauges at both ends and the middle, aiming for uniformity (33 mm nominal wall, ± 0.3 mm).
- OD is continuously verified with micrometres, with runout checked at steady intervals to ensure < 0.2 mm TIR along the full hollow length.

Quality Notes:

- The transition shoulder and blend fillet between the hollow section and the solid taper are rough machined per the provided section, facilitating subsequent finishing.
- Chip evacuation and tool wear are monitored closely due to Duplex 2205's hardening behaviour; fresh inserts are used for each shaft when required.

At the completion of this step, the shaft presents a concentric, cylindrical hollow section; dimensioned to 126 –127 mm OD with finished shoulders, correct transitional blends, and wall thickness suitable for final machining and balance operations. This precise alignment and rough dimensioning form the foundation for successful taper machining, spline operation, and final tolerance achievement in all subsequent steps.

08.5 Taper and Shoulder Machining

Following the rough OD turning of the hollow section, the shaft's solid end, measuring approximately 300 mm, is prepared for taper machining a critical interface region responsible for transmitting torque to the propeller.

08.5.1 Setup and Mounting

The shaft is securely mounted on the CNC lathe, with the inner bore referenced using the expanding mandrel to maintain axial and radial concentricity between the bore and external features. The taper turning attachment or CNC interpolation cycles are programmed using the shaft's detailed CAD model and drawings for accuracy.



Figure 87 - Taper turning from CNC lathe

08.5.2 Machining the Taper

- The taper is machined to a 1:16 ratio (or roughly 3.58° angle) taper as per marine shaft standards, reducing the shaft diameter smoothly from approximately 125 mm at the shoulder to 106 mm at the tip over the specified 300 mm length.
- During taper cutting, a light finish allowance of about 0.5 mm is maintained for later finishing operations.
- This taper size conforms to marine industry standards and class society requirements for propeller fittings.
- Rough passes remove bulk material with depths of cut around 1–2 mm and feeds slowed to 0.3–0.5 mm/rev to minimize vibration and maintain dimensional control.
- Final passes employ slower feeds and cutting speeds (0.1–0.2 mm/rev feeds, 60–80 m/min speeds), targeting tight tolerances within ± 0.02 mm and a fine surface roughness of $R_a \leq 0.8$ μm . [94]

08.5.3 Machining Shoulders and Blend Radii

- The shoulder defines the transition zone between the hollow cylindrical section and the solid taper, designed for smooth stress flow and to serve as an assembly reference.
- A fillet radius or blend radius of about 5 mm is machined at this junction to reduce stress concentrations during torsional loading.
- These radii are critical to preventing fatigue cracks and ensuring the shaft withstands operational stresses over time.
- Machining tools with polished carbide inserts and appropriate nose radii are used to achieve a smooth contour linking the cylindrical and tapered sections.

08.5.4 Quality Assurance

- Dimensional validation is conducted using high-precision contact or optical measurement instruments to verify taper angles, diameters, and shoulder geometries.

- Prussian blue or similar materials are applied to the taper for contact pattern assessment to ensure a uniform fit with the propeller hub.
- Excessive clearance or localized high points are corrected through minimal controlled material removal, often involving hand lapping or fine polishing.

08.5.5 Heat Management and Cutting Conditions

- Duplex 2205's work-hardening and thermal sensitivity necessitate flood coolant use and conservative cutting parameters during taper machining.
- Insert selection prioritises heat-resistant coatings (TiAlN or similar) to maintain tool life and surface quality.

Throughout taper and shoulder machining, in-process inspection with contact gauges and Prussian blue checking ensures uniform contact and proper taper angle for the desired propeller fit.

The taper and shoulder machining steps are pivotal to the shaft's performance, requiring precision and control to guarantee secure torque transmission and fatigue resistance under marine operating conditions.

08.6 Heat Treatment

08.6.1 Solution Annealing and Straightening

Following rough machining of the outer diameter and taper, the Duplex 2205 marine propeller shaft undergoes solution annealing a critical heat treatment step to restore the metallurgical balance and mechanical properties essential for marine applications.

Heat treatment is an essential step in the manufacturing of Duplex 2205 marine propeller shafts to restore and optimize the metallurgical structure after machining and forging operations. Machining can introduce residual stresses and localized work hardening, while hot forging and thermal cycles may produce undesirable phases or microstructural imbalances. Without proper heat treatment, the shaft would suffer from reduced toughness, non-uniform mechanical properties, and compromised corrosion resistance, which are critical for long-term marine durability and performance.

Solution annealing is the preferred heat treatment for Duplex 2205 stainless steel because it restores the alloy's balanced two-phase microstructure approximately equal parts ferrite and austenite. Heating the shaft to 1040 –1100 °C dissolves harmful intermetallic phases such as sigma phase and chromium nitrides, which can embrittle the material and reduce corrosion resistance. The subsequent rapid water quench "freezes" the desirable microstructure in place.

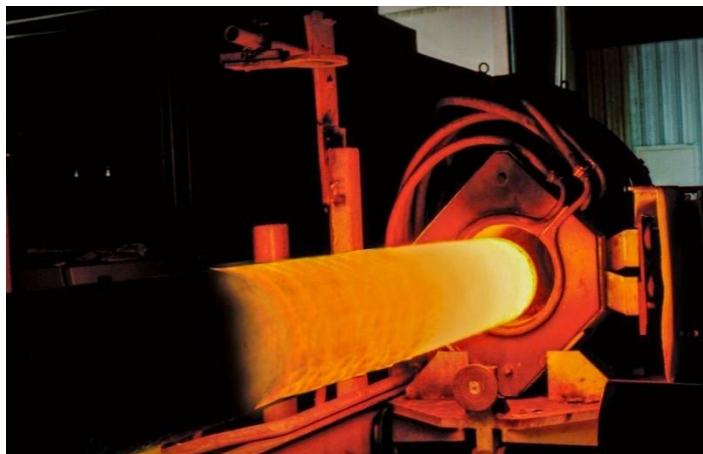


Figure 88 - Solution Annealing

This treatment also relieves internal stresses from previous manufacturing steps, improves ductility, and maximizes corrosion resistance especially against chloride-induced pitting and stress corrosion cracking. Overall, solution annealing unlocks the superior strength-to-corrosion resistance ratio of Duplex 2205, ensuring structural integrity and long service life in harsh marine environments.

The shaft is placed in a controlled atmosphere furnace and heated uniformly to a temperature range between 1040 °C and 1100 °C. This temperature ensures that the duplex stainless steel dissolves any unwanted intermetallic phases, such as sigma phase or chromium nitrides, and achieves an optimized balance between the ferritic and austenitic phases (approximately 50% each), which impart high strength and excellent corrosion resistance.[95]

The soaking time is calculated based on the shaft's cross-sectional thickness typically 1 hour per 25 mm thickness to ensure full homogenization. After annealing, the shaft is rapidly water-quenched to lock in the desired microstructure and avoid the precipitation of harmful phases.

Post-quench, residual stresses developed during machining and heat treatment might induce distortion. Therefore, the shaft undergoes cold straightening using a hydraulic press and roller supports. This process incrementally applies controlled mechanical forces to bend the shaft back to the required straightness tolerance, ensuring alignment critical to dynamic balance and bearing fitment in the final assembly. Because cold straightening is performed at ambient temperature, it preserves the duplex microstructure, maintaining the enhanced mechanical and corrosion-resistant properties achieved through annealing.

During straightening, contact dial gauges reference the shaft's bore and OD to maintain runout within 0.1 mm across the entire length. Straightening operations are incremental and monitored carefully to prevent over-correction that might induce microstructural damage.

The heat treatment and straightening process ensures the shaft achieves optimal mechanical properties, including tensile strength, yield strength, toughness, and high resistance to pitting and stress corrosion cracking critical for long-term marine service reliability.

The benefits of solution annealing for the microstructure of Duplex 2205 stainless steel are:

- **Restoration of balanced duplex microstructure:** Solution annealing dissolves undesirable intermetallic phases (e.g., sigma phase, chromium nitrides) formed during prior processing, restoring the desired approximately 50% ferrite and 50% austenite phases.
- **Enhanced mechanical properties:** This rebalancing improves tensile strength, yield strength, ductility, and impact toughness, providing resistance to embrittlement and maintaining toughness even at low temperatures.
- **Improved corrosion resistance:** By removing harmful phases and maintaining a protective chromium oxide layer, solution annealing greatly enhances resistance to pitting, crevice corrosion, and chloride-induced stress corrosion cracking critical for marine environments.
- **Relief of residual stresses:** The process alleviates stresses induced by machining, forging, or welding, thereby reducing distortion and preventing premature failure.
- **Optimized fabrication performance:** It prepares the alloy for subsequent fabrication steps or service conditions by ensuring a uniform and stable microstructure.

08.7 Finish Turning

After the solution annealing and straightening treatment, the marine propeller shaft undergoes precision finish turning on a CNC lathe. This step brings the shaft to its final dimensional tolerances and surface finish standards critical for optimal performance and fitment in marine propulsion systems.

The shaft is remounted using the expanding mandrel inside the inner bore to ensure concentricity between the internal and external diameters. This precise referencing system minimizes runout and ensures all critical diameters share the same central axis, essential for smooth rotation and minimal vibration during operation.

Finish turning involves machining the uniform hollow section OD to the exact final dimension of 125.00 ± 0.02 mm, removing the finish allowance previously left after rough turning. The taper section is also finished to the final specified diameters and taper angle, with careful control to achieve smooth, taper-compliant surfaces.

In the final turning of Duplex 2205 stainless steel, the cutting speed is deliberately reduced compared to roughing operations. Typically, cutting speeds range from 70 to 90 meters per minute. This reduction helps to enhance surface finish quality and prevents accelerated tool wear, which is crucial given the alloy's tendency to work harden and generate heat.

The feed rate during finish turning is also slowed to between 0.1 and 0.3 millimetres per revolution. This slower feed minimizes surface roughness and maintains close control over dimensional tolerances. Controlled feed rates help avoid tool marks and ensure that the tight specifications necessary for marine shaft applications are met.

Depths of cut in finishing are light, generally between 0.3 and 0.5 millimetres per pass. These light cuts are essential for achieving fine surface integrity and refinement of the shaft's external diameter after prior rough machining stages. Smaller depths of cut reduce cutting forces and thermal impact on the workpiece.

Tooling selection plays a significant role in finish turning Duplex 2205. Fine-grain carbide inserts, or ceramic-tipped tools are preferred. These tools typically feature polished rake faces and a nose radius sized between 0.4 and 0.8 millimetres. Such configurations reduce the risk of built-up edge formation, improving surface quality and consistent tool life in machining this highly corrosion-resistant alloy.

Continuous application of flood coolant throughout the finish turning process is mandatory. Coolant not only prevents overheating, which can tarnish the surface and alter metallurgy, but also stabilizes the dimensional accuracy of the shaft and prolongs tool life. A well-managed coolant flow helps flush away chips and dissipate heat effectively.

During the finishing turning of the taper, a blueing contact check is performed using Prussian blue or a similar transfer medium. This non-destructive technique verifies uniform contact between the machined taper and the mating propeller hub, ensuring complete load transmission without gaps or stress concentrations.

08.8 Spline Taper Machining

After the final turning and taper machining operations, spline cutting is performed on the coupling end of the shaft to enable torque transmission between the shaft and the propeller hub. This step is crucial to ensure proper power transfer, alignment, and fitment with the mating component.

The exact spline section on the solid tapered end is marked using inspection jigs, referencing engineering drawings and CAD geometry. The taper profile is confirmed for length, angle, and

diameter per product specification. Accurate referencing through the shaft bore, using an expanding mandrel, ensures that the spline machining will be perfectly concentric with the established shaft axis.



Figure 89 - Spline Shaft Milling

The spline machining is carried out on a horizontal milling machine using a form cutter, which is the most suitable and economical method for producing splines on medium-length shafts. Milling with a form cutter allows accurate control of tooth width, spacing, and depth, and produces a smooth finish that minimizes stress concentration.

Alternatively, broaching or slotting methods may also be used for the spline formation. However, for this design, milling with a form cutter is preferred due to better dimensional control and surface finish. The spline profile is designed in accordance with the ISO 4156 standard, with a nominal width of 10 mm and a tolerance grade of JS9, ensuring proper fit with the mating hub.

With each pass of the cutter, a longitudinal groove is created to the specified width, depth, and pressure angle. Several light passes are typically used to achieve final dimensions, with roughing passes followed by fine finishing cuts. Attention is then given to the flank radii and fillet geometry at the root of each tooth; these curved transitions are polished using abrasive sticks or finishing tools to minimise tool marks, remove microcracks, and counteract stress concentration. All grooves are inspected for uniformity, correct depth, and taper alignment.

Following primary machining, the entire splined region undergoes deburring using rotary brushes and abrasive stones to eliminate any sharp edges or residual burrs left from the cutter. This stage is essential for safety, fit, and fatigue resistance. For surface verification, a contact or laser profilometer is used, logging roughness values against the project's required Ra finish, frequently targeting $Ra \leq 0.8 \mu\text{m}$ for optimal service in marine environments.[88]

The newly machined spline is finally checked with go/no-go spline gauges, coordinate measuring machines (CMM), and optical comparators to confirm all aspects of geometry, such as pitch diameter, tooth profile, and angular orientation, meet both the design drawings and industry standards. Meticulous documentation of the process, tool settings, and inspection results is maintained as part of the shaft's quality record, ensuring traceability and compliance with marine propulsion requirements. This comprehensive approach ensures the spline taper delivers robust, fatigue-resistant, and perfectly concentric coupling with the propeller hub over the shaft's operational life.

08.9 Thread Machining of the M95 × 4 Left-Hand Shaft Thread

After completion of the taper section, a 100 mm long left-hand external thread is machined to accommodate the M95 × 4 ISO metric nut. This threaded region acts as a retaining mechanism that prevents the propeller hub from loosening under hydrodynamic thrust and reverse torque effects. The thread conforms to ISO 68-1 geometry and ISO 965-1 tolerance specifications. [86]

08.9.1 Thread Specification

Table 19 - Calculated vs. ISO standard values of thread

Parameter	Value (Calculated)	ISO Standard Range
Thread designation	M95 × 4 LH	M95 × 4 LH
Major diameter (d)	95.000 mm	94.940 – 94.465 mm
Pitch (P)	4 mm	4 mm
Pitch diameter (d ₂)	92.402 mm	92.342 – 92.092 mm
Minor diameter (d ₃)	90.672 mm	90.610 – 89.628 mm
Tolerance class	6g	ISO 965-1

(LH = left-hand thread)

08.9.2 Purpose of Thread

Once the propeller is seated on the 1:16 taper, a **counter-clockwise nut** is tightened onto this thread to mechanically lock the propeller in position. Marine propulsion loads naturally tend to rotate the nut in the tightening direction (left-hand), thereby preventing any loosening during operation.

08.9.3 Machining Process for the M95 × 4 LH Thread

Preparation and Setup

1. The shaft is mounted on the CNC lathe using the expanding mandrel referencing the bore to maintain concentricity.
2. The threaded region (100 mm length) is rough-turned to 94.80–94.90 mm OD, providing a uniform cylindrical base for the final thread form.
3. A carbide insert designed for ISO metric external threading (60° V-profile) is installed.
4. Cutting parameters are selected for Duplex 2205 stainless steel:
 - Cutting speed: 40–55 m/min
 - Feed: 4 mm per revolution (equal to pitch)
 - Depth of cut (initial): 0.1–0.2 mm
 - Flood coolant to control temperature and prevent work-hardening.

CNC Thread Turning Procedure

1. **Set tool orientation for left-hand cutting**

The CNC lathe is programmed for left-hand threading, meaning the tool moves from the shaft shoulder toward the free end while the spindle rotates clockwise (M03).

2. **Multiple-pass threading**

The thread is cut in several radial passes, each removing $\sim 0.05\text{--}0.15$ mm, until reaching the target minor diameter (≈ 90.67 mm).

3. **Control of Thread Geometry**

The following are continuously verified:

- flank angle (60°)
- thread profile shape
- pitch accuracy (4 mm)
- root radius in compliance with ISO 68
- cumulative error within tolerance class 6g

4. **Final spring passes**

One or two “spring passes” are made at zero additional depth to eliminate elastic deflection and achieve consistent flank contact.

Inspection and Quality Verification

1. **Pitch diameter check**

Using a Three-Wire Method or CNC measuring probe to achieve 92.40 ± 0.02 mm.

2. **Go/No-Go gauge testing**

An $M95 \times 4$ LH ring gauge or calibrated nut confirms compliance with ISO geometry.

3. **Surface finish**

Target surface roughness $R_a \leq 3.2$ μm , suitable for large marine threads.

4. **Deburring**

A fine file or ceramic deburring tool removes sharp edges to avoid crack initiation.

08.9.4 Final Result

A precision-machined $M95 \times 4$ left-hand threads, concentric with the shaft axis and compliant with ISO tolerances, is produced over 100 mm length. This threaded section reliably accepts the stainless-steel retaining nut and ensures secure axial locking of the propeller during marine operation.



Figure 90 - Thread of the shaft

08.10 Machining of the Flange Section for Shaft–Intermediate Shaft Connection

A forged flange section is incorporated immediately before the 125 mm diameter portion of the shaft to allow rigid connection between the propeller shaft and the intermediate shaft. The flange is machined to a final outside diameter of 275 mm, a thickness of 45 mm, and contains six equally spaced bolt holes on a 200 mm pitch-circle diameter (PCD). This flange ensures safe torque transmission between shaft segments and maintains axial alignment under marine operating loads.

08.10.1 Initial Preparation

1. The shaft blank is rough turned to establish the correct outer reference surfaces.
2. The region designated for the flange is left with sufficient stock ($\approx 4 - 6$ mm per side) to allow accurate finish machining after heat treatment and straightening.
3. The shaft is mounted between centers or using a precision expanding mandrel to ensure the flange is perfectly concentric with the shaft axis.

08.10.2 Rough Machining of the Flange Profile

1. **Turning the Outside Diameter (OD)**
The CNC lathe rough-turns the flange OD to approximately 272 – 273 mm, leaving a small allowance for finishing to 275 mm.
2. **Face Roughing**
Both faces of the flange are rough-faced to about 46 – 47 mm thickness.
This allows precise finishing after heat treatment.
3. **Fillet and Blend Radius Formation**
A smooth radius is machined at the root of the flange to reduce stress concentration at the shoulder.

08.10.3 Post-Heat-Treatment Finish Machining

After solution annealing and straightening (as described earlier in the shaft process), the flange is finished-machined.

08.10.3.1 Finish Turning

- OD is finished to 275.00 ± 0.10 mm.
- Thickness is finished to 45.00 ± 0.05 mm.
- Both faces are skim-cut to achieve a surface finish of $Ra \leq 1.6$ μm .

08.10.3.2 Facing and Perpendicularity

A final facing operation ensures:

- Face runout ≤ 0.05 mm
- Perfect perpendicularity between the flange face and shaft axis

These tolerances are required to avoid coupling misalignment.

08.10.4 Machining of the Bolt Circle

The workpiece is transferred from the lathe to a CNC vertical milling machine or horizontal machining center.

Centre Datum and Tool Setup

1. The shaft axis is aligned with the machine spindle using a coaxial indicator.
2. The machine is zero at the center of the flange.
3. A carbide drill and reamer set is prepared for accurate bolt-hole formation.

CNC Drilling of Bolt Holes

The bolt-hole pattern consists of 6 holes equally spaced at 60° intervals on a 200 mm PCD.

Bolt Hole Coordinates

- For a 200 mm PCD:
 - Radius = 100 mm
- The CNC drilling cycle indexes to each angle:
0°, 60°, 120°, 180°, 240°, 300°.

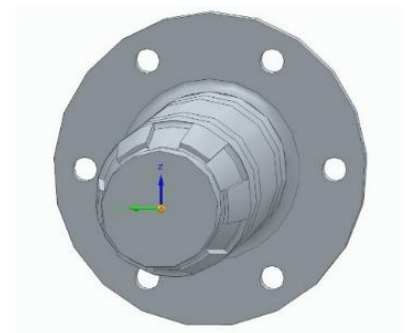


Figure 91 - Flange of the shaft

Procedure

1. **Spot Drilling**
A 120° carbide spot drill is used to prevent wandering of the final drill.
2. **Pilot Drilling**
Pilot hole drilled to the required depth.
3. **Final Drilling**
Holes are drilled to the bolt diameter (e.g., Ø18, Ø20, or as per coupling design).
Clearance hole tolerance: H13 (typical for flanged connections).
4. **Reaming (if required)**
If fitted bolts are used, reaming ensures precise hole alignment.
5. **Chamfering**
All hole edges are chamfered to remove burrs and avoid stress concentrations.

08.10.5 Machining of Threaded Holes (if applicable)

If the design requires threaded holes in the flange instead of clearance holes (depending on the selected coupling design. e.g., intermediate shaft end uses clearance holes, flange uses tapped holes):

Procedure for Tapped Holes

1. Drill the correct core diameter (tap drill size).
2. Use rigid CNC tapping or synchronized tapping cycle.
3. Apply cutting oil suitable for Duplex 2205.

4. Verify thread gauge fit (Go/No-Go).

Typical thread class: 6H.

Note – If design uses nuts on the intermediate shaft side, then our flange holes are not threaded.

08.10.6 Final Finishing and Quality Checks

Dimensional Inspection

- OD (D_2): 275.00 mm
- Thickness: 45.00 mm
- PCD (D_1): 200.00 ± 0.05 mm
- Hole position error: ≤ 0.05 mm

Face Runout

- Measured with dial indicator
- Target: ≤ 0.05 mm

Surface Roughness

- Flange faces: $R_a \leq 1.6 \mu\text{m}$
- Bolt hole edges deburred and chamfered

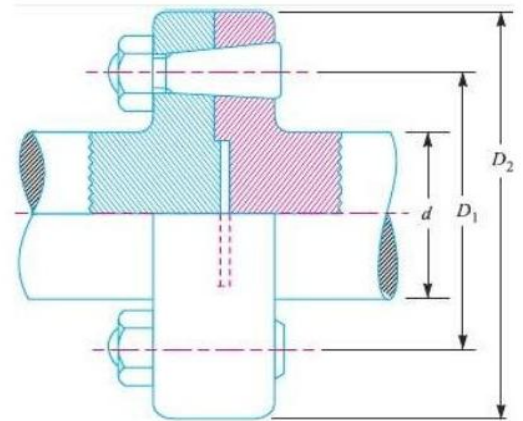


Figure 92 - Connection of the two flanges of the propeller shaft and intermediate shaft

08.10.7 Functional Outcome

The finished flange provides:

- High-accuracy bolt alignment
- Perfect concentricity with shaft axis
- Low runout for smooth power transmission
- Structural rigidity under torsion and thrust loads
- Reliable coupling between propulsion shaft and intermediate shaft

This ensures safe, vibration-free operation of the marine drive system.

08.11 Finish Grinding and Polishing

In this crucial phase, the outer diameters (OD) of key shaft sections and the tapered surfaces that interface directly with bearings and the propeller hub undergo finish grinding. This precision grinding operation eliminates any residual machining marks or minor surface inconsistencies, ensuring compliance with tight dimensional tolerances. Grinding helps achieve the desired surface quality and contributes to the geometric accuracy essential for reliable shaft performance.

Given the Duplex 2205 alloy's high strength and work hardening tendencies, specialized grinding wheels made from aluminium oxide or silicon carbide with appropriate grain sizes are selected to balance efficient material removal and surface quality. Cooling fluids, typically flood coolant with

additives for corrosion prevention, are continuously applied to minimize heat buildup that could otherwise induce thermal distortion or affect microstructure at the machined surface.

The target surface roughness value (Ra) on bearing journals and tapered surfaces is set to 0.8 micrometres or better to reduce friction, wear, and stress concentration. The finish grinding is complemented by manual or automated polishing, particularly on taper surfaces, to ensure high visual and functional quality, enabling optimal mating with the propeller hub and bearings.

08.12 Non-Destructive Examination (NDE) and Final Dimensional /Runout Checks

In the final step propeller shaft undergoes rigorous non-destructive examination (NDE) coupled with precise dimensional and runout inspections to confirm the structural integrity and geometric accuracy of the complete component.

Ultrasonic Testing (UT) is employed on the shaft's main body. Using high-frequency sound waves, UT detects internal discontinuities such as cracks, voids, and inclusions that might compromise the shaft's mechanical performance. This test ensures that the entire volumetric structure of the shaft meets the high safety standards expected in marine propulsion applications.[96]

For the critical taper and splined areas, where stresses are typically concentrated, Dye Penetrant Inspection (DPI) or Magnetic Particle Inspection (MPI) is conducted. DPI involves applying a penetrant liquid to the surface, allowing it to seep into fine cracks, and then highlighting these with a developer. MPI utilizes magnetic fields and iron particles to reveal surface and near-surface flaws specific to the ferromagnetic properties of Duplex 2205. These surface NDE methods are essential for detecting defects that are not visible to the naked eye but could be potential failure initiation sites.

Alongside NDE, comprehensive dimensional and runout measurements are taken. Precision metrology tools, including coordinate measuring machines (CMM), dial indicators, and runout gauges, are used to verify that the shaft maintains concentricity and minimum eccentricity through its full length. Runout measurements are particularly emphasized on bearing journals and the splined taper since these areas must meet exact tolerance limits to prevent vibration, undue bearing wear, and potential operational failures.[97]

Any deviations from the standard dimensional specifications or detection of defects during NDE trigger detailed investigations, corrective machining, or rejection of the shaft. Only shafts passing all inspection criteria proceed to final surface treatment (if applicable), packaging, and shipment.

All test results, measurement data, and inspection certificates are meticulously documented to ensure traceability and compliance with marine classification societies. This comprehensive NDE and inspection step guarantees that the marine propeller shaft possesses the necessary strength, durability, and precision geometry for safe, long-term operation in demanding maritime environments.

08.13 Discussion

08.13.1 Spline Machining, Thread Machining and flange Machining Should NOT Be Done Before Heat Treatment

✓ Duplex 2205 heat treatment causes dimensional changes

Solution annealing at 1040–1100 °C, followed by rapid quenching, produces:

- thermal expansion
- contraction during quench
- residual stress redistribution

Even with perfect control, a long shaft (1.6 m) will experience:

- slight bending
- local diameter variation
- taper shift
- changes in pitch diameter

If you cut precise splines before heat treatment, the thermal cycle destroys their accuracy.

This is the same reason you never grind turbine blades or gear teeth before heat treatment.

✓ Splines require final-stage dimensional accuracy

Splines depend on:

- pitch accuracy
- tooth profile
- flank angle
- root radius
- concentricity with the bore

Heat treatment would distort all of these, ruining:

- fit with the coupling
- torque load distribution
- fatigue performance

So spline machining must happen after heat treatment, and ideally after finish turning.

✓ Surface hardening is not done here

We are using Duplex 2205 (not carburized or nitride steel).

So there is no hardening after spline cutting that might risk cracking the teeth.

That makes post-heat-treatment spline machining the safest sequence.

08.13.2 Heat Treatment Should NOT Be Done After Grinding or Polishing

✓ Polished surfaces lose their finish during heat treatment

Solution annealing scale forms at $>1000\text{ }^{\circ}\text{C}$:

- heavy oxide
- discoloration
- roughening
- distortion

Grinding/polishing done before heating is wasted effort.

✓ Heat treatment destroys dimensional precision

If you grind first, then heat treat:

- roundness is lost
- surface waviness increases
- Bearing fits loosen
- taper angle shifts
- runout increases

Grinding and fine polishing must be the last machining operations.

✓ Straightening is done after heat treatment

After quenching, shafts almost always require:

- cold straightening
- checks for TIR (total indicated runout)
- local correction using presses

This would ruin a polished surface.

So polishing must always come after straightening.

09. Quality Assurance & Safety

09.1 Quality Assurance Strategy

The quality assurance (QA) plan ensures the marine propulsion shaft is manufactured and tested to meet all design and service requirements under seawater conditions. Emphasis is placed on thorough inspections, adherence to industry standards, defined acceptance criteria, and full traceability of materials and processes. The shaft material (e.g., duplex stainless steel) and components must satisfy strict criteria to withstand high salinity (~25 °C seawater) and cyclic torsional loads without premature failure. Key elements of the QA strategy include:

09.2 Inspection Protocols

09.2.1 Dimensional Inspection

All critical dimensions of the shaft are verified against engineering drawings and tolerance specifications. Using calibrated measuring tools (micrometers, calipers) or CMM (coordinate measuring machine) as needed, the shaft's overall length, journal diameters, fillet radii, keyway geometry, and threaded sections are inspected. This ensures proper fit with bearings, couplings, and propeller. Any threaded elements (e.g. coupling bolts or propeller nut) are gauged to confirm compliance with thread standards (e.g. ASME B1.1 for Unified screw threads) so that form, pitch, and tolerances are within allowable limits [98]. Surface finish on sealing areas or bearing journals is also checked.

09.2.2 Material Certificate Verification

The material is confirmed to be of the specified grade and quality via mill certificates. For example, if a duplex stainless steel is used, the certificate is reviewed for chemical composition (to ensure the correct alloy, e.g. UNS S32205 or S32750, with PREN sufficient for pitting resistance) and mechanical properties (yield strength, tensile strength, elongation, hardness, toughness) per the relevant ASTM/ASME material standard. Particular attention is given to properties critical for marine service, such as Charpy V-notch impact toughness in the heat-treated condition. (Classification societies often require a minimum impact energy, e.g. 27 J at –10 °C [99], to ensure adequate toughness in critical components.) The certificate is checked for compliance with standards like ASTM A182 or ASTM A276 (if applicable) and supplementary requirements for marine hardware. Material traceability is verified by matching the heat/lot number on the certificate with the identification on the shaft (stamped or tagged). This ensures the shaft's material is fully traceable to the original melt and heat treatment batch [99].

09.2.3 Non-Destructive Testing (NDT)

After preliminary machining and final heat treatment, the shaft undergoes NDT to detect any internal or surface defects that could lead to failure:

- **Ultrasonic Testing (UT):** A 100% volumetric UT scan of the shaft is performed to check for internal flaws such as shrinkage cavities, cracks, or inclusions. Testing is done after the shaft is machined to a smooth cylindrical form (suitable for UT) and heat treated, as required by class guidelines [99]. Both axial and radial scan orientations are used, covering high-stress regions (e.g. near fillets, keyways, and changes in section) to ensure no harmful subsurface indications. UT acceptance criteria are based on standards such as ASTM A388 or EN 10228, or classification society rules, which typically allow no reportable indications above a certain echo size/level in critical zones. For instance, *Det Norske Veritas* rules mandate that forgings like propeller shafts be free from internal defects prejudicial to their use [99]; any UT

indication exceeding the class acceptance level (e.g. equivalent to a flat-bottom hole of specified diameter) would lead to rejection or repair.

- **Magnetic Particle Inspection (MPI) or Dye Penetrant Inspection (DPI):** All machined surfaces, especially at stress concentrators (fillet transitions, keyway corners, thread roots), are inspected for surface cracks. If the shaft material is ferromagnetic (duplex stainless has sufficient ferrite for magnetic inspection), MPI is employed using visible or fluorescent magnetic particles. Non-ferromagnetic areas or austenitic welds (if any) are checked with liquid penetrant. This inspection is typically performed after final machining/polishing. No linear indications or crack-like defects are permissible. Any minor imperfections (e.g. superficial marks) are evaluated against acceptance standards (ASTM E1444 for MPI or ASTM E165 for DPI) and repaired or polished out as needed. Acceptance: The shaft must have zero surface cracks; even fine indications are cause for investigation and rework in such a critical rotating component. Notably, class rules prohibit repair welding on certain highly stressed shaft areas (like in-situ welding on a propeller shaft) due to fatigue sensitivity [99], so initial forging and machining quality must be high to avoid repairs.

09.2.4 Specialized Testing for Duplex Stainless Steel

If the shaft is made from a duplex stainless steel, additional inspections are conducted to ensure proper metallurgical quality. Duplex stainless is susceptible to the formation of detrimental intermetallic phases (sigma, chi, etc.) if improperly heat treated. To guard against this, the QA plan invokes ASTM A923 testing on samples from the shaft or same heat. ASTM A923 provides methods to detect harmful intermetallic phase precipitation in duplex stainless steels [100]. For example, Method A is metallographic examination, Method B is Charpy impact testing at low temperature, and Method C is a ferric chloride corrosion test – collectively these tests indicate the presence of embrittling phases [100]. The acceptance criterion is that the material shows no evidence of deleterious intermetallics: e.g., toughness remains above the minimum specified for the grade (no significant drop in impact energy), and corrosion test weight loss is within limits for “phase-free” material. By requiring ASTM A923 compliance, we ensure the duplex microstructure has the correct austenite-ferrite balance and has not been embrittled by sigma or other phases, which could severely reduce corrosion resistance and toughness [100].

09.2.5 Inclusion Assessment

For high-strength steel shafts, the cleanliness of the steel (non-metallic inclusion content) can be critical to fatigue performance. ASTM E45 microscopic inclusion rating may be employed as a QA check on the steel quality. ASTM E45 classifies inclusions (sulfides, oxides, silicates, etc.) by type, shape, and severity [101]. A low inclusion rating (indicating very few inclusions of small size) is desirable, as inclusions can act as initiation sites for fatigue cracks. Many manufacturers order electroslag remelted (ESR) or vacuum-arc remelted steel for critical shafts to improve cleanliness. If an inclusion rating is specified (e.g., “thin series ≤ 1.5 , heavy series 0” by ASTM E45 Method A), the mill certificate or lab analysis must show compliance. This helps ensure the shaft material’s high cycle fatigue strength is not compromised by internal cleanliness issues [101].

09.3 Compliance with Standards and Certifications

The QA process is aligned with relevant international standards and classification society requirements to guarantee quality and consistency:

- **ISO 9001 Quality Management:** The shaft manufacturer operates under an ISO 9001-certified quality management system. This ensures systematic control of processes, calibration of inspection equipment, trained personnel qualifications, and documentation. In particular, ISO 9001:2015 Clause 8.5.2 on *Identification and Traceability* is implemented – every material and

part is identified and tracked through all stages to maintain conformity [102]. An internal audit and quality plan are in place so that all inspection and test activities (ITPs) are performed and recorded.

- **Material and Testing Standards:** The shaft material and testing procedures comply with recognized standards:
 - **ASTM A923** – Detection of Intermetallic Phases in Duplex Stainless: Ensures duplex stainless steel quality (absence of sigma, etc.) as discussed above [100]. This is critical for corrosion resistance and toughness in seawater service.
 - **ASTM E45** – Inclusion Content of Steel: Verifies steel cleanliness, classifying non-metallic inclusions that could impact fatigue performance [101]. This supplements manufacturer certifications to ensure a high-purity forging.
 - **ASTM A370 / ASTM E8/E8M** – Mechanical testing standards for tensile and impact tests, to verify that yield strength, tensile strength, elongation, and Charpy impact values meet specification and any class requirements.
 - **DNVGL-CG-0039** – DNV GL Class Guideline for Propeller and Shaft Calculations: The design and manufacturing adhere to class guideline recommendations. For instance, DNVGL-CG-0039 highlights that fatigue properties must account for surface roughness and notch sensitivity [103]; accordingly, the shaft's surface finish and fillet radii are controlled in QA. Additionally, the material's S-N curves and fatigue strength for design were chosen per class rules, and the manufacturing QA ensures the actual product meets the assumed quality (free of harmful defects or stress raisers).
 - **DNV/ABS Classification Rules:** The shaft is produced under the surveillance of a classification society (like DNV or ABS) if required for class certification. Class rules (e.g., DNV RU SHIP Pt.2) mandate certain inspections and documentation: material must come from an approved forge, a 3.2 inspection certificate (certified by an independent surveyor) is issued, and NDT is witnessed or verified. The manufacturing process meets *IACS* (International Association of Classification Societies) unified requirements for ship shafts – for example, IACS UR W7 for steel forgings requires adequate forging reduction ratios and heat treatment, 100% visual and ultrasonic inspection, and defect-free quality [99]. The shaft forging and testing thus comply with these class standards, ensuring it is eligible for use in a classed vessel.
 - **ASME B1.1** – Thread Standards: Any threaded sections (such as the shaft end thread for a propeller nut or threads for coupling bolts) conform to ASME B1.1 (Unified inch screw threads) or the relevant ISO metric thread standard. ASME B1.1 defines the thread form, series (UNC, UNF, etc.), class of fit, allowances and tolerances [98]. The QA inspectors verify thread gauges (e.g., 2A/2B or 3A/3B class fit) to ensure the threads are precise and compatible with mating hardware, preventing looseness or galling. This guarantees that critical fasteners will develop proper preload and not back off in service.
 - **Welding Standards:** If any welding is involved (for example, weld-on flanges, or attachment of wear sleeves), procedures and qualifications adhere to ASME Section IX and ISO welding qualifications. However, in a one-piece forged shaft, welding is generally avoided except for attaching sacrificial sleeves or repair cladding. Any weld repairs would be subject to class approval and follow standards like ASTM A488 (steel casting/forging weld repair) and subsequent NDT.

- **Industry and Military Standards:** Other standards are referenced as needed for marine hardware. For instance, certification of **cathodic protection anodes** may follow standards like *ISO 9351:2025*, which specifies chemical composition and electrochemical performance for aluminum and zinc galvanic anodes [104]. Bolting materials might follow ASTM A193/A320 (for high-strength stainless or alloy bolts) with proper ASTM A194 nuts, etc., all with appropriate corrosion-resistant grades for seawater. Coating processes (if applied to any part of the shaft assembly) would follow ISO 12944 or NORSOK M-501 for marine coatings (see below). By complying with this network of standards and guidelines, the project ensures the shaft meets both **academic and industrial criteria**, bridging theoretical design with practical, class-approved construction.

09.4 Acceptance Criteria and Traceability

Clear acceptance criteria are defined for all inspections and tests to decide if the shaft is acceptable or if rework is needed. All results are documented and traceable:

- **Material Acceptance:** The chemical composition must match the specified alloy grade within tolerances, and all mechanical properties must meet or exceed the minimum requirements of the governing standard or design. For duplex stainless, this includes meeting minimum impact energy at low temperature and proof of phase correctness (per ASTM A923 tests). For example, if ASTM A890 Grade 5A (super duplex) was specified, the actual yield and tensile strength must meet that standard's values, and the material should pass intermetallic tests (no cracking in bend tests or acceptable Charpy toughness drop per ASTM A923 Method B) as evidence of proper heat treatment [100].
- **Dimensional Tolerances:** All measured dimensions are checked against the drawing tolerance bands. Typical acceptance might be ± 0.1 mm on diameters critical to fit, tighter if press-fit or bearing surfaces are involved. Geometric tolerances (runout, straightness) are also checked – the shaft runout should be minimal (often < 0.05 mm at midpoint) to avoid vibration. If any dimension is out of tolerance, the piece is rejected or re-machined if possible. A final alignment check of coupling flanges (flange face runout, concentricity of bolt holes, etc.) is included to ensure proper installation.
- **NDT Acceptance:** The shaft must exhibit no injurious defects:
 - UT must show no internal discontinuities beyond acceptable size. Any indication echo amplitude above the reference level (calibrated to a flat-bottom hole per ASTM standards) in a critical zone is cause for rejection. Typically, class rules call for *Level 1 quality* for propulsion shafts, meaning essentially no detectable large defects; smaller indications may be acceptable if far from highly stressed areas, but those are subject to engineering critical assessment. In practice, a clean UT (no indications) is expected for a high-quality forging. Manufacturer UT reports (scans) are reviewed and, if required, a class surveyor may witness or audit the UT results.
 - MPI/DPI must find zero cracks. Any linear indication (no matter how small) on a high-stress area like a fillet or thread is unacceptable. Non-relevant indications (from surface geometry or grinding marks) are evaluated; if in doubt, the area can be re-polished and re-tested. Only if the indication is verified as non-crack (e.g., a scratch that is removed) can the part be accepted. The acceptance criteria might invoke standards like ASTM E125 for visual severity of indications, but practically it's a go/no-go criterion for cracks.

- If any repair is attempted (e.g., blending out a small flaw), the repair process must be approved and then the area re-inspected with NDT to ensure defect removal [99]. Weld repair on shafts is generally discouraged; if absolutely needed (for a superficial defect outside highly stressed zones), it would require approved procedures and re-heat-treatment. The QA plan sets the expectation that no repair welds are to be done on critical sections (to avoid any metallurgical notch or new residual stresses).
- **Surface Finish:** Surface roughness requirements are specified for various areas of the shaft, and these are verified by inspection (see next subsection for details). In brief, an acceptable surface finish (e.g. $R_a \leq 3.2 \mu\text{m}$ on shaft journals and fillets) is required to avoid micro-notches that could initiate fatigue. The finish is checked by a profilometer and visual inspection. Any areas not meeting the finish criterion (e.g. tool marks or scoring) are re-polished. Special attention is given to the transition radii and any change in section, where a smooth blend and fine finish are needed for fatigue strength [103].
- **Documentation and Traceability:** A robust traceability system is maintained throughout. Each shaft (and any test coupons from the same heat) is stamped or hard-marked with a unique identification number (often the heat number or a code linking to the heat number). All inspection and test records (dimensional check reports, NDT reports, material certs, heat treatment records, etc.) are indexed by this ID. According to class rules, the manufacturer must be able to trace any finished forging back to its original cast (heat) and processing history [99]. The QA documentation includes a Material Test Certificate package (usually EN 10204 Type 3.1 or 3.2 format) that contains:
 - Chemical analysis of the heat.
 - Mechanical test results (tensile, hardness, Charpy).
 - NDT results (UT report showing no significant indications, MPI/DPI report showing no defects).
 - Heat treatment chart (time/temperature of quench and temper or solution anneal).
 - Dimensional inspection report.
 - Coating or surface treatment reports (if any applied).
 - Certificate of compliance to applicable standards (e.g., ASTM/ASME, DNV, ABS).

Each step is signed off by quality engineers and, where required, witnessed by third-party inspectors (e.g., a class surveyor stamp for the 3.2 certificate). This documentation provides full traceability and accountability, which is critical for both academic thoroughness and industrial practice.

09.5 Surface Finish and Roughness Inspection

Surface finish is an important quality aspect for a marine shaft because it influences both fit and fatigue performance. Rough or scored surfaces can harbor stress concentrations or initiate corrosion pits, reducing the shaft's fatigue life [103]. The QA plan therefore specifies surface finish requirements and verification methods:

- **Specified Finish:** Critical surfaces of the shaft are finished to a low roughness. Typically:
 - Bearing journal surfaces and seal areas: finished by fine machining or polishing to around $R_a 0.8\text{--}1.6 \mu\text{m}$ (Grade N5-N6 finish) for smooth rotation and to prevent excessive wear on bearings or seals.

- Fillet radii and change-of-section transitions: polished to remove machining marks, target 3.2 μm . A smooth fillet is essential because as noted by DNV, notch sensitivity and surface roughness affect fatigue strength [103]. Polishing or grinding these areas yields compressive surface stresses and reduces micro-notches, improving fatigue resistance.
- Threaded regions: typically a fit and finish per standard, with any burrs removed. Sometimes critical threads are rolled (which also improves surface finish and fatigue strength by cold working). Visual inspection ensures threads are free of nicks, and thread gauges ensure no tight spots due to surface imperfections.
- If any coating is to be applied on the shaft (not common for stainless, but possibly for carbon steel shafts), the surface is prepared per coating spec. In that case, surface profile (peak-to-valley) is measured rather than R_a , to ensure the coating will adhere properly (usually 50–75 μm profile for epoxy coatings per ISO 8503).
- **Inspection Tools:** Surface roughness is quantitatively measured using a calibrated profilometer (surface roughness tester). Multiple measurements are taken along and around the shaft in critical areas. These are checked against the acceptance criteria (e.g., R_a value). Additionally, inspectors perform visual and tactile checks – the surface should feel smooth with no detectable ridges or tool marks. In critical areas like fillets, a visual magnifier or borescope might be used to ensure no circumferential scoring from machining. Any region that does not meet the required finish is reworked (re-machined with a finishing pass or polished with fine abrasive) and re-inspected.
- **Criteria:** A common criterion for marine shafting is that there should be *no sharp transitions or scratches*. All machining marks should be longitudinal (not circumferential) where possible, or smoothly blended. The QA criteria may call for specific roughness grades (e.g., “machined finish 3.2 μm R_a or better on all fatigue critical surfaces”). If the design or class requires, a compare to a surface roughness standard (like ISO 1302 or using roughness comparison specimens) is done to validate the finish. The influence of roughness is considered in design fatigue calculations, so meeting the specified finish is mandatory to achieve the expected fatigue life [103].
- **Surface Defects:** During surface inspection, any minor defects (e.g. burrs, dents, or tool chatter marks) are noted. Burrs from threading or keyway cutting are removed by deburring. Dents or gouges (which can act as stress raisers) would be ground out and blended, if within allowable material removal limits, followed by DPI to ensure no crack. The final surface is also inspected for corrosion protection: for a stainless shaft, it should be passivated (often shafts are acid pickled and rinsed after machining to restore a uniform passive oxide layer). A water-break test or ferroxyl test can be done on stainless surfaces to check for iron contamination that might lead to rust. For carbon steel shafts that will be coated, the surface cleanliness is inspected (per ISO 8501 visual standards, e.g. Sa 2.5 blast cleanliness, no mill scale or rust before painting).

In summary, the QA strategy combines rigorous inspections with adherence to standards and defined acceptance criteria. By inspecting dimensions, verifying material certificates, performing NDT (UT/MPI) per standards, and controlling surface quality, the shaft is ensured to meet design specifications and reliability targets. All findings are documented and traceable, establishing a QA trail that would satisfy both academic scrutiny and industrial quality requirements.

09.6 Safety Plan

The safety plan addresses both the safe manufacturing/installation of the shaft and the operational safety throughout the shaft's service life. It covers handling of the heavy shaft, identification of potential failure modes (corrosion, fatigue) and their mitigation, in-service monitoring to preempt issues, and precautions during fabrication and cathodic protection (CP) system installation. The aim is to minimize the risk of accidents, component failure, or environmental hazards, thereby protecting personnel, equipment, and the vessel.

09.6.1 Handling, Installation, and Operational Safety

09.6.1.1 Handling and Lifting

Marine propulsion shafts are large, heavy components (often hundreds of kilograms or more), so safe handling is paramount. During manufacturing and assembly:

- A detailed *lifting plan* is used whenever the shaft is moved. Certified lifting equipment (overhead cranes, slings, shackles) with adequate capacity (SWL well above the shaft weight) are employed. The shaft should be slung at balanced lift points (often a shaft will have eye bolts or a pallet for lifting) to avoid sudden shifts. Tag lines are used to guide the load and keep personnel clear of pinch points.
- Only trained riggers operate the crane; all personnel in the vicinity must stand clear of the load path. Hard hats, steel-toe shoes, and gloves are mandatory. The shaft is never to be lifted over or near personnel – the area is barricaded during lifts.
- When setting the shaft down, ensure *chocks or V-blocks* are in place to prevent it from rolling. The shaft is secured on wooden or rubber-covered V-blocks both in storage and during machining to avoid any unexpected movement.
- During machining or any rotational operation (like turning on lathe), machine *guards and interlocks* are in place. Operators ensure no loose clothing or items that could entangle. A *lock-out/tag-out (LOTO)* procedure is followed for the machining equipment during tool changes or inspections – i.e., power is cut and the spindle is verified stationary before approaching the shaft in the machine.

09.6.1.2 Installation Safety

Installing the shaft into the vessel (or test rig) involves careful alignment and bolting:

- Prior to installation, a Job Hazard Analysis is done to identify risks (such as heavy lifts in the engine room, working in confined spaces, or high locations at the stern). Appropriate measures like ventilation (for confined stern tube spaces), lighting, and communication methods are set.
- The vessel's stern tube or bearing supports must be ready to receive the shaft. Edges are checked to be free of burrs that could snag. When inserting the shaft through bearings or seals, come-alongs or hydraulic jacks are often used to pull the shaft slowly – this avoids sudden movements. Hands are kept clear of any gap between shaft and hull; tools like pushers are used instead of fingers to maneuver the shaft.
- Alignment: During coupling to the engine or gearbox, alignment pins and jacks are used to line up the flanges. Bolts are inserted and tightened sequentially. This is done with the engine locked out (no chance of rotation) to ensure safety. If any on-site machining (like final facing of coupling flange) is needed, portable machines are used with proper guarding.

- Personal safety: Technicians use PPE including gloves (for sharp edges on keys or threads), eye protection (in case a tool slips or fragments), and hearing protection if pneumatic tools are used for bolting. Heavy components like coupling hubs or propellers are also handled with cranes and proper rigging.
- Once installed, rotating shaft safety is important. On ships, shaft lines often run in tunnels or are exposed in machinery spaces – these areas must have guards or cover plates to prevent crew from touching the spinning shaft. The safety plan requires that all exposed rotating parts are shielded and warning signs posted (e.g. “Danger – rotating machinery”). Before any maintenance near the shaft (e.g. in dry dock or inspecting a bearing), the engine is to be immobilized and tagged out to prevent inadvertent start.

09.6.1.3 Operational Safety

In service, the shaft’s reliable operation is critical to vessel safety (loss of propulsion can be a hazard, and a fractured shaft could cause secondary damage):

- The shaft is designed with a safety factor and complies with class rules to minimize the chance of failure. For instance, the torsional vibration analysis ensures that vibratory stresses are within allowable limits to avoid fatigue cracking [105]. The safety plan includes adherence to operating limits: not exceeding rated engine RPM or torque, as excessive torsional vibration at certain critical speeds can sharply increase fatigue damage.
- Regular monitoring (discussed in the next section) is part of operational safety – this includes checking bearing temperatures (overheating could indicate misalignment or oil issues), listening for abnormal noises (which might indicate a crack or damage), and ensuring the cathodic protection system is functioning (to prevent unseen corrosion).
- Crew training is also a safety element: crew are trained to recognize signs of shaft problems (vibration, noise, leaking seals) and to respond promptly (for example, reducing engine power if severe vibration is detected to avoid a catastrophic failure). Emergency procedures are in place in case of shaft failure at sea (e.g., how to secure a broken shaft or use emergency propulsion).

By rigorously controlling handling and installation and by enforcing operational safeguards, the risk of injury and the risk of shaft damage due to human error are minimized. The heavy shaft is treated with respect throughout its life cycle, and all interactions with it are governed by safety protocols.

09.7 Risk of Failure Modes and Mitigation Strategies

The marine shaft in seawater is susceptible to several failure modes. The safety plan identifies these potential failure mechanisms and implements design or operational mitigations to reduce their likelihood or consequences:

09.7.1 Cyclic Fatigue (Torsional/Bending Fatigue)

Under the alternating torsional loads from engine firing and propeller excitation, the shaft experiences cyclic stress. Over time, this can initiate a fatigue crack, especially at stress concentrators (keyways, fillets) or if vibrations are severe. Fatigue is a well-documented cause of propulsion shaft failures [106].

09.7.1.1 Mitigations

- **Design & Material:** The shaft is designed for infinite-life (high-cycle fatigue) under expected loading, with a factor of safety. Fillet radii are generous and surface finish is smooth to reduce notch effects. Material with high toughness and cleanliness is used, and residual stresses are minimized via proper heat treatment. Sometimes, shot peening or cold rolling of fillets is done to introduce compressive surface stress and further improve fatigue strength.
- **Torsional Vibration Control:** A torsional vibration analysis (TVA) is performed and the shafting system is tuned such that natural frequencies are away from operating RPM ranges. Torsional dampers or flexible couplings may be used to absorb shock loads. By limiting vibration amplitudes, the cyclic stress range is kept low.
- **Operational Limits:** The crew is instructed not to operate for long durations at any resonance RPM (if passing through a critical speed). Class rules and the engine manufacturer set acceptable vibration limits; adherence prevents excessive fatigue damage.
- **Inspection:** As part of maintenance, the shaft is periodically inspected for early signs of fatigue cracking (e.g., via MPI at the propeller end or coupling end at dry-dock intervals). Early detection prevents catastrophic breakage.
- These measures collectively address fatigue safety – noting that geometric stress concentrators and torsional vibrations are the primary fatigue culprits [106], the design and monitoring focus on those.

09.7.2 Pitting Corrosion

In stagnant seawater or crevices, even stainless steel can develop pitting. Chloride ions can break down the passive film on steel, leading to small pits. Pitting is dangerous as it can act as an initiation site for fatigue or stress corrosion cracking.

09.7.2.1 Mitigations

- **Material Selection:** The duplex stainless steel chosen has high pitting resistance (PREN>35 with adequate Mo and N content). Its critical pitting temperature (per ASTM G48 or similar tests) exceeds the service temperature, providing a margin against pit initiation.
- **Microstructure QA:** Ensuring the material is free from detrimental phases (via ASTM A923) is crucial because intermetallic phases can reduce pitting resistance [100]. A proper 50/50 duplex phase balance yields good localized corrosion resistance.
- **Cathodic Protection:** The shaft is integrated into the vessel's cathodic protection system. Sacrificial anodes (or impressed current) maintain the shaft at a negative potential, suppressing anodic dissolution (thus preventing pitting). Even if small coating holidays or crevices exist, CP can dramatically reduce the corrosion rate inside them by making the shaft cathodic.
- **Coatings:** If practical, susceptible areas can be coated. Often shafts are unpainted in way of bearings (and stainless may be left bare), but any carbon steel coupling or flange might have a coating system applied. A marine-grade epoxy or metallizing can prevent seawater contact. Coating systems would follow ISO 12944/NORSOK M-501 guidelines for durability in immersion service [107]. Inspection of coatings for damage and prompt touch-up prevents exposure of bare metal.
- **Maintenance:** Regular inspection for any pitting is scheduled (diver inspections or during dry dock). If pits are found early, they can be ground out or the component replaced before they

grow. The safety plan ensures that even minor pitting is documented and assessed by engineering to decide if remediation is needed.

09.7.3 Galvanic Corrosion

Galvanic corrosion can occur when the shaft is coupled to dissimilar metal components in seawater (common case: a stainless or steel shaft coupled to a bronze propeller). The more anodic metal will preferentially corrode.

09.7.3.1 Mitigations

- **Isolation:** Wherever feasible, electrical isolation is introduced between dissimilar metals. For example, some designs use non-conductive bushings or gaskets in the propeller coupling or mounting bolts to break the galvanic circuit [108]. An insulating sleeve or coating on the shaft in the propeller hub area can reduce galvanic interaction.
- **Sacrificial Anodes:** The CP system is the primary defense. Zinc or aluminum alloy anodes, which are highly active metals, are attached to the hull (or sometimes as a shaft collar anode) and electrically connected to the shaft and propeller. These anodes will corrode preferentially, protecting the shaft and propeller [109]. The anodes are sized according to CP design standards (e.g., DNV-RP-B401) to provide sufficient current for the whole service life [110]. By maintaining the shaft at a cathodic potential, galvanic currents are diverted to the anodes.
- **Material Pairing:** The material combination itself is considered – using more similar metals reduces galvanic potential. In some cases, high-grade stainless props or composite material shafts have been used to avoid large potential differences. In our case, if the shaft is duplex stainless and the propeller Ni-Al-bronze, the CP system covers the difference.
- **Monitoring & Renewal:** Anode depletion is monitored (divers check anodes or ICCP readouts). Worn anodes are replaced well before they are consumed to ensure continuous protection [111]. The safety plan requires logging anode status at each hull inspection.
- These steps ensure galvanic corrosion is mitigated such that the shaft does not become the sacrificial element in the galvanic couple.

09.7.4 Crevice Corrosion

This is related to pitting, often occurring under gaskets, deposits, or in keyways where water stagnates. The shaft design minimizes crevices (e.g., using tight-fitting seals and promptly removing marine growth). Cathodic protection also helps here, as crevices maintained at CP potential will not suffer acidification and crevice attack. Using duplex stainless (which has better crevice corrosion resistance than standard stainless) adds safety. During assembly, applying an assembly paste or corrosion-inhibiting compound on fitted connections (like the propeller taper) can exclude seawater from the interface, preventing crevice attack inside the fitting.

09.7.5 Hydrogen Embrittlement (Hydrogen-Induced Stress Cracking):

A specific safety concern when using high-strength steels or duplex stainless in combination with cathodic protection is hydrogen embrittlement. Overprotection (excessively negative CP potentials) can evolve hydrogen on the metal surface; hydrogen atoms can diffuse into the steel and cause embrittlement or cracking (known as HISC in duplex stainless steels).

09.7.5.1 Mitigations

- **Control of CP Potential:** Impressed current CP systems are set not to exceed the recommended potential (typically around -0.8 to -1.0 V vs Ag/AgCl for steel). For duplex stainless, DNV

recommends not going more negative than about -1.1 V vs Ag/AgCl. If using sacrificial anodes, aluminum anodes have an open-circuit potential of around -1.05 V which is usually safe; avoid anode materials that drive more negative potential than necessary. In practice, the shaft's potential is monitored to ensure it stays in the protective range but not excessively cathodic.

- **Material and Microstructure:** The duplex stainless is chosen with a yield strength and hardness in a range proven resistant to HISC. DNV's recommended practice DNV-RP-F112 provides guidance – for example, it notes that duplex materials with fine microstructure (austenite spacing $< 30\ \mu\text{m}$) are satisfactory in resisting hydrogen cracking under CP[26]. Our QA already ensures the microstructure is appropriate (solution annealed, no embrittling phases, moderate hardness). Also, keeping stresses below certain thresholds (e.g., < 0.8 of yield in presence of CP, as per DNV-RP-F112) reduces HISC risk.
- **Coatings/Isolation:** Another approach is to coat or electrically isolate duplex components so they are not directly exposed to cathodic current. However, since the shaft must be protected from corrosion, completely isolating it is not practical; thus controlling the CP and using a qualified duplex grade is the strategy.
- **Monitoring:** Inspections look for any unusual cracking on the shaft especially around high-stress locations, which could hint at hydrogen effects. If the vessel switches to an impressed current system, hydrogen probes or polarization monitoring can be implemented to ensure potentials are correct.

09.7.6 Mechanical Overload and Misalignment

Though not a “mode” in the same sense, sudden overload (due to a propeller impact or severe misalignment) can fracture a shaft. Mitigations include designing the shaft with a margin for shock loads (e.g., a torque spike from a propeller strike) and using a shearable element (some shafts include a shear pin or coupling that fails in a controlled manner to save the shaft or gearbox in extreme overload). Proper alignment of the shaft with the engine and bearings is ensured during installation; misalignment can cause bending stresses and local heating. Flexible couplings or alignment checks at regular intervals are used to prevent misalignment from developing (e.g., due to foundation shifting).

Each of these potential failure modes is addressed with engineering and procedural controls. By combining material selection, protective systems (CP, coatings), design provisions, and operational monitoring, the safety plan significantly reduces the risk of shaft failure in service. Furthermore, classification society guidance (like ABS steel vessel rules or DNV Marine Machinery rules) on shafting is followed, which inherently addresses these failure modes with required safety factors and inspections.

09.8 In-Service Monitoring and Maintenance

Continuous monitoring and periodic inspections are essential to detect early signs of degradation and to ensure the shaft's long-term safe operation. The safety plan recommends the following in-service monitoring and maintenance actions:

09.8.1 Cathodic Protection Monitoring

For shafts protected by cathodic systems, it's important to verify that protection is effective. If an *Impressed Current Cathodic Protection (ICCP)* system is used, the shaft's polarization is monitored via reference electrodes and control circuitry. Readouts of hull/shaft potential are periodically checked to ensure they remain in the target range (e.g. around -0.85 V vs Ag/AgCl). Alarms are configured if

the potential drifts too positive (loss of protection) or too negative (overprotection, which could cause hydrogen issues). In the case of sacrificial anodes, divers or ROVs inspect the anodes at set intervals (typically annually or during hull surveys). They measure anode wastage and the ship's *overall potential* using a portable silver chloride reference cell in water. If potentials are above -0.8 V or anodes are significantly consumed, maintenance is scheduled (anodes replaced at next drydock). This monitoring ensures the shaft continuously receives adequate corrosion protection, thereby preventing unnoticed pitting or galvanic attack.

09.8.2 Vibration and Condition Monitoring

Vibration analysis is a powerful tool for early warning of shaft or machinery problems. The propulsion shaft is part of a larger drivetrain, and changes in vibration signature can indicate issues:

- Accelerometers or permanently installed vibration sensors on the engine, gearbox, and bearings can log vibration levels. The data is trended over time. An increase in torsional vibration amplitude or the emergence of new vibration frequencies could signal a developing crack or imbalance in the shaft.
- Many class societies offer *condition-based monitoring* schemes where, if vibration and other parameters are monitored, inspection intervals can be extended. Our plan suggests installing a torsional vibration monitoring system (often a pair of optical or magnetic pickups reading a phonic wheel on the shaft to measure twist). This real-time torsional measurement can detect if the shaft stiffness changes (a crack would reduce stiffness and alter the twist angle at a given torque). While not common on all ships, such technology is increasingly used for critical machinery.
- At a simpler level, engineers should periodically perform a vibration survey at different loads. If there is a notable increase in vibration or noise, it triggers further inspection (such as checking alignment, bearing wear, or performing an on-site MPI on accessible shaft areas). As noted in literature, constant monitoring and data collection of fatigue indicators greatly help in managing shaft fatigue life [106].

09.8.3 Regular Inspections (Planned Maintenance):

The safety plan includes scheduled inspections:

- **Dry-Dock Surveys:** Typically every 5 years (or as mandated by class), the shaft and propeller are inspected in dry dock. The propeller is removed at certain survey intervals (unless a condition monitoring program justifies not removing it) to allow close examination of the shaft. The exposed shaft end, keyway, and coupling are examined visually and with NDT (MPI or dye penetrant) for any cracks or corrosion wastage. Classification rules often require a tailshaft inspection at designated intervals to maintain class notation; our maintenance schedule complies with this.
- **Stern Tube and Bearing Monitoring:** If the shaft runs in a stern tube with oil, the oil is sampled periodically for water content and metal particles. Rising metal content in oil can warn of bearing wear or shaft rubbing. Temperature sensors in the stern tube bearings are monitored continuously; an abnormal temperature rise could indicate misalignment or friction, prompting shutdown and inspection.
- **Corrosion Inspections:** Besides anodes, other parts like the shaft coupling flange, the exposed portion between stern tube and propeller (if any) are inspected underwater by divers every year or two. Any biofouling is cleaned and the metal surface checked for damage. If the shaft is

coated, inspectors look for coating disbondment or damage (which would be repaired in water if minor or scheduled for dry dock).

- **Operational Checks:** Crew regularly check for unusual vibration, listen for noises (a rhythmic knocking could indicate a crack propagating or a loose coupling bolt), and observe the shaft rotation through inspection ports. Any anomaly triggers an investigation.
- **Documentation:** All monitoring readings and inspection findings are logged. Trends (for vibration, potential, temperature) are reviewed by engineers to predict if a problem is emerging. For example, if a vibration peak at a certain harmonic is slowly rising over months, it may indicate a propagating crack, allowing a planned outage to replace the shaft before a failure. This proactive approach uses the data to enhance safety.
- **Emerging Technologies:** The plan remains open to newer monitoring methods such as *acoustic emission sensors* on the shaft or couplings that can detect the high-frequency noise of a crack growing. While not yet commonplace in commercial shipping, these could be employed in critical or experimental cases to get immediate alerts of crack initiation. Also, installation of a shaft torque meter (strain-gauge based telemetry) can provide continuous measurement of torque and detect any sudden drops in stiffness (potential crack indicator) or overload events, enabling condition-based maintenance.

Through these in-service monitoring strategies, the safety plan ensures that the shaft's condition is known throughout its life, and any onset of damage is caught early. This not only protects the ship from an unexpected breakdown but also enhances safety by avoiding scenarios like a thrown propeller or loss of propulsion in dangerous situations.

09.9 Safety During Fabrication and Cathodic Protection Installation

Safety considerations extend to the manufacturing/fabrication phase and the installation of corrosion protection systems:

09.9.1 Fabrication and Machining Safety

Producing a large shaft involves heavy industrial processes (forging, heat treating, machining) that carry inherent hazards:

- **Forging and Heat Treatment:** These steps are conducted at specialist facilities. Workers follow high-temperature safety protocols (heat-resistant PPE, face shields, and tongs for handling hot metal). Large furnace doors are operated remotely or with guardrails to avoid exposure to heat. Quenching operations (if any) are planned to avoid splatter and steam explosions (e.g., ensuring shafts are dry before quenching in polymer solution). From a QA/safety standpoint, the manufacturer must also ensure uniform heat treatment to avoid residual stresses – while not a direct safety hazard to personnel, uneven quenching could warp the shaft or cause cracking, leading to later failure. Thus, controlled cooling and proper quench agitation (with safety covers to contain any splash) are implemented.
- **Machining:** Turning a long shaft on a lathe has pinch and entanglement hazards. Machine tools are equipped with emergency stop buttons easily accessible to the operator. Chip breakers and shields are used to prevent long razor-sharp swarf from forming (long stringy chips from machining can be dangerous). Chips are removed with tools (never by hand) and only when the machine is stopped. Given the shaft's size, slow rotational speeds are used during initial machining – this reduces the risk of a catastrophic imbalance. Lathes have chuck guards and the shaft is properly centered and secured to prevent it from coming loose. Lifting the shaft in/out of machines is done with cranes as described earlier.

- **Welding and Grinding:** If any welding (such as tack welding temporary lifting lugs, or if repair welding small defects is allowed) is performed, standard hot-work precautions are taken. Welders wear appropriate PPE (helmet, gloves, leathers) and ensure no flammable materials are nearby. For duplex stainless, welding must be carefully controlled (interpass temperature, filler material) for quality – the welding procedure qualification itself addresses safety by preventing defects that could later cause failure. Grinding and cutting operations (for keyways or surface finishing) produce sparks and dust; eye protection and dust masks are used, and adequate ventilation/extraction is provided (especially if any surface treatment like pickling is done, see below).
- **Chemical Handling:** Duplex stainless shafts are often *pickled* in a nitric acid + hydrofluoric acid bath after fabrication to remove scale and passivate the surface. This is a hazardous chemical process – it is carried out in a controlled area with acid-resistant equipment. Workers handling the acids wear acid-resistant suits, gloves, and face protection. Safety showers and neutralizing agents (lime or soda ash) are on hand in case of spills. The process is done per ASTM A380/A967 for stainless cleaning, and waste acid is treated to neutralize heavy metals before disposal, in compliance with environmental regulations. By thoroughly cleaning the shaft surface, this step also improves corrosion performance (removing embedded iron that could rust) – indirectly a safety benefit for the product.
- **Testing:** Non-destructive testing like MPI sometimes uses magnetic fields and inks – operators wear gloves to avoid skin contact with ink (which can contain hydrocarbons). Ultrasonic testing couplants can make floors slippery, so areas are roped off and cleaned after. If any radiographic testing (X-ray) were used for weld inspections, strict radiation safety protocols would be enforced (barricading area, using personal dosimeters). In our case, radiography is unlikely on a solid shaft, but we consider all contingencies.

09.9.2 Cathodic Protection System Installation

Installing sacrificial anodes or impressed current systems on the vessel involves its own hazards:

- **Sacrificial Anodes:** These are often bolted or welded onto the hull or shaft support structure. They are heavy (an aluminum anode for a shaft might weigh tens of kg). During dry-dock, workers fit anode assemblies around the shaft or on the hull. Proper rigging must be used to hold an anode in place while bolting, to prevent it from dropping on a worker's hands or feet. If welding pads for anodes: this is hot work on the hull – a fire watch is posted on the opposite side of the plate, and the ship's safety protocols for hot work are followed (gas free certification if in tanks, etc.).
- **When installing a shaft-mounted anode (common on smaller vessels:** half-shell anodes bolted around the shaft), ensure the shaft is locked to prevent rotation. The installation crew uses correct torque on anode clamp bolts to secure it – if too loose, it could fling off in service, which is a serious safety hazard (a thrown heavy object in water or in the machinery space). Thus, double-nuts or lock-nuts are used and the assembly is cotter-pinned or safety-wired as needed.
- **Impressed Current (ICCP) System:** This involves mounting anodes and reference electrodes usually in the hull, not on the shaft, but there may be shaft grounding brushes or slip rings to ensure the shaft is bonded into the circuit. Electrical safety is key – the power units are kept off (LOTO) while workers handle the anode cables. Cables must be routed properly to avoid mechanical damage. If divers are installing or checking ICCP anodes underwater, the system is switched off to avoid any risk of electric shock or evolved gas near the diver. Although ICCP

voltages are low (few volts) and pose little direct shock risk, it is standard practice for diver safety.

- **ICCP anodes generate hydrogen gas as part of protection** – normally this just dissipates in seawater. In enclosed tanks or spaces, hydrogen accumulation can be a concern, but on an open hull it's not an issue. However, during testing of ICCP in dry dock (when anodes are air-exposed), any hydrogen should be vented and ignition sources controlled if the system is inadvertently energized.
- **Hazardous Materials:** Historically, some sacrificial anodes contained Cadmium (e.g., US MIL-spec Zn anodes had ~0.1% Cd to activate them). Cadmium fumes from welding or grinding are toxic. In modern practice, anode alloys are low in cadmium, but any grinding or cutting of old anodes requires respiratory protection and local fume extraction. Spent anodes are treated as hazardous waste due to metal content; the safety plan covers proper disposal procedures (environmental safety).
- **Coating Application:** If part of the shaft or hardware is to be coated (painted), the painting operation in the shipyard must follow safety guidelines. Blasting to prepare surfaces creates dust – operators wear blast helmets and others are kept away; dust collection keeps visibility. Painting in confined areas (like inside a shaft tunnel) requires ventilation and respirators due to solvent fumes. Only approved marine coatings (with documented MSDS) are used, and curing times are respected to avoid solvent entrapment (which could lead to health issues for crew later or fire hazard).
- **Personnel Safety:** Workers installing CP components often work under the vessel or in odd positions. The plan requires use of proper scaffolding or man-lifts for any overhead work, fall protection harnesses if above ground, and adherence to confined space protocols if working in interior hull spaces for ICCP wiring.

By addressing these fabrication and installation hazards, the safety plan ensures that the process of building and outfitting the shaft does not endanger workers or result in latent defects. Each step – from forging to final assembly on the vessel – is conducted with both safety and quality in mind, reflecting standards of both academia (thoroughness) and industry (practical risk management).

09.10 Compliance with Safety Standards and Class Guidance

Finally, it's worth noting that the safety measures above are in line with marine industry standards and class society guidance:

Classification Society Rules: Rules from DNV, ABS, or Lloyd's Register for machinery and shafts inherently include safety factors and inspection regimes. For example, class rules mandate minimum fracture toughness for shafts, periodic survey of tailshafts, and proper alignment procedures. Complying with these rules (as we have outlined) ensures the design and maintenance meet a proven safety baseline.

Standards for Risk Management: The project follows ISO 31000 risk management principles in identifying hazards (like those in fabrication and operation) and mitigating them. A formal risk assessment (FMEA or HAZID) was done on the shaft design, and the results (such as risk of fatigue failure, risk of corrosion) drove the countermeasures described.

Occupational Safety Regulations: During manufacturing, OSHA (or local workplace safety laws) guidelines are adhered to for machinery operation, crane lifts, chemical handling, etc. This not only protects workers but also reduces the chance of quality incidents (since accidents can damage the part).

Environmental Safety: Preventing oil or chemical spills (stern tube oil, pickling acids, etc.) and proper disposal of wastes (used anodes, contaminated blast media) are included, aligning with MARPOL and local environmental regulations.

By incorporating these standards and guidelines, the QA and Safety section provides a comprehensive plan. It ensures the shaft is manufactured to high quality, inspected to stringent criteria, and operated with robust safeguards against the harsh marine environment challenges (saline corrosion, fatigue, galvanic coupling). All references – from ASTM material tests to DNV class notes – have been considered to produce a propulsion shaft that is safe, reliable, and compliant with both academic standards and real-world marine industry requirements.

10. Discussion

The results of this project highlight the complex relationship between **mechanical design**, **material behavior**, and **environmental degradation** in marine propulsion applications. The selection of Duplex Stainless Steel 2205 emerged as a pivotal decision, providing significantly improved resistance to pitting, crevice corrosion, and stress corrosion cracking compared to alternatives such as SS316L. This conclusion is supported by the PREN-based corrosion screening and Granta property analysis contained within the report.

The mechanical design calculations demonstrate that the combined torsion–bending–axial loading condition produces stress levels well within the allowable limits for Duplex 2205, even under conservative assumptions. Fatigue calculations based on the axial amplitude at 10^7 cycles show that safety factors remain comfortably above unity. Likewise, torsional vibration and lateral whirling analyses indicate that the shaft’s natural frequencies remain safely separated from operational speeds, reducing the risk of dynamic instability. This aligns with class-rule interpretations from IACS UR M68 and DNV-CG-0038.

This project also revealed a key challenge: **corrosion protection must be integrated, not treated as an isolated step**. Cathodic protection (CP) alone cannot fully protect the shaft due to splash-zone wet–dry cycling, crevice regions near couplings, and MIC (microbial influenced corrosion) effects. Conversely, coatings alone may degrade under mechanical wear or turbulent flow. The hybrid strategy adopted in Section 05 integrating multi-layer epoxy/PU/antifouling coatings with controlled CP potential addresses these limitations and reduces long-term maintenance demands.

From a manufacturing perspective, the sequencing of machining and heat-treatment steps proved crucial. Duplex 2205 undergoes dimensional distortion during solution annealing (as detailed in the report), necessitating that spline machining, threading, and critical taper finishing occur *after* heat treatment to preserve geometric accuracy. This aligns with industry standards and avoids premature tolerance deviations that compromise fit and fatigue life.

Class-rule alignment (DNV, ABS, LR) also shaped the design. Requirements for bearing alignment, runout limits, cyclic loading margins, and vibration separation ratios were referenced to validate the design. The chosen geometry satisfies these criteria, demonstrating compatibility with practical marine-industry requirements beyond purely theoretical design.

Finally, the surface treatment workflow universal preparation, localized polishing, dry-film lubrication on splines, and a final multilayer coating—provides a coherent, realistic protection pathway. The NDE and QA process ensures traceability and quality assurance, reflecting the real-world expectations of classification societies and shipyards.

Overall, the project successfully integrates mechanical integrity, corrosion science, manufacturability, and regulatory compliance. The design represents a balanced, optimized solution, though further refinements are possible through deeper dynamic simulation or operational monitoring.

11. Conclusion and Recommendations

11.1 Conclusion

❖ Material Selection

- Duplex Stainless Steel **2205** was validated as the optimal material after screening >1900 candidates using ANSYS Granta, meeting requirements for yield strength ($\approx 460\text{--}510$ MPa), fatigue strength, and $\text{PREN} \geq 30$.
- Exhibited superior resistance to pitting, crevice corrosion, MIC, and chloride attack compared to SS316L.
- Mechanical and corrosion properties satisfy IACS UR M68 and DNV-CG-0038 requirements.

❖ Shaft Geometry and Structural Performance

- Final shaft geometry (OD = 125 mm, ID = 60 mm) satisfies torsion–bending–axial loading criteria with adequate safety margins.
- Combined stress levels remain below allowable limits for Duplex 2205 ($\sigma_{\text{eq}} < \sigma_y / \text{safety factor}$).
- Fatigue assessment shows stresses remain below the conservative endurance limit (~ 262 MPa axial amplitude), ensuring long service life ($>10^7$ cycles).

❖ Vibration and Dynamic Behavior

- Lateral whirling analysis confirms the first critical speed remains >20% away from the operating speed (225 rpm), meeting class separation-margin criteria.
- Torsional vibration analysis indicates no resonance with dominant engine orders, and vibratory shear amplitude is well below fatigue thresholds.
- Jeffcott and continuous-beam models both confirm stable dynamic behaviour across the expected operating envelope.

❖ Corrosion Protection Effectiveness

- The selected multi-layer coating system (epoxy + polyurethane + antifouling) provides durable resistance to seawater immersion and splash exposure.
- Coating breakdown factor remains within acceptable limits for long-term marine operation.
- Cathodic protection (CP), when combined with coatings, reduces corrosion current demand and stabilizes shaft potential without hydrogen embrittlement risks (in line with DNV-RP-F112).
- Hybrid system effectively mitigates pitting, galvanic coupling effects, and biofouling-driven crevice corrosion.

❖ Manufacturing Reliability and QA Compliance

- CNC machining followed by controlled solution annealing ensures consistent metallurgical and dimensional quality.
- Sequencing of machining steps spline, taper, and thread machining performed *after heat treatment* prevents distortion and tolerance loss.
- NDE (UT + MPI/DPI) verifies internal and surface integrity; final runout checks confirm concentricity and bearing-journal accuracy.
- Full QA plan aligns with DNV, ABS, and ISO 31000 risk-management guidelines.

❖ **Operational Safety and Compliance**

- Shaft design meets or exceeds requirements in IACS UR M68, DNV-CG-0038, ABS alignment notes, and LR/BV alignment and vibration criteria.
- Safety factors for static and dynamic loading remain within class-approved ranges.
- Integrated corrosion protection prolongs service life and reduces maintenance cost and frequency.

❖ **Overall Technical Outcome**

- The final shaft design achieves an optimal balance between mechanical performance, corrosion resistance, manufacturability, class-rule compliance, and lifecycle durability.
- The integrated approach results in a propulsion shaft that is structurally robust, dynamically stable, and corrosion-resistant under the aggressive conditions of seawater immersion and splash zones.
- The solution is scalable and adaptable for higher power ratings or different vessel classes with minimal redesign effort.

11.2 Recommendation

❖ **Implement continuous condition monitoring**

Shaft potential measurements, CP current logs, and vibration data should be periodically collected to detect coating degradation, misalignment, or bearing wear.

❖ **Perform periodic dry-dock inspections**

Although the design is robust, splash-zone coating wear and MIC risk warrant scheduled surface inspections every 2–3 years.

❖ **Consider HVOF or ceramic coatings for higher-load future applications**

For vessels operating at higher power or in abrasive seabeds, advanced coatings may further enhance wear and corrosion resistance.

❖ **Conduct full 3D finite element fatigue assessment if shaft geometry changes**

Should the vessel's operational envelope or coupling design change, a detailed FEA-based stress analysis is recommended to verify fatigue margins.

❖ **Align installation with class-society guidance**

Final alignment, runout checks, torquing of couplings, and CP commissioning should strictly follow DNV/ABS specifications to preserve design integrity.

12. References

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13. Appendices

Annex A – ANSYS Granta Selection Report

Annex B – ANSYS Datasheet for DSS 2205

Annex C – 2D Manufacturing drawing

Annex D – Propeller Shaft standard dimensions

SELECTION REPORT

SUMMARY

[Stage Details](#)

1. SELECTION DATA

Database	Metals plus
Table	MaterialUniverse
Subset	Metals
Reference	

2. SELECTION CRITERIA (SUMMARY)

Stage	Attribute	Constraints
1		
2	Yield strength (elastic limit) (ksi)	≥ 50
	AND Fatigue strength at 10^7 cycles (ksi)	≥ 30
	AND Metal hot forming	Acceptable, Excellent
	AND Weldability	Good, Excellent
	AND Water (salt)	Acceptable, Excellent
	AND Pitting resistance equivalent number (PREN)	≥ 30
	AND Stress corrosion cracking	Susceptible, Slightly susceptible, Not susceptible
3	Mass per unit of strength	
	Cost per unit of strength	
4	Pitting resistance equivalent number (PREN)	
	Cost per unit of strength	

3. SELECTION RESULTS

Records passing: All Stages 25 of 1923

Ranked by: Pitting resistance equivalent number (PREN)

Ranked order: Low to high

Rank	Material	Pitting resistance equivalent number (PREN)
1	 Stainless steel, ferritic, AISI 446, annealed	23 - 31
2	 Stainless steel, austenitic, AISI 316LVM, cold worked	24.4 - 31.2
3	 Nickel-Fe-Cr alloy, HAYNES HR-120, annealed	25.4 - 44.2

SELECTION REPORT

Rank	Material	Pitting resistance equivalent number (PREN)
4	 Stainless steel, super-austenitic, alloy 20, annealed	25.6 - 30.9
5	 Stainless steel, duplex, AISI 329, annealed	26.3 - 34.6
6	 Stainless steel, duplex, UNS S32003 (AL 2003), annealed	26.7 - 32.3
7	 Nickel-Fe-Cr alloy, INCOLOY 825, annealed	27.8 - 35.1
8	 Stainless steel, austenitic, AISI 216, annealed	28.1 - 39.9
9	 Stainless steel, austenitic, Nitronic 50, XM-19, cold drawn, wire (nitrogen strengthened)	28.7 - 39.8
10	 Stainless steel, austenitic, Nitronic 50, XM-19, annealed (nitrogen strengthened)	28.7 - 39.8
11	 Stainless steel, duplex, AISI 2205, annealed	30.5 - 37.8
12	 Nickel-Fe-Cr alloy, NIMONIC 942, bar	32.3
13	 Nickel-Fe-Cr alloy, PYROMET 860, bar	32.4
14	 Nickel-Fe-Cr alloy, D-979, bar	32.6 - 38.3
15	 Nickel-Fe-Cr alloy, NIMONIC PE11, bar	32.7 - 38
16	 Iron-base-superalloy, N-155, solution treated	33.2 - 42.2
17	 Stainless steel, duplex, UNS S32550, annealed	35.2 - 43.9
18	 Stainless steel, duplex, UNS S44660, annealed	35.5 - 41.4
19	 Stainless steel, duplex, UNS S44635, annealed	35.9 - 41.8
20	 Stainless steel, ferritic, UNS S44700, annealed	38.8 - 45.2
21	 Stainless steel, duplex, UNS S44800, annealed	38.8 - 45.2
22	 Stainless steel, duplex, UNS S32750, annealed	39.5 - 46.1
23	 Stainless steel, ferritic, AL 29-4C, annealed	39.9 - 44.6
24	 Stainless steel, austenitic, AL-6XN, annealed	42.7 - 49.1
25	 Stainless steel, austenitic, UNS S31254, annealed	44.1 - 51.4

SELECTION REPORT

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PROJECT SELECTION SETTINGS:

Pass records with no data: Off

Fail estimated records: Off

[Change number of records to display...](#)

[Stage Details](#)

SELECTION REPORT

STAGE 1:	TREE	Summary
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Linked Records	Pass
 Metals and alloys	1799

Records passing: 1799 of 1923

SELECTION REPORT

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STAGE 2:	SCREENING	Summary
----------	-----------	-------------------------

CONSTRAINTS

Attribute	Constraints
Yield strength (elastic limit) (ksi)	≥ 50
AND Fatigue strength at 10^7 cycles (ksi)	≥ 30
AND Metal hot forming	Acceptable, Excellent
AND Weldability	Good, Excellent
AND Water (salt)	Acceptable, Excellent
AND Pitting resistance equivalent number (PREN)	≥ 30
AND Stress corrosion cracking	Susceptible, Slightly susceptible, Not susceptible

STAGE SELECTION SETTINGS:

Pass when: Any part of record within selection

Records passing: 25 of 1923

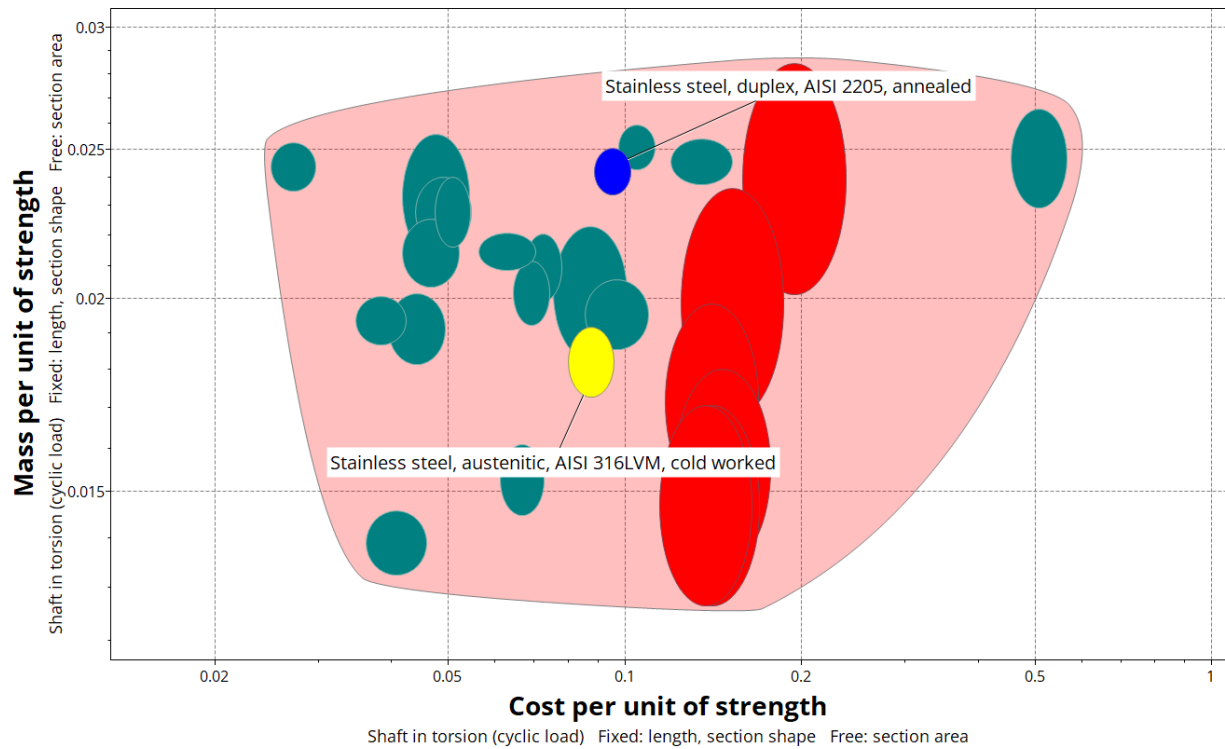
SELECTION REPORT

Page 6 of 15

STAGE 3:

COST PER UNIT OF STRENGTH, MASS PER UNIT OF STRENGTH

[Summary](#)



Axis attributes or formula

Mass per unit of strength

$[\text{Density}] / ([\text{Fatigue strength at } 10^7 \text{ cycles}]^{2/3})$

Cost per unit of strength

$[\text{Price}] * [\text{Density}] / ([\text{Fatigue strength at } 10^7 \text{ cycles}]^{2/3})$

Stage selection settings:

Show results from all enabled stages: On

Pass when: Any part of record within selection

Records passing:

1912 of 1923

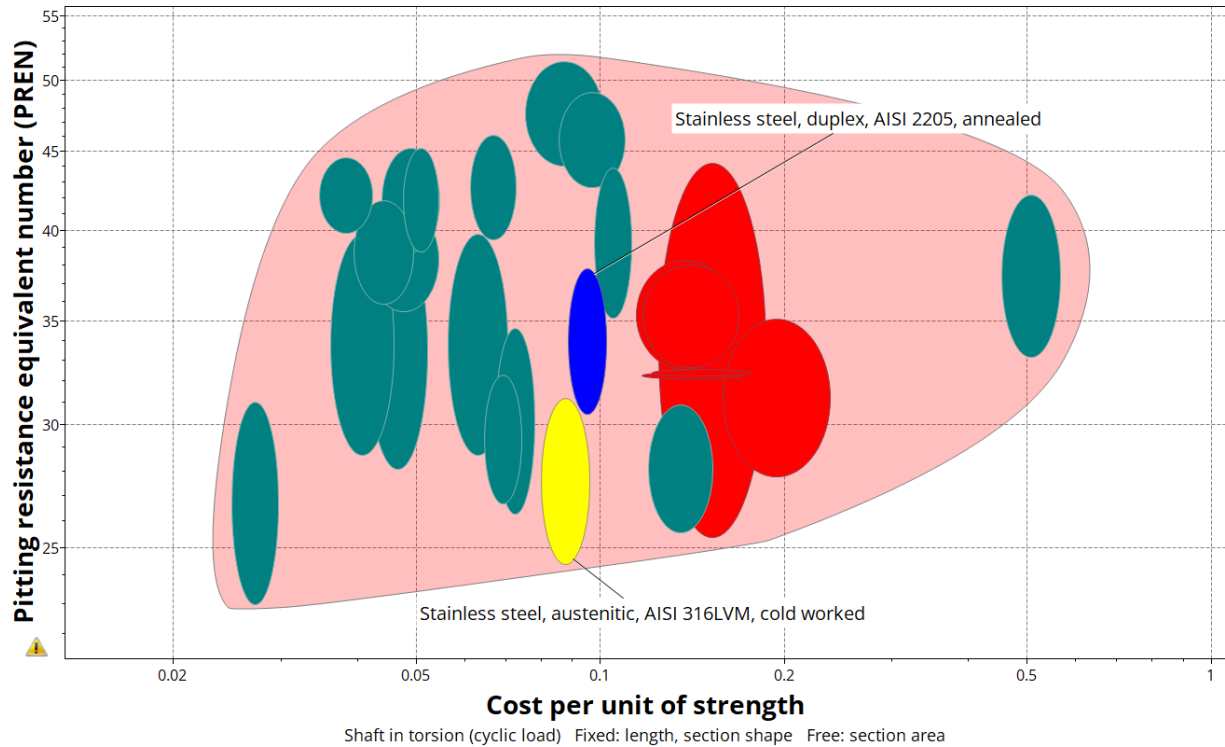
Generated by: prema, 20120530

Friday, October 3,
2025

SELECTION REPORT

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STAGE 4:	COST PER UNIT OF STRENGTH, PITTING RESISTANCE EQUIVALENT NUMBER (PREN)	Summary
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Axis attributes or formula

Cost per unit of strength

$[Price] * [Density] / ([Fatigue\ strength\ at\ 10^7\ cycles]^{2/3})$

Stage selection settings:

Show results from all enabled stages: On

Pass when: Any part of record within selection

Records passing:

235 of 1923

SELECTION REPORT

4. COMPARISON TABLE

	Stainless steel, austenitic, AISI 316LVM, cold worked	Stainless steel, duplex, AISI 2205, annealed	Nickel-Cr-Mo alloy, ILLIUM 98, as cast
Computed Properties			
Mass per unit of strength	0.0173 - 0.0192	0.0234 - 0.025	0.0314 - 0.0426
Cost per unit of strength	0.0802 - 0.0956	0.0888 - 0.102	0.429 - 0.598
General information			
Condition	50% cold worked	Solution annealed	As cast
UNS number	S31673	S31803 - S32205	
US name	S31673 to ASTM F139, ASTM F138	ASTM S32205, ASTM S31803, ASTM F60, ASTM F51	
EN name		X2CrNiMoN22-5-3	
EN number	1.4441 - X2CrNiMo18- 15-3, BS 7252	1.4462	
ISO name		X2CrNiMoN22-5-3	
GB (Chinese) name	00Cr18Ni14Mo3, 00Cr18Ni15Mo3N to GB/T 4234	0Cr21Ni5Ti to GB 3281, 00Cr18Ni5Mo3Si2 to GB/T 14976, GB/T 14975, GB 4237, GB 1220	
JIS (Japanese) name	SUS 316L	SUS329J3L	
Included in Materials Data for Simulation		Yes	
Materials Data for Simulation name		Stainless steel, 2205, annealed	
Composition overview			
Material family	Metal (ferrous)	Metal (ferrous)	Metal (non-ferrous)
Base material	Fe (Iron)	Fe (Iron)	Ni (Nickel)
Composition detail (metals, ceramics and glasses)			
C (carbon) (%)	0 - 0.03	0 - 0.03	0 - 0.07
Cr (chromium) (%)	17 - 19	21 - 23	26 - 30
Cu (copper) (%)	0 - 0.5	0	4 - 6.5
Fe (iron) (%)	59.4 - 67.8	63.7 - 71.9	0 - 1.5
Mn (manganese) (%)	0 - 2	0 - 2	0 - 1.5
Mo (molybdenum) (%)	2.25 - 3.2	2.5 - 3.5	7.5 - 9

SELECTION REPORT

N (nitrogen) (%)	0 - 0.1	0.08 - 0.2	0
Ni (nickel) (%)	13 - 15	4.5 - 6.5	50.9 - 62.5
P (phosphorus) (%)	0 - 0.025	0 - 0.03	0
S (sulfur) (%)	0 - 0.01	0 - 0.02	0
Si (silicon) (%)	0 - 0.75	0 - 1	0 - 0.5
Price			
Price (USD/lb)	4.47 - 5.17	3.7 - 4.18	12.9 - 14.9
Price per unit volume (USD/ft ³)	2220 - 2590	1800 - 2040	6680 - 7820
Physical properties			
Density (lb/in ³)	0.287 - 0.29	0.282 - 0.283	0.3 - 0.303
Mechanical properties			
Young's modulus (10 ⁶ psi)	26.9 - 27.5	28.3 - 29.7	23.9 - 28.3
Specific stiffness (lb _f .ft/lb)	7.76e6 - 7.93e6	8.36e6 - 8.77e6	6.62e6 - 7.83e6
Yield strength (elastic limit) (ksi)	107 - 141	66.7 - 74	37 - 46.4
Tensile strength (ksi)	130 - 172	92.8 - 138	47.9 - 59.5
Specific strength (lb _f .ft/lb)	30900 - 40800	19700 - 21800	10200 - 12800
Elongation (% strain)	12 - 21.3	20 - 25	10 - 30
Tangent modulus (ksi)	290	306	98.8
Compressive modulus (10 ⁶ psi)	27 - 27.4		
Compressive strength (ksi)	107 - 141	66.7 - 74	37 - 46.4
Flexural modulus (10 ⁶ psi)	26.9 - 27.5	28.3 - 29.7	23.9 - 28.3
Flexural strength (modulus of rupture) (ksi)	105 - 127	66.7 - 74	37 - 46.4
Shear modulus (10 ⁶ psi)	9.71 - 10.7	10.9 - 12.3	8.7 - 11.6
Shear strength (ksi)	97.8 - 130		
Bulk modulus (10 ⁶ psi)	26.4 - 26.8		16.7 - 22.5
Poisson's ratio	0.33	0.295 - 0.305	0.27 - 0.29
Shape factor	26	47	28
Hardness - Vickers (HV)	306 - 357	248 - 310	150 - 190
Hardness - Rockwell B (HRB)		99 - 108	
Hardness - Rockwell C (HRC)	32 - 37	22 - 31	
Hardness - Brinell (HB)	278 - 324	278 - 308	

SELECTION REPORT

Elastic stored energy (springs) (ft.lbf/in ³)	17.8 - 30.1	6.39 - 7.87	2.19 - 3.46
Fatigue strength at 10 ⁷ cycles (ksi)	58.5 - 68.2	37.9 - 41.9	18.9 - 29.7
Fatigue strength model (stress amplitude) (ksi) #	48 - 83.1	27.6 - 57.4	17.3 - 32.3
Impact & fracture properties			
Fracture toughness (ksi.in ^{0.5})	76.4 - 92.8	131 - 177	95.2 - 106
Toughness (G) (ft.lbf/in ²)	18 - 26.2	50.4 - 89	28.3 - 36.7
Ductility index (mil)	0.00494 - 0.00544	0.0179 - 0.0219	0.0269 - 0.0329
Thermal properties			
Melting point (°F)	2730	2550 - 2640	2250 - 2520
Maximum service temperature (°F)	572 - 594	635 - 689	1500 - 2010
Minimum service temperature (°F)	-328	-76 - -58	-459
Thermal conductivity (BTU/ft.hr.°F)	8.09 - 9.42	8.32 - 9.01	4.62 - 6.93
Specific heat capacity (BTU/lb. °F)	0.116	0.115 - 0.119	0.0955 - 0.107
Thermal expansion coefficient (μstrain/°F)	9.17	7.5 - 10.1	5.56 - 8.33
Thermal shock resistance (°F)	460 - 599	270 - 358	216 - 329
Thermal distortion resistance (BTU/hr.ft)	882000 - 1.03e6	853000 - 1.16e6	631000 - 1.09e6
Latent heat of fusion (BTU/lb)	112 - 123	163 - 197	120 - 129
Electrical properties			
Electrical resistivity (μohm.in)	29.1	30.9 - 32.1	47.2 - 55.1
Electrical conductivity (%IACS)	2.33	2.11 - 2.2	1.23 - 1.44
Galvanic potential (V)	-0.15 - -0.07	-0.26 - -0.18	-0.12 - -0.04
Magnetic properties			
Magnetic type	Non-magnetic	Magnetic	Non-magnetic
Optical, aesthetic and acoustic properties			
Transparency	Opaque	Opaque	Opaque
Softness to touch	6.21 - 6.47	6.33 - 6.71	7.3 - 7.87
Warmth to touch	12.6 - 13.6	12.9 - 13.5	15.2 - 18.9
Acoustic velocity (in/s)	190000 - 192000	197000 - 202000	175000 - 191000

SELECTION REPORT

Mechanical loss coefficient (tan delta)	0.00095 - 0.0013	0.0006 - 0.00074	0.0006 - 0.001
Healthcare & food			
Food contact	Yes	Yes	Yes
Medical grades? (USP Class VI, ISO 10993)	Yes		
Healthcare applications	Bone fixation and repair, Catheters and cannulas, Implantable pacemakers and defibrillators, Joint replacement, Spinal devices, Stents, Surgical instruments, Wound and tissue closure		
Sterilizability (ethylene oxide)	Excellent		
Sterilizability (radiation)	Excellent		
Sterilizability (steam autoclave)	Excellent		
Guidance for MRI Safety	High Risk for Potential Interaction	Caution - Strong Interaction	Low Risk for Potential Interaction
Restricted substances risk indicators			
RoHS 2 (EU) compliant grades?	Yes	Yes	Yes
EU REACH Candidate List indicator (0-1, 1 = high risk)	0	0	0
SIN List indicator (0-1, 1 = high risk)	0	0	0
Critical materials risk			
Abundance risk level	Medium	Medium	Medium
Sourcing and geopolitical risk level	Low	Low	Low
Environmental country risk level	Low	Low	Low
Price volatility risk level	Medium	Medium	Low
Conflict material risk level	None	None	Caution
Processing properties			
Metal casting	Unsuitable	Unsuitable	Acceptable
Metal cold forming	Excellent	Limited use	Unsuitable
Metal hot forming	Excellent	Acceptable	Unsuitable
Metal press forming	Excellent	Limited use	Unsuitable

SELECTION REPORT

Metal deep drawing	Excellent	Limited use	Unsuitable
Machining speed (sfm)	36	52	25
Weldability	Excellent	Good	Good
Weldability_Notes	Preheating is not required, post weld heat treatment is required	Preheating and post weld heat treatments are not required	Preheating and post weld heat treatments may be required
Weldability - MIG	Excellent	Good	
Weldability - plasma	Excellent	Good	
Weldability - SAW	Excellent	Good	
Weldability - TIG	Excellent	Good	
Brazeability	Good	Fair	
Carbon equivalency	1.22 - 1.59	1.29 - 1.66	
Durability			
Water (fresh)	Excellent	Excellent	Excellent
Water (salt)	Excellent	Excellent	Excellent
Weak acids	Excellent	Excellent	Excellent
Strong acids	Excellent	Acceptable	Excellent
Weak alkalis	Excellent	Excellent	Excellent
Strong alkalis	Excellent	Excellent	Excellent
Organic solvents	Excellent	Excellent	Excellent
Oxidation at 500C	Excellent	Excellent	Excellent
UV radiation (sunlight)	Excellent	Excellent	Excellent
Galling resistance (adhesive wear)	Acceptable	Limited use	Acceptable
Flammability	Non-flammable	Non-flammable	Non-flammable
Corrosion resistance of metals			
Pitting resistance equivalent number (PREN)	24.4 - 31.2	30.5 - 37.8	
Pitting and crevice corrosion resistance	Medium (20-30)	High (30-40)	
Stress corrosion cracking	Slightly susceptible	Not susceptible	Not susceptible
Stress corrosion cracking_Notes	Rated in chloride; Other susceptible environments: Hydrogen sulfide	Rated in chloride; Other susceptible environments: Hydrogen sulfide	Rated in chloride; Other susceptible environments: Hydroxide
Intergranular (weld line) corrosion resistance	Good	Excellent	

SELECTION REPORT

Inorganic acids	Moderate	Good	
Organic acids	Good	Good	
Alkalies	Moderate	Moderate	
Humidity / water	Excellent	Excellent	
Sea water	Good	Good	
Sour oil and gas	Moderate	Good	
Environmental indicators - material production			
Climate change (CO2-eq) (virgin grade production) (lb/lb)	7.59 - 8.37	6.22 - 6.85	15.6 - 17.2
Climate change (CO2-eq) (typical grade production) (lb/lb)	5.65 - 6.23	4.77 - 5.25	10 - 11.1
Embodied energy (virgin grade production) (BTU/lb)	38400 - 42400	29500 - 32500	113000 - 125000
Embodied energy (typical grade production) (BTU/lb)	29200 - 32200	23000 - 25400	70000 - 77100
Water usage (in ³ /lb)	4210 - 4650	3990 - 4430	8280 - 9130
Environmental indicators - processing			
Metal casting, climate change (CO2-eq) (lb/lb)			3.22 - 3.81
Metal casting, embodied energy (BTU/lb)			19400 - 22700
Metal rolling and forging, climate change (CO2-eq) (lb/lb)	1.24 - 1.78	0.812 - 0.985	
Metal rolling and forging, embodied energy (BTU/lb)	8180 - 11700	5340 - 6460	
Metal extrusion, hot, climate change (CO2-eq) (lb/lb)	2.41 - 3.47	1.56 - 1.89	
Metal extrusion, hot, embodied energy (BTU/lb)	15900 - 22800	10200 - 12400	
Wire drawing, climate change (CO2-eq) (lb/lb)	0.287 - 0.317	0.287 - 0.317	
Wire drawing, embodied energy (BTU/lb)	1880 - 2080	1880 - 2080	
Metal powder forming, climate change (CO2-eq) (lb/lb)	5.92 - 6.54	5.48 - 6.52	
Metal powder forming, embodied energy (BTU/lb)	31200 - 34500	28900 - 34400	

SELECTION REPORT

Chemical vapor deposition (CVD), climate change (CO2-eq) (lb/lb)	793 - 960	1150 - 1540	854 - 1010
Chemical vapor deposition (CVD), embodied energy (BTU/lb)	4.14e6 - 5.02e6	6.02e6 - 8.05e6	4.46e6 - 5.28e6
Machining, coarse, climate change (CO2-eq) (lb/lb)	1.31 - 1.57	1.18 - 1.33	1.07 - 1.22
Machining, coarse, embodied energy (BTU/lb)	8610 - 10300	7760 - 8730	7050 - 7980
Machining, fine, climate change (CO2-eq) (lb/lb)	2.02 - 2.59	1.63 - 1.88	1.31 - 1.54
Machining, fine, embodied energy (BTU/lb)	13300 - 17000	10700 - 12300	8590 - 10100
Machining, grinding, climate change (CO2-eq) (lb/lb)	2.73 - 3.62	2.08 - 2.43	1.54 - 1.86
Machining, grinding, embodied energy (BTU/lb)	17900 - 23700	13700 - 16000	10100 - 12200
Machining, laser, CO2, climate change (CO2-eq) (lb/ft)	0.0634 - 0.0768	0.0922 - 0.123	0.0683 - 0.0808
Machining, laser, CO2, embodied energy (BTU/in)	27.6 - 33.4	40.1 - 53.7	29.7 - 35.2
Recycling and end of life			
Recycle	Yes	Yes	Yes
Functional recycle	Yes	No	No
Climate change (CO2-eq), recycling (lb/lb)	2.05 - 2.26	1.68 - 1.85	2.03 - 2.24
Embodied energy, recycling (BTU/lb)	11900 - 13100	9140 - 10100	7910 - 8730
Recycle fraction in current supply (%)	35	32	41
Combust for energy recovery	No	No	No
Landfill	Yes	Yes	No
Biodegrade	No	No	No
Geo-economic data for principal component			
Principal component	Iron	Iron	Nickel
Typical exploited ore grade (%)		45.1 - 49.9	0.997 - 1.1
Minimum economic ore grade (%)		25 - 70	0.1 - 2

SELECTION REPORT

Abundance in Earth's crust (ppm)	41000 - 63000	41000 - 63000	59 - 90
Abundance in seawater (ppm)	0.0025 - 0.003	0.0025 - 0.003	0.0005 - 0.002
Annual world production, principal component (ton/yr)	2.19e9	2.19e9	2.33e6
Reserves, principal component (l. ton)	1.56e11	1.56e11	8.38e7
Links			
Elements in this material	0	5	5
Global Metals similar alloys	110	163	3
JAHM Curve data	113	16	0
ProcessUniverse	127	137	112
Producers	34	30	18
Reference	38	44	48
Shape	24	24	24
Full Datasheet			

# FUNCTIONAL PARAMETERS	
Stress Ratio	-1
Number of Cycles (cycles)	1e7

General information

Designation

Stainless steel, duplex, AISI 2205, annealed, wrought	
Condition	Solution annealed
UNS number	S31803 - S32205
US name	ASTM S32205, ASTM S31803, ASTM F60, ASTM F51
EN name	X2CrNiMoN22-5-3
EN number	1.4462
ISO name	X2CrNiMoN22-5-3
GB (Chinese) name	0Cr21Ni5Ti to GB 3281, 00Cr18Ni5Mo3Si2 to GB/T 14976, GB/T 14975, GB 4237, GB 1220
JIS (Japanese) name	SUS329J3L

Tradenames

Eastern Stainless; Krupp; Prodec; Slater; Springflex
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Typical uses

Oil and gas exploration, processing equipment, transport, storage and chemical processing, high chloride and marine environments, paper machines, liquor tanks and pulp and paper digesters

Included in Materials Data for Simulation	✓
Materials Data for Simulation name	Stainless steel, 2205, annealed

Composition overview

Compositional summary

Fe64-72 / Cr21-23 / Ni4.5-6.5 / Mo2.5-3.5 / N0.08-0.2 (impurities: Mn<2, Si<1, C<0.03, P<0.03, S<0.02)	
Material family	Metal (ferrous)
Base material	Fe (Iron)

Composition detail (metals, ceramics and glasses)

C (carbon)	0	-	0.03	%
Cr (chromium)	21	-	23	%
Fe (iron)	* 63.7	-	71.9	%
Mn (manganese)	0	-	2	%
Mo (molybdenum)	2.5	-	3.5	%

N (nitrogen)	0.08	-	0.2	%
Ni (nickel)	4.5	-	6.5	%
P (phosphorus)	0	-	0.03	%
S (sulfur)	0	-	0.02	%
Si (silicon)	0	-	1	%

Price

Price	* 3.7	-	4.18	USD/lb
Price per unit volume	1.8e3	-	2.04e3	USD/ft^3

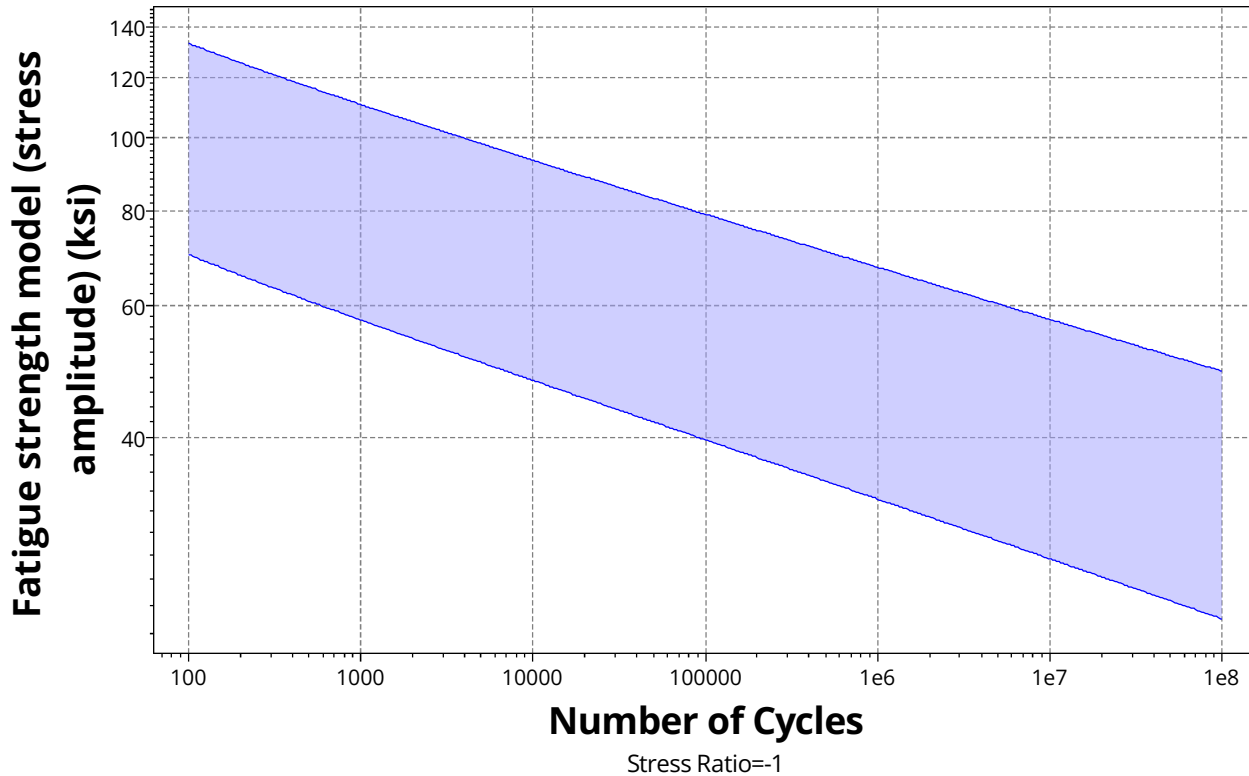
Physical properties

Density	0.282	-	0.283	lb/in^3
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Mechanical properties

Young's modulus	28.3	-	29.7	10^6 psi
Specific stiffness	8.36e6	-	8.77e6	lbf.ft/lb
Yield strength (elastic limit)	66.7	-	74	ksi
Tensile strength	92.8	-	138	ksi
Specific strength	1.97e4	-	2.18e4	lbf.ft/lb
Elongation	20	-	25	% strain
Tangent modulus	306			ksi
Compressive strength	* 66.7	-	74	ksi
Flexural modulus	* 28.3	-	29.7	10^6 psi
Flexural strength (modulus of rupture)	* 66.7	-	74	ksi
Shear modulus	* 10.9	-	12.3	10^6 psi
Poisson's ratio	0.295	-	0.305	
Shape factor	47			
Hardness - Vickers	* 248	-	310	HV
Hardness - Rockwell B	* 99	-	108	HRB
Hardness - Rockwell C	22	-	31	HRC
Hardness - Brinell	278	-	308	HB
Elastic stored energy (springs)	6.39	-	7.87	ft.lbf/in^3
Fatigue strength at 10^7 cycles	* 37.9	-	41.9	ksi
Fatigue strength model (stress amplitude)	* 27.6	-	57.4	ksi

[Parameters:](#) Stress Ratio = -1, Number of Cycles = 1e7cycles



Impact & fracture properties

Fracture toughness	* 131	-	177	ksi.in ^{0.5}
Toughness (G)	50.4	-	89	ft.lbf/in ²
Ductility index	0.0179	-	0.0219	mil

Thermal properties

Melting point	* 2.55e3	-	2.64e3	°F
Maximum service temperature	* 635	-	689	°F
Minimum service temperature	* -76	-	-58	°F
Thermal conductivity	8.32	-	9.01	BTU/ft.hr.°F
Specific heat capacity	* 0.115	-	0.119	BTU/lb.°F
Thermal expansion coefficient	7.5	-	10.1	μstrain/°F
Thermal shock resistance	270	-	358	°F
Thermal distortion resistance	* 8.53e5	-	1.16e6	BTU/hr.ft
Latent heat of fusion	* 163	-	197	BTU/lb

Electrical properties

Electrical resistivity	30.9	-	32.1	μohm.in
Electrical conductivity	2.11	-	2.2	%IACS
Galvanic potential	* -0.26	-	-0.18	V

Magnetic properties

Magnetic type	Magnetic
---------------	----------

Optical, aesthetic and acoustic properties

Transparency	Opaque
Softness to touch	* 6.33 - 6.71
Warmth to touch	* 12.9 - 13.5
Acoustic velocity	1.97e5 - 2.02e5 in/s
Mechanical loss coefficient (tan delta)	* 6e-4 - 7.4e-4

Healthcare & food

Food contact	Yes
Guidance for MRI Safety	Caution - Strong Interaction

Restricted substances risk indicators

RoHS 2 (EU) compliant grades?	✓
EU REACH Candidate List indicator (0-1, 1 = high risk)	0
SIN List indicator (0-1, 1 = high risk)	0

Critical materials risk

Abundance risk level	Medium
Highest risk elements	
Molybdenum, Nickel	
Sourcing and geopolitical risk level	Low
Environmental country risk level	Low
Price volatility risk level	Medium
Highest risk elements	
Iron	
Conflict material risk level	None

Processing properties

Metal casting	Unsuitable
Metal cold forming	Limited use
Metal hot forming	Acceptable
Metal press forming	Limited use
Metal deep drawing	Limited use
Machining speed	52 sfm
Weldability	Good
Notes	Preheating and post weld heat treatments are not required

Weldability - MIG

	Good
Weldability - plasma	Good
Weldability - SAW	Good
Weldability - TIG	Good
Brazeability	Fair
Carbon equivalency	1.29 - 1.66

Durability

Water (fresh)	Excellent
Water (salt)	Excellent
Weak acids	Excellent
Strong acids	Acceptable
Weak alkalis	Excellent
Strong alkalis	Excellent
Organic solvents	Excellent
Oxidation at 500C	Excellent
UV radiation (sunlight)	Excellent
Galling resistance (adhesive wear)	Limited use

Notes

Aluminum bronze is the most suitable mating material to minimize galling.

Flammability	Non-flammable
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Corrosion resistance of metals

Pitting resistance equivalent number (PREN)	30.5 - 37.8
Pitting and crevice corrosion resistance	High (30-40)
Stress corrosion cracking	Not susceptible

Notes

Rated in chloride; Other susceptible environments:
Hydrogen sulfide

Intergranular (weld line) corrosion resistance	Excellent
Inorganic acids	Good
Organic acids	Good
Alkalis	Moderate
Humidity / water	Excellent
Sea water	Good
Sour oil and gas	Good

Environmental indicators - material production

Climate change (CO2-eq) (virgin grade production)	* 6.22 - 6.85 lb/lb
---	---------------------

Sources

ecoinvent v3.10, market for cast iron//[GLO]; ecoinvent v3.10, market for nickel, class 1//[GLO]; ecoinvent v3.10, market for nitrogen, liquid//[RER]; ecoinvent v3.10, market for nitrogen, liquid//[RoW]; ecoinvent v3.10, market for pig iron//[RER]; ecoinvent v3.10, market for pig iron//[RoW]; Hammond and Jones, 2008; Nickel Institute, 2003; Norgate et al., 2007; Voet, van der and Oers, van, 2003

Climate change (CO2-eq) (typical grade production)	* 4.77	-	5.25	lb/lb
--	--------	---	------	-------

Sources

Calculated by an Ansys internal method using the virgin grade data sources and the recycled content data.

Embodied energy (virgin grade production)	* 2.95e4	-	3.25e4	BTU/lb
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Sources

ecoinvent v3.10, market for cast iron//[GLO]; ecoinvent v3.10, market for nickel, class 1//[GLO]; ecoinvent v3.10, market for nitrogen, liquid//[RER]; ecoinvent v3.10, market for nitrogen, liquid//[RoW]; ecoinvent v3.10, market for pig iron//[RER]; ecoinvent v3.10, market for pig iron//[RoW]; Hammond and Jones, 2008; Nickel Institute, 2003; Norgate et al., 2007; Sullivan and Gaines, 2010

Embodied energy (typical grade production)	* 2.3e4	-	2.54e4	BTU/lb
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Sources

Calculated by an Ansys internal method using the virgin grade data sources and the recycled content data.

Water usage	* 3.99e3	-	4.43e3	in^3/lb
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Environmental indicators - processing

Metal rolling and forging, climate change (CO2-eq)	* 0.812	-	0.985	lb/lb
Metal rolling and forging, embodied energy	* 5.34e3	-	6.46e3	BTU/lb
Metal extrusion, hot, climate change (CO2-eq)	* 1.56	-	1.89	lb/lb
Metal extrusion, hot, embodied energy	* 1.02e4	-	1.24e4	BTU/lb
Wire drawing, climate change (CO2-eq)	* 0.287	-	0.317	lb/lb
Wire drawing, embodied energy	* 1.88e3	-	2.08e3	BTU/lb
Metal powder forming, climate change (CO2-eq)	* 5.48	-	6.52	lb/lb
Metal powder forming, embodied energy	* 2.89e4	-	3.44e4	BTU/lb
Chemical vapor deposition (CVD), climate change (CO2-eq)	* 1.15e3	-	1.54e3	lb/lb
Chemical vapor deposition (CVD), embodied energy	* 6.02e6	-	8.05e6	BTU/lb
Machining, coarse, climate change (CO2-eq)	* 1.18	-	1.33	lb/lb
Machining, coarse, embodied energy	* 7.76e3	-	8.73e3	BTU/lb
Machining, fine, climate change (CO2-eq)	* 1.63	-	1.88	lb/lb
Machining, fine, embodied energy	* 1.07e4	-	1.23e4	BTU/lb
Machining, grinding, climate change (CO2-eq)	* 2.08	-	2.43	lb/lb
Machining, grinding, embodied energy	* 1.37e4	-	1.6e4	BTU/lb
Machining, laser, CO2, climate change (CO2-eq)	* 0.0922	-	0.123	lb/ft
Machining, laser, CO2, embodied energy	* 40.1	-	53.7	BTU/in

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 1.68	-	1.85	lb/lb
Embodied energy, recycling	* 9.14e3	-	1.01e4	BTU/lb

Recycle fraction in current supply	32	%
Combust for energy recovery	✗	
Landfill	✓	
Biodegrade	✗	

Possible substitutes for principal component

Iron is the least expensive and most widely used metal. In most applications, iron and steel compete either with less expensive nonmetallic materials or with more expensive materials having a property advantage. Iron and steel compete with lighter materials, such as aluminum and plastics, in the motor vehicle industry; aluminum, concrete, and wood in construction; and aluminum, glass, paper, and plastics in containers.

Geo-economic data for principal component

Principal component	Iron			
Typical exploited ore grade	45.1	-	49.9	%
Minimum economic ore grade	25	-	70	%
Abundance in Earth's crust	4.1e4	-	6.3e4	ppm
Abundance in seawater	0.0025	-	0.003	ppm
Annual world production, principal component	2.19e9			ton/yr
Reserves, principal component	1.56e11			l. ton

Main mining or production areas

Australia, 37%
 Brazil, 16%
 China, 14%
 India, 10%
 Russia, 4%
 South Africa, 3%
 Ukraine, 3%
 Canada, 2%
 United States, 2%
 Sweden, 1%
 Iran, 1%
 Kazakhstan, 1%
 Turkey, 1%
 Peru, 1%
 Chile, 1%
 Other countries, 3%

Notes

Other notes

This is a second generation of duplex.

Standards with similar compositions

- China:

0Cr21Ni5Ti to GB 3281, 00Cr18Ni5Mo3Si2 to GB/T 14976, 00Cr18Ni5Mo3Si2 to GB/T 14975, 00Cr18Ni5Mo3Si2 to GB 4237, 00Cr18Ni5Mo3Si2 to GB 1220

- International:

X2CrNiMoN22-5-3 to ISO 16143-1, X2CrNiMoN22-5-3 to ISO 16143-2, X2CrNiMoN22-5-3 to ISO 16143-3, X2CrNiMoN22-5-3 to ISO 9327-5, X2CrNiMoN22-5-3 to ISO 9328-7

- Japan:

SUS329J3L to JIS G4303, SUS329J3L to JIS G4304, SUS329J3L to JIS G4305, SUS329J3LTB to JIS G3463, SUS329J3LTP to JIS G3459, SUS329J3LTPY to JIS G3468, SM22CR-65, SM22CR-125, SM22CR-110, SANYO QS2205

- South Korea:

STS 329J3L to KS D 3706, STS329J3L to KS D 3698, STS329J3L to KS D 3705

- Sweden:

2377 to SS 142377

- UK:

318S13 to BS 1501/3, X2CrNiMoN22-5-3 to BS EN 10088-1, X2CrNiMoN22-5-3 to BS EN 10088-2, X2CrNiMoN22-5-3 to BS EN 10088-3, X2CrNiMoN22-5-3 to BS EN 10216-5, X2CrNiMoN22-5-3 to BS EN 10222-5, X2CrNiMoN22-5-3 to BS EN 10250-4, X2CrNiMoN22-5-3 to BS EN 10272, X2CrNiMoN22-5-3 to BS EN 10296-2, X2CrNiMoN22-5-3 to BS EN 10297-2

- USA:

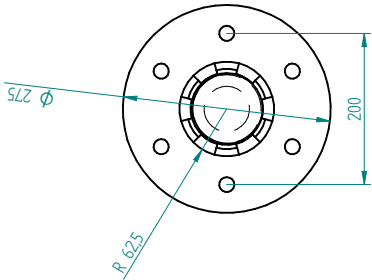
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Links

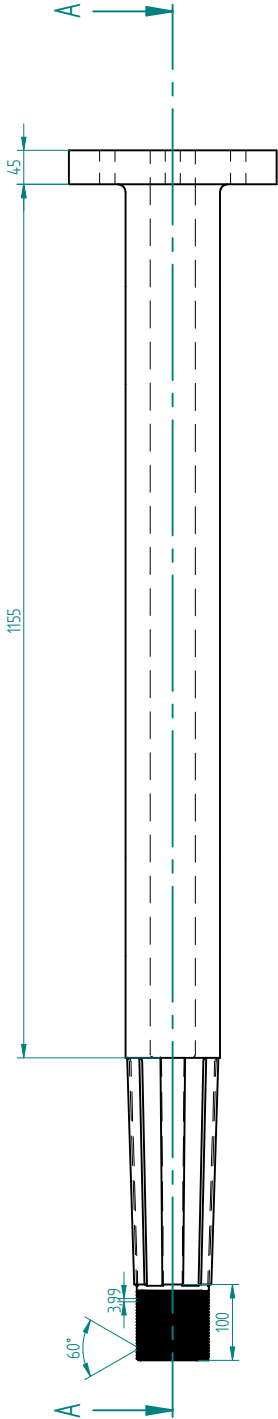
[Elements in this material](#)
[Global Metals similar alloys](#)
[JAHM Curve data](#)
[ProcessUniverse](#)
[Producers](#)
[Reference](#)
[Shape](#)

Annex C

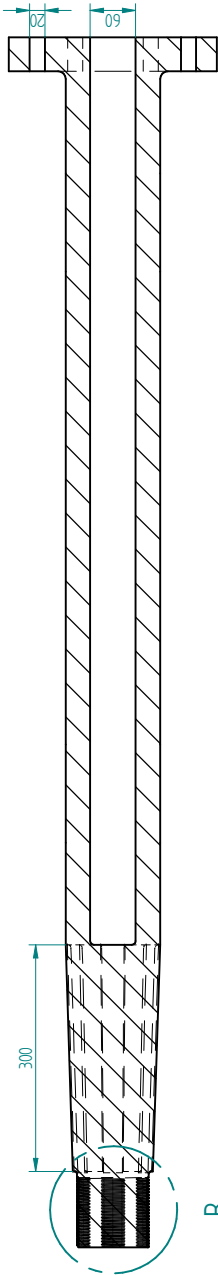
REVISION HISTORY			DATE	APPROVED
REV	DESCRIPTION			



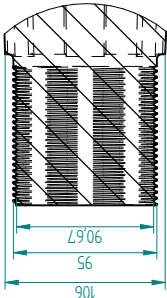
END ELEVATION



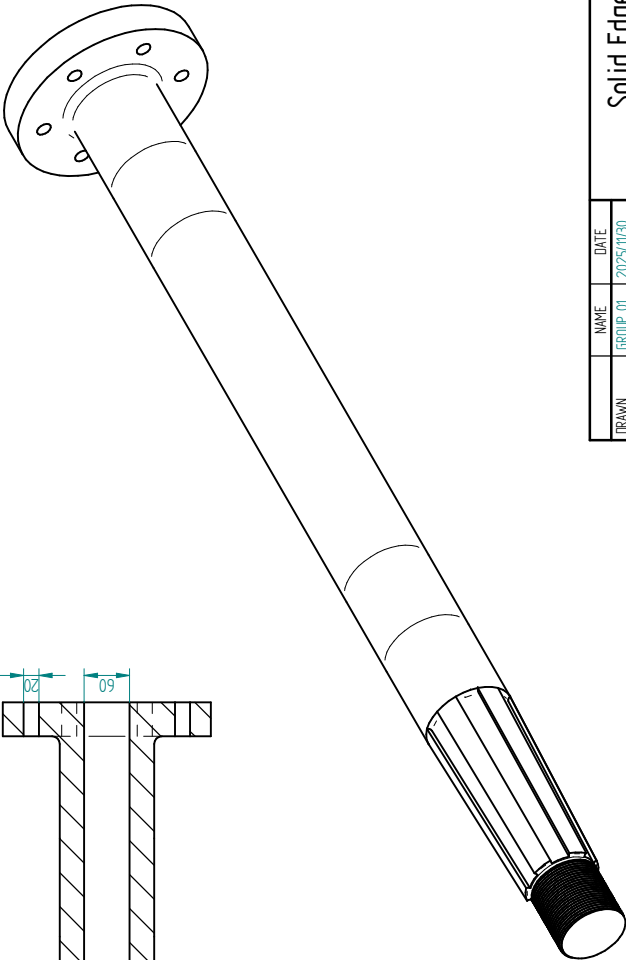
FRONT ELEVATION



SECTION A-A



DETAIL B



NAME		DATE		Solid Edge	
DRAWN	GROUP 01	2025/11/30			
CHECKED					
ENG APPR				PROPELLOR SHAFT	
MGR APPR					
DIMENSIONS ARE IN MILLIMETERS ANGLES IN DEGREES				SIZE	REV
				A2	1
				FILE NAME: DART001	
				SCALE: 1:1	SHEET 1 OF 1

Annex D MARINE PROPELLER SHAFT END DIMENSIONS

Nom Shaft Diameter	Diameter Small End		Taper Length	Keyway Width			Keyway Side Depth (i)		Keyway Fillet Radius (ii)	Thread (iii)		End of Taper to End of Thread	Ext. Beyond taper	Undercut		Dia. of Pin End	Length of Pin End	Cotter-Pin Hole		Cotter-Pin		Nuts (iv)	Keyway Length
	Min.	Max.		Min	Max		Min	Max		R	F			J	K			N	(Drill) P	Q	Length	Plain Thick	
A			C	Min	Max		Min	Max				G	H				M					T	X
3/4	0.624	0.626	2	3/16	0.1865	0.1875	3/32	0.095	0.097	1/32	13	1-1/16	1-5/16	25/64	1/8	3/8	1/4	1-9/64	9/64	1/8	3/4	1/2	1-1/2
7/8	0.726	0.728	2-3/8	1/4	0.249	0.250	1/8	0.125	0.127	1/32	5/8	1-1/4	1-1/2	31/64	1/8	7/16	1/4	1-21/64	9/64	1/8	3/4	5/8	1-25/32
1	0.827	0.829	2-3/4	1/4	0.249	0.250	1/8	0.125	0.127	1/32	3/4	1-7/16	1-3/4	19/32	1/8	1/2	5/16	1-33/64	9/64	1/8	1	3/4	2-1/8
1-1/8	0.929	0.931	3-1/8	1/4	0.249	0.250	1/8	0.125	0.127	1/32	3/4	1-7/16	1-3/4	19/32	1/8	1/2	5/16	1-33/64	9/64	1/8	1	3/4	2-1/8
1-1/4	1.030	1.032	3-1/2	5/16	0.3115	0.3125	5/32	0.157	0.160	1/16	7/8	2	2-3/32	1/8	5/8	3/8	1-1/2	1-23/32	11/64	1/8	1-1/4	7/8	2-13/16
1-3/8	1.132	1.134	3-7/8	3/8	0.374	0.375	5/32	0.189	0.192	1/16	1	1-13/16	2-1/4	13/16	1/8	3/4	7/16	1-29/32	11/64	5/32	1-1/2	1	3-3/16
1-1/2	1.233	1.235	4-1/4	3/8	0.374	0.375	5/32	0.189	0.192	1/16	1	2	2-7/16	1/8	3/4	7/8	1/2	2-3/32	11/64	5/32	1-1/2	1	3-3/16
1-3/4	1.437	1.439	5	7/16	0.4365	0.4375	7/32	0.219	0.222	1/16	1-1/4	2-5/8	3-1/8	1-1/32	3/16	1	1/2	2-23/64	13/64	3/16	1-1/4	3/4	4-7/32
2	1.640	1.642	5-3/4	1/2	0.499	0.500	1/4	0.251	0.254	1/16	1-1/2	3-1/8	4-3/8	1-1/4	3/16	1-1/4	1/2	2-47/64	13/64	3/16	2	7/8	4-15/16
2-1/4	1.843	1.845	6-1/2	9/16	0.5610	0.5625	9/32	0.281	0.284	3/32	1-3/4	3	3-1/2	1-3/8	3/16	1-3/8	1/2	3-9/64	17/64	1/4	2-1/4	1	5-5/8
2-1/2	2.046	2.048	7-1/4	5/8	0.6235	0.625	5/16	0.312	0.315	3/32	1-3/4	3	3-1/2	1-7/16	3/16	1-7/16	1/2	3-9/64	17/64	1/4	2-1/4	1	6-3/32
2-3/4	2.254	2.259	7-7/8	5/8	0.6235	0.625	5/16	0.313	0.316	3/32	2	3-1/2	4-3/8	1-1/16	1/4	1-1/16	1/2	3-41/64	17/64	1/4	2-1/2	1-1/8	6-21/32
3	2.460	2.462	8-5/8	3/4	0.7485	0.750	5/16	0.311	0.314	3/32	2-1/4	3-7/8	4-3/8	1-5/16	1/4	1-5/16	1/2	4-1/64	17/64	1/4	3	2-1/4	7-11/32

DIMENSIONS OF SHAFTS FROM 3-1/4 TO 8 INCHES IN DIAMETER

Nom Shaft Diameter	Diameter Small End		Taper Length	Keyway Width			Keyway Side Depth			Keyway Fillet Radius	Thread		End of Taper to End of Thread	Ext. Beyond taper	Undercut		Dia. of Pin End	Length of Pin End	Cotter-Pin Hole	Cotter-Pin		Nuts		Sleeve Drive (v)		Clearance	Keyway Length																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																										
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* 6" through 8" shaft has 1 inch per foot taper; 1/12" per inch taper. Angle with centerline is 2° 23' 9".

(i) Keyways shall be cut parallel to taper.

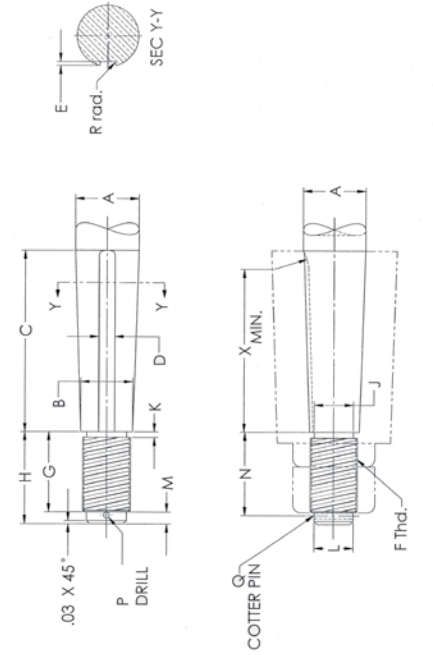
(ii) Fillets are recommended for keyways in shafts through 2" in diameter. Fillets are mandatory for shafts above 2" in diameter.

(iii) Threads are Unified and American Standard, Class 3A.

(iv) Nuts are to be semi-finished stock, American Standard B18.2.

(v) The shaft sleeve shown is recommended practice, but the use of a sleeve is optional.

(vi) The shaft sleeve shown is recommended practice, but the use of a sleeve is optional.



PROPELLER BORING

To insure retention of inherent factory accuracy, order your propeller factory-bored whenever possible. When bored in the field, propellers should be bored to the pilot hole, NOT to the hub or blade edges.

*6" through 8" shaft has 1 inch per foot taper, 1/12" per inch taper.

Angle with centerline is 2° 23' 9".

Overseas specifications on request.